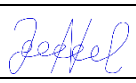
138 HA FEZ 1,
PHASE 2.1 & PHASE 4
AREA OF ALAT FREE ECONOMIC ZONE

PUMP STATION
SPECIFICATION
(STRUCTURAL)

	Prepared by	Checked by	Approved by
Position	Leading Architect	Leading Architect	Design Manager
Name, surname	Ziya Suleyman	Ziya Suleyman	Ali Mammadov
Signature			
Date	20.10.2023	20.10.2023	20.10.2023

1.

SPECIFICATIONS

1. **GENERAL REQUIREMENTS**
2. **CONCRETE SPECIFICATION**
3. **STEEL SPECIFICATION**

2.

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GENERAL REQUIREMENTS

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SECTION 01100 - SUMMARY

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Identification: Project comprises Alat Free Economic Zone.

1. Works location: Alat Free Economic Zone, Azeribijan, as more particularly shown on the Location Plan included in the Contract Drawings.
2. Employer: Refer to Clause 1.1.31, Definitions of the General Conditions of Contract, Part I.

B. Engineer Identification: Refer to Clause 1.1.35, Definitions in the Conditions of Particular Application, Part II

C. The Works in this Contract briefly consists of: Refer to Article 1, Description of the Contract, in the Instructions to Tenderers.

D. The Works include: Refer to Scope of the Works.

1.3 CONTRACT

A. Project will be constructed under an amended version of the FIDIC Conditions of Contract for Construction (Second Ed. 2017 Red Book) for Building and Engineering Works Designed by the Employer.

1.4 WORK SEQUENCE

A. The Work shall be executed and completed as shown in the drawings.

1.5 USE OF SITE

A. General: The Contractor shall have full use of the site of the Works during construction period. However, the Contractor's use of the Site is limited only by, the Employer's right to perform work or to retain other contractors to do so.

1. Driveways and Entrances: Keep driveways and entrances serving the Site clear and available to the Employer the Engineer and their employees, other contractors engaged in work on the Site and emergency vehicles at all times. Do not use these areas for parking or storage of materials.

a. Schedule deliveries to minimize use of driveways and entrances.
b. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

B. Partial Employer Occupancy: The Employer reserves the right to occupy and to place and install equipment in completed areas of building, before substantial completion, provided such occupancy does not interfere with the Contractor's completion of the Work. Such placement of equipment and partial occupancy shall not, by itself, constitute completion or acceptance, nor Taking-Over of any part of the Work.

C. Client Foundation Stone Ceremony – The Client shall organize a Foundation Stone Laying Ceremony in Q4 2020 which will mark the International Marketing Launch of the Zone. The contractor shall facilitate this event in every way.

1.6 WORK UNDER OTHER CONTRACTS

A. Separate Contract: The Employer may award a separate contract for performance of certain construction operations at the Site, particularily, the Authority Building. Those operations are scheduled to be conducted concurrently with work under this Contract but not by way of limitation.

B. Cooperate fully and coordinate with separate contractors and programme the Works so that work on those contracts may be carried out efficiently and effectively, without interfering with or delaying work under this Contract.

1.7 FUTURE WORK

A. Future Contracts: Employer reserves the right to perform additional work and/or award further separate contracts for performance of additional work at the Project site; whether concurrently with, or after substantial completion of, Work under this Contract. Performance and completion of such additional work may depend on successful completion of preparatory work under this Contract.

B. Cooperate fully and coordinate with any future construction forces and/or separate contractors as necessary, and programme the works so far as reasonably possible so that work on those contracts may be carried out efficiently and effectively, without interfering with or delaying work under this Contract.

1.8 SPECIFICATION FORMATS AND CONVENTIONS

A. Specification Format: The Specifications are organized into specific materials.

B. Specifications etc for the Design, Construction and Operation (two (2) Years) are included in a separate section

1. Section Identification: The Specifications use section numbers and titles to help cross-referencing in the Contract Documents. Sections in the Specifications are in numeric sequence; consult the table of contents at the beginning of the Specifications to determine numbers and names of the Sections contained in the Specifications.

C. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:

1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be inferred, as the sense requires. Singular words shall be interpreted as plural and plural words shall be interpreted as singular where applicable as the context of the Contract Documents indicates.

2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood are to be performed by the Contractor. Occasionally, the indicative or subjunctive mood may be used in the Section Text for clarity to describe responsibilities that must be fulfilled indirectly by the Contractor or by others when so noted.

    a. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.

1.9 CONTRACTORS KEY PERSONNEL

The Contractor shall provide for the following key personnel on the project and provide CV's for same for the Engineers agreement:

A. Contractors Representative
B. Construction Manager
C. Commercial Manager
D. Site Supervisor
E. Services Co-Ordinator
F. Safety Officer
G. Project Controls / Programme Lead

PART 2 – CONTRACTOR DESIGN ITEMS

The contract requires the following 'Contractor Design' elements:

1. Pile Foundations
2. Ground Improvement Geotextile

PART 3 – BASELINE DOCUMENTS

The following documents are contractual docuemnts and are to be used as baseline docuemnts for the project:

A. Environmental Impact Assessement

   The contractor shall fully comply with the the requirements set out in the Environmental Impact Assessment for this project.

B. Site Investigation Report

C. Site Topography Drawings

D. Tensar Feasibility Studies

END OF SECTION 01100

SECTION 01140 - WORK RESTRICTIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 USE OF SITE

A. General: The Contractor shall have full use of the site of the Works, during construction period. However, the Contractor's use of site is limited only by the Employer's right to perform work or to retain other contractors to do so.

B. Use of Site: Limit work and activities to the area of the Site as defined on Drawings in areas indicated. Do not disturb areas outside the Site or in which the work is indicated.
   1. Limits: Confine constructions operations to the Phase 1 of Alat Free Economic Zone Site as more particularly shown on the Location Plan included in the Contract Drawings.
   2. Employer Occupancy: Allow for Employer occupancy of Site and use by the public.
   3. Driveways and Entrances: Keep driveways and entrances serving premises clear and available to the Employer, the Engineer and their employees, other contractors engaged in work on the Site and emergency vehicles at all times. Do not use these areas for parking or storage of materials.

      a. Schedule deliveries to minimize use of driveways and entrances.
      b. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

1.3 OCCUPANCY REQUIREMENTS

A. Partial Employer Occupancy: Employer reserves the right to occupy and to place and install equipment in completed areas of building, before substantial completion, provided such occupancy does not interfere with Contractor's completion of the Work. Such placement of equipment and partial occupancy shall not, by itself, constitute completion or acceptance, nor Taking-Over of any part of the Work.

   1. The Employer will not maintain the facilities during liability period; the Contractor is responsible for full maintenance during Defects Liability Period.
   2. The Engineer will issue a Taking-Over Certificate for each specific Section or part of the Works occupied by the Employer before substantial completion of the whole of the Works in accordance with Clauses 48.3 of the General Conditions of the Contract, Part-I.
   3. If necessary, obtain a Certificate of Occupancy from authorities having jurisdiction before Employer occupancy.
   4. Before partial the Employer occupancy, mechanical and electrical systems shall be fully operational, and required tests and inspections shall be successfully completed.

On occupancy, the Employer will provide, operate, and maintain mechanical and electrical systems serving occupied portions of building.

5. On occupancy, Employer will assume responsibility for maintenance and custodial service for occupied portions of building.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01140

SECTION 01200 - QUALITY ASSURANCE/QUALITY CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

1. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 QUALITY ASSURANCE PROGRAM

1. Provide and maintain an effective Quality Assurance Program that complies with Clause 4.9 of the Conditions of the Contract.

1.3 SCOPE OF PROGRAM

1. The Contractor shall establish a Quality Assurance Program to perform sufficient inspection and tests of all items of work, including that of his suppliers and subcontractors, to ensure conformance to applicable Technical Specifications and Drawings with respect to the materials, workmanship, construction, finish, functional performance, and identification.

1.4 PLAN DEFINITION

1. The PQAP (Project Quality Assurance Plan): This Plan is the means to ensure that construction complies with the requirements of the Contract Documents. The PQAP shall be adequate to cover all construction operations, including both on-site and off-site fabrication and installation works by subcontractors and will be keyed to the proposed Contract Schedule. The PQAP shall include as a minimum, at least three phases of inspection for all definable phases of the Works, which are subsequently described in paragraphs 2, 3 and 4.

2. Preparatory Inspection: This inspection shall be performed prior to beginning any work on any activity on any definable phase of the Works and as shown in the Contract Schedule. It shall include a review of contract requirements; a check to assure that all materials and/or equipment have been tested, submitted, and approved by the Engineer; a check to assure that provisions have been made to provide required control testing; and plan mock-ups when appropriate; examination of work area to ascertain that all preliminary work has been completed; and a physical examination of materials and equipment to assure that they conform to approved shop drawings or submittal data and that all materials and/or equipment are on hand. The Engineer shall be notified at least 24 hours in advance of the preparatory inspections and such inspection shall be made a matter of record in the Contractor's Quality Control Reports as required by paragraph 3.3.

3. Initial Inspection: This inspection shall be performed as shown in the Contract Schedule. This inspection shall occur as soon as a representative portion of the particular phase of construction has been accomplished. This shall establish the acceptable standard of workmanship, including a review of control testing for compliance with contract requirements, review of mock-up, use of defective or damaged materials, omissions, and

dimensional requirements. The Engineer shall be notified at least 24 hours in advance of the initial inspection and such inspection shall be made a matter of record in the Quality Control Reports as required by paragraph 3.3.

4. Follow-up Inspections: Inspections shall be performed daily to assure continuing compliance with Contract requirements, including control testing, until completion of the particular phase of construction. Such inspections shall be made a matter of record in the Quality Control Reports as required by paragraph 3.3.

1.5 CONDITIONS OF PROJECT QUALITY ASSURANCE PLAN (PQAP)

1. Furnish to the Engineer, a PQAP which shall include the personnel, instructions, procedures, records to be used, document controls and quality assurance overview. This document will include as a minimum:

    a. The Quality Control Organization Chart including major reporting lines and relationships.
    b. Reporting relationships within and external to the organization. Duties and responsibilities within the said organization shall be stated.
    c. The name of the Quality Control Manager (QCM). This individual shall report directly to senior management, independent of manufacturing/ construction.
    d. Names and Qualifications of Quality Control Personnel.
    e. Authority and area of responsibilities of Quality Control Personnel.
    f. An explanation as to how the Quality Control Organization relates to other staff elements as regards Shop drawing submittals, As-Built drawings, revisions to the contract and safety.
    g. Methods proposed to control the use of the various design documents, shop drawings, procedures, etc. to assure that only the latest reviewed documents are used and are distributed to the individuals performing the Work. Recall of documents which have been superseded or revised shall be implemented. Describe the process used to determine what submittals are required by the Contract and the system used to track these submittals and their current status.
    h. A narrative discussion of how the Quality Control staff will accomplish the tasks assigned to quality control.
    i. A copy of a letter of direction to each representative responsible for Quality Control, outlining his duties and responsibility and signed by a responsible officer.
    j. Identification and description of all mock-up items.
    k. Methods for accomplishing quality control inspections addressing the how, what, where, when and why of these inspections, including that for subcontractor's work (see paragraph 3.).
    l. Method of documenting quality control operations, inspections and testing. This shall include how various records originated and how they are maintained, received, filed, protected, and authenticated. Quality Records required for submittal to the Engineer shall be described.
    m. Detailed listing by Specification Section and paragraph designating all Quality Control requirements and tests to be performed and their frequency. Further indicate which tests will be performed by technicians or by an approved testing laboratory.

n. Methods to be employed when any Work is found not to meet Technical Specifications and the means to be implemented to document the items (non-conformance reports) and provide resolution.

o. A matrix of all Contract and Technical Specification requirements to enable monitoring of all items from start until acceptance and completion. This is to facilitate Contract closeout.

2. To facilitate contract management and project administration the Contractor shall acquire and implement a project control system that is compatible with the Microsoft Project System. The Contractor shall provide information on tracking and control of documentation extracted from the System as and when required by either the Employer or Engineer.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 GENERAL REQUIREMENTS

3.1 Prior to submittal of the PQAP for acceptance, meet with the Engineer or his representative and discuss the PQAP. The meeting shall develop mutual understanding relative to details of the system, including the forms to be used for recording the Quality Control operations, inspection, administration of the program, and the interrelationship and Engineer's inspection. Requirements for Hold Point Release sign-offs on Quality Control checklists/ inspection sheets by Engineer shall also be discussed and established.

3.2 QUALITY CONTROL ORGANIZATION

1. The PQAP shall be implemented by the establishment of the Quality Control Organization the sole duty of which will be to ensure conformance to the Contract of all Quality Assurance/Quality Control (QA/QC) activities.

2. The Quality Control Manager (QCM) must be an individual with demonstrated experience implementing a Quality Control program and supervising Inspectors.

    1. This experience shall include at least 5 years of Quality Control background in any combination of the following areas:

        a. Field inspection.
        b. Construction phase experience relevant to the scope of work.
        c. Previous experience as a Quality Control Manager.
        d. Fabrication/Manufacturing experience.

    2. In addition, the QCM shall have the following managerial experience:

        a. Supervised at least 2 people.
        b. Experience on a government regulated project.

c. Implemented a Quality Control program.
d. Experience on a comparably sized project of similar complexity.

3. Submit the résumé of the proposed QCM for acceptance by the Engineer. If, for any reason, the QCM is replaced, the résumé of the proposed replacement QCM shall be submitted and accepted by the Engineer prior to the replacement taking effect.

4. Due to the complexity and nature of this Contract a full time QCM is required.

3. Quality Control Organization shall be sufficiently staffed to perform the following tasks:

1. Conduct phased inspections (Preparatory, Initial and Follow-up) in accordance with paragraph herein before.
2. Perform all testing required under the Technical Specifications, including all listed Codes and Standards.
3. Prepare daily and monthly Quality Control Reports in accordance with paragraph 3.3 hereinafter.
4. Review and endorse all shop drawing submittals for compliance with Contract requirements prior to submission to the Engineer.
5. Inspect materials as they are delivered on site to ensure compliance with reviewed shop drawings and contract specifications.
6. Conduct off-site inspections of supplies and services to be incorporated into the work. Provide monthly report of off-site Quality Control activities.
7. Maintain record of all Quality Control activities. These records shall be available for the Engineer's use.

4. Where applicable and as a minimum, the Quality Control staff shall include the following suitably qualified personnel:

1. Structural Quality Control Inspector.
2. Civil Quality Control Inspector.
3. Mechanical/Electrical Quality Control Inspector.

5. If at any time during the contract period the Engineer determines that the Contractor's Quality Control staff is not capable of performing all the tasks listed above, then the Engineer may direct to revise and/or supplement the present organization structure at the Contractor's own cost.

3.3 QUALITY CONTROL REPORTS

1. Daily Quality Control Report: Submit a daily Quality Control Report. The report shall contain a record of inspections and tests for all work accomplished subsequent to the previous report and shall include the following information:

1. Phase(s) of construction underway during the time frame of the report. (i.e. earthwork, concrete work, structural steel erection, mechanical and electrical installations, architectural finishes etc.)
2. Phase Inspection (preparatory, initial, or follow-up), phase of construction and location of inspections and/or tests that were made.
3. Results of inspection, including nature of deficiencies observed and corrective actions taken or to be taken.
4. Report of tests performed, including those specified, with the results of the tests, including failures and remedial action to be taken. Test results, including all computations shall be attached to the report form. Where test results cannot be

completed by the time the report is submitted, a notation shall be made that the test was performed and the approximate date test results will be available. Delayed test results shall be submitted with the report form on the date received.

5. Results of inspection of materials and equipment upon arrival at the site and prior to incorporation into the work for compliance with submittal approvals, damage and proper storage.
6. Offsite QA/QC activities.
7. Instructions received from the Engineer.
8. All non-conformances, either in tests or inspections, shall be specifically listed under a separate heading.
9. In all cases, the report must be verified and signed by the QCM. The verification is to contain the statement that all supplies and materials incorporated in the work are in compliance with the Contract except as noted.

2. Monthly Quality Control Report: Submit a Monthly Quality Control Report. This report shall consist of an executive summary of all QA/QC activities executed in the previous month. The report shall contain the following information:

1. Phase(s) of construction underway during the time frame of the report. (i.e. earthwork, concrete work, structural steel erection, mechanical and electrical installations, architectural finishes, etc.)
2. Confirmation that all necessary Quality Control testing has been performed in accordance with the Technical Specifications and other Contract documents, with respect to quantity and quality.
3. Summary of the tests performed during the month, including number passing and failing.
4. Summary of all non-conformances, including actions taken or actions proposed.
5. Details of significant trends in test results.
6. In all cases, the report must be verified and signed by QCM. The verification is to contain the statement that all supplies and materials incorporated in the work are in compliance with the Contract except as noted.

3.4 CHANGES TO QUALITY ASSURANCE PLAN

1. Notify the Engineer or his authorized representative in writing of any proposed change to the Contractor's Quality Assurance Plan. Any proposed changes must not be implemented until prior approval has been received from the Engineer.

END OF SECTION 01200

SECTION 01305 - DATA REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes requirements for submittals for the following:

1. Construction.

1.3 SUBMITTALS FOR – CONSTRUCTION

A. Shop Drawings:

1. Initial Submittal: Submit 1 opaque blue/black line print and 1 correctable, reproducible transparency. The transparency will be processed and returned. After review, print the processed transparency for job use and distribution.
2. Final Submittal: After review and approval submit 3 prints; if the drawing is required for maintenance manuals submit a total of 5 prints; with final submittal, include additional prints as necessary for job use and distribution. Engineer will retain 2 prints and will return the remainder. Maintain 1 print as mark-up copy for record drawings.

B. Product Data: Submit manufacturer's catalog data for materials and equipment, showing manufacturer, type, model number, capacity, color, and related spatial, power supply and utilities/facility interface support information. Mark each copy to indicate the actual product to be provided; show selections from among options in the manufacturer's printed product data. Submit 2 copies to the Engineer; submittal is for information and record purposes only. Where the product data is required for operation and maintenance manuals, submit 2 additional copies, which will be returned with the Engineer’s review comments. Maintain 1 additional copy at the Project site for reference purposes. Do not proceed with the installation of manufactured products until a copy of related product data is in the installer's possession at the Project site.

C. Samples: Submit 3 sets of samples; 1 set will be returned. Provide 3 or more samples in each set where variations in color, pattern or texture are observable; show average condition and extreme range of variations. Submit full documentation with each set. Sample submittals are for the Engineer's review of color, texture, pattern, and kind. Maintain returned set at the Project site for purposes of quality control comparisons.

D. Miscellaneous Submittals: Provide copies of miscellaneous submittals as follows:

1. Project Photographs: 2 prints plus negatives. Submit photographs at regular intervals during construction as determined by the Engineer.

2. Field Records: 4 copies, including 1 copy, which will be returned for inclusion in the submittal of record documents.
3. Operation and Maintenance (O&M) Manuals and recommended / provided Spare Parts Lists: 2 bound copies. O&M manuals shall include warranty documents.
4. As-Built and Record Drawings: Original maintained marked-up prints. Prepare a complete set of all engineering / architectural drawings which have been conformed (modified) to reflect the as-constructed / installed (as-built) configuration, layout, details, and dimensional information. These As-Built and Record Drawings shall be clearly identified and labeled (denoted) as such in order to avoid confusion between this set of drawings and the preliminary or final design documents.
5. Inspection and Test Reports: 2 copies.

    a. Structural Reports: Licensed structural engineer responsible for design shall make a minimum of 3 visits to the Project site to perform structural observations and submit statements in accordance with Uniform Building Code (UBC) Section 3107 to ensure conformance of foundations, walls, and roof construction to the approved drawings.
    b. HVAC, both air and hydronics, testing and balancing reports.
    c. Fluid system hydrotest/pressure test reports.
    d. Plumbing piping system pressure testing reports.
    e. Domestic hot and cold water cleaning and disinfection reports.
    f. Software used and As-Built program configurations of all programmable systems or equipment controllers.

PART 2 - PRODUCTS

(Not Applicable)

PART 3 - EXECUTION

(Not Applicable)

END OF SECTION 01305

SECTION 01310 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:

1. General project coordination procedures.
2. Coordination Drawings.
3. Administrative and supervisory personnel.
4. Project meetings.

B. Related Sections: The following Sections contain requirements that relate to this Section:

1. General Requirements Section "Summary" for identity of Project Coordinator and a description of the division of Work among separate contracts and responsibility for coordination activities not in this Section.
2. General Requirements Section "Construction Progress Documentation" for preparing and submitting the Contractor's Construction Program.
3. General Requirements Section "Execution Requirements" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
4. General Requirements Section "Closeout Procedures" for coordinating Contract closeout.

1.3 COORDINATION

A. Coordination: Coordinate sequencing and programming of construction operations included in various Sections of the Specifications to ensure efficient and orderly installation of each part of the Works. Coordinate construction operations, included in different Sections of the Specifications, which depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Works depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components with other contractors, if any, to ensure maximum accessibility for required maintenance, service, and repair.
3. Make adequate provisions to accommodate items scheduled for later installation.
4. Coordinate shared access to Site, staging areas and workspaces.

5. Coordinate provision of temporary facilities and controls.
6. Coordinate and schedule interruptions of temporary utilities, including those necessary to make connections for permanent services.
7. Coordinate sequence of activities to accommodate tests and inspections, and coordinate schedule of tests and inspections.
8. Coordinate progress cleaning of common areas.
9. Coordinate protection and security of the Works.
10. Coordinate preparation and submittal of Record (As-Built) Documents.
11. Coordinate preparation and submittal of operation and maintenance manuals.

B. If necessary, prepare memoranda for distribution to other parties involved, outlining special procedures required for coordination. Include such items as required notices, reports, and attendance at meetings.

1. Prepare similar memoranda for the Engineer if coordination of the work of any separate contractors is required.

C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of any other contractors to avoid conflicts and to ensure orderly progress of the Works. Such administrative activities include, but are not limited to, the following:
1. Preparation of Contractor's Construction Program.
2. Installation and removal of temporary facilities and controls.
3. Delivery and processing of submittals.
4. Progress meetings.
5. Pre-installation conferences.
6. Closeout activities.

D. Coordination Drawings: Prepare Coordination Drawings if, in the Engineer's opinion, limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.

1. Indicate the relationship of components shown on separate Shop Drawings.
2. Indicate required installation sequences.

E. Staff Names: Within seven (7) days of the Commencement Date, submit a list of principal staff assignments in attendance at Site. Identify individuals and their job title, duties and responsibilities; list office and home telephone numbers. Provide names and telephone numbers of individuals assigned as replacements in the absence of individuals assigned to Project.
1. Attach curricula vitae for individuals assigned to the following or equivalent staff positions. Include: details of education and qualifications; a brief career history including employment, experience and present position; note special skills and abilities.

a. Project Manager
b. Construction Manager.
c. Coordinator.
d. QA/QC Manager.
e. Procurement Manager.

f. Administration Manager.
g. Document Control Supervisor.
h. Design/Engineering Manager.
i. Safety/Security Manager.

j. Site Superintendent(s).
k. Deputy Superintendent(s).
l. Supervising Engineers.
m. Construction Supervisors.

1.4 ADMINISTRATIVE AND SUPERVISORY PERSONNEL

1.5 PROJECT MEETINGS

A. Generally: Unless otherwise indicated, the Engineer will schedule and conduct meetings and conferences. The party responsible for scheduling and conducting a meeting or conference shall also:

1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date, time and location of each meeting. The Engineer will notify the Employer and design consultants, if necessary, of scheduled meeting dates and times.
2. Agenda and Minutes: Set meeting agendas; record significant discussions, commitments and agreements achieved; and prepare and distribute meeting minutes to everyone concerned, including the Employer, within two (2) days of the meeting.

B. Preconstruction Conference: Engineer will schedule and conduct a preconstruction conference prior to commencement of construction activities at a location and time convenient to the Employer.

1. Attendees: Authorized representatives of the Employer, Engineer and design consultants if required; Contractor's Project Manager, major subcontractors, manufacturers and suppliers; and other concerned parties. All participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Works.
2. Agenda: To review responsibilities and personnel assignments, discuss items of significance that could affect progress and examine administrative and procedural requirements; including but not necessarily limited to the following, as applicable:

a. Distribution of Contract Documents.
b. Site mobilization.
c. Possession of Site.
d. Tentative construction program.
e. Phasing of Works and Sectional completion.
f. Critical work sequencing.
g. Designation of responsible personnel and lines of communication.
h. Procedures for processing field decisions and Variations.
i. Procedures for processing Applications for Payment.
j. Submittal procedures.
k. Preparation of Record Documents.
l. Use of the site premises.
m. Responsibility for temporary facilities and controls.
n. Parking availability.

o. Office, work, and storage areas.
p. Equipment deliveries and priorities.
q. Safety and health.
r. Security.
s. Progress cleaning.
t. Working hours.

C. Pre-installation Conferences: Where individual Specification Sections indicate such a requirement, conduct a pre-installation conference at Project site before construction activity related to that Section.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Notify the Engineer in advance of scheduled meeting dates. The Engineer may or may not attend pre-installation conferences.

2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following, as applicable:

a. Options.
b. Related Variations.
c. Purchases.
d. Deliveries.
e. Submittals.
f. Review of mockups.
g. Possible conflicts.
h. Compatibility problems.
i. Time schedules.
j. Weather limitations.
k. Manufacturer's written recommendations.
l. Warranty requirements.
m. Compatibility of materials.
n. Acceptability of substrates.
o. Temporary facilities and controls.
p. Space and access limitations.
q. Regulations of authorities having jurisdiction.
r. Testing and inspecting requirements.
s. Required performance results.
t. Protection of construction and personnel.
u. Safety and health.
v. Security.

3. Minutes: Record significant discussions, commitments, agreements and disagreements. Prepare and distribute minutes to everyone concerned, including Engineer, within two (2) days of the conference.

4. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.

D. Progress Meetings: The Engineer will schedule and conduct progress meetings at weekly intervals, or more frequently whenever he considers it to be necessary.

1. Agenda Input: Submit request to the Engineer not later than 24 hours before scheduled meetings of any particular requirements for agenda discussion.

2. Attendees: Attend all progress meetings and secure attendance of representatives of each subcontractor, supplier, and other entity concerned with or having a significant part in current progress or involved in planning, coordination, or performance of future activities. All participants at these meetings shall be familiar with Project and authorized to conclude matters relating to the Works.

3. Agenda: The Engineer will set the agenda for each meeting to: review and correct or approve minutes of previous progress meeting; review and record current progress and items of significance that could affect future progress; and include topics for discussion, appropriate to status of Project.

   a. Construction Program: Review and report on progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Construction Program. Determine how construction behind schedule will be expedited and secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Time for Completion.
   b. Review present and future needs of each entity present, including but not necessarily limited to the following, as applicable:

      1) Interface requirements.
      2) Sequence of operations.
      3) Status of submittals.
      4) Status of clarification requests.
      5) Procurement and deliveries.
      6) Off-site fabrication.
      7) Access.
      8) Site utilization.
      9) Temporary facilities and controls.
      10) Work hours.
      11) Hazards and risks.
      12) Safety and health.
      13) Security.
      14) Progress cleaning.
      15) Quality and work standards.
      16) Variations.

4. Program Updating: Update Construction Program immediately after each progress meeting where updates and/or revisions to the program have been made or recognized. Issue update program to Engineer within two (2) days of meeting, for inclusion with the Engineer's report.

   a. Incorporate into update programs, the effect of all Variations issued and agreed as at time of program update.

5. Reporting: The Engineer will record, prepare and distribute minutes of the meeting to each party present and to the Employer, together with a brief report summary in narrative form, of progress, actions required and other significant matters arising since the previous meeting and report.

E. Coordination Meetings: The Engineer will schedule and conduct Project coordination meetings at regular intervals, or whenever he considers it to be necessary. Project coordination meetings are in addition to specific meetings held for other purposes, such as progress meetings and pre-installation conferences.

1. Attendees: In addition to Employer and Engineer representatives, the Contractor and other separate contractors, subcontractors, suppliers, and other entities concerned with or having a significant part in current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the meeting shall be familiar with Project and authorized to conclude matters relating to their work.

2. Agenda: The Engineer will set the agenda for each meeting to: review and correct or approve minutes of the previous coordination meeting; review and record other items of significance that could affect progress; and include topics for discussion, appropriate to status of Project.

   a. The Engineer will review progress since the last coordination meeting. Determine whether each separate contract is on time, ahead of schedule, or behind schedule, and the impact thereof on the Contractor's construction programme. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Time for Completion.

   b. The Engineer will review present and future coordination needs of each contractor present, including but not necessarily the following, as applicable:

      1) Interface requirements.
      2) Sequence of operations.
      3) Status of submittals.
      4) Status of clarification requests.
      5) Progress and deliveries.
      6) Off-site fabrication.
      7) Access.
      8) Site utilization.
      9) Temporary facilities and controls.
      10) Work hours.
      11) Hazards and risks.
      12) Safety and health.
      13) Security.
      14) Progress cleaning.
      15) Quality and work standards.
      16) Variations.

3. Program Updating: The Contractor’s Coordinator will update the Contractor's own construction program immediately after each coordination meeting where updates and/or revisions to the program have been made or recognized, and forward such updated program to the Engineer so that it can be issued concurrently with the meeting report.

4. Reporting: The Engineer will record, prepare and distribute minutes of the meeting to each party present and to the Employer and others affected by decisions or actions resulting from the meeting.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01310

SECTION 01320 - CONSTRUCTION PROGRESS DOCUMENTATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1. Refer to Conditions of Contract and Agreement for definitions and specific dates of Contract Time.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:

1. Preliminary Construction Program.
2. Construction Program.
3. Submittals Schedule.
4. Daily construction reports.
5. Monthly progress reports.
6. Material location reports.
7. Field condition reports.
8. Accident reports.
9. Special reports.
10. Wage book and time sheet records.
11. Construction photographs.

B. Related Sections include the following:

1. General Requirements Section "Summary of Multiple Contracts" for preparing a Combined Contractor's Construction Program.
2. General Requirements Section "Project Management and Coordination" for distributing meeting and conference minutes.
3. General Requirements Section "Submittal Procedures" for submitting schedules and reports.
4. General Requirements Section "Quality Requirements" for submitting a schedule of tests and inspections.
5. All other Specifications Sections for specific submittal requirements.

1.3 DEFINITIONS

A. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.

B. Critical Path: The longest continuous chain of activities through the network schedule that establishes the minimum overall Project duration and contains no float.

C. Network Diagram: A graphic diagram of a network schedule, showing activities and activity relationships.

D. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction program consume time and resources.

1. Critical activities are activities on the critical path. They must start and finish on the planned early start and finish times.
2. Predecessor activity is an activity that must be completed before a given activity can be started.

E. Event: The starting or ending point of an activity.

F. Float: The measure of leeway in starting and completing an activity.

1. Float time is not for the exclusive use or benefit of either Party, but is a jointly owned, expiring Project resource available to both parties as needed to meet milestones and Contract completion date.
2. Free float is the amount of time an activity can be delayed without adversely affecting the early start of the following activity or activities.
3. Total float is the measure of leeway in starting or completing an activity without adversely affecting the planned Project completion date.

G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

H. Major Area: A story of construction, a separate building, or a similar significant construction element.

I. Milestone: A key or critical point in time for reference or measurement, including for example, but not necessarily limited to:

1. Commencement Date.
2. Completion dates for specific Sections or parts of the Works.
3. Completion date for the whole of the Works.

1.4 SUBMITTALS

A. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

B. Submittals Schedule: Submit three (3) copies of the schedule. Arrange the following information in a tabular format, identifying corresponding program activity or event number for each scheduled date:

1. Scheduled date for first submittal.

2. Specification Section number and title.
3. Submittal type and category (action or informational).
4. Name of manufacturer and/or subcontractor, as applicable.
5. Description of the work covered.
6. Scheduled date for the Engineer's final release or approval.
7. Scheduled dates for Purchase Order and first delivery to Project site.
8. Scheduled date for commencement of installation.

C. Update Submittals Schedules: Submit two (2) copies of update schedules.

D. Construction Program: Submit two (2) blue- or black-line print copies of program, large enough to show entire program for entire construction period. In title block indicate "Initial", "Revised" or "Update" as applicable, and date of issue.

1. Submit an electronic copy of program, using software indicated, on CD-ROM labeled to comply with requirements for submittals. Indicate type of program (Initial, Revised or Update) and date on label.

E. CPM Reports: Concurrent with each CPM schedule submittal, submit three (3) printed copies of each of the following corresponding, computer-generated reports. Format for each activity in reports shall contain activity number, activity description, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float.

1. Activity Report: List of all activities sorted by major area, then by activity number and then early start date, or actual start date if known.
2. Logic Report: List of preceding and succeeding activities for all activities, sorted in ascending order by activity number and then early start date, or actual start date if known.
3. Total Float Report: List of all activities sorted in ascending order of total float.
4. Resource Allocation and Loading Report: List of all resources allocated to schedule activities, sorted in ascending order by activity number and then early start date, or actual start date if known, including but not necessarily limited to:

    p. Number and trade classification of workmen.
    q. Quantities of materials and products.
    r. Contractor's Equipment.

5. Monetary Value Summaries.

F. Construction Photographs: Submit two (2) print sets of each photographic view within five (5) days of taking photographs.

1. Format: 200 mm x 250 mm smooth-surface matte color prints on single-weight commercial-grade stock, enclosed back to back in clear plastic sleeves that are punched for standard ring binder.
2. Identification: On back of each print, provide an applied label or rubber-stamped impression with the following information:

    s. Name of Project.
    t. Name of Employer.
    u. Name of Engineer.
    v. Name of Contractor.
    w. Date photograph was taken.

x. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction, as applicable.

3. Negatives: Submit a complete set of corresponding photographic negatives, in protective envelopes, with each submittal of prints. Identify date photographs were taken.

y. Negatives shall be for the Employer's free and unrestricted use.

G. Daily Construction Reports: Submit two (2) copies weekly, and no later than 4:00 pm on the day immediately following date of report.

H. Monthly Progress Reports: Submit two (2) copies at monthly intervals. Report progress status coincidental with cutoff date associated with each Monthly Statement payment application.

1. Submit an electronic copy, using Microsoft Office software, on 90 mm diskette(s) labeled to comply with requirements for submittals.

I. Material Location Reports: Submit two (2) copies at weekly intervals.

J. Field Condition Reports: Submit two (2) copies at time of discovery of differing conditions.

K. Accident Reports: Submit two (2) copies at time of accident event.

L. Special Reports: Submit two (2) copies at time of unusual event.

1.5 QUALITY ASSURANCE

A. Scheduling Consultant Qualifications: An experienced specialist in CPM scheduling and reporting.

B. Photographer Qualifications: An individual who has been regularly engaged in the taking of construction progress photographs for not less than three (3) years.

C. Computer Software: Use a professional, high-end, reputable project management software program, acceptable to the Engineer, that has been developed specifically to manage CPM construction programming, scheduling and reporting indicated for Project.

D. Pre-scheduling Conference: Conduct conference at Project site to comply with requirements in General Requirements Section "Project Management and Coordination." Review methods and procedures related to the Contractor's Construction Program, including, but not necessarily limited to the following, as appropriate:

1. Review software limitations and content and format for reports.
2. Verify availability of qualified personnel needed to develop and update program.
3. Discuss constraints, including phases, sections, work sequences, area separations, interim milestones, interface with other contractors, statutory agencies and authorities having jurisdiction; staged completion, partial Employer occupancy and the like, as applicable.
4. Review delivery dates for Employer-furnished products.
5. Review program for work of Employer's separate contracts.
6. Review time required for review of submittals and re-submittals.

7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review and finalize list of construction activities to be included in program.
10. Review submittal requirements and procedures.
11. Review procedures for updating program.

1.6 COORDINATION

A. Coordinate preparation and processing of programs, schedules and reports with performance of construction activities and with programming, scheduling and reporting of separate contractors.

B. Coordinate Construction Program with the list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.

1. Secure time commitments for performing critical elements of the Works from parties involved.
2. Coordinate each construction activity in the network with other activities and schedule them in proper sequence.

C. Auxiliary Services: Cooperate with photographer and provide auxiliary services requested, including access to Site and use of temporary facilities including temporary lighting.

PART 2 - PRODUCTS

2.1 SUBMITTALS SCHEDULE

A. Preparation: Arrange Submittals Schedule in chronological order by dates required by construction program. Include time required for review, resubmittal, ordering, manufacturing, fabrication, and delivery when establishing dates.

1. Coordinate Submittals Schedule with list of subcontracts and Contractor's Construction Program.
2. Submittal: Submit concurrently with the submittal of completed Contractor's Construction Program. List those required to maintain orderly progress of the Work and those required early because of long lead-time for manufacture or fabrication.
3. Update Submittals: Continuously update Submittals Schedule to reflect actual and on-going current status, including additions to and changes in timing of submittals. Submit an Update Submittals Schedule on the day before each regular Progress Meeting.

2.2 CONTRACTOR'S CONSTRUCTION PROGRAM, GENERAL

A. Arrangement: Arrange activities on program by Section and/or major area, as applicable.

B. Activities: Treat each building story and/or separate area as a separate numbered activity for each principal element of the Work. Comply with the following:

1. Activity Duration: Define activities so that no activity is longer than twenty (20) days, unless specifically allowed by the Engineer.

2. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 days, as separate activities in the program. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.
3. Submittal Review Time: Include review and resubmittal times indicated in General Requirements Section "Submittal Procedures", in program. Coordinate submittal review times in Contractor's Construction Program with Submittals Schedule.
4. Startup and Testing Time: Indicate and include appropriate time allowances for startup and testing of each equipment and system installation.
5. Substantial Completion: Indicate substantial completion in advance of date established for Taking-Over, and allow time for Engineer's inspection and administrative procedures necessary for certification of Taking-Over.

C. Constraints: Include constraints and work restrictions indicated in the Contract Documents and as follows in the program, and show how the sequence of the Work is affected.

1. Work under Multiple Contracts: Include a separate activity for each portion of work performed by each separate contract.
2. Work by the Employer: Include a separate activity for each portion of work performed by the Employer.
3. Products Ordered in Advance: Include a separate activity for each product. Include delivery date indicated in General Requirements Section "Summary." Delivery dates indicated stipulate the earliest possible delivery date.
4. Employer-Furnished Products: Include a separate activity for each product. Include delivery date indicated in General Requirements Section "Summary." Delivery dates indicated stipulate the earliest possible delivery date.
5. Work Restrictions: Show the effect of the following items on the program:

    a. Coordination with existing construction.
    b. Limitations of continued occupancies.
    c. Uninterruptible services.
    d. Partial occupancy before substantial completion.
    e. Use of premises restrictions.
    f. Provisions for future construction.
    g. Seasonal variations.
    h. Environmental control.

6. Work Stages: Indicate important stages of construction for each major portion of the Works, including, but not limited to, the following:

    a. Subcontract awards.
    b. Submittals.
    c. Purchases.
    d. Mockups.
    e. Fabrication.
    f. Sample testing.
    g. Deliveries.
    h. Installation.
    i. Tests and inspections.
    j. Adjusting.
    k. Curing.
    l. Startup and placement into final use and operation.

7. Area Separations: Identify each major area of construction for each major portion of the Works. Indicate where each construction activity within a major area must be sequenced or integrated with other construction activities to provide for the following:

    a. Structural completion.
    b. Permanent space enclosure.
    c. Completion of mechanical installation.
    d. Completion of electrical installation.
    e. Substantial completion.

8. Other Constraints:

D. Sectional Completions and Milestones: Include in program, milestones and any and all sectional completion requirements, indicated in Contract Documents, including but not necessarily limited to Commencement Date, dates for Taking-Over, sections or parts of the Works, and date for Taking-Over the whole of the Works.

E. Cash Flow Correlation: At the head of program, provide a cash flow correlation line, indicating on the line, monetary value volume of the Works as planned and as actually performed.

1. Tabulate cash flow correlation, coincidental with cutoff dates associated with Monthly Statement payment applications, and use as a basis for preparation of Cash Flow Estimate submittal.

F. Contract Variations: If and when requested by the Engineer in connection with any proposed or instructed Contract Variation, prepare and submit a time-impact analysis, using fragnets if necessary, to demonstrate the effect of the proposed or instructed Variation on the overall construction program.

G. Computer Software: Prepare and update programs and schedules using the following software program that has been developed specifically to manage construction programming and scheduling.

1. Microsoft Project, Latest Version for Windows 10 operating system.

2.3 PRELIMINARY CONSTRUCTION PROGRAM (BAR-CHART SCHEDULE)

A. Preliminary Program: Submit preliminary horizontal bar-chart-type or network-type construction program with Tender in accordance with the requirements of the "Instructions to Tenderers".

B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line. Outline significant construction activities for first [**60**] 60 days of construction. Include skeleton diagram for the remainder of the Works and a cash flow prediction based on indicated activities.

2.4 CONTRACTOR'S CONSTRUCTION PROGRAM (GANTT CHART)

A. Gantt-Chart Program: Submit a comprehensive, fully developed, horizontal Gantt bar-chart-type Contractor’s Construction Program within fourteen (14) days after date of the letter of

Acceptance. Base program on the Preliminary Construction Program submitted with Tender and whatever updating and feedback received to date.

B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line.

1. For construction activities that require 3 months or longer to complete, indicate an estimated completion percentage in ten (10) percent increments within time bar.

2.5 CONTRACTOR'S CONSTRUCTION PROGRAM (CPM SCHEDULE)

A. General: Prepare network diagrams using AON (activity-on-node) format.

B. CPM Schedule: Prepare and submit a Contractor's Construction Program using a CPM network analysis diagram. Base program on the Preliminary Construction Program submitted with Tender and whatever updating and feedback received to date. Follow procedures and produce CPM schedule and reports in such form and detail as specified. Incorporate additional requirements, as the Engineer shall reasonably prescribe.

1. Develop, finalize and submit completed CPM schedule and reports for the Engineer's review and acceptance no later than fourteen (14) days after the date of the Letter of Acceptance.
2. Conduct educational workshops to train and inform key personnel, including subcontractors' personnel, in proper methods of providing data and using CPM schedule information.
3. Establish procedures for monitoring and updating CPM schedule and for reporting progress. Coordinate procedures with regular Progress Meeting, Monthly Progress Report and Monthly Statement payment application dates.
4. Use "one workday" as the unit of time and indicate number of shifts per workday.
5. Use calendar to identify and indicate holidays and other non-work days.

C. CPM Schedule Preparation: Prepare a list of all activities required to complete the Works.

1. Activities: Indicate for each activity, the estimated time duration, sequence requirements, resource requirements, relationship to other activities, and monetary value. Include dates and estimated time frames for the following activities:

    a. Preparation and processing of submittals.
    b. Purchase of materials.
    c. Delivery.
    d. Fabrication.
    e. Installation.

2. Processing: Process data to produce output data or a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the CPM schedule within the limitations of the Contract Time.

    a. Use resource-leveling method in computing the CPM schedule.

3. Format: Mark the critical path. Locate the critical path near center of network; locate paths with most float near the edges.

a. Sub-networks on separate sheets are permissible for activities clearly off the critical path.

D. Initial Issue of Schedule: Prepare initial network diagram from a list of straight "early start-total float" sort. Identify critical activities. Prepare tabulated reports showing the following:

1. Contractor or subcontractor and the work or activity.
2. Description of activity.
3. Principal events and constraints of activity.
4. Immediate preceding and succeeding activities.
5. Early and late start dates.
6. Early and late finish dates.
7. Activity duration in workdays.
8. Calendar for each activity.
9. Total float or slack time.
10. Resource allocation and loading of activity.
11. Monetary value of activity (coordinated with Bill of Quantities values).
12. Average size of workforce.
13. Holidays and non-work days.

E. Schedule Updating: Concurrent with making updates and/or revisions to schedule, prepare (in addition to other specified CPM reports) tabulated reports showing the following:

1. Identification of activities that have changed.
2. Changes in early and late start dates.
3. Changes in early and late finish dates.
4. Changes in activity durations in workdays.
5. Changes in the critical path.
6. Changes in total float or slack time.
7. Changes in usage or availability of resources.
8. Changes in the Contract Time.
9. Indicate actual versus original planned progress.

F. Monetary Value Summaries: Prepare cash flow predictions based on indicated activities and include two cumulative value lists, sorted by finish dates.

1. In first list, tabulate the following:

b. Activity number.
c. Early finish date.
d. Monetary value.
e. Cumulative monetary value.

2. In second list, tabulate the following:

a. Activity number.
b. Late finish date.
c. Monetary value.
d. Cumulative monetary value.

3. Prepare lists for ease of comparison with Cash Flow Estimates and Monthly Statement payment applications.

4. In subsequent issues of both lists, substitute actual finish dates for activities completed as of list date.
5. Submit update value summary printouts with each Monthly Statement payment application.

    a. In both value summary lists, tabulate "actual percent complete" and "cumulative value completed" with total at bottom.

2.6 REPORTS

A. Daily Construction Reports: Prepare a daily construction report, recording the following information concerning events at Project site:

1. List of construction work activities and operations performed on the Works, referenced to locations and CPM schedule activity numbers.
2. List of subcontractors on the Works.
3. List of separate contractors at Site.
4. Count and trade classification of workmen on the Works, allocated to CPM schedule activity numbers.
5. Count, category and status (working, idle, under maintenance) of Contractor's Equipment at Site, allocated to CPM schedule activity numbers.
6. High and low temperatures and general weather conditions.
7. Accidents (refer to accident reports).
8. Meetings and significant decisions.
9. Unusual events (refer to special reports).
10. Stoppages, delays, shortages, and losses.
11. Meter readings and similar recordings.
12. Emergency procedures.
13. Orders and requests of authorities having jurisdiction.
14. Variation Orders received and implemented.
15. Site Instructions received.
16. Services connected and disconnected.
17. Equipment or system tests and startups.
18. Partial Completions and occupancies.
19. Engineer's Inspections.
20. Taking-Over Certificates issued.
21. Any other general or specific information, requested by the Engineer to be reported.

B. Monthly Progress Reports: Prepare and submit monthly progress reports, summarizing activities, achievements, advancements and significant events at Site in month preceding, and reporting current progress status of Works; including but not necessarily limited to the following:

1. Summary of the information included in Daily Construction Reports for the month.
2. Summary of the information included in Material Location Reports for the month.
3. Latest update Submittals Schedule.
4. Progress Review Statement: Refer to latest Update Contractor's Construction Program; include:

    a. Listing of all current month programmed activities and report whether each is on time, ahead of schedule or behind schedule in relation to original Contractor's Construction Program.

b. On-going comparative line plot of Cash Flow Estimate against Interim Payment Certificates.
c. Description of means, methods and actions to be taken in following month to expedite construction behind schedule and ensure current and subsequent activities will be completed within Time for Completion.

5. Any other general or specific information, requested by the Engineer to be reported.

C. Material Location Reports: At intervals indicated, prepare and submit a comprehensive listing, in tabular form, of materials delivered to and stored at Site. List shall be cumulative, showing materials previously reported plus items delivered since previous report. Coordinate reports with information in Product List submitted under General Requirements Section "Product Requirements".

1. Form: Tabulate information for each material delivery under the following column headings:

a. Delivery date.
b. Specification Section number and title.
c. Generic name used in the Contract Documents.
d. Proprietary name, model number, and similar designations.
e. Manufacturer's name and address.
f. Supplier's name and address.
g. Delivered quantity.

D. Field Condition Reports: Immediately on discovery of a difference between field conditions and the Contract Documents, prepare and submit a detailed report. Include a detailed description of the differing conditions, together with a request for instruction, if applicable.

E. Accident Reports: On the occurrence of any accident which causes personal injury, or damage to the Works or any other property, prepare a detailed accident report and submit directly to the Engineer within one day of occurrence. Include details of date and time, location, prevailing conditions, chain of events and causes leading up to accident, persons involved, response by the Contractor's personnel and result or effect of accident. Obtain and attach witness statements, photographs and sketches as applicable.
1. On the occurrence of an accident or event that causes serious injury to any person, summon emergency services, notify appropriate authorities and inform the Engineer immediately.

2.7 SPECIAL REPORTS

A. General: Prepare and submit special reports, in an acceptable format, directly to the Engineer within one day of an occurrence. Distribute copies of report to parties affected by the occurrence.

B. Reporting Unusual Events: When an event of an unusual and significant nature occurs at Site, whether or not related directly to the Works, prepare and submit a special report. List chain of events, persons participating, response by the Contractor's personnel, evaluation of results or effects, and similar pertinent information. Advise the Engineer in advance when these events are known or predictable.

2.8 WAGE BOOK AND TIME SHEET RECORDS

A. Wage Books and Time Sheets: Keep and maintain accurate and proper wage books and time sheets of wages paid to and time worked by all workmen employed by the Contractor and subcontractors at Site. Comply with regulations and requirements, if any, of authorities having jurisdiction. Store wage books and time sheets in field office and make available for the Engineer's inspection and reference during normal working hours. If requested, produce and submit photocopies to the Engineer.

PART 3 - EXECUTION

3.1 CONTRACTOR'S CONSTRUCTION PROGRAM

A. Scheduling Consultant: Engage a consultant to provide planning, evaluation, and reporting using CPM scheduling.

1. In-House Option: Engineer may waive the requirement to engage and retain a consultant if the Contractor can demonstrate, to the Engineer's satisfaction, that he employs skilled and competent personnel with appropriate experience in CPM scheduling and reporting techniques. Submit a request, together with qualification data to demonstrate in-house employee capabilities and experience.
2. Meetings: Scheduling consultant shall attend all meetings related to Project progress, alleged delays, and time impact.

B. Update Contractor's Construction Programs: Continuously monitor performance and progress of programmed activities, including submittals, procurement and deliveries, supply and availability of allocated resources, etc. At monthly intervals, update program to reflect actual progress of construction and activities. Issue and submit update program and corresponding CPM reports coincidental with each Monthly Statement payment application, unless otherwise directed by the Engineer.

1. As work progresses, indicate Actual Completion percentage for each activity.
2. Revise program immediately after each meeting or other activity where revisions have been recognized and accepted by the Engineer. Issue and submit revised program within 3 days of such meetings or activities.
3. Include a report with revised program that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.

C. Distribution: Distribute copies of accepted initial program to the Employer, separate contractors, and other parties identified by the Contractor with a need-to-know program responsibility.

1. Post copies in meeting rooms and temporary field offices.
2. Distribute accepted revised programs to the same parties and post in the same locations. Delete parties from distribution when they have completed their assignments and are no longer involved in performance of construction activities.

3.2 CONSTRUCTION PHOTOGRAPHS

A. Photographer: Engage a person capable of taking photographs of a standard acceptable to the Engineer. In the event that the photographs submitted are not of an acceptable standard the Contract shall, on the instructions of the Engineer and at his own cost, engage the services of a qualified commercial photographer to take the required construction photographs.

B. Photographic Film: Medium-format, 60 mm x 70 mm.

C. Date Stamp: Unless otherwise indicated, date and time stamp each photograph as it is being taken so stamp is integral to photograph.

D. Preconstruction Photographs: Before starting construction, take 24 photographs of Site and surroundings from different vantage points, as directed by the Engineer.

1. Take additional photographs in sufficient number and detail to:

a. Show conditions of existing buildings, structures, site improvements and features on and/or adjacent to Site, accurately recording any physical condition, that might otherwise be misconstrued as damage caused by subsequent demolition or construction operations.
b. Show conditions of existing buildings, structures, assemblies, features or other components scheduled for selective demolition or dismantling, accurately recording physical condition, for subsequent reconstruction, restoration or rehabilitation.

E. Construction Progress Photographs: Take a minimum of 24 photographs monthly, adjusted to coincide with cutoff date associated with each payment application. Photographer shall select vantage points, in consultation with the Engineer and the Contractor, to best show status of construction and rate of progress since last photographs were taken.

1. Field Office Prints: Retain an additional print set of all preconstruction and construction progress photographs in field office at Site, and make available at all times for reference by the Engineer. Identify photographs the same as for those submitted to the Engineer.

F. Final Construction Photographs: Take 24 photographs immediately after date of Taking-Over for the whole of the Works. The Engineer will direct photographer for desired vantage points.

END OF SECTION 01320

SECTION 01322 - PHOTOGRAPHIC DOCUMENTATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for the following:

1. Preconstruction photographs.
2. Construction progress photographs.
3. Final construction photographs.
4. Preconstruction video.
5. Construction progress video.
6. Time-lapse sequence construction live video.
7. Demonstration and training video.

1.3 SUBMITTALS

A. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

B. Key Plan: Submit key plan of Project site and building with notation of vantage points marked for location and direction of each phtograph. Indicate elevation or story of construction. Include the same label information as the corresponding set of photographs.

C. Construction Photographs: Submit two (2) print sets of each photographic view within five (5) days of taking photographs.

1. Format: 200 x 250 mm smooth-surface matte color prints on single-weight commercial-grade stock, enclosed back to back in clear plastic sleeves that are punched for standard ring binder.
2. Identification: On back of each print, provide an applied label or rubber-stamped impression with the following information:

a. Name of Project.
b. Name of Employer.
c. Name of Engineer.
d. Name of Contractor.
e. Name and address of photographer.
f. Date photograph was taken.
g. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction, as applicable.

3. Negatives: Submit a complete set of corresponding photographic negatives, in protective envelopes, with each submittal of prints. Identify date photographs were taken.

   a. Negatives shall be for the Engineer's and Employer's free and unrestricted use.

4. Digital Images: Submit a complete set of corresponding digital image electronic files with each submittal of prints. Identify electronic media with date photographs were taken. Submit images that have the same aspect ratio as the sensor, uncropped.

   b. Digital image files shall be for the Engineer's and Employer's free and unrestricted use.

D. Videotapes: Submit two (2) copies of each videotape within five (5) days of recording. Videotapes shall be for the Engineer's and Employer's free and unrestricted use.

1. Identification: On each copy provide an applied label with the following information:

   a. Name of Project.
   b. Name of Employer.
   c. Name of Engineer.
   d. Name of Contractor.
   e. Name and address of photographer.
   f. Date videotape was taken.
   g. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction, as applicable.

2. Transcript: Prepared on (A4) paper, punched and bound in heavy-duty, vinyl-covered, ring binders. Mark appropriate identification on front and spine of each binder. Include a cover sheet with the same label information as the corresponding videotape. Include name of Contract and date of videotape on each page.

1.4 QUALITY ASSURANCE

A. Photographer Qualifications: An individual of established reputation who has been regularly engaged as a professional photographer for not less than three years.

1.5 COORDINATION

A. Auxiliary Services: Cooperate with photographer and provide auxiliary services requested, including access to Site and use of temporary facilities including temporary lighting.

PART 2 - PRODUCTS

2.1 PHOTOGRAPHIC MEDIA

A. Photographic Film: Medium-format, 100 mm x 1500 mm.

B. Digital Images: Provide images in JPEG format, with minimum sensor size of 10 megapixels.

PART 3 - EXECUTION

3.1 PHOTOGRAPHS, GENERAL

A. Photographer: Engage a qualified commercial photographer to take construction photographs.

B. Date Stamp: Unless otherwise indicated or directed, date and time stamp each photograph as it is being taken so stamp is integral to photograph.

C. Field Office Prints: Retain an additional print set of all preconstruction and construction progress photographs in field office at Site, and make available at all times for reference by the Engineer. Identify photographs the same as for those submitted to the Engineer.

3.2 CONSTRUCTION PHOTOGRAPHS

A. Preconstruction Photographs: Before starting construction, take twenty four (24) photographs of Site and surroundings from different vantage points, as directed by the Engineer.

1. Take additional photographs in sufficient number and detail to:

a. Show conditions of existing buildings, structures, site improvements and features on and/or adjacent to Site, accurately recording any physical condition, that might otherwise be misconstrued as damage caused by subsequent demolition or construction operations.
b. Show conditions of existing buildings, structures, assemblies, features or other components scheduled for selective demolition or dismantling, accurately recording physical condition, for subsequent reconstruction, restoration or rehabilitation.

B. Construction Progress Photographs: Take a minimum of twenty four (24) photographs monthly, adjusted to coincide with date associated with each payment application. Photographer shall select vantage points, in consultation with the Engineer and the Contractor, to best show status of construction and rate of progress since last photographs were taken.

C. Time-Lapse Sequence Construction Photographs: Take twenty four (24) color photographs as indicated, to best show status of construction and rate of progress since the last photographs were taken.

1. Frequency: Take photographs monthly adjusted to coincide with date associated with each payment application.
2. Vantage Points: Photographer shall select vantage points, in consultation with the Engineer and the Contractor. During each of the following construction phases, take

not less than two (2) of the required shots from the same vantage point each time to create a time-lapse sequence as follows:

a. Commencement of the Work, through completion of subgrade construction.
b. Above-grade structural framing.
c. Exterior building enclosure.
d. Interior Work, through to date of Substantial Completion.

D. Final Construction Photographs: Take seventy two (72) photographs immediately after date of Taking-Over for the whole of the Works. The Engineer will direct photographer for desired vantage points.

1. Do not include date stamp.

E. Additional Photographs: The Engineer may issue requests for additional photographs, in addition to periodic photographs specified. Such additional photographs shall be included in the Contract Price.

1. Photographer will be given two days' notice, where feasible.
2. In emergency situations, photographer shall take additional photographs within 24 hours of request.
3. Circumstances that could require additional photographs include, but are not limited to, the following:

a. Special events planned at Site.
b. Immediate follow-up when on-site events result in construction damage or losses.
c. Photographs to be taken at fabrication locations away from Site.
d. Substantial completion of a Section or part of the Works.
e. Extra record photographs at time of Taking-Over.
f. Employer's requirement for special publicity photographs.

3.3 CONSTRUCTION VIDEOTAPES, GENERAL

A. Videotape Photographer: Engage a qualified commercial photographer to record construction videotapes.

B. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of construction. Display continuous running time. At start of each videotape, record weather conditions from local newspaper or television and the actual temperature reading at Site.

C. Narration: Describe scenes on videotape by dubbing audio narration off-site after videotape is recorded. Include description of items being viewed, recent events, and planned activities. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

D. Transcript: Provide a typewritten transcript of the narration. Display images and running time captured from videotape opposite the corresponding narration segment.

3.4 CONSTRUCTION VIDEOTAPES

A. Preconstruction Videotape: Before starting construction, record videotape of Site and surroundings from different vantage points, as directed by the Engineer. Minimum recording time shall be one (1) hour.

1. Show conditions of existing buildings, structures, site improvements and features on and/or adjacent to Site, accurately recording any physical condition, that might

otherwise be misconstrued as damage caused by subsequent demolition or construction operations.

2. Show conditions of existing buildings, structures, assemblies, features or other components scheduled for selective demolition or dismantling, accurately recording physical condition, for subsequent reconstruction, restoration or rehabilitation.

B. Construction Progress Videotapes: Record videotape monthly, adjusted to coincide with the date associated with each payment application. Photographer shall select vantage points, in consultation with the Engineer and the Contractor, to best show status of construction and rate of progress since last videotape was recorded. Minimum recording time shall be one (1) hour.

C. Time-Lapse Sequence Construction Videotapes: Record videotape as indicated, to best show status of construction and rate of progress.

1. Frequency: During each of the following construction phases, set up videotape recorder to automatically record one frame of videotape every five (5) minutes, from the same vantage point each time, to create a time-lapse sequence as follows:

   a. Commencement of the Work, through completion of subgrade construction.
   b. Above-grade structural framing.
   c. Exterior building enclosure.

2. Timer: Provide timer to automatically start and stop videotape recorder so recording occurs only during daylight construction work hours.
3. Vantage Points: Photographer shall select vantage points, in consultation with the Engineer and the Contractor.

D. Demonstration and Training Videotapes: Record instruction of the Employer's personnel in the operation and maintenance of equipment and systems. Edit videotape to remove non-instructional conversation. Photographer shall select vantage points to best show equipment, systems, and procedures demonstrated. Minimum recording time, unless otherwise indicated or directed, shall be sixteen (16) hours.

END OF SECTION 01322

SECTION 01330 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for submitting:

1. Shop Drawings.
2. Product Data.
3. Samples.
4. Other miscellaneous submittals.

B. Related Sections include the following:

1. General Requirements Section "Construction Progress Documentation" for submitting schedules and reports, including Contractor's Construction Program and the Submittals Schedule and construction photographs.
2. General Requirements Section "Photographic Documentation" for submitting construction photographs.
3. General Requirements Section "Quality Requirements" for submitting test and inspection reports and Delegated-Design Submittals and for erecting mockups.
4. General Requirements Section "Product Requirements" for submitting Product List.
5. General Requirements Section "Closeout Procedures" for submitting warranties project record documents and operation and maintenance manuals.
6. General Requirements Section "Project Record (As-Built) Documents" for submitting record drawings, documents and data.
7. General Requirements Section "Operation and Maintenance Data" for operation and maintenance manual requirements.
8. All other Sepcification Sections for specific submittal requirements.

1.3 DEFINITIONS

A. Action Submittals: Written and graphic information that requires the Engineer's responsive action.

B. Informational Submittals: Written information that does not require the Engineer's approval. Submittals may be rejected for not complying with requirements.

C. Shop Drawings: Include, but are not limited to, the following:

1. Fabrication drawings.
2. Installation drawings.
3. Setting diagrams.
4. Shopwork manufacturing instructions.

5. Templates and patterns.
6. Design mix formulas.
7. Schedules.

(i) Standard information prepared without specific reference to the Works is not Shop Drawings.

D. Coordination Drawings show the relationship and integration of different construction elements that require careful coordination during fabrication or installation to fit in the space provided or to function as intended.

1. Preparation of Coordination Drawings is specified in General Requirements Section "Project Management and Coordination" and may include components previously shown in detail on Shop Drawings or Product Data.

E. Product Data: Include, but are not limited to, the following:

1. Manufacturer's product specifications.
2. Manufacturer's installation instructions.
3. Standard color charts.
4. Catalog cuts.
5. Roughing-in diagrams and templates.
6. Standard wiring diagrams.
7. Printed performance curves.
8. Operational range diagrams.
9. Mill reports.
10. Standard product operating and maintenance manuals.

F. Samples: Include, but are not limited to, the following:

1. Complete units or partial sections of manufactured or fabricated components.
2. Small cuts or containers of materials.
3. Complete units of repetitively used materials.
4. Swatches showing color, texture, and pattern.
5. Color range sets.
6. Components used for independent inspection and testing.
7. Field samples.

G. Mockups are full-size physical assemblies; they are not Samples.

1.4 SUBMITTAL PROCEDURES

A. General: Electronic copies of CAD Drawings of the Contract Drawings will not be provided by the Engineer for the Contractor's use in preparing submittals.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.

1. Transmit each submittal sufficiently in advance of performance of related procurement and construction activities, allowing ample time for review and resubmittal if necessary, in order to prevent delays to the Works.
2. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.

3. Coordinate transmittal of different types of submittals for related parts of the Works so processing will not be delayed because of need to review submittals concurrently for coordination.

    (i) Engineer reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

C. Submittals Schedule: Comply with requirements in General Requirements Section "Construction Progress Documentation" for list of submittals and time requirements for scheduled performance of related construction activities.

D. Processing Time: Allow enough time for submittal review, including time for resubmittals, as follows. Time for review shall commence on the Engineer's receipt of submittal.

1. Initial Review: Allow thirty five (35) days for initial review of each submittal. Allow additional time if processing must be delayed to permit coordination with subsequent submittals. The Engineer will advise the Contractor when a submittal being processed must be delayed for coordination.
2. If intermediate submittal is necessary, process it in same manner as initial submittal.
3. Allow twenty one (21) days for processing each resubmittal.

E. Identification: Place a permanent label or title block on each submittal for identification.

1. Indicate name of firm or entity that prepared each submittal on label or title block.
2. Provide a space approximately 100 x 125 mm on label or beside title block to record the Contractor's review and approval markings and action taken by the Engineer.
3. Include the following information on label for processing and recording action taken:

    (i) Contract name.
    (ii) Employer's name.
    (iii) Date.
    (iv) Name and address of Engineer.
    (v) Name and address of Contractor.
    (vi) Name and address of subcontractor.
    (vii) Name and address of supplier.
    (viii) Name and address of manufacturer.
    (ix) Unique identifier, including revision number.
    (x) Number and title of appropriate Specification Section.
    (xi) Drawing number and detail references, as appropriate.
    (xii) Other necessary identification.

F. Deviations: Highlight, encircle, or otherwise indicate and identify on submittals, deviations from the Contract Documents.

G. Additional Copies: Unless additional copies are required for final submittal, and unless the Engineer observes noncompliance with provisions of the Contract Documents, initial submittal may serve as final submittal.

1. For submittals requiring concurrent review, submit one extra copy in addition to specified number of copies to the Engineer.
2. Additional copies submitted for maintenance manuals will not be marked with action taken and will be returned.

H. Transmittal: Package each submittal individually and appropriately for transmittal and handling. Transmit each submittal using a transmittal form attached to a cover letter. The Engineer will discard, without review, submittals received from sources other than the Contractor.

1. Cover Letter: On attached, numbered, separate sheet(s), prepared on the Contractor's letterhead, record relevant information, requests for data, revisions other than those requested by the Engineer on previous submittals, and deviations from requirements of the Contract Documents, including minor variations and limitations. Include the same label information as the related submittal.

(i) Include Contractor's certification stating that information submitted complies with requirements of the Contract Documents.

2. Transmittal Form: Use a form acceptable to and approved by the Engineer. Provide locations on form for the following information:

(i) Contract name.
(ii) Employer's name.
(iii) Date.
(iv) Destination (To:).
(v) Source (From:).
(vi) Names of subcontractor, manufacturer, and supplier, as applicable.
(vii) Category and type of submittal.
(viii) Submittal purpose and description.
(ix) Submittal and transmittal distribution record.
(x) Remarks.
(xi) Signature of transmitter.

I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.

J. Use for Construction: Permit only final submittals with appropriate approved stamp, or other mark indicating action taken by the Engineer, to be used in connection with construction.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

A. General: Prepare and submit Action Submittals required by individual Specification Sections.

1. Number of Copies: Submit copies of each submittal, as follows, unless otherwise indicated:

(i) Initial Submittal: Submit a preliminary single copy of each submittal where selection of options, color, pattern, texture, or similar characteristics is required. The Engineer will return submittal with options selected.
(ii) Final Submittal: Submit four (4) copies, unless otherwise indicated. Submit additional copies where copies are required for operation and maintenance manuals. The Engineer will retain two (2) copies;

remainder will be returned. Mark up and retain one returned copy as a Record Document (shall be used for preparation of the As-Built).

B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. Provide only copies of original manufacturer published or printed product information, catalogs, data, color charts and the like.
2. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
3. Mark each copy of each submittal to show which products and options are applicable.
4. Include the following information, as applicable:

(i) Manufacturer's written recommendations.
(ii) Manufacturer's product specifications.
(iii) Manufacturer's installation instructions.
(iv) Standard color charts.
(v) Manufacturer's catalog cuts: original publication or digitally-scanned electronic copy in JPEG format.
(vi) Wiring diagrams showing factory-installed wiring.
(vii) Printed performance curves.
(viii) Operational range diagrams.
(ix) Mill reports.
(x) Standard product operating and maintenance manuals.
(xi) Compliance with recognized trade association standards.
(xii) Compliance with recognized testing agency standards.
(xiii) Application of testing agency labels and seals.
(xiv) Notation of coordination requirements.

C. Shop Drawings: Produce newly prepared, Contract-specific, information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.

1. Preparation: Include the following information, as applicable:

(i) Dimensions; in SI units unless otherwise indicated or directed.
(ii) Identification of products.
(iii) Fabrication and installation drawings.
(iv) Roughing-in and setting diagrams.
(v) Wiring diagrams showing field-installed wiring, including power, signal, and control wiring.
(vi) Shopwork manufacturing instructions.
(vii) Templates and patterns.
(viii) Schedules.
(ix) Design calculations.
(x) Compliance with specified standards.
(xi) Notation of coordination requirements.
(xii) Notation of dimensions established by field measurement.

2. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring.
3. Sheet Size: Except for templates, patterns, and similar full-size drawings, submit Shop Drawings on sheets at least A4 size, and no larger than A1 size, unless otherwise approved.

4. Number of Copies: Submit copies of each submittal, as follows:

    (i) Initial Submittal: Submit one correctable, translucent, reproducible print and one (1) blue- or black-line print. Engineer will return the reproducible print.
    (ii) Final Submittal: Submit four (4) blue- or black-line prints, unless otherwise indicated. Submit additional prints where prints are required for operation and maintenance manuals. The Engineer will retain three (3) prints; remainder will be returned. Mark up and retain one returned print as a Record Drawing (shall be used for preparation of the As-Built).

D. Coordination Drawings: Comply with requirements in General Requirements Section "Project Management and Coordination."

E. Samples: Prepare physical units of materials or products, including the following:

1. Comply with requirements in General Requirements Section "Quality Requirements" for mockups.
2. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
3. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from the same material to be used for the Works, cured and finished in manner specified, and physically identical with the product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
4. Preparation: Mount, display, or package Samples in manner specified to facilitate review of qualities indicated. Prepare Samples to match the Engineer's sample where so indicated. Attach label on unexposed side that includes the following:

    (i) Generic description of Sample.
    (ii) Product name.
    (iii) Name and address of manufacturer.
    (iv) Sample source.

5. Additional Information: On an attached separate sheet, prepared on the Contractor's letterhead, provide the following:

    (i) Size limitations.
    (ii) Compliance with recognized standards.
    (iii) Availability and delivery time.

6. Submit Samples for review of kind, color, pattern, and texture for a final check of these characteristics with other elements and for a comparison of these characteristics between final submittal and actual component as delivered and installed.

    (i) If variation in color, pattern, texture, or other characteristic is inherent in the product represented by a Sample, submit at least three sets of paired units that show approximate limits of the variations.

(ii) Refer to individual Specification Sections for requirements for Samples that illustrate workmanship, fabrication techniques, details of assembly, connections, operation, and similar construction characteristics.

7. Number of Samples for Initial Selection: Submit one full set of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. The Engineer will return submittal with options selected.
8. Number of Samples for Verification: Submit two (2) sets of Samples. The Engineer will retain one (1) Sample set; remainder will be returned, marked with action taken.

(i) Submit a single Sample where assembly details, workmanship, fabrication techniques, connections, operation, and other similar characteristics are to be demonstrated.

9. Disposition: Maintain sets of approved Samples in field offices at Project site, in a secure, lockable store room, provided and equipped specifically for that purpose. Make available for the Engineer's reference during normal working hours, for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.

(i) Samples that may be incorporated into the Works are indicated in individual Specification Sections. Indicate such as special requests on transmittal and obtain approval for disposition in the Works. Such Samples must be in an undamaged condition at time of use.
(ii) Samples not incorporated into the Works, or otherwise designated as the Employer's property, are the property of the Contractor.

F. Product List: Comply with requirements in General Requirements General Requirements Section "Product Requirements."

G. Delegated-Design Submittal: Comply with requirements in General Requirements Section "Quality Requirements."

H. Contractor's Construction Program: Comply with requirements in General Requirements Section "Construction Progress Documentation".

I. Submittals Schedule: Comply with requirements in General Requirements Section "Construction Progress Documentation."

J. Subcontract List: Prepare and submit a list identifying subcontractor individuals or firms proposed for principal portions of the Works, including those who are to fabricate products or equipment to a special design. Include the following information in tabular form:

1. Name, address, and telephone number of entity performing subcontract.
2. Number and title of related Specification Section(s) covered by subcontract.
3. Drawing number and detail references, as appropriate, covered by subcontract.

2.2 INFORMATIONAL SUBMITTALS

A. General: Prepare and submit Informational Submittals required by other Specification Sections.

1. Number of Copies: Submit two (2) copies of each submittal, unless otherwise indicated. The Engineer will not return copies.
2. Certificates and Certifications: Provide a notarized statement that includes signature of the Contractor, testing agency, or design professional responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of the company.
3. Test and Inspection Reports: Comply with requirements in Divisi General Requirements Section "Quality Requirements."

B. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and employers, and other information specified.

C. Product Certificates: Prepare written statements on manufacturer's letterhead certifying that product complies with requirements.

D. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on AWS forms. Include names of firms and personnel certified.

E. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying that Installer complies with requirements and, where required, is authorized for this specific Project.

F. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying that manufacturer complies with requirements. Include evidence of manufacturing experience where required.

G. Material Certificates: Prepare written statements on manufacturer's letterhead certifying that material complies with requirements.

H. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements.

I. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements.

J. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.

K. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements.

L. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.

M. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements in General Requirements Section " Operation and Maintenance Data."

N. Design Data: Prepare written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.

O. Manufacturer's Instructions: Prepare written or published information that documents manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer. Include the following, as applicable:

1. Preparation of substrates.
2. Required substrate tolerances.
3. Sequence of installation or erection.
4. Required installation tolerances.
5. Required adjustments.
6. Recommendations for cleaning and protection.

P. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. Include the following, as applicable:

1. Name, address, and telephone number of factory-authorized service representative making report.
2. Statement on condition of substrates and their acceptability for installation of product.
3. Statement that products at Site comply with requirements.
4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
6. Statement whether conditions, products, and installation will affect warranty.
7. Other required items indicated in individual Specification Sections.

Q. Insurance Certificates and Bonds: Prepare written information indicating current status of insurance or bonding coverage. Include name of entity covered by insurance or bond, limits of coverage, amounts of deductibles, if any, and term of the coverage.

R. Construction Photographs and Videotapes: Comply with requirements in General Requirements Sections Construction Progress Documentation and Photographic Documentation.

PART 3 – EXECUTION

3.1 CONTRACTOR'S REVIEW AND APPROVAL

A. Review each submittal and check for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to the Engineer.

B. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Contract name and location, submittal number, Specification Section title and number, name of

reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, coordinated and approved for compliance with the Contract Documents.

3.2 ENGINEER'S ACTION

A. General: The Engineer will not review submittals that do not bear the Contractor's approval stamp and will return them without action.

B. Engineer's Action: The Engineer's review is limited only to checking conformance with information given and the design concept expressed in the Contract Documents. It is not conducted for the purpose of determining the accuracy and completeness of details, dimensions or quantities, nor substantiating integrity or compatibility, nor confirming instructions for installation or performance. The Engineer's approval does not in any way relieve the Contractor of responsibility for compliance with specified provisions and the Contract Document requirements.

C. Action Submittals: The Engineer will review each submittal, make marks to indicate corrections or modifications required, and return it to the Contractor. The Engineer will stamp each submittal with an action stamp and will mark stamp appropriately to indicate action taken, as follows:

1. Final Unrestricted Release: Where the submittal is marked "APPROVED", the work covered by the submittal may proceed provided it complies with the Contract Documents. Final acceptance of the work will depend on that compliance.
2. Final-but-Restricted Release: Where the submittal is marked "APPROVED AS NOTED", the work covered by the submittal may proceed provided it complies with both the Engineer's notations and corrections on the submittal and the Contract Documents. Final acceptance of the work will depend on that compliance.
3. Returned for Resubmittal: Where the submittal is marked "NOT APPROVED, REVISE AND RESUBMIT", do not proceed with the work covered by the submittal, including purchase, fabrication, delivery, or other activity for the product submitted. Revise or prepare a new submittal according to the Engineer's notations and corrections.
4. Rejected: Where the submittal is marked "NOT APPROVED, RESUBMIT" or "REJECTED", do not proceed with the work covered by the submittal. Prepare a new submittal for a product that complies with the Contract Documents.

D. Informational Submittals: The Engineer will review each submittal and will not return it, or will reject and return it if it does not comply with requirements.

E. Submittals not required by the Contract Documents will not be reviewed and may be discarded.

END OF SECTION 01330

SECTION 01400 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for quality assurance and quality control, including the erection of mockups generally.

B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve the Contractor of responsibility for compliance with the Contract Document requirements.

1. Specific quality-control requirements for individual construction activities are specified in the Sections that specify those activities. Requirements in those Sections may also cover production of standard products.
2. Specified tests, inspections, and related actions do not limit the Contractor's quality-control procedures that facilitate compliance with the Contract Document requirements.
3. Requirements for the Contractor to provide quality-control services required by the Engineer, the Employer, or authorities having jurisdiction are not limited by provisions of this Section.

C. Related Sections include the following:

1. General Requirements Section "Cutting and Patching" for repair and restoration of construction disturbed by testing and inspecting activities.
2. All Specification Sections for specific test and inspection requirements.
3. All SpecificationSections for specific mockup requirements not indicated in this Section.

1.3 DEFINITIONS

A. Quality-Assurance Services: Activities, actions, and procedures performed before and during execution of the Work to guard against defects and deficiencies and ensure that proposed construction complies with requirements.

B. Quality-Control Services: Tests, inspections, procedures, and related actions performed during and after execution of work, by the Contractor, testing agencies or authorities having jurisdiction, to evaluate that completed construction complies with requirements. Services do not include contract enforcement activities performed by the Engineer.

C. Mockups: Full-size, physical example assemblies to illustrate finishes and materials. Mockups are used to verify selections made under Sample submittals, to demonstrate aesthetic effects and, where indicated, qualities of materials and execution, and to review construction,

coordination, testing, or operation; they are not Samples. Mockups may establish the standard by which the work will be judged.

D. Testing Agency: An entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

1.4 DELEGATED DESIGN

A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of the Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.

1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to the Engineer.

1.5 SUBMITTALS

A. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in the form of a recent report on the inspection of the testing agency by a recognized authority.

B. Delegated-Design Submittal: In addition to Shop Drawings, Product Data, and other required submittals, submit a statement, signed and sealed by the responsible design professional, for each product and system specifically assigned to the Contractor to be designed or certified by a design professional, indicating that the products and systems are in compliance with performance and design criteria indicated. Include list of codes, loads, and other factors used in performing these services.

C. Schedule of Tests and Inspections: Prepare in tabular form and include the following:

1. Specification Section number and title.
2. Description of test and inspection.
3. Identification of applicable standards.
4. Identification of test and inspection methods.
5. Number of tests and inspections required.
6. Time schedule or time span for tests and inspections.
7. Entity responsible for performing tests and inspections.
8. Requirements for obtaining samples.
9. Unique characteristics of each quality-control service.

D. Tests and Inspection Reports: Prepare and submit certified written reports that include the following:

1. Date of issue.
2. Project title and number.
3. Name, address, and telephone number of testing agency.
4. Dates and locations of samples and tests or inspections.
5. Names of individuals making tests and inspections.
6. Description of the work and test and inspection method.
7. Identification of product and Specification Section.
8. Complete test or inspection data.
9. Test and inspection results and an interpretation of test results.

10. Ambient conditions at time of sample taking and testing and inspecting.
11. Comments or professional opinion on whether tested or inspected work complies with the Contract Document requirements.
12. Name and signature of laboratory inspector.
13. Recommendations on retesting and reinspecting.

E. Permits, Licenses, and Certificates: For Employer's records, submit copies of permits, licenses, certifications, inspection reports, releases, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of the work.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: A firm experienced in producing products similar to those indicated in the Contract Documents and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

B. Factory-Authorized Service Representative Qualifications: An authorized representative of the manufacturer who is trained and approved by the manufacturer to inspect installation of the manufacturer's products that are similar in material, design, and extent to those indicated in the Contract Documents.

C. Installer Qualifications: A firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent to that indicated in the Contract Documents, whose work has resulted in construction with a record of successful in-service performance.

D. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated in the Contract Documents and with a record of successful in-service performance.

E. Professional Engineer Qualifications: A professional engineer who is legally qualified and licensed to practice in the engineering discipline indicated, in jurisdiction where the Works are located and who is experienced in providing engineering services of the kind indicated. Engineering discipline and engineering services are defined as those related to and performed for installations of the system, assembly, or product that are similar to those indicated in the Contract Documents in material, design, and extent.

F. Specialists: Certain sections of the Specifications may require that specific construction activities shall be performed by entities that are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for the activities indicated.

G. Testing Agency Qualifications: An agency with the experience and capability to conduct testing and inspecting indicated, as documented by ASTM E 548, and that specializes in types of tests and inspections to be performed.

H. Preconstruction Testing: Testing agency shall perform preconstruction testing for compliance with specified requirements for performance and test methods.

1. Contractor responsibilities include the following:

d. Provide test specimens and assemblies representative of proposed materials and construction. Provide sizes and configurations of assemblies to adequately demonstrate capability of product to comply with performance requirements.
e. Submit specimens in a timely manner with sufficient time for testing and analyzing results to prevent delaying the Works.
f. When testing is complete, remove assemblies; do not reuse materials on the Works.

2. Testing Agency Responsibilities: Submit a certified written report of each test, inspection, and similar quality-assurance service to the Engineer, with copy to the Contractor. Interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from the Contract Documents.

1.7 QUALITY CONTROL

A. Contractor Responsibilities: Unless otherwise indicated, provide quality-control services specified and required by authorities having jurisdiction.

1. Where services are indicated as the Contractor's responsibility, engage a qualified testing agency, acceptable to the Engineer, to perform these quality-control services.
2. Notify testing agencies at least twenty four (24) hours in advance of time when work that requires testing or inspecting will be performed.
3. Where quality-control services are indicated as the Contractor's responsibility, submit a certified written report, in duplicate, of each quality-control service.
4. Testing and inspecting requested by the Contractor and not required by the Contract Documents are the Contractor's responsibility.

B. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.

C. Retesting/Reinspecting: Regardless of whether original tests or inspections were the Contractor's responsibility, provide quality-control services, including retesting and reinspecting, for construction of revised or replaced work that failed to comply with requirements established by the Contract Documents.

D. Testing Agency Responsibilities: Cooperate with the Engineer and the Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify the Engineer and the Contractor promptly of irregularities or deficiencies observed in the work during performance of its services.
2. Interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from requirements.
3. Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through the Contractor.
4. Do not release, revoke, alter, or increase requirements of the Contract Documents or approve or accept any portion of the Works.
5. Do not perform any duties of the Contractor.

E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide the following:

1. Access to the Site/ work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
4. Facilities for storage and field-curing of test samples.
5. Preliminary design mix proposed for use for material mixes that require control by testing agency.
6. Security and protection for samples and for testing and inspecting equipment at Project site.

F. Coordination: Coordinate sequence of activities to accommodate required quality-assurance and quality-control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

G. Schedule of Tests and Inspections: Prepare a schedule of tests, inspections, and similar quality-control services required by the Contract Documents. Submit schedule within twenty eight (28) days of Commencement Date.

1. Distribution: Distribute schedule to the Employer, the Engineer, testing agencies, and each party involved in performance of portions of the Works where tests and inspections are required.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 REPAIR AND PROTECTION

A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Provide materials and comply with installation requirements specified in other Sections of these Specifications. Restore patched areas and extend restoration into adjoining areas in a manner that eliminates evidence of patching.

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are the Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01400

SECTION 01420 - REFERENCE STANDARDS AND DEFINITIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 DEFINITIONS

A. General: Basic contract definitions are included in the Conditions of the Contract.

B. "Approved": The term "approved", when used in conjunction with the Engineer's action on the submittals, applications, and requests, is limited to the Engineer's duties and responsibilities as stated in the Conditions of the Contract.

C. "Directed": Terms such as "directed", "requested", "authorized", "selected", "approved", "required", and "permitted" mean directed by the Engineer, requested by the Engineer, and similar phrases.

D. "Furnish": The term "furnish" means to supply and deliver to the Site, ready for unloading, unpacking, assembly, installation, and similar operations.

E. "Indicated": The term "indicated" refers to graphic representations, notes, or schedules on the Drawings; or to other paragraphs or schedules in the Specifications and similar requirements in the Contract Documents. Terms such as "shown", "noted", "scheduled", and "specified" are used to help the user locate the reference. Location is not limited.

F. "Install": The term "install" describes operations at the Site including the actual unloading, temporary storage, unpacking, assembling, erecting, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning, and similar operations.

G. "Installer": The term "installer" means the Contractor or another entity engaged by the Contractor, either as an employee, subcontractor, or contractor of lower tier, who performs a particular construction activity including installation, erection, application, or similar operations. Installers shall be experienced in the operations they are engaged to perform.

PART 2 -

1. "Experienced": The term "experienced", when used with the term "installer", means having successfully completed a minimum of 5 previous projects similar in size and scope to the Works; being familiar with the special requirements indicated; and having complied with requirements of authorities having jurisdiction.
2. Trades: Using terms such as "carpentry" does not imply that certain construction activities shall be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to tradespeople of the corresponding generic name.
3. Assigning Specialists: Certain Sections of the Specifications require that specific construction activities will be performed by specialists who are recognized experts in those operations. The specialists shall be engaged for those activities, and their assignments are requirements over which the Contractor has no option. However, the ultimate responsibility for fulfilling contract requirements remains with the Contractor.

a. This requirement shall not be interpreted to conflict with enforcing building codes and similar regulations governing the Works. It is also not intended to interfere with local trade-union jurisdictional settlements and similar conventions.

H. "Provide": The term "provide" means to furnish and install, complete and ready for the intended use.

I. "Regulations": The term "regulations" includes laws, ordinances, statutes, and lawful orders issued by authorities having jurisdiction, as well as rules, conventions, and agreements within the construction industry that control performance of the Works.

J. "Testing Agencies": The term "testing agency" means an independent entity engaged to perform specific inspections or tests, either at the Site or elsewhere, and to report on and, if required, to interpret results of those inspections or tests.

k. 'Law': The term 'Law' refers to the Alat FEZ Law.

1.3 SPECIFICATION FORMAT AND CONTENT EXPLANATION

A. Specification Format: These Specifications are organized into sections based on material types.

B. Specification Content: These Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:

1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be interpolated as the sense requires. Singular words shall be interpreted as plural and plural words interpreted as singular where applicable as the context of the Contract Documents indicates.
2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood shall be performed by the Contractor. At certain locations in the Text, subjective language is used for clarity to describe responsibilities that shall be fulfilled indirectly by the Contractor or by others when so noted.

a. The words "shall", "shall be", or "shall comply with", depending on the context, are implied where a colon (:) is used within a sentence or phrase.
b. The word "per" means "in accordance with", "according to", "in compliance with", "complying with", and similar phrases.

1.4 INDUSTRY STANDARDS

A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into the Contract Documents to the extent referenced. Such standards are made a part of the Contract Documents by reference.

B. If requested by the Engineer, the Contractor shall submit the equivalent to any standard referred to in the Specifications and the Contractor shall convert the parameters mentioned in the specifications to be comparable with the equivalent standard.

C. Publication Dates: Unless otherwise indicated, comply with the standards in effect as of the date of the Contract Documents.

D. Conflicting Requirements: Where compliance with 2 or more standards is specified and the standards establish different or conflicting requirements for minimum quantities or quality levels, comply with the most stringent requirement. Refer uncertainties and requirements that are different but apparently equal to the Engineer for a decision before proceeding.

1. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified shall be the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of the requirements. Refer uncertainties to the Engineer for a decision before proceeding.

E. Copies of Standards: Each entity engaged in construction on the Project shall be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with the Contract Documents.

1. Where copies of standards are needed to perform a required construction activity, the Contractor shall obtain copies directly from the publication source and make them available on request.

1.5 SUBMITTALS

A. Permits, Licenses, and Certificates: For the Employer's records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of the Works.

PART 2 - PRODUCTS
(Not Applicable)

PART 3 - EXECUTION
(Not Applicable)

END OF SECTION 01420

SECTION 01500 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes requirements for temporary facilities and controls, including temporary utilities, support facilities, and security and protection facilities.

B. Temporary utilities include, but are not limited to, the following:

1. Sewers and drainage.
2. Water service and distribution.
3. Sanitary facilities, including toilets, wash facilities, and drinking-water facilities.
4. Heating and cooling facilities.
5. Ventilation and humidity control.
6. Electric power service and distribution.
7. Lighting.
8. Telephone service and facilities.

C. Support facilities include, but are not limited to, the following:

1. Temporary roads and paving.
2. Dewatering facilities and drains.
3. Identification of the Works and temporary signs.
4. Waste disposal facilities.
5. Temporary Site offices and site laboratory.
6. Storage and fabrication sheds.
7. Temporary stairs.
8. Construction aids and miscellaneous services and facilities.

D. Security and protection facilities include, but are not limited to, the following:

1. Environmental protection.
2. Stormwater control.
3. Tree and plant protection.
4. Site enclosure fence.
5. Security enclosure and lockup.
6. Barricades, warning signs, and lights.
7. Covered walkways.
8. Temporary enclosures.
9. Temporary partitions.
10. Fire protection.

E. Related Sections include the following:

1. General Requirements Section "Summary of Multiple Contracts" for responsibilities for temporary facilities and controls.
2. General Requirements Section "Submittal Procedures" for procedures for submitting copies of implementation and termination schedule and utility reports.
3. General Requirements Section "Execution Requirements" for progress cleaning requirements.

1.3 DEFINITIONS

A. Permanent Enclosure: As determined by the Engineer, permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and all openings are closed with permanent construction or substantial temporary closures.

1.4 USE CHARGES

A. Water Service: Pay water service use charges, whether metered or otherwise, for water used by all entities engaged in construction activities at Site.

B. Electric Power Service: Pay electric power service use charges, whether metered or otherwise, for electricity used in construction activities at Site.

C. Telephone Service: Pay telephone service use charges, for telephones installed at Site.

1.5 SUBMITTALS

A. Temporary Utility Reports: Submit reports of tests, inspections, meter readings, and similar procedures performed on temporary utilities.

B. Temporary Utilities: Submit detailed proposals including drawings and product data as requested; obtain the Engineer's approval prior to purchase, delivery, installation or implementation, as applicable, of the following:

1. Sewers and drainage.
2. Water service and distribution.
3. Sanitary facilities, including toilets, wash facilities, and drinking-water facilities.
4. Electric power service.
5. Telephone service.

C. Support Facilities: Submit detailed proposals including drawings and product data as requested; obtain the Engineer's approval prior to purchase, delivery, installation or implementation, as applicable, of the following:

1. Temporary roads and paving.
2. Dewatering facilities and drains.
3. Identification of the Works and temporary signs.
4. Temporary Site offices and site laboratory, including laboratory equipment.
5. Storage and fabrication sheds.
6. Construction aids and miscellaneous services and facilities:

a. Exterior scaffold system, including ladders, walkways, platforms and guardrails.
b. Protective screens and chutes.
c. Tower cranes.

a. Lifts and hoists for materials and workmen.

D. Security and Protection Facilities: Submit detailed proposals including drawings and product data as requested; obtain the Engineer's approval prior to purchase, delivery, installation or implementation, as applicable, of the following:

1. Site enclosure fence.
2. Covered walkways.
3. Fire protection.

1.6 QUALITY ASSURANCE

A. Regulations: Comply with industry standards and Alat FEZ Law and regulations of authorities having jurisdiction including, but not limited to, the following:

1. Building code requirements.
2. Health and safety regulations.
3. Utility company regulations.
4. Police and fire department regulations.
5. Environmental protection regulations.

B. Standards: Comply with ANSI A10.6, NECA's "Temporary Electrical Facilities," and NFPA 241.

1. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.

C. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.7 CONDITIONS

A. Temporary Utilities: At earliest feasible time, when acceptable to the Engineer, change over from use of temporary service to use of permanent service.

B. Temporary Use of Permanent Facilities: The Contractor shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Taking-Over, regardless of previously assigned responsibilities.

1. Contractor's personnel and workmen, including subcontractors, are not permitted to use permanent toilet facilities or wash facilities.

C. Conditions of Use: The following conditions apply to use of temporary services and facilities by all parties engaged in the Works:

1. Keep clean and neat.
2. Operate in a safe and efficient manner.
3. Do not overload or permit them to interfere with progress.
4. Take appropriate fire prevention measures.
5. Do not allow hazardous, dangerous, or unsanitary conditions, or public nuisance to develop or persist on Site.
6. Relocate as required by progress of the Works.

PART 2 - PRODUCTS

2.1 MATERIALS

A. General: Provide new materials suitable for use intended. Undamaged, previously used materials in serviceable condition may be used for temporary facilities if approved by the Engineer.

B. Construction Materials and Products: Comply with standards and applicable requirements in appropriate Sections of the Specification.

C. Tarpaulins: Waterproof, fire-resistive, with low flame-spread rating.

D. Temporary Enclosures: Translucent, fire-retardant, nylon-reinforced laminated polyethylene or polyvinyl chloride tarpaulins.

E. Water: Clean, fresh, potable.

2.2 EQUIPMENT

A. General: Provide new equipment suitable for use intended. Undamaged, previously used equipment in serviceable condition may be used for temporary facilities if approved by the Engineer.

B. Temporary Site Offices: Prefabricated, mobile units or job-built construction, with lockable entrances, operable windows, serviceable finishes and robust and secure fittings; insulated, weathertight, heated and air-conditioned.

1. Incorporate sanitary facilities to the extent required for the number and gender of personnel accommodated.
2. Support on foundations adequate for loading.
3. Provide and maintain all necessary and required services and utilities, including but not necessarily limited to:

    a. Sewers and drainage.
    b. Water service and distribution.
    c. Sanitary facilities, including toilets, wash facilities, and drinking-water facilities.
    d. Air-conditioner heating and cooling.
    e. Ventilation.
    f. Electric power service.
    g. Lighting.
    h. Telephone service.
    i. Firefighting appliances.
    j. Janitorial services and facilities, and general attendance.

C. Fire Extinguishers: Hand carried, portable, UL rated. Provide class and extinguishing agent as indicated or a combination of extinguishers of NFPA-recommended classes for exposures.

1. Comply with NFPA 10 and NFPA 241 for classification, extinguishing agent, and size required by location and class of fire exposure.

D. Temporary Site Toilets: Prefabricated, self-contained, single-occupant units of chemical, aerated recirculation, or combustion type; vented; fully enclosed with a glass-fiber-reinforced polyester shell or similar nonabsorbent material.

E. Drinking-Water Fixtures: Containerized, tap-dispenser, bottled-drinking-water units, including paper cup supply.

1. Where power is accessible, provide electric water coolers to maintain dispensed water temperature at 7.2 to 12.7 deg. C (45 to 55 deg. F).

F. Electrical Outlets: Properly configured, NEMA-polarized outlets to prevent insertion of 110- to 120-V plugs into higher-voltage outlets; equipped with ground-fault circuit interrupters, reset button, and pilot light.

G. Power Distribution System Circuits: Where permitted and overhead and exposed for surveillance, wiring circuits, not exceeding 125-V ac, 20-A rating, and lighting circuits may be nonmetallic sheathed cable.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Locate facilities where they will serve the Works adequately and result in minimum interference with performance of the Works. Relocate and/or modify facilities as often as required by progress of the Works.

B. Provide each facility ready for use when needed to avoid delay. Maintain and modify as required. Do not remove facilities until permitted or directed to do so by the Engineer, or until replaced, to the satisfaction of the Engineer, by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

A. General: Engage appropriate local utility company to install temporary service or connect to existing service. Where utility company provides only part of the service, provide the remainder with matching, compatible materials and equipment. Comply with utility company requirements and recommendations.

1. Arrange with utility company, the Engineer, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
2. Provide for adequate capacity and peak-demand or special load requirements at each stage of construction. Before temporary utility is available, provide trucked-in services if necessary.
3. Obtain any necessary easements to bring temporary utilities to Site.
4. Charges and Costs; Pay for all connection, service and user charges and costs arising in connection with the provision of temporary utilities.

B. Sewers and Drainage: If sewers are available, provide temporary connections to remove effluent that can be discharged lawfully. If sewers are not available or cannot be used, provide drainage ditches, dry wells, stabilization ponds, and similar facilities. If neither sewers nor drainage facilities can be lawfully used for discharge of effluent, provide containers to remove and dispose of effluent off-site in a lawful manner.

1. Filter out excessive soil, construction debris, chemicals, oils, and similar contaminants that might clog sewers or pollute waterways before discharge.
2. Connect temporary sewers to municipal system as directed by sewer department officials.
3. Maintain temporary sewers and drainage facilities in a clean, sanitary condition. After heavy use, restore normal conditions promptly.

C. Water Service: Install water service and distribution piping in sizes and pressures adequate for construction.

1. Sterilize temporary water piping before use.
2. Provide storage tanks and/or rubber hoses as necessary to serve Site.
3. If at any time, temporary water service supply is interrupted or inadequate for requirements, provide back-up storage and truck-in water supplies as necessary.
4. If water pressure is inadequate, provide pumps to supply a minimum of 200-kPa static pressure at highest point. Equip pumps with surge and storage tanks and automatic controls to supply water uniformly at reasonable pressures.

D. Water Service: If use of the Employer's existing or permanent water service facilities is permitted, service, clean and maintain in a condition acceptable to the Engineer. At Substantial Completion, restore these facilities to condition existing before initial use.

E. Sanitary Facilities: Provide temporary toilets, wash facilities, and drinking-water fixtures. Comply with regulations, health codes and the Engineer's directions for type, number, location, operation, and maintenance of facilities and fixtures. Provide separate toilet facilities for male and female personnel.
1. Generally: Install where facilities will best serve needs of the Works, including relocation whenever necessary.

    b. Service and maintain, and keep clean, sanitary and orderly for use.
    k. Provide toilet tissue, paper towels, paper cups, and similar disposable materials for each facility. Maintain adequate supply. Provide covered waste containers for disposal of used material.

2. Toilets: Install toilet facilities connected to local water and sewer lines. Provide lavatories, mirrors, and water closets. Provide only potable-water connections. Provide individual compartments for water closets. Provide suitable enclosure with nonabsorbent sanitary finish materials and adequate ventilation and lighting.
3. Wash Facilities: Install wash facilities supplied with potable water at convenient locations for personnel who handle materials that require wash-up. Dispose of drainage properly. Supply cleaning compounds appropriate for each type of material handled.
4. Drinking-Water Facilities: Install containerized, tap-dispenser, bottled-drinking-water units, including paper cup supply.

F. Cooling: Provide temporary cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment from that specified that will not have a harmful effect on completed installations or elements being installed.

G. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment from that specified that will not have a harmful effect on completed installations or elements being installed.

Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.

H. Electric Power Service: Provide weatherproof, grounded electric power service and distribution system of appropriate and sufficient size, capacity, and power characteristics required for use during construction, including necessary start-up, testing and commissioning of building operating systems and equipment. Include meters, transformers, overload-protected disconnecting means, automatic ground-fault interrupters, and main distribution switchgear.

1. Install electric power service underground, unless overhead service must be used.
2. Install power distribution wiring overhead and rise vertically where least exposed to damage.
3. If at any time, temporary electric power service supply is interrupted or inadequate for requirements, provide and operate back-up electric generators as necessary.

I. Electric Power Service: If use of the Employer's existing or permanent electric power service is permitted, maintain equipment in a condition acceptable to the Engineer. At Substantial Completion, restore equipment to condition existing before initial use.

J. Electric Distribution: Provide receptacle outlets adequate for connection of power tools and equipment.

1. Provide waterproof connectors to connect separate lengths of electrical power cords if single lengths will not reach areas where construction activities are in progress. Do not exceed safe length-voltage ratio.

K. Lighting: Provide temporary lighting with appropriate control and local switching to facilitate safe and proper execution of the Works. Provide adequate illumination to internal areas during finishing operations and for inspections.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
2. Install exterior site lighting that will provide adequate illumination for construction operations, traffic conditions, and signage visibility when the Work is being performed.
3. Install lighting for Works identification sign.

L. Telephone Service: Provide and maintain adequate temporary telephone facilities and service for the duration of the Contract including Defects Liability Period, for all personnel accommodated in temporary field offices.

3.3 SUPPORT FACILITIES INSTALLATION

A. General: Comply with the following:

1. Locate field offices, site laboratory, storage and fabrication sheds, workshops, sanitary facilities, and other temporary construction and support facilities for easy access.
2. Maintain support facilities until near Substantial Completion. Unless otherwise directed, remove before Substantial Completion, but retain adequate support facilities to accommodate personnel remaining on Site after Substantial Completion, including relocating if necessary and under conditions acceptable to the Employer / Engineer.

B. Contractor's Site Offices: Provide and maintain temporary offices and facilities of sufficient size, number and type to accommodate required office personnel, including sub-contractors, at the Site. Furnish and equip as necessary. Keep clean and orderly for use.

C. Site Laboratory: Provide temporary site laboratory facility of sufficient size to accommodate, conduct, perform and report on-site sampling and testing of materials as prescribed and specified in various Sections of Divisions 2 through 16 of the Specification. Furnish and equip as necessary. Keep clean and orderly for use.

1. Employ qualified laboratory technicians familiar with specified industry standards and skilled in the testing of construction materials and associated administrative procedures.

D. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate to support loads and to withstand exposure to traffic during construction period. Submit proposals and obtain prior approval to locate temporary roads and paved areas coincidental with and in same location as permanent roads and paved areas. In such cases review any proposed modifications to permanent paving with the Engineer, and:

1. Comply with applicable requirements in appropriate Sections of the Specification for sub-grade, sub-base and base course construction of temporary paving.
2. Coordinate elevations and development of temporary roads and paved areas with subgrade grading, compaction, installation and stabilization of subbase, and installation of base and finish courses of permanent paving.
3. Install temporary roads and paved areas to minimize the need to rework the installations and to result in permanent roads and paved areas without damage or deterioration when occupied by the Employer.
4. Recondition and repair base after temporary use, including removing contaminated material, regrading, proof-rolling, compacting and testing.
5. Delay installation of the final course of permanent pavement until immediately before Substantial Completion. Coordinate with weather conditions to avoid unsatisfactory results.
6. Extend temporary roads and paved areas, within construction limits indicated, as necessary for construction operations, including delivery and storage of materials, equipment usage, administration and supervision.

E. Traffic Controls: Provide temporary traffic controls at junction of temporary roads with public roads. Include warning signs for public traffic and "STOP" signs for entrance onto public roads. Comply with requirements of authorities having jurisdiction.

F. Dewatering Facilities and Drains: Comply with requirements for temporary drainage and dewatering facilities and operations not directly associated with construction activities included in individual Sections. Where feasible, use same facilities. Maintain Site, excavations, and construction free of water.

1. Dispose of rainwater in a manner that will not result in flooding of Site or adjoining property nor endanger Permanent Works or temporary facilities.
2. Before connection and operation of permanent drainage piping system, provide temporary drainage where roofing or similar waterproof deck construction is completed.

G. Works Identification and Temporary Signs: Provide Works identification and other signs of appropriate sizes and types. Install signs to inform public generally and persons seeking entrance to Site. Do not permit installation of unauthorized signs.

1. Design, fabricate and erect **two (2)** Works identification signs in approved locations. Support on posts or framing of paint finished, preservative-treated wood or galvanized steel.
2. Engage an experienced sign manufacturer to apply graphics for Works identification signs. Write sign in **both** English **and Azerbaijani** and include:

    c. Works name.
    l. Employer's name, style and logo.
    m. Engineer's name, style and logo
    n. Contractor's name, style and logo.
    o. Such other names, details and information as permitted and/or directed.

3. Provide other appropriate temporary signs to provide directional and safety information to construction personnel and visitors.

H. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Containerize and clearly label hazardous, dangerous, or unsanitary waste materials separately from other waste. Comply with General Requirements Section "Execution Requirements" for progress cleaning requirements.

1. If required by authorities having jurisdiction, provide separate containers, clearly labeled, for each type of waste material to be deposited.

I Janitorial Services: Provide janitorial cleaning services on a daily basis for all temporary offices, toilets, wash facilities, and similar areas.

J. Storage and Fabrication Sheds: Provide sheds sized, furnished, and equipped to accommodate materials and equipment involved, including temporary utility services. Sheds may be open shelters or fully enclosed spaces within building or elsewhere on site.

K. Construction Aids and Miscellaneous Facilities: Provide all necessary aids and miscellaneous facilities incidental and ancillary to the safe and efficient execution of construction operations and the carrying-out of work in a proper workmanlike manner, including but not necessarily limited to:

1. Scaffolds, ladders, walkways, platforms and guardrails.
2. Protective screens, chutes, coverings and roofs.
3. Small tools; hand and power operated.
4. Mechanical plant, machinery and equipment; stationary and mobile.
5. Tower cranes.
6. Truck mounted cranes.
7. Transportation equipment and vehicles; on and off Site.
8. Lifts and hoists for materials and workmen.
9. Protective clothing, including helmets, goggles, ear protectors and steel-toed safety boots.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects. Avoid using tools and equipment that produce harmful noise. Restrict use of

noisemaking tools and equipment to hours that will minimize complaints and nuisance to persons or establishments near Site.

B. Stormwater Control: Provide earthen embankments and similar barriers in and around excavations and subgrade construction, sufficient to prevent flooding by runoff of stormwater from heavy rains.

C. Tree and Plant Protection: Unless otherwise indicated, install temporary fencing located outside the drip line of trees to protect vegetation from construction damage. Protect tree root systems from damage, flooding, and erosion.

D. Pest Control: Before deep foundation work has been completed, retain a local exterminator or pest-control company to recommend practices to minimize attraction and harboring of rodents, roaches, and other pests. Engage this pest-control service to perform extermination and control procedures at regular intervals so that the Site will be free of pests and their residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials.

E. Site Enclosure Fence: Before construction operations begin install an enclosure fence with lockable entrance gates. Locate where directed. Enclose entire Site or portion determined sufficient to accommodate construction operations. Install in a manner that will prevent people and animals from easily entering site except by entrance gates.
Provide gates in sizes and at locations necessary to accommodate delivery vehicles and other construction operations. Maintain security by limiting number of keys and restricting distribution to authorized personnel.

F. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.

1. Secure Storage: Where materials and/or equipment are of high value or attractive for theft, provide a secure lockup. Enforce discipline in connection with the installation and release of such materials and/or equipment to minimize the opportunity for theft.

G. Barricades, Warning Signs, and Lights: Comply with standards, code requirements and directions of the Engineer for erecting structurally adequate barricades. Paint with appropriate colors, graphics, and warning signs to inform personnel and public of possible hazard. Where appropriate and needed, provide lighting, including flashing red or amber lights.

H. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, inclement weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, provide insulated temporary enclosures. Coordinate enclosure with ventilating and material drying or curing requirements to avoid dangerous conditions and effects.
2. Vertical Openings: Close openings of 2.5 m^2 or less with plywood or similar materials.
3. Horizontal Openings: Close openings in floor or roof decks and horizontal surfaces with load-bearing, wood-framed construction.
4. Install tarpaulins securely using fire-retardant-treated wood framing and other materials.

I. Temporary Partitions: Where necessary, construct, erect and maintain dustproof partitions and temporary enclosures to limit dust and dirt migration, and to separate areas from fumes and noise.

J. Temporary Fire Protection: Until fire-protection needs are supplied by permanent facilities, install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241. Strictly observe all requirements and recommendations pertaining to fire safety on construction sites as contained in the latest edition of the Fire Instruction Manual produced by the Fire Service Department of the Department of Civil Aviation.

1. Provide fire extinguishers, installed on walls on mounting brackets, visible and accessible from space being served, with sign mounted above.

    a. Field Offices: Class A stored-pressure water-type extinguishers.
    b. Other Locations: Class ABC dry-chemical extinguishers or a combination of extinguishers of NFPA-recommended classes for exposures.
    c. Locate fire extinguishers where convenient and effective for their intended purpose; provide not less than one extinguisher on each floor at or near each usable stairwell.

2. Store combustible materials in containers in fire-safe locations.
3. Maintain unobstructed access to fire extinguishers, fire hydrants, temporary fire-protection facilities, stairways, and other access routes for firefighting. Prohibit smoking in hazardous fire-exposure areas.
4. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition.
5. Permanent Fire Protection: At earliest feasible date in each area of Works, complete installation of permanent fire-alarm and fire-protection facilities, including connected services, and place into operation and use. Instruct key personnel on use of facilities.
6. Develop and supervise an overall fire-prevention and first-aid fire-protection program for personnel at Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
7. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.5 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal. Protect from damage caused by extreme temperatures and similar elements.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.
2. Maintain markers for underground lines. Protect from damage during excavation operations.

C. Termination and Removal: At the Engineer's sole discretion, remove each temporary facility either when need for its service has ended, or when it has been replaced by authorized use of a permanent facility, or at Substantial Completion. Complete, or if necessary, restore permanent construction that may have been delayed because of interference with a temporary facility. Repair damaged work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Unless otherwise specified, materials, products, equipment and facilities that constitute temporary facilities are the property of the Contractor.
2. The Employer reserves right to take possession of Works identification signs.
3. Remove temporary paving not intended for or acceptable for integration into permanent paving. Where area is intended for landscape development, remove soil and aggregate fill that do not comply with requirements for fill or subsoil. Remove materials contaminated with road oil, asphalt and other petrochemical compounds, and other substances that might impair growth of plant materials or lawns. Repair or replace street paving, curbs, and sidewalks at temporary entrances, damaged by construction operations, as required by authorities having jurisdiction.
4. At Substantial Completion, clean and renovate permanent facilities used during construction period. Comply with final cleaning requirements in General Requirements Section "Closeout Procedures."

3.6 SAFETY PROGRAM

A. In addition to the requirements of the provisions of the General Conditions of Contract and other Contract requirements, the Contractor shall submit for the Engineer's approval, within 15 days after the effective date of the notice to proceed for the Works, a proposed Safety Program covering the following items:

1. Health and Safety Plan:

   a. The Health and Safety Plan shall contain but not be limited to the following:

      1) Safety Policy and Strategy.
      2) Statutory and Contractual Framework for Safety.
      3) Management Safety Responsibilities.
      4) Safety Training and awareness.
      5) Safety Reviews.
      6) Safety Method Statements.
      7) Reporting and Investigation.
      8) Contingency Arrangements.
      9) Safety Inspections and Audits.
      10) Accident Procedures.
      11) Site cleanliness and removal of rubbish.
      12) Labor messing facilities.

   b. The Contractor shall take into account that the works under this Contract may be undertaken simultaneously with other Package Contracts. The Contractor is to coordinate safety procedures with the Engineer and Safety Representatives from other Package Contracts to implement and maintain a common safety strategy.
   c. Site Cleaning and Housekeeping: Each Package Contractor shall be responsible for cleaning and housekeeping in their own work areas. The provision of waste

chutes, garbage skips and disposal shall be provided by Superstructure Civil Works Package Contractor.

2. Health and Safety Representatives:

   a. The Contractor shall include within the Health and Safety Plan the name of the company's qualified Safety Officer. He shall also submit the name and qualification of a Health and Safety representative on site responsible for the implementation of the Health and Safety Plan on site. The site safety representative shall remain on site during operational working hours and shall coordinate with the safety representatives of other package Contractors.
   b. Appointment of the site Health and Safety representative will be subject to the Engineer's approval.
   c. If in the opinion of the Engineer there is an infringement or breach in the approved Health and Safety Plan, the Engineer reserves the right to instruct the Contractor to undertake immediate corrective action without any liability for any additional costs or time implication incurred by the Contractor in undertaking the corrective action.

B. Upon the Engineer's approval of the Safety Program the Contractor shall, for the full term of the Contract, operate the Safety Program, maintain accurate records of safety activities and accident, and submit safety and accident reports to the Engineer on the approved forms.

END OF SECTION 01500

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SECTION 01600 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 PROPRIETARY NAMES

A. In accordance with statutory requirements, unless unavoidable, no proprietary products or manufacturers have been named within this Specification.

1.3 SUMMARY

A. This Section includes administrative and procedural requirements relating to products to be used in the Works, including the following:

1. Selection of products for the Works.
2. Quality assurance.
3. Product delivery, storage, and handling.
4. Product warranties.
5. Product substitutions.
6. Comparable products.

B. Related Sections include the following:

1. General Requirements Section "References Standards and Definitions" for applicable industry standards for products specified.
2. General Requirements Section "Execution Requirements" for general procedural requirements governing execution of the Work.
3. General Requirements Section "Closeout Procedures" for submitting warranties for contract closeout.
4. All Specification Sections for specific requirements for warranties on products and installations specified to be warranted.

1.4 DEFINITIONS

A. Products: Items purchased for incorporating into the Works, whether purchased for the Works or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.

1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation, shown or listed in manufacturer's published product literature, that is current as of date of the Contract Documents.
2. New Products: Items that have not previously been incorporated into another project or facility. Products salvaged or recycled from other projects are not considered new products.

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3. Comparable Product: Product that is demonstrated and approved through submittal process, to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed the requirements of the Specification.
4. Local Products: Naturally occurring materials found in Azerbaijan that satisfy the required standards, products manufactured in Azerbaijan from naturally occurring materials found in Azerbaijan that satisfy the required standards, or products manufactured in Azerbaijan that satisfy the required standards.
5. Other Products: All other products required for the Works that satisfy the required standards and which are excluded from the category of "Local Products".

B. Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents and proposed by the Contractor after the date of issue of the Letter of Acceptance. The following are not Substitutions:

1. Revisions to Contract Documents required by the Engineer or Employer.
2. Options for products, materials, equipment and methods of construction, already specified in Contract Documents, including "or approved equal" and “but are not limited to” options.
3. Contractor's compliance with governing regulations or lawful orders issued by authorities having jurisdiction.

C. Manufacturer's Instructions and Manufacturer's Recommendations mean latest published or printed version of 'manufacturer's written instructions' and 'manufacturer's written recommendations.

D. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for a particular product and specifically endorsed by manufacturer to the Employer.

E. Special Warranty: Written warranty required by or incorporated into the Contract Documents, whether to extend time limit provided by manufacturer's warranty, or to provide for joint and several liability, or to provide more rights for the Employer.

1.5 SUBMITTALS

A. Product List: Prepare and submit a list, in tabular form, identifying those products that the Contractor intends to propose for the Works and include the generic names of such products. Include manufacturer's and supplier's name and proprietary product name for each product.

1. Coordinate product list with Contractor's Construction Program, the Submittals Schedule, and the Subcontract List.
2. Form: Tabulate information for each proposed product under the following column headings:

a. Specification Section number and title.
b. Generic name used in the Contract Documents.
c. Specified warranty requirements, if any.
d. Proprietary name, model number, and similar designations.
e. Manufacturer's name and address.
f. Supplier's name and address.
g. Installer's name and address.
h. Total quantity required for the Works.
i. Projected delivery date or, if multiple deliveries, first and last delivery dates.

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j. Identification of items that require early submittal approval for scheduled delivery date.

3. Initial Submittal: Within 14 days after Commencement Date, submit 3 copies of initial product list. Include a written explanation for omissions of data and for variations from Contract requirements.

a. At Contractor's option, initial submittal may be limited to product selections and designations that must be established early in Contract period.

4. Completed List: Within 42 days after Commencement Date, submit 3 copies of completed product list. Include a written explanation for omissions of data and for variations from Contract requirements.
5. Engineer's Action: The Engineer will respond in writing to the Contractor within 28 days of receipt of completed product list. The Engineer's response will include a list of unacceptable product selections and a brief explanation of reasons for this action. The Engineer's response, or lack of response, does not constitute a waiver of requirement that products comply with the Contract Documents.

B. Substitution Requests: Submit three copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.

1. Substitution Request Form: Use a form acceptable to the Engineer.
2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:

a. Statement indicating why specified material or product could not be provided.
b. Coordination information, including a list of changes or modifications needed to other parts of the Works and to construction performed by the Employer and separate contractors, which will be necessary to accommodate proposed substitution.
c. Detailed comparison of significant qualities of proposed substitution with those of the product or work specified. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
e. Samples, where applicable or if requested.
f. List of similar installations for completed projects with project names and addresses and names and addresses of architects/engineers and owners.
g. Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.
h. Detailed comparison of Contractor's Construction Program using proposed substitution with products specified for the Works, including effect, if any, on the overall Time for Completion. If specified product or method of construction cannot be provided within the Time for Completion, include letter from manufacturer, stating reason for non-availability and/or delays in delivery.
i. Cost information, including a proposal of change, if any, in the Contract Sum.
j. Contractor's certification that proposed substitution complies with requirements in the Contract Documents and is appropriate for applications indicated.

3. Engineer's Action: If necessary, the Engineer will request additional information or documentation for evaluation within one week of receipt of a request for substitution.

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The Engineer will notify Contractor of acceptance or rejection of proposed substitution within twenty eight (28) days of receipt of request, or within fourteen (14) days of receipt of additional information or documentation, whichever is later.

a. Form of Acceptance: Variation Order.
b. Use product specified if the Engineer cannot make a decision on use of a proposed substitution within time allocated.

C. Comparable Product Requests: Submit three copies of each request for consideration. Identify product or fabrication to be replaced. Include Specification Section number and title and Drawing numbers and titles.

1. Request Form: Use a form acceptable to the Engineer.
2. Documentation: Show compliance with requirements for approval.
3. Engineer's Action: If necessary, the Engineer will request additional information or documentation for evaluation within one week of receipt of a request. The Engineer will notify the Contractor of acceptance or rejection of proposed comparable product within twenty eight days of receipt of request, or within fourteen (14) days of receipt of additional information or documentation, whichever is later.

a. Form of Acceptance: Variation Order, without additional compensation.
b. Use product specified if the Engineer cannot make a decision on use of a proposed comparable product within time allocated.

1.6 QUALITY ASSURANCE

A. Alternative Equivalent Standards: Product references to industry standards establish quantitative and qualitative attributes, characteristics and properties, required to be inherent in the specified product. Standard products not conforming to referenced standards, but otherwise exhibiting the same essential properties and conforming to another recognized equivalent industry standard may be used, subject to approval and compliance with other specified requirements.

1. Recognized standards are those which are listed in General Requirements Section " References Standards and Definitions ".
2. Comply with provisions in "Comparable Products" Article to obtain approval for use of a product conforming to an alternative equivalent standard.

B. Compatibility of Options: If the Contractor is given the option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1. Each contractor is responsible for providing products and construction methods compatible with products and construction methods of other contractors.
2. If a dispute arises between contractors over concurrently selectable but incompatible products, the Engineer will determine which products shall be used.

C. Source Limitations: To the fullest extent possible, provide products of the same type and kind from a single source.

1. If a specified product is available only from a source that cannot produce or supply sufficient quantities to execute and/or complete requirements in a timely manner, request the Engineer to determine the most essential qualities, attributes and

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characteristics required in the specified product, in order to select a comparable product possessing the same relevant, significant properties.

2. Comply with provisions in "Comparable Products" Article to obtain approval.

1.7 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.

1. Schedule delivery to minimize long-term storage at Site and to prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to ensure compliance with the Contract Documents and to ensure that products are undamaged and properly protected.
5. Store products to allow for inspection and measurement of quantity or counting of units.
6. Store materials in a manner that will not endanger the structure of the Permanent Works.
7. Store products that are subject to damage by the elements, under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation.
8. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
9. Protect stored products from damage.

B. Storage: Provide a secure location and enclosure at Site for storage of materials and equipment, if any, to be furnished by the Contractor. Coordinate location with the Engineer.

1.8 PRODUCT WARRANTIES

A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve the Contractor of any obligations under requirements of the Contract Documents.

B. Warranty Requirements:

1. Related Damages and Losses: When correcting failed or damaged warranted construction, remove and replace construction that has been damaged as a result of such failure or must be removed and replaced to provide access for correction of warranted construction.
2. Reinstatement of Warranty: When work covered by a warranty has failed and been corrected by replacement or rebuilding, reinstate the warranty by written endorsement. The reinstated warranty shall be equal to the original warranty with an equitable adjustment for depreciation.
3. Replacement Cost: Upon determination that work covered by a warranty has failed, replace or rebuild the work to an acceptable condition complying with requirements of the Contract Documents. The Contractor is responsible for the cost of replacing or

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rebuilding defective work regardless of whether the Employer has benefited from use of the work through a portion of its anticipated useful service life.

4. Employer's Recourse: Expressed warranties made to the Employer are in addition to implied warranties and shall not limit the duties, obligations, rights, and remedies otherwise available under the Alat FEZ law. Expressed warranty periods shall not be interpreted as limitations on the time in which the Employer can enforce such other duties, obligations, rights, or remedies.

C. Rejection of Warranties: The Employer reserves the right to reject warranties and to limit selection to products with warranties not in conflict with requirements of the Contract Documents.

D. Where the Contract Documents require a special warranty, or similar commitment on the Works or part of the Works, the Employer reserves the right to refuse to accept the Works, until the Contractor presents evidence that entities required to countersign such commitments are willing to do so.

E. Manufacturer's and Special Warranties: Prepare a written document, ready for execution, that contains appropriate terms and conditions, and identifies commencement date and warranty period. Submit a draft and obtain the Engineer's written approval before final product selection.

1. Manufacturer's Standard Form: Modified to include requirements and information specific to the Works, and properly executed; or
2. Special Form: Individually prepared and specially drafted to include requirements and information specific to the Works, and properly executed.
3. Refer to Divisions 2 through 16 Sections for specific content requirements and particular requirements for submitting special warranties.

F. Submittal of Warranties: Comply with requirements in General Requirements Section "Closeout Procedures."

PART 2 - PRODUCTS

2.1 SELECTION OF PRODUCTS WHERE PROPRIETARY NAMES ARE NOT USED

A. All products shall be to a standard not less than the minimum requirements specified.

B. In addition to compliance with the Specification, the Contractor shall submit such further documentation and warranties as the Engineer may request to demonstrate compliance.

C. The products selected and submitted by the Contractor shall fulfill the following criteria:

1. In selecting and submitting products for approval, the Contractor shall only submit products where the stated performance criteria are fully adequate for the operation intended and the Contractor in making the submittal warrants the fitness of purpose of the products.
2. Product manufacturers shall be fully qualified to manufacture the respective products and they shall possess a proven track record of not less than 5 (five) years (preferably 10 years) manufacturing experience in their field. Those with the maximum experience shall be given preference in approval by the Engineer.
3. "Other Products" shall be either manufactured in, or be a naturally occurring material found in, any of the following geographical locations: Western Europe, North America,

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Australia or Japan. However, Manufacturers originating from the stated geographical locations who operate in a geographical location other than the stated ones are also acceptable provided that the Contractor can demonstrate that the parent manufacturing company still operates in one of the stated geographical locations.

4. Submittals for "Other Products" assembled in Azerbaijan must include evidence that demonstrates that the assembly is approved by the original manufacturer.
5. Submittals shall only include products where manufacturers can certify ‘forward compatibility’ (as applicable) of current models of their products with future models.
6. Submittals shall only include products produced by manufacturers who have received ISO certification for a period of not less than 3 years (preferably 7 years) and preference shall be given by the Engineer to those possessing the longest periods of certification.
7. Submittals shall only include products produced by manufacturers who have had not less than 3 years commercial experience (preferably 7 years) and preference shall be given by the Engineer to those possessing the longest periods.
8. Submittals shall include evidence to demonstrate that the various components of the products are certified by the product manufacturer with respect to specific suitability, country of origin, quality and reliability.
9. Submittals shall include type test certificates (as applicable) issued by an Internationally reputable and accredited testing facility.
10. Product submittals (as applicable) shall include a re-submittal of the following information as originally submitted by the Contractor for all products included in his Tender lists for "Local Products" and "Other Products":

    a. Guarantee particulars of the Product.
    b. Original catalog of the proposed product manufacturer.
    c. Schedule of users (including names of contact persons) for the previous 5 years.
    d. Schedule of recommended minimum spares based on normal running/usage in the environment proposed projected over a ten year period together with current price lists (major equipment).
    e. Life cycle cost for the product covering the cost of the equipment, spare parts, maintenance and energy consumption in the environment proposed for the lifetime of the product.
    f. Manufacturers' website address.

2.2 PRODUCT OPTIONS

A. General Product Requirements: Provide products that comply with the Contract Documents, that are undamaged, and unless otherwise indicated, that are new at time of installation.

1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
3. The Employer reserves the right to limit selection to products with warranties not in conflict with requirements of the Contract Documents.
4. Where products are accompanied by the term "as selected," the Engineer will make selection.
5. Where products are accompanied by the term "match sample," sample to be matched is the Engineer's.
6. Descriptive, performance, and reference standard requirements in the Specifications establish "salient characteristics" of products.

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7. Or Approved Equal: Where products are specified by name and accompanied by such terms as "or other equal and approved", "or approved equal" "or equal", or “but are not limited to” comply with provisions in "Comparable Products" Article to obtain approval for use of an unnamed product.

    a. The Engineer reserves the right to limit selection to named products specified.
    b. Unless otherwise agreed, contract unit rates and prices will be deemed to be based on the use of named products specified in the Contract Documents.

B. Product Selection Procedures: Procedures for product selection include the following:

1. Product: Where Specification paragraphs or subparagraphs titled "Product" name a single product and manufacturer, provide the product named, unless otherwise indicated.
2. Manufacturer/Source: Where Specification paragraphs or subparagraphs titled "Manufacturer" or "Source" name single manufacturers or sources, provide a product by the manufacturer or from the source named that complies with requirements, unless otherwise indicated.
3. Products: Where Specification paragraphs or subparagraphs titled "Products" introduce a list of names of both products and manufacturers, provide one of the products listed that complies with requirements, unless otherwise indicated.
4. Manufacturers: Where Specification paragraphs or subparagraphs titled "Manufacturers" introduce a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements, unless otherwise indicated.
5. Available Products: Where Specification paragraphs or subparagraphs titled "Available Products" introduce a list of names of both products and manufacturers, provide one of the products listed or another product that complies with requirements. Comply with provisions in "Comparable Products" Article to obtain approval for use of an unnamed product.
6. Available Manufacturers: Where Specification paragraphs or subparagraphs titled "Available Manufacturers" introduce a list of manufacturers' names, provide a product by one of the manufacturers listed or another manufacturer that complies with requirements. Comply with provisions in "Comparable Products" Article to obtain approval for use of a product by another manufacturer.
7. Product Options: Where Specification paragraphs titled "Product Options" indicate that size, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide either the specific product or system indicated or a comparable product or system by another manufacturer. Comply with provisions in "Comparable Products" Article to obtain approval for use of an unnamed product.
8. Basis-of-Design Products: Where Specification paragraphs or subparagraphs titled "Basis-of-Design Product" are included and also introduce or refer to a list of manufacturers' names, provide either the specified product or a comparable product by one of the other named manufacturers. Comply with provisions in "Comparable Products" Article to obtain approval for use of an unnamed product.
9. Visual Matching Specification: Where Specifications require matching an established Sample, select a product (and manufacturer) that complies with requirements and matches the Engineer's sample. The Engineer's decision will be final on whether a proposed product matches satisfactorily.
10. Visual Selection Specification: Where Specifications include the phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product (and manufacturer) that complies with other specified requirements.

    a. Standard Range: Where Specifications include the phrase "standard range of colors, patterns, textures" or similar phrase, the Engineer will select color,

7.

pattern, or texture from manufacturer's product line that does not include premium items.

b. Full Range: Where Specifications include the phrase "full range of colors, patterns, textures" or similar phrase, the Engineer will select color, pattern, or texture from manufacturer's product line that includes both standard and premium items.

2.3 PRODUCT SUBSTITUTIONS

A. Timing: The Engineer will consider requests for substitution if received within sixty (60) days after date of Letter of Acceptance. Requests received after that time may be considered or rejected at discretion of the Engineer.

B. Conditions: The Engineer will consider the Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, the Engineer will return requests without action, except to record noncompliance with these requirements:

1. Requested substitution offers the Employer a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities the Employer must assume. The Employer's additional responsibilities may include compensation to the Engineer for redesign and evaluation services, increased cost of other construction by the Employer, and similar considerations.
2. Requested substitution does not require extensive revisions to the Contract Documents.
3. Requested substitution is consistent with the Contract Documents and will produce indicated results.
4. Substitution request is fully documented and properly submitted.
5. Requested substitution will not adversely affect the Contractor's Construction Program.
6. Requested substitution has received necessary approvals of authorities having jurisdiction.
7. Requested substitution is compatible with other portions of the Works.
8. Requested substitution has been coordinated with other portions of the Works.
9. Requested substitution provides specified warranty.
10. If requested substitution involves other contractors, requested substitution has been coordinated with the work of such other contractors, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

2.4 COMPARABLE PRODUCTS

A. Where products are specified by name or industry standard, submit the following, in addition to other required submittals, to obtain approval of an unnamed or alternative equivalent industry standard, product:

1. Evidence that the proposed product does not require extensive revisions to the Contract Documents, that it is consistent with the Contract Documents and will produce the indicated results, and that it is compatible with other portions of the Works.
2. Detailed comparison of significant qualities of proposed product with those named in the Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
3. Evidence that proposed product provides specified warranty, if any.
4. List of similar installations for completed projects with project names and addresses and names and addresses of architects/engineers and employers, if requested.
5. Samples, if requested.

7.

PART 3 - EXECUTION (Not Used)

END OF SECTION 01600

8.

SECTION 01700 - EXECUTION REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes general procedural requirements governing execution of the Works including, but not limited to, the following:

1. Construction layout.
2. Field engineering and surveying.
3. General installation of products.
4. Progress cleaning.
5. Starting and adjusting.
6. Protection of installed construction.
7. Correction of work.

B. Related Sections include the following:

1. General Requirements Section "Project Management and Coordination" for procedures for coordinating field engineering with other construction activities.
2. General Requirements Section "Submittal Procedures" for submitting surveys.
3. General Requirements Section "Product Requirements" for procedural requirements relating to products to be used in the Work.
4. General Requirements Section "Closeout Procedures" for final cleaning.
5. General Requirements Section "Cutting and Patching" for procedural requirements for cutting and patching necessary for the installation or performance of other components of the Work.

1.3 SUBMITTALS

A. Method Statement: Submit method statement to be used for construction layout.

B. Landfill Receipts: If Contract Documents contain requirement for hazardous waste disposal, submit copy of receipts issued by a landfill facility, licensed to accept hazardous materials.

PART 2 - PRODUCTS (Not Used)

8.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Existing Conditions: The existence and location of site improvements, utilities, and other construction indicated as existing are not guaranteed. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Works.

1. Before construction, verify the location and points of connection of utility services.

B. Existing Utilities: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify the existence and location of underground utilities and other construction affecting the Works.

1. Before construction, verify the location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.
2. Furnish location data for work that must be performed by public utilities serving the Site.

C. Project Site Information: A sub-surface site investigation report has been prepared for this project and has been made available for baseline information only. The report is not part of the Contract. The Contractor is entirely responsible for his own assessment, interpretation, use and conclusions drawn from the information, data, tests, analyses and opinions contained in the report.

1. Make additional test borings and conduct other exploratory operations as deemed necessary, at own cost and expense.

3.2 PREPARATION

A. Existing Utility Information: Furnish information to the Engineer that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate, where necessary, with authorities having jurisdiction.

B. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by the Employer or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

1. Notify the Engineer and Employer not less than two (2) days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without the Engineer's written permission.

C. Field Measurements: Take field measurements as required to fit the construction properly. Recheck measurements before installing each product. Where portions of the Works are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Works.

8.

D. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

E. Review of Contract Documents and Field Conditions: Immediately on discovery of any discrepancy or the need for a clarification of the Contract Documents, submit a request for information or instruction to the Engineer. Include a detailed description of discrepancy or problem encountered, together with recommendations.

3.3 CONSTRUCTION LAYOUT

A. Method Statement: Before verifying layout information shown on Drawings, prepare a detailed method statement comprising a descriptive narrative and drawings, clearly describing and identifying means and methods to be used for construction layout including but not limited to:

1. Establishing benchmarks and control points to set lines and levels at each story of construction and elsewhere as needed to locate each element of the Works.
2. Locating offsets for gridlines.
3. Locating and sizing penetrations in structures as needed to transfer lines and levels horizontally and vertically.

B. Verification: Before proceeding to set out the Works, verify layout information shown on Drawings, in relation to the Contractor's site survey and existing benchmarks. If discrepancies are discovered, notify the Engineer promptly.

C. General: Set out the Works using accepted surveying techniques and practices.

1. Establish benchmarks and control points to set lines and levels at each story of construction and elsewhere as needed to locate each element of the Works.
2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
3. Inform installers of lines and levels to which they must comply.
4. Check the location, level and plumb, of every major element as work progresses.
5. Notify the Engineer when deviations from required lines and levels exceed allowable tolerances.
6. Close site surveys with an error of closure equal to or less than the standard indicated or otherwise as acceptable to the Engineer.

D. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.

E. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

F. Record Log: Maintain a log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make the log available for reference by the Engineer.

1. Advise the Engineer when deviations, exceeding indicated or recognized tolerances, are detected.

8.

2. Record deviations that are accepted and not corrected on Record (As-Built) Drawings.

G. Auxiliary Services: Cooperate with the Engineer and provide, when requested, auxiliary services to enable and assist the Engineer to check the Contractor's site survey, layout and control work, including means of access to Site, use of instruments and tapes, and supply of survey crew.

3.4 FIELD ENGINEERING

A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning the Works. Preserve and protect permanent benchmarks and control points during construction operations.

1. Do not change or relocate existing benchmarks or control points without prior written approval of the Engineer. Report lost or destroyed permanent benchmarks or control points promptly. Report the need to relocate permanent benchmarks or control points to the Engineer before proceeding.
2. Replace lost or destroyed permanent benchmarks and control points promptly. Base replacements on the original survey control points.

B. Benchmarks: Establish and maintain a minimum of two (2) permanent benchmarks on Site, referenced to data established by survey control points. Comply with the Engineer's instructions for type and size of benchmark.

1. Record benchmark locations, with horizontal and vertical data, on Record Drawings (to be used for As-Built drawings).
2. Where the actual location or elevation of layout points cannot be marked, provide temporary reference points sufficient to locate the Works.
3. Remove temporary reference points when no longer needed. Restore marked construction to its original condition.

C. Record Log: Maintain a log of field engineering work. Include dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make the log available for reference by the Engineer.

3.5 INSTALLATION

A. General: Locate the Works and components of the Works accurately, in correct alignment and elevation, as indicated.

1. Make vertical work plumb and make horizontal work level.
2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.

B. Comply with the manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions that will ensure the best possible results. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Works is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

8.

E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.

F. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of the Work.

1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by the Engineer.
2. Allow for building movement, including thermal expansion and contraction.

G. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for the best visual effect. Fit exposed connections together to form hairline joints.

H. Hazardous Materials: Use products, cleaners, and installation materials that are not considered hazardous.

3.6 EMPLOYER-INSTALLED PRODUCTS

None

3.7 PROGRESS CLEANING

A. General: Clean Site and work areas daily, including common areas. Coordinate and contribute to progress cleaning for Site and work areas jointly-used with other separate contractors. Enforce requirements strictly. Dispose of materials lawfully.

1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.
2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 27 deg. C (81 deg. F).
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.

B. Site: Maintain Site free of waste materials and debris.

C. Work Areas: Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the Works.

1. Remove liquid spills promptly.
2. Where dust would impair proper execution of work, broom-clean or vacuum the entire work area, as appropriate.

D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and that will not damage exposed surfaces.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at time of Substantial Completion.

8.

G. Cutting and Patching: Clean areas and spaces where cutting and patching are performed. Completely remove paint, mortar, oils, putty, and similar materials.

1. Thoroughly clean piping, conduit, and similar features before applying paint or other finishing materials. Restore damaged pipe covering to its original condition.

H. Waste Disposal: Burying or burning waste materials on site will not be permitted. Washing waste materials down sewers or into waterways will not be permitted.

I. During handling and installation, clean and protect construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

J. Clean and provide maintenance on completed construction as frequently as necessary through the remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

K. Limiting Exposures: Supervise construction operations to assure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period. Where applicable, such exposures include, but are not limited to, the following:

1. Excessive static or dynamic loading.
2. Excessive internal or external pressures.
3. Excessively high or low temperatures.
4. Thermal shock.
5. Excessively high or low humidity.
6. Pollution and air contamination.
7. Water.
8. Chemicals and solvents.
9. Light.
10. Radiation.
11. Puncture.
12. Abrasion.
13. Heavy traffic.
14. Soiling, staining, and corrosion.
15. Bacteria.
16. Rodent and insect infestation.
17. Combustion.
18. Electrical current.
19. High-speed operation.
20. Improper lubrication.
21. Unusual wear or other misuse.
22. Contact between incompatible materials.
23. Destructive testing.
24. Misalignment.
25. Excessive weathering.
26. Unprotected storage.
27. Improper shipping or handling.
28. Theft or vandalism.

3.8 STARTING AND ADJUSTING

8.

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.

C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

D. Manufacturer's Field Service: If a factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in General Requirements Section "Quality Requirements."

3.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed work is without damage or deterioration at time of Substantial Completion.

B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.10 CORRECTION OF WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in General Requirements Section "Cutting and Patching."

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Restore permanent facilities used during construction to their original condition, unless otherwise specified.

C. Remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence of repair.

D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01700

9.

SECTION 01731 - SELECTIVE DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Demolition and removal of selected portions of a building or structure.
2. Demolition and removal of selected site elements.
3. Repair procedures for selective demolition operations.

B. Related Sections include the following:
1. General Requirements Section "Photographic Documentation" for preconstruction photographs taken before selective demolition.
2. General Requirements Section "Temporary Facilities and Controls" for temporary construction and environmental-protection measures for selective demolition operations.
3. General Requirements Section "Cutting and Patching" for cutting and patching procedures for selective demolition operations.

1.3 DEFINITIONS

A. Remove: Detach items from existing construction and legally dispose of them off-site, unless indicated to be removed and salvaged or removed and reinstalled.

B. Remove and Salvage: Detach items from existing construction and deliver them to the Employer ready for reuse.

C. Remove and Reinstall: Detach items from existing construction, prepare them for reuse, and reinstall them where indicated.

D. Existing to Remain: Existing items of construction that are not to be removed and that are not otherwise indicated to be removed, removed and salvaged, or removed and reinstalled.

1.4 MATERIALS OWNERSHIP

A. Except for items or materials indicated to be reused, salvaged, reinstalled, or otherwise indicated to remain the Employer's property, demolished materials shall become the Contractor's property and shall be removed from Project site.

9.

B. Historic items, relics, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, antiques, and other items of interest or value to the Employer that may be encountered during selective demolition remain Employer's property. Carefully remove and salvage each item or object in a manner to prevent damage and deliver promptly to the Employer.

1. Coordinate with the Employer's archaeologist or historical adviser, who will establish special procedures for removal and salvage.

1.5 SUBMITTALS

A. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

B. Proposed Dust-Control and Noise-Control Measures: Submit statement or drawing that indicates the measures proposed for use, proposed locations, and proposed time frame for their operation. Identify options if proposed measures are later determined to be inadequate.

C. Schedule of Selective Demolition Activities: Indicate the following:

1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity. Ensure that the Employer's building manager's and other tenants' on-site operations are uninterrupted.
2. Interruption of utility services.
3. Coordination for shutoff, capping, and continuation of utility services.
4. Use of elevator and stairs.
5. Locations of temporary partitions and means of egress, including for other tenants affected by selective demolition operations.
6. Coordination of Employer's continuing occupancy of portions of existing building and of Employer's partial occupancy of completed Work.

D. Inventory: After selective demolition is complete, submit a list of items that have been removed and salvaged.

E. Predemolition Photographs or Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by selective demolition operations. Submit before Work begins.

F. Landfill Records: Indicate receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Demolition Firm Qualifications: An experienced firm that has specialized in demolition work similar in material and extent to that indicated for this Project.

C. Professional Engineer Qualifications: Comply with General Requirements Section "Quality Requirements."

9.

D. Regulatory Requirements: Comply with governing Environmental Protection Agency (EPA) notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

E. Standards: Comply with ANSI A10.6 and NFPA 241.

F. Predemolition Conference: Conduct conference at Project site to comply with requirements in General Requirements Section "Project Management and Coordination."

1. Inspect and discuss condition of construction to be selectively demolished.
2. Review structural load limitations of existing structure.
3. Review and finalize selective demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
4. Review requirements of work performed by other trades that rely on substrates exposed by selective demolition operations.

1.7 PROJECT CONDITIONS

A. Employer will occupy portions of the building immediately adjacent to selective demolition area. Conduct selective demolition so that the Employer's operations will not be disrupted. Provide not less than seventy two (72) hours' notice to the Employer of activities that will affect the Employer's operations.

B. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities.

1. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from authorities having jurisdiction.

C. The Employer assumes no responsibility for condition of areas to be selectively demolished.

1. Conditions existing at time of inspection for bidding purpose will be maintained by the Employer as far as practical.

D. Hazardous Materials: It is not expected that hazardous materials will be encountered in the Work, but if they are:

1. Hazardous materials will be removed by the Employer before start of the Work.
2. If materials suspected of containing hazardous materials are encountered, do not disturb; immediately notify the Engineer and the Employer. Hazardous materials will be removed by the Employer under a separate contract.

E. Storage or sale of removed items or materials on-site will not be permitted.

F. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.

1. Maintain fire-protection facilities in service during selective demolition operations.

PART 2 - PRODUCTS

9.

2.1 REPAIR MATERIALS

A. Use repair materials identical to existing materials.

1. If identical materials are unavailable or cannot be used for exposed surfaces, use materials that visually match existing adjacent surfaces to the fullest extent possible.
2. Use materials whose installed performance equals or surpasses that of existing materials.

C. Comply with material and installation requirements specified in individual Specification Sections.

PART 3 - EXECUTION

1.1 EXAMINATION

A. Verify that utilities have been disconnected and capped.

B. Survey existing conditions and correlate with requirements indicated to determine extent of selective demolition required.

C. Take inventory and record the condition of items to be removed and reinstalled and items to be removed and salvaged.

D. When unanticipated mechanical, electrical, or structural elements that conflict with intended function or design are encountered, investigate and measure the nature and extent of the conflict. Promptly submit a written report to the Engineer.

E. Engage a professional engineer to survey the condition of the building to determine whether removing any element might result in structural deficiency or unplanned collapse of any portion of structure or adjacent structures during selective demolition operations.

F. Perform surveys as the Work progresses to detect hazards resulting from selective demolition activities.

1.2 UTILITY SERVICES

A. Existing Utilities: Maintain services indicated to remain and protect them against damage during selective demolition operations.

B. Do not interrupt existing utilities serving occupied or operating facilities unless authorized in writing by the Employer and authorities having jurisdiction. Provide temporary services during interruptions to existing utilities, as acceptable to the Employer and to authorities having jurisdiction.

1. Provide at least seventy two (72) hours' notice to the Employer if shutdown of service is required during changeover.

C. Utility Requirements: Locate, identify, disconnect, and seal or cap off indicated utilities serving areas to be selectively demolished.

9.

1. The Employer/Building manager will arrange to shut off indicated utilities when requested by the Contractor.

D. Utility Requirements: Refer to MEP Specifications.

1.3 PREPARATION

A. Dangerous Materials: Drain, purge, or otherwise remove, collect, and dispose of chemicals, gases, explosives, acids, flammables, or other dangerous materials before proceeding with selective demolition operations.

B. Pest Control: Employ a certified, licensed exterminator to treat building and to control rodents and vermin before and during selective demolition operations.

C. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

1. Do not close or obstruct streets, walks, walkways, or other adjacent occupied or used facilities without permission from the Employer and authorities having jurisdiction. Provide alternate routes around closed or obstructed trafficways if required by governing regulations.
2. Erect temporary protection, such as walks, fences, railings, canopies, and covered passageways, where required by authorities having jurisdiction.
3. Protect existing site improvements, appurtenances, and landscaping to remain.
4. Erect a plainly visible fence around drip line of individual trees or around perimeter drip line of groups of trees to remain.

D. Temporary Facilities: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.

1. Provide protection to ensure safe passage of people around selective demolition area and to and from occupied portions of building.
2. Provide temporary weather protection, during interval between selective demolition of existing construction on exterior surfaces and new construction, to prevent water leakage and damage to structure and interior areas.
3. Protect walls, ceilings, floors, and other existing finish work that are to remain or that are exposed during selective demolition operations.
4. Cover and protect furniture, furnishings, and equipment that have not been removed.

E. Temporary Enclosures: Provide temporary enclosures for protection of existing building and construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, provide insulated temporary enclosures. Coordinate enclosure with ventilating and material drying or curing requirements to avoid dangerous conditions and effects.

1.4 POLLUTION CONTROLS

1.5 SELECTIVE DEMOLITION

9.

A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:

1. Proceed with selective demolition systematically, from higher to lower level. Complete selective demolition operations above each floor or tier before disturbing supporting members on the next lower level.
2. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping, to minimize disturbance of adjacent surfaces. Temporarily cover openings to remain.
3. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
4. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations.
5. Maintain adequate ventilation when using cutting torches.
6. Remove decayed, vermin-infested, or otherwise dangerous or unsuitable materials and promptly dispose of off-site.
7. Remove structural framing members and lower to ground by method suitable to avoid free fall and to prevent ground impact or dust generation.
8. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.
9. Dispose of demolished items and materials promptly.
10. Return elements of construction and surfaces that are to remain to condition existing before selective demolition operations began.
11. Existing Facilities: Comply with the building manager's requirements for using and protecting elevators, stairs, walkways, loading docks, building entries, and other building facilities during selective demolition operations.

B. Removed and Salvaged Items: Comply with the following:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in a secure area until delivery to the Employer.
4. Transport items to the Employer's storage area as indicated on the Drawings.
5. Protect items from damage during transport and storage.

C. Removed and Reinstalled Items: Comply with the following:

1. Clean and repair items to functional condition adequate for intended reuse. Paint equipment to match new equipment.
2. Pack or crate items after cleaning and repairing. Identify contents of containers.
3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.

D. Concrete: Demolish in sections. Cut concrete full depth at junctures with construction to remain and at regular intervals, using power-driven saw, then remove concrete between saw cuts.

9.

E. Masonry: Demolish in small sections. Cut masonry at junctures with construction to remain, using power-driven saw, then remove masonry between saw cuts.

F. Concrete Slabs-on-Grade: Saw-cut perimeter of area to be demolished, then break up and remove.

G. Resilient Floor Coverings: Remove floor coverings and adhesive according to recommendations in RFCI-WP and its Addendum.

1. Remove residual adhesive and prepare substrate for new floor coverings by one of the methods recommended by RFCI.

H. Air-Conditioning Equipment: Remove equipment without releasing refrigerants.

3.6 PATCHING AND REPAIRS

A. Repairs: Where repairs to existing surfaces are required, patch to produce surfaces suitable for new materials.

3.7 DISPOSAL OF DEMOLISHED MATERIALS

A. General: Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.

3.8 SELECTIVE DEMOLITION SCHEDULE

END OF SECTION 01731

10.

SECTION 01732 - CUTTING AND PATCHING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes procedural requirements for cutting and patching.

B. Related Sections include the following:
   1. Divisions 2 through 16 Sections for specific requirements and limitations applicable to cutting and patching individual parts of the Work.

1.3 DEFINITIONS

A. Cutting: Removal of existing construction necessary to permit installation or performance of other Work.

B. Patching: Fitting and repair work required to restore surfaces to original condition after installation of other Work.

1.4 SUBMITTALS

1. Utilities: List utilities that cutting and patching procedures will disturb or affect. List utilities that will be relocated and those that will be temporarily out of service. Indicate how long service will be disrupted.
2. Engineer's Approval: Obtain approval of cutting and patching proposal before cutting and patching. Approval does not waive right to later require removal and replacement of unsatisfactory work.

1.5 QUALITY ASSURANCE

A. Structural Elements: Do not cut and patch structural elements in a manner that could change their load-carrying capacity or load-deflection ratio.

B. Operational Elements: Do not cut and patch the following operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety.

   1. Primary operational systems and equipment.
   2. Air or smoke barriers.
   3. Fire-protection systems.
   4. Control systems.
   5. Communication systems.

10.

6. Conveying systems.
7. Electrical wiring systems.
8. Operating systems of special construction in MEP Specifications.

C. Miscellaneous Elements: Do not cut and patch the following elements or related components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.

1. Water, moisture, or vapor barriers.
2. Membranes and flashings.
3. Exterior curtain-wall construction.
4. Equipment supports.
5. Piping, ductwork, vessels, and equipment.
6. Noise- and vibration-control elements and systems.

D. Visual Requirements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch construction exposed on the exterior or in occupied spaces in a manner that would, in the Engineer's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

b. Processed concrete finishes.
c. Stonework and stone masonry.
d. Ornamental metal.
e. Preformed metal panels.
f. Roofing.
g. Firestopping.
h. Window wall system.
i. Stucco and ornamental plaster.
j. Terrazzo.
k. Fluid-applied flooring.
l. Aggregate wall coating.
m. Swimming pool finishes.
n. HVAC enclosures, cabinets, or covers.

1.6 WARRANTY

Not Applicable

PART 2 - PRODUCTS

2.1 MATERIALS

A. General: Comply with requirements specified in other Sections of the Specification.

B. Existing Materials: Use materials identical to existing materials. For exposed surfaces, use materials that visually match existing adjacent surfaces to the fullest extent possible.

1. If identical materials are unavailable or cannot be used, use materials that, when installed, will match the visual and functional performance of existing materials.

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PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.

1. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
2. Proceed with installation only after unsafe or unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Temporary Support: Provide temporary support of Work to be cut.

B. Protection: Protect existing construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

C. Adjoining Areas: Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

D. Existing Services: Where existing services are required to be removed, relocated, or abandoned, bypass such services before cutting to avoid interruption of services to occupied areas.

3.3 PERFORMANCE

A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.

1. Cut existing construction to provide for installation of other components or performance of other construction, and subsequently patch as required to restore surfaces to their original condition.

B. Cutting: Cut existing construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer; comply with original Installer's written recommendations.

1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
2. Existing Finished Surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
3. Concrete and Masonry: Cut using a cutting machine, such as an abrasive saw or a diamond-core drill.

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C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams that are as invisible as possible. Provide materials and comply with installation requirements specified in other Sections of the Specification.

1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate integrity of installation.
2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.

END OF SECTION 01732

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SECTION 01770 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for Contract closeout, including, but not limited to, the following:

1. Substantial Completion procedures.
2. Final Completion procedures.
3. Warranties.
4. Record/As-Built Documents.
5. Operation and maintenance manuals.
6. Instruction of Employer's personnel.
7. Final cleaning.

B. Related Sections include the following:
1. General Requirements Section "Photographic Documentation" for submitting final construction photographs and negatives.
2. General Requirements Section "Product Requirements" for general requirement for warranties.
3. General Requirements Section "Execution Requirements" for progress cleaning of Site.
4. General Requirements Section "Demonstration and Training" for requirements for instruction of Employer's personnel.
5. All Specification Sections for specific closeout and special cleaning requirements for products of those Sections.

1.3 SUBSTANTIAL COMPLETION

A. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion for the whole or any Section or part of the Works, complete the following. List on request, items below that are incomplete.

1. Prepare, in a tabular format acceptable to the Engineer, a list of items to be completed and corrected (punch list). Include the value for each item on the list, and reasons why the work is not complete.
2. Submit a written undertaking in a form acceptable to the Engineer, to complete and/or correct all items contained in the punch list, to the satisfaction of the Engineer, within 28 days from date of issue of the Taking-Over Certificate.
3. Advise the Employer, with copy to the Engineer, of impending insurance changeover requirements.
4. Submit warranties, bonds, maintenance service agreements, final certifications, and similar documents.

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5. Obtain and submit releases permitting the Employer unrestricted use of the Works and access to services and utilities. Include occupancy permits, operating certificates, and similar releases as applicable.
6. Prepare and submit Record/As-Built Documents, and operation and maintenance manuals.
7. Submit final construction photographs and photographic negatives, and similar final record information.
8. Deliver tools, spare parts, extra materials, and similar items to locations designated by the Employer. Label with manufacturer's name and model number where applicable.
9. Make final changeover of permanent locks and deliver keys to the Employer. Advise the Employer's personnel of changeover in security provisions.
10. Complete startup testing of systems.
11. Submit test/adjust/balance records.
12. Terminate and remove temporary facilities from Site, together with mockups, unused samples, construction tools, and similar elements.
13. Advise the Employer, with copy to the Engineer, of impending changeover in utilities.
14. Submit changeover information related to the Employer's occupancy, use, operation, and maintenance.
15. Complete final cleaning requirements, including touchup painting.
16. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.
17. Commence instruction of the Employer's personnel in operation, adjustment, and maintenance of systems and equipment, and maintenance of products.

B. Inspection: Submit a written request for inspection for Substantial Completion. On receipt of request, the Engineer will either proceed with inspection or notify Contractor of unfulfilled requirements. The Engineer will issue a Taking-Over Certificate after satisfactory inspection or will notify the Contractor of items, either on the Contractor's list or additional items identified by the Engineer, which must be completed or corrected before Certificate will be issued.

1. Reinspection: Request reinspection when work identified in previous inspections as incomplete is completed or corrected.
2. Results of completed inspection will form the basis of requirements for Final Completion.

1.4 FINAL COMPLETION

A. Preliminary Procedures: Before requesting inspection for determining Final Completion, complete the following:

1. Complete instruction of the Employer's personnel in operation, adjustment, and maintenance of systems and equipment, and maintenance of products. Submit demonstration and training videotapes, if applicable.
2. Submit certified copy of the Engineer's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by the Engineer. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
3. Submit evidence of final, continuing insurance coverage complying with insurance requirements.
4. Submit pest-control final inspection report.

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B. Inspection: Submit a written request for inspection for Final Completion. On receipt of request, the Engineer will either proceed with inspection or notify the Contractor of unfulfilled requirements. The Engineer will issue a Defects Liability Certificate after satisfactory inspection or will notify the Contractor of work that must be completed or corrected before Certificate will be issued.

1. Reinspection: Request reinspection when work identified in previous inspections as incomplete is completed or corrected.

1.5 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

A. Preparation: Submit three (3) copies of list. Include name and identification of each building space and site area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by the Contractor that are outside the limits of construction.

1. Organize list of spaces in sequential order, starting with exterior areas first and proceeding from lowest floor to highest floor.
2. Organize items applying to each space by major element, including categories for ceiling, individual walls, floors, equipment, and building systems.
3. Include the following information at the top of each page:

o. Works name.
p. Date.
q. Name of Engineer.
r. Name of Contractor.
s. Page number.

1.6 WARRANTIES

A. Form of Submittal: Produce two (2) original copies of each required warranty, written in the form approved by the Engineer, and properly executed as specified.

B. Organize warranty documents into an orderly sequence based on the divisions and sections in the table of contents of the Specifications.

1. Bind warranties and bonds in heavy-duty, vinyl-covered, loose-leaf ring binders, thickness as necessary to accommodate contents, and sized to receive A4 paper.
2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation. Provide a typed description of the product or installation, including the name of the product and the name, address, and telephone number of Installer.
3. Provide a typed index and table of contents list at the front of each binder.
4. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Works name, and name of Contractor.

C. Submittal Time: Submit warranties prior to requesting inspection for determining date of Substantial Completion for the whole of the Works.

1. Submit warranties on request of the Engineer for designated parts or portions of the Works where commencement of warranties other than date of Substantial Completion is indicated.

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D. Provide additional copies of each warranty required, for operation and maintenance manuals.

1.7 RECORD/AS-BUILT DOCUMENTS

A. General: Store Record Documents in field office, apart from the Contract Documents used for construction. Do not use Record Documents for construction purposes. Maintain Record Documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Provide access to Record Documents for Engineer's inspection and reference during normal working hours.

B. Recording: Mark up Record Documents as the Works are progressively accomplished and completed relative to each Record Document. Record changes and modifications as they occur; do not wait until Substantial Completion.

C. Record Drawings: During construction period, print and maintain blue- or black-line white prints of all Contract Drawings, approved Shop Drawings and newly prepared Drawings, for Record Drawing purposes.

1. Preparation: Mark record prints to indicate all changes and field adjustments and to show the actual installation where installation varies from that shown originally.

a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
b. Accurately record information in an understandable drawing technique.
c. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.
d. Mark record prints of Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. If Shop Drawings are marked, show cross-reference on the Contract Drawings.
e. Mark record prints with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of work at the same location.
f. Mark important additional information that is either shown only schematically or omitted from original Drawings.
g. Note Variation numbers, Site Instruction numbers, and similar identification, where applicable.

2. Identify and date each Record Drawing; include the designation "RECORD/AS-BUILT DRAWING" in a prominent location. Organize into manageable sets; bind each set with durable paper cover sheets. Include identification and list contents on cover sheets.

D. Record Specifications: During construction period, print and maintain one copy of the Specifications, including addenda and contract modifications issued, for Record Specification purposes.

1. Preparation: Mark record specifications to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.

a. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.

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b. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
c. Record the name of the manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
d. Note related Variation numbers and Record Drawing numbers where applicable.

E. Miscellaneous Record Submittals: Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

Submit copies of record/As-Built documents in the numbers and formats specified in General Requirements Section “Record/As-Built Documents”

1.8 OPERATION AND MAINTENANCE MANUALS

A. Assemble a complete set of operation and maintenance data indicating the operation and maintenance of each system, subsystem, and piece of equipment not part of a system. Include operation and maintenance data required in individual Specification Sections and as follows:

1. Operation Data:

a. Emergency instructions and procedures.
b. System, subsystem, and equipment descriptions, including operating standards.
c. Operating procedures, including startup, shutdown, seasonal, and weekend operations.
d. Description of controls and sequence of operations.
e. Piping diagrams.

2. Maintenance Data:

a. Manufacturer's information, including list of spare parts.
b. Name, address, and telephone number of Installer or supplier.
c. Maintenance procedures.
d. Maintenance and service schedules for preventive and routine maintenance.
e. Maintenance record forms.
f. Sources of spare parts and maintenance materials.
g. Copies of maintenance service agreements.
h. Copies of warranties and bonds.

B. Organize operation and maintenance manuals into suitable sets of manageable size. Bind and index data in heavy-duty, vinyl-covered, loose-leaf ring binders, in thickness necessary to accommodate contents, with pocket inside the covers to receive folded oversized sheets. Identify each binder on front and spine with the printed title "OPERATION AND MAINTENANCE MANUAL," Works name, and subject matter of contents.

1. Submit copies of Operation and Maintenance Manuals in the number and format described in General Requirements Section “Operation and Maintenance Data”.

PART 2 - PRODUCTS

11.

2.1 MATERIALS

A. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

PART 3 - EXECUTION

3.1 DEMONSTRATION AND TRAINING

A. Instruction: Instruct the Employer's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

1. Provide instructors experienced in operation and maintenance procedures.
2. Provide instruction at mutually agreed-on times. For equipment that requires seasonal operation, provide similar instruction at the start of each season.
3. Schedule training with the Employer, through the Engineer, with at least seven days' advance notice.
4. Coordinate instructors, including providing notification of dates, times, length of instruction, and course content.

B. Program Structure: Develop and submit for the Engineer's approval, two copies of an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections. For each training module, develop a learning objective and teaching outline. Include instruction for the following:

1. System design and operational philosophy.
2. Review of documentation.
3. Operations.
4. Adjustments.
5. Troubleshooting.
6. Maintenance.
7. Repair.

3.2 FINAL CLEANING

A. General: Provide final cleaning. Conduct cleaning and waste-removal operations to comply with Alat FEZ laws and ordinances and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete the following cleaning operations before requesting inspection for determining date of Substantial Completion for the whole or any Section or part of the Work, as applicable:

a. Clean Site and grounds in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.

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b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
c. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.
d. Remove tools, construction equipment, machinery, and surplus material from Project site.
e. Remove temporary coverings and casings installed during construction to protect completed work during remainder of construction period, and remove from Project site.
f. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
h. Sweep concrete floors broom clean in unoccupied spaces.
i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo if visible soil or stains remain.
j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision-obscuring materials. Replace chipped or broken glass and other damaged transparent materials. Polish mirrors and glass, taking care not to scratch surfaces.
k. Remove labels that are not permanent.
l. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

1) Do not paint over "UL" and similar labels, including mechanical and electrical nameplates.

a. Wipe surfaces of mechanical and electrical equipment, elevator equipment if applicable, and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
b. Replace parts subject to unusual operating conditions.
c. Clean food preparation areas and surfaces, and food service equipment to a sanitary and hygienic condition, ready and acceptable for intended use.
d. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.
e. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
f. Clean ducts, blowers, and coils if units were operated without filters during construction.
g. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out bulbs, and those noticeably dimmed by hours of use, and defective and noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
h. Leave Works clean and ready for immediate occupancy and intended use.

C. Pest Control: Make a final inspection and rid site of rodents, insects, and other pests. Prepare a report.

D. Comply with safety standards for cleaning. Do not burn waste materials. Do not bury debris or excess materials on the Employer's property. Do not discharge volatile, harmful, or

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dangerous materials into drainage systems. Remove waste materials from Site and dispose of lawfully.

END OF SECTION 01770

13.

SECTION 01781 – RECORD/AS-BUILT DOCUMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for Record/As-Built Documents, including the following:

1. Record Drawings.
2. Record Specifications.
3. Miscellaneous Records.

B. Related Sections include the following:
1. General Requirements Section "Operation and Maintenance Data" for operation and maintenance manual requirements.
2. All Specification Sections for specific requirements for Record/As-Built Documents in those Sections.
3. All Specification Sections for specific requirements for Miscellaneous Record keeping and submittal in those Sections.

1.3 SUBMITTALS

A. Record Drawings: Submit copies of Record Drawings as follows:

1. Initial Submittal: Submit two sets of plots from Record CAD Drawing files and the original marked-up Record Prints. The Engineer will initial and date one set of plots and mark whether general scope of changes, additional information recorded, and quality of drafting are acceptable. The Engineer will return one set of plots and Record Prints together with review comments, for completing, printing, binding, and final submittal.
2. Final Submittal: After incorporating the Engineer's initial submittal review comments, submit:

a. Original marked-up Record Prints set.
b. Sets of As-Built Drawings as follows:

1) One (1) Set electronic format: (in CD-ROM).

2) Two (2) bound sets of prints (A2 Size).

3) Three (3) sets of O&M manuals, including a complete list, showing the actual number of volumes and files of O&M Manuals for each AX. A4 size paper shall be used, filed with good quality files and proper labeling.

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4) One (1) set of loose copy (blue print/black line print). Size should be the same as the original.

5) One (1) set of O&M manuals in electronic format (in CD-ROM), including tile index, procedures and all related catalogs/brochures.

6) One (1) set of any other document/report about the project, Test results and any other information/documents which are not included in the O&M Manuals and As-Built drawings.

B. Record Specifications: Submit two (2) copies of Record Specifications, including addenda and contract modifications.

C. Miscellaneous Records: Submit two (2) sets of original miscellaneous records.

PART 2 - PRODUCTS

2.1 RECORD/AS-BUILT DRAWINGS

A. Record Prints: During construction period, print and maintain one set of blue- or black-line white prints of all Contract Drawings, approved Shop Drawings and newly prepared Drawings, for Record Print purposes.

1. Preparation: Mark Record Prints to indicate all changes and field adjustments and to show the actual installation where installation varies from that shown originally.

a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
b. Accurately record information in an understandable drawing technique.
c. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.

2. Content: Types of items requiring marking include, but are not limited to, the following:

a. Dimensional changes to Drawings.
b. Revisions to details shown on Drawings.
c. Depths of foundations.
d. Locations and depths of underground utilities.
e. Revisions to routing of piping and conduits.
f. Revisions to electrical circuitry.
g. Actual equipment locations.
h. Duct size and routing.
i. Locations of concealed internal utilities.
j. Changes made by Variation or Site Instruction directives.
k. Changes made following Engineer's written orders and the Engineer's acceptance of substitutions or alternatives, etc.
l. Details not on the original Contract Drawings.
m. Field records for variable and concealed conditions.
n. All coordinates shown on the drawings shall be in Universal Transverse Meractor (UTM).

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3. Mark record prints of Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. If Shop Drawings are marked, show cross-reference on the Contract Drawings.
4. Mark record prints with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of work at the same location.
5. Mark important additional information that is either shown only schematically or omitted from original Drawings.
6. Note Variation numbers, Site Instruction numbers, and similar identification, where applicable.

B. Newly Prepared Record Drawings: Prepare new Drawings where the Engineer determines that neither the original Contract Drawings nor Shop Drawings are suitable to show actual installation.

1. New Drawings may be required when a Variation is issued or as a result of the Engineer accepting an alternate, substitution, or other modification.
2. Consult with the Engineer for proper scale and scope of detailing and notations required to record the actual physical installation and its relation to other construction. Integrate newly prepared Record Drawings into Record Drawing sets; comply with procedures for formatting, organizing, copying, binding, and submitting.

C. Record CAD Drawings: Before requesting inspection for determining date of Substantial Completion, review marked-up final Record Prints with the Engineer. When authorized, prepare a full set of CAD Drawings of all Contract Drawings, Shop Drawings and newly prepared Drawings, whether or not changes or additional information are recorded:

1. Incorporate changes and additional information previously marked on Record Prints. Delete, redraw, and add details and notations where applicable.
2. Refer instances of uncertainty to the Engineer for resolution.
3. Identify and date each Record Drawing; include the designation "RECORD/AS-BUILT DRAWING" in a prominent location.

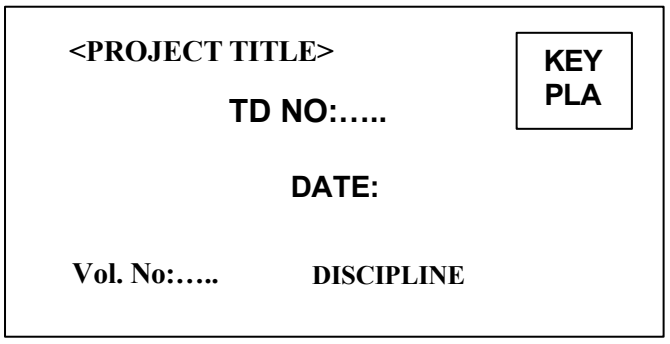
D. Record CAD Drawing Plots: Produce a full set of Record Drawing Plots from completed Record CAD Drawing files.

E. As-Built Final Submittal Format:

1. As-Built Prints: Organize As-Built Prints and newly prepared Record Drawings into manageable sets. Bind each set with thick cover sheets. Include identification and list contents on cover sheets.
2. As-Built CAD Drawings: Organize CAD information into separate electronic files that correspond to each sheet of the Record Prints. Produce "List of Drawings" sheet. Name each file with the sheet identification number. Include identification in each CAD file. All as follows:
    a. All As-Built drawings files to be submitted in UTM coordinate system.
    b. The master map unit for As-Built drawings files should be in meter (m) only. (For Microstation drawings, sub unit should be 1000 (mm) and positional unit should be 80).
    c. All As-Built CAD drawings files to be submitted in digital formats on CD’s media.
    d. As-Built CAD formats are both required for any As-Built design / entity submission.
    e. All As-Built CAD drawings should be submitted in original Microstation format, not converted, unless special permission is obtained for other format.

f. In the event of such special permission being granted, for AutoCAD drawings, pen width details for plotting should also be provided in each drawing. (Outside the drawing area).
g. In Microstation files for 2D drawings always use 2D seed files.
h. All reference files (if used) should be copied in a directory called 'reference'.
i. If any of the files/drawings are using fonts other than the normal fonts provided by the software, then those fonts shall also be copied in a separate directory called 'fonts'.
j. Drawing should be neat and perfect. None of the items shall be outside the drawing sheet area.
k. Drawings of different disciplines (i.e. Architectural, Structural, Electrical, Mechanical etc.) shall be kept in separate directories with suitable names.
l. One file shall contain only one drawing.
m. Drawing shall be drawn/organized in suitable/proper layers/levels. Same kind of items shall always be in the same layer/level.
n. Drawing shall be copied into CD's in such a way that, when we open one drawing from the CD, it must be complete and all reference files, fonts & libraries (if used) should attach automatically.
o. The index of drawings shall also be provided in the original format of the drawing (DGN or DWG) and it must be kept in a directory called "general".
p. Total number of As-Built drawings in the index and in the CD shall be the same.
q. Change notice drawings shall submit along with all other drawings of that AX in a new CD.

3. Copy Prints: Organize into bound sets matching As-Built CAD Drawing Plots. The Engineer will deliver the Employer copy sets. All as follows:
a. They must be in standard in A2 size.
b. They must be properly bound using thick covering sheets.
c. Laminated labeling must be as in the following format:



d. The scale of the drawings in the bound book shall match with the scale shown in the title block of the drawing of bound book.

4. General (for Soft copy & Hard copy)

a. Each document should be verified thoroughly by the consultant/contractor before submission.
b. "As-Built" stamp/seal with date and signature of the verified person shall be provided in each drawing.

c. "As-Built" stamp/seal (item 4. b above) shall be exactly like below:

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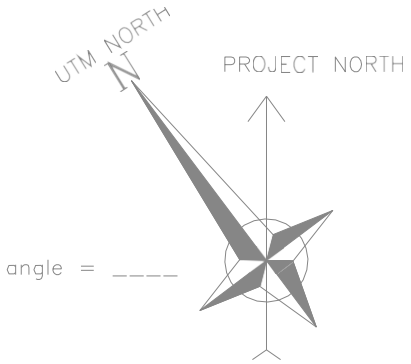
CHECKED AND CERTIFIED BY

CONTRACTOR

d. Drawing should be in proper scale and AX number, its description & scale of the drawing shall also be mentioned in the title block along with other details.

e. Key plan & north symbol shall be provided in the upper right corner of the drawings (wherever it is applicable).

f. North symbol (item 4. e above) shall be as sample below:



g. Provide hard copy of list (index) of drawings as in the following format:

Discipline (Architectural, Structural, Mechanical, etc.)					
SI. No.	Sheet no.	Drg. No.	Rev. No.	Drg. File name	Description

h. All the drawings must be numbered serially (sheet number), starting from 001 (drawing index as sheet no. 001) and must be listed in the drawing index. (Total number of drawings in one AX should be same as the sheet number of the last drawing).

i. One file shall contain only one drawing. (This point is also incorporated under item 2.1. F.2 above).

j. All the drawings, sketches or diagrams inside the O&M manuals shall also be stamped as As-Built and certified by the contractor and consultant.

k. All As-Built drawings files to be submitted in UTM coordinate system.

l. All As-Built CAD drawings files to be submitted in digital formats on CD's media.

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m. As-Built CAD formats are both required for any As-Built design / entity submission.
n. All naming, numbering and related attribute information for As-Built to be matched with Project standards.

2.2 RECORD SPECIFICATIONS

A. Record Specifications: During construction period, print and maintain two copies of the Specifications, including addenda and contract modifications issued, for Record Specification purposes.

B. Preparation: Mark Record Specifications to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.

1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
3. Record the name of the manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
4. Note related Variation numbers and Record Drawings where applicable.

2.3 MISCELLANEOUS RECORD SUBMITTALS

A. Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference. Miscellaneous records include, but are not limited to, the following:

1. Field records on excavations and foundations.
2. Field records on underground construction and similar work.
3. Surveys showing locations and elevations of underground lines.
4. Invert elevations of drainage piping.
5. Surveys establishing building lines and levels.
6. Special and authorized measurements.
7. Certifications received in lieu of labels on bulk products.
8. Batch mixing and bulk delivery records.
9. Testing and qualification of tradesmen.
10. Documented qualification of installation firms.
11. Tests and inspections.
12. Inspections and certifications by governing authorities.
13. Final inspection and correction procedures.
14. Any other information / documents which are not included on As-Built Drawings and Operation and Maintenance Manuals.

PART 3 - EXECUTION

3.1 RECORDING AND MAINTENANCE

13.

A. Recording: Mark up Record Documents as the Work is progressively accomplished and completed relative to each Record Document. Record changes and modifications as they occur; do not wait until the end of Project.

B. Maintenance of Record Documents: Store Record Documents in field office, apart from the Contract Documents used for construction. Do not use Record Documents for construction purposes. Maintain Record Documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Make Record Documents available for the Engineer's inspection and reference during normal working hours.

END OF SECTION 01781

16.

SECTION 01782 - OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for Operation and Maintenance (O&M) Manuals, including the following:

1. Preparing and submitting O&M Manuals for operating systems and equipment.
2. Preparing and submitting instruction manuals covering the care, preservation, and maintenance of engineered components and systems and architectural products and finishes.
3. Instruction of the Employer's operating personnel in the operation and maintenance of systems and equipment.

B. Related Sections: The following Sections contain requirements that relate to this Section:

1. General Requirements "Data Requirements" for preparation of Shop Drawings and Product Data.
2. All Specification Sections for special O&M data requirements for specific pieces of equipment or building operating systems.

1.3 QUALITY ASSURANCE

A. O&M Manual Preparation: In preparation of O&M Manuals, use personnel thoroughly trained and experienced in operation and maintenance of equipment or system involved.

1. Where O&M Manuals require written instructions, use personnel skilled in technical writing where necessary for communication of essential data.
2. Where O&M Manuals require drawings or diagrams, use CAD operators or drafters capable of preparing drawings clearly in a clear, understandable industry standard format.
3. System-Level O&M Manuals for mechanical and electrical systems shall be developed specifically for this Project. Compilation of equipment vendor catalogs and O&M Manuals is not acceptable.

B. Instructions for the Employer's Personnel: Use experienced instructors thoroughly trained and experienced in operation and maintenance of equipment or system involved to instruct the Employer's O&M personnel.

1.4 SUBMITTALS

A. Submittal Schedule: Comply with the following schedule for submitting O&M Manuals:

16.

1. Not later than 60 days prior to initial startup of the subject work scope equipment, when each installation that requires O&M Manuals is nominally complete, submit 2 draft copies of each manual to the Engineer for review and approval. Include a complete index or table of contents of each manual.
2. The Engineer will return 1 copy of the draft with comments within 30 days of receipt.
3. After receipt of the Engineer's approval, make any required corrections or modifications to comply with the Engineer's comments. After approval, and at least 30 calendar days in advance of initial startup of an item (i.e., within 30 days of receipt of the Engineer's final comments and approval), submit the O&M Manuals, in final form, to the Engineer for distribution and use by the Employer, as follows:
    a) Three (3) bound sets of O&M Manuals.
    b) One (1) loose set (blue line / black line prints) of O&M Manuals.
    c) One (1) set of O&M Manuals in electronic format (in CD-ROM).

B. Form of Submittal: Prepare O&M Manuals in the form of an instructional manual for use by the Employer's operating personnel. Organize into suitable sets of manageable size. Where possible, assemble instructions for similar equipment into a single binder.

1. Complete Listing: Provide a complete list showing the actual number of volumes and files of O&M Manuals, for the project. Also provide a list showing the number of volumes and files included in a particular submittal.

2. Binders: For each manual, provide heavy-duty, commercial-quality, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to receive Metric-Size "A4" (210 mm by 297 mm) paper. Provide a clear plastic sleeve on the spine to hold labels describing contents. Provide pockets on the inside of the covers to receive folded sheets.

    a. Where 2 or more binders are necessary to accommodate data, correlate data in each binder into related groupings for similar items. Establish a logical, well-organized table of contents and O&M Manual binders layout. Cross-reference other binders where necessary to provide essential information for proper operation or maintenance of the piece of equipment or system.
    b. Identify each binder on front and spine, with the printed title "OPERATION AND MAINTENANCE MANUAL", Works title or name, and subject matter covered. Indicate volume number for multiple-volume manuals. Indicate applicable 5-digit specification section number for reference as appropriate on the front of each major O&M Manual section.
    c. They must be filled with good quality files and proper labeling.

3. Organize each O&M Manual into separate sections for each piece of related equipment. Each manual section shall contain a title page; a table of contents; copies of pertinent product data, supplemented by drawings and written text; and copies of each warranty, bond, and service contract issued.

    a. Title Page: Provide a title page in a transparent, plastic envelope as the first sheet of each manual. Provide the following:

        1) Subject matter covered by the manual section.
        2) Name and address of the Works.
        3) Contract Number (e.g., AX….).
        4) Date of submittal.
        5) Name, address, and telephone number of the Contractor.

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6) Name and address of the Engineer.
7) Cross-reference to related systems in other O&M Manuals or sections of the subject O&M Manual.

b. Table of Contents: After title page, include a typewritten table of contents for each volume, arranged systematically. Include a list of each product included, identified by product name or other appropriate identifying symbol and indexed to the content of the volume. Where a system requires more than one volume to accommodate data, provide a comprehensive table of contents for all volumes in each volume of the set.
c. General Information: Provide a general information section immediately following table of contents, listing each product included in the manual, identified by product name. Under each product, list the name, address, and telephone number of the subcontractor or installer and the maintenance contractor. Clearly delineate the extent of responsibility of each of these entities
d. Product Data: Where the manuals include manufacturer's standard printed data, include only sheets that are pertinent to the part or product installed. Mark each sheet to identify each part or product included in the installation. Where the Project includes more than one item in a tabular format, identify each item, using appropriate references from the Contract Documents. Identify data that is applicable to the installation, and delete references to information that is not applicable.
e. Written Text: Prepare written text to provide necessary information where manufacturer's standard printed data is not available, and the information is necessary for proper operation and maintenance of equipment or systems. Prepare written text where it is necessary to provide additional information or to supplement data included in the manual. Organize text in a consistent format under separate headings for different procedures. Where necessary, provide a logical sequence of instruction for each operation or maintenance procedure.
f. Drawings: Original project record documents as part of O&M Manuals.
g. Warranties, Bonds, and Service Contracts: Original documentation.

4. Dividers: Provide heavy paper dividers with celluloid-covered tabs for each separate Section. Mark each tab to indicate contents. Provide a typed description of the product and major parts of equipment included in the section on each divider.
5. Protective Plastic Jackets: Provide protective, transparent, plastic jackets designed to enclose diagnostic software for computerized electronic equipment.
6. Text Material: Where O&M Manuals require written material, use the manufacturer's standard printed material. If manufacturer's standard printed material is not available, provide specially prepared data, typewritten, on Metric Size "A4" (210 mm by 297 mm), 75-g/sq. m, white bond paper.
7. Drawings: Provide reinforced, punched binder tabs on drawings and bind in with text.

a. Where oversize drawings are necessary, fold drawings to the same size as text pages and use as a foldout.
b. If drawings are too large to be used practically as a foldout, place the drawing, folded, in front or rear pocket of binder. Insert a typewritten page indicating drawing title, description of contents, and drawing location at the appropriate location in the manual.

8. Loose Copy: Shall be in blue / black line prints. Enclosed drawings shall be same sign as original mylar / polyester films.

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9. Electronic Format: Provide one (1) set of O&M Manuals in CD – ROM format including file index, procedures and all related catalogs and brochures.
10. All O&M Manuals shall be stamped as "Final Copy" on the front page assuring that such are final.
11. All drawings, sketches and diagrams included in the O&M Manuals shall also be stamped as "As-Built" and certified accordingly by the Contractor.
12. Other Documents / Reports: Provide one list of full report about the project, Test results and any other information / documents, which are not included in the O&M manuals and As-Built drawings.

1.5 O&M MANUAL CONTENT

A. Each O&M Manual shall include the information specified in the applicable individual Specification Section and, as a minimum, shall include the following general sections/subsections and information. The O&M Manual shall address each component/equipment/item and its controls and each major facility included within the scope of work.

1. How to Use the Manual. This section shall be a guide to the contents, structure, and layout of the manual. It shall enable the reader to comprehend the scope of the document and identify readily where specific information can be obtained.
2. Overall Purpose. This section shall provide a general overview of the original design intent.
3. Description. This section shall include the following:

    a. Type of system.
    b. System location and what it serves.
    c. System inputs/interfaces; i.e., what the system depends upon in order to function.
    d. Design data, basic design parameters, and basic assumptions made during design. The appropriate Employer's Design Criteria documents will be made available for the Contractor's use and information and to assist in the preparation of this section of the O&M Manual.
    e. Rationale for selecting particular plant processes/equipment.
    f. Expected service life (where available).
    g. Planned operational efficiencies.

4. Equipment Schedule and Identification. This section shall include the following for all components within each system:

    a. Type of equipment.
    b. Name of manufacturer.
    c. Equipment model number of each component.
    d. Equipment serial number of each component.
    e. Items within each system shall be listed, together with the names of their respective manufacturers or suppliers.

5. Parts Identification and Recommended Spares. This section shall include the following for all components within each system:

    a. A parts identification list detailing and identifying replaceable assemblies, sub-assemblies, and components. It shall include suppliers' recommendation for both spares and "running spares" for the First year period and Second year period

as a minimum; i.e., parts required for scheduled replacement due to wear or deterioration.

b. Items normally held in stock locally by the supplier or for which a refurbishment service is available shall be clearly identified.
c. Recommended stocking levels of spare parts.
d. Manufacturing and shipping time required for imported spare parts.

6. Commissioning Data. This section shall include the following:

a. Measured data.
b. Measurement points.
c. Test equipment used.
d. Calibration certificate details.
e. Test certificates.
f. Safety and fire certificates (where applicable).
g. A statement of whether the design requirements were achieved.

7. Operation. This section shall include instructions for the safe and efficient operation, under both normal and emergency conditions. These will be in addition to manufacturers' literature for all plant items and shall include the following:

a. A recommended strategy for operation and control.
b. An outline of the general operating mode.
c. Control data (location, effect, object, sequence, limits of capability, modes, set points).
d. For electric and electronic systems, provide complete circuit directories of control panels, local panels/racks, programmable logic controller units, panelboards, etc. including the following:

1) Electric service.
2) Controls.
3) Communications.

e. Charts of valve-tag numbers, with the location and function of each valve.
f. Procedure and sequences for startup, running, and shutdown, under both normal and emergency conditions.
g. Interlocks between various plant items.
h. Operating procedure for standby operations and plant.
i. Precautions necessary to overcome known hazards.
j. The means by which any potentially hazardous condition may be made safe.
k. Target figures for both energy consumption and energy costs.
l. Forms for recording plant running hours, energy consumption, and energy costs.

8. Maintenance. This section shall include the following:

a. Manufacturers' recommendations and instructions for maintenance for each item of plant and equipment. A clear distinction shall be made between planned tasks (preventative maintenance) and work done on a corrective basis. Instructions shall be given on each of the following, as appropriate:

1) The isolation and return to service of systems and equipment.
2) Adjustments, calibration, and testing.
3) Dismantling and re-assembly.

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4) The exchange of components and assemblies.
5) Dealing with hazards that might arise during maintenance.
6) The nature of expected deterioration and the types of defects anticipated.
7) Special tools, test equipment, and ancillary services.

b. Maintenance schedules shall be provided for preventative maintenance tasks and shall be based on manufacturers' recommendations and other authoritative sources and shall include the following:

1) Inspections.
2) Examinations.
3) Tests.
4) Adjustments.
5) Calibration.
6) Lubrication.
7) Periodic overhaul.

c. Procedures for the logical diagnosis and correction of faults.
d. A schedule for all systems/components requiring lubrication shall be provided together with manufacturers' recommendations on the type of lubricant and the method and frequency of application. Where the type of lubricant is identified by product name, a generic reference (e.g., BS) shall also be given. Special requirements for the handling and storage of lubricants shall be provided. Where applicable, names of locally available lubricants shall be provided.

9. Modification Information. This section shall include information on permitted plant or system modifications allowed for by the manufacturers or system designers for each system. Space shall be provided in the manual for the recording of all modifications and changes as they occur.
10. Disposal Instructions. This section shall include the following:

a. Any known dangers likely to arise during the disposal of specific items of systems or equipment, together with the necessary precautions and safety measures.
b. Methods for safely disposing of or destroying the equipment or any parts thereof, including packaging, insulation, and fluids.
c. Sources from which further advice on safe disposal can be obtained.

11. Names and Addresses of Manufacturers. This section shall include the following:

a. Name of each equipment/item.
b. Name of manufacturer.
c. Address of manufacturer.
d. Telephone number and telex/fax number of manufacturer. Include E-Mail address if available.
e. Name of local supplier/agent.
f. Address of local supplier/agent.
g. Telephone number and telex/fax number of local supplier/agent. Include E-Mail address if available.

12. Index of Plans and Drawings. This section shall include the following:

16.

a. An index shall be provided of all "As Fitted" drawings supplied during the course of the installation work, included by number and title.
b. The index shall also include a schedule of all drawings issued by each manufacturer and supplier during the course of the installation work; e.g., control panel wiring diagrams.
c. In addition to manufacturer-prepared documentation and drawings as specified below, the O&M Manual shall contain copies of Contractor-prepared Coordination Drawings that accurately represent the final as-built condition of the facility/system.

13. Emergency Information. This section shall include the following:

a. Names, addresses, telephone, and telex/fax numbers of the appropriate contacts in the event of fire, theft, burglary, and fuel, gas, electricity or water failure/leaks. It shall also list those firms or staff to contact in the event of the failure or breakdown of such facility items as elevators, security gates, and miscellaneous mechanical and electrical equipment.
b. Where applicable, location of firefighting equipment, hydrants, and rising mains shall be identified and described.

14. Manufacturers' Literature. This section shall include the following:

a. A complete set of all manufacturers' literature shall be provided for each of the systems and equipment installed and assembled. As a minimum, the following document types shall be provided:

1) Product data shall be manufacturer's original printed data (not copies).
2) Copies of applicable Shop Drawings and Product Data.
3) Assembly drawings and diagrams required for maintenance.
4) Wiring diagrams.
5) Inspection and test procedures.
6) Repair instructions including spare parts listing.
7) Maintenance procedures and schedules.
8) Precautions against improper use and maintenance.
9) Sources of required maintenance materials and related services.
10) Description of the sequence of operation and as-installed control diagrams by the control manufacturer for systems requiring controls.
11) Specially prepared drawings where necessary to supplement manufacturer's printed data to illustrate the relationship of component parts of equipment or systems or to provide control or flow diagrams. Coordinate these drawings with information contained in Project Record Drawings to assure correct illustration of the completed installation.

b. This literature shall provide the following:

1) Description of the product as purchased.
2) The cost and date of purchase.
3) Performance - behavioral characteristics of the equipment in use (e.g., performance curves for pumps).
4) Equipment or system function.
5) Operating characteristics.
6) Limiting conditions.
7) Engineering data and tests.

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8) Complete nomenclature and number of replacement parts.
9) Applications - suitability for use.
10) Operation and Maintenance details.
11) Resources of labor, plant, material, and space required.
12) Methods of operation and control.
13) Cleaning and maintenance requirements.
14) Protective measures.
15) Labor safety and welfare associated with the equipment.
16) Public safety considerations.

15. Manufacturers' Guarantees and Warranties. This section shall include copies of each warranty, bond, or service contract in the appropriate manual for the information of the Employer's operating personnel. Provide written data outlining procedures to follow in the event of product failure. List circumstances and conditions that would affect validity of warranty or bond.

1.6 INSTRUCTIONS FOR THE EMPLOYER'S PERSONNEL

A. Prior to final inspection, instruct the Employer's personnel in operation, adjustment, and maintenance of products, equipment, and systems. Provide instruction at mutually agreed upon times.

1. Use O&M Manual sections for each piece of equipment or system as the basis of instruction. Review contents in detail to explain all aspects of operation and maintenance.

PART 2 - PRODUCTS
(Not Applicable)

PART 3 - EXECUTION
(Not Applicable)

END OF SECTION 01782

18.

SECTION 01820 - DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and other Specification Sections, apply to this Section.

1.2 SUMMARY

A. This section covers the requirements for preparing, administering, and conducting a training program in Azerbijani for the Employer's personnel - nominated operator, maintenance, supervisor, and instructor trainees in operation, maintenance, and computer software programming of the system. Training shall be of sufficient scope and depth to ensure that all operations, maintenance, and computer software trainees who complete the training program will be fully qualified, certified and capable of operating and/or maintaining the system/subsystems as installed under this Contract. The program shall provide the training, orientation, and familiarization necessary to ensure a smooth transition between system installation and operations activities.

B. Related Sections include the following:

1. General Requirements Section "Project Management and Coordination".
2. General Requirements Section "Photographic Documentation" for preparing and submitting demonstration and training videotapes.
3. General Requirements Section "Closeout Procedures" for Contract closeout.
4. General Requirements Section "Operation and Maintenance Data" for procedural requirements relating to Operation and Maintenance Manuals.

C. Two copies of submittals required by this Section shall be supplied to Authorized Representative for review not later, than ninety (90) days prior to scheduled start of each formal training course. Thirty (30) days shall be allowed for such review. Training shall be performed only in accordance with such reviewed and approved training program submittals. All training aids proposed for use in the program shall be available for inspection prior to commencement of training. Upon completion of the training program, the training aids, tools, test equipment, training manuals and other materials relating to the program shall be handed over to the Employer.

D. When short familiarization courses are offered to the Contractor by the equipment manufacturer/vendor as a common industry practice at no extra cost to the Contractor, then the Contractor shall make arrangements for the Employer nominated staff to attend such courses at no additional cost to the Employer.

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1.3 SUBMITTALS

The following submittals will be required:

A. Instruction Program: Submit for the Engineer's approval, two copies of outline of instructional program for demonstration and training, including a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1. Include for providing instruction at start of each season for equipment that requires seasonal operation and maintenance.

B. A training program stating all program courses and activities, and their individual duration, sequence, and interrelationships.

C. A narrative description of each course, including its purpose, technical scope, and the number of hours from start of course to trainee certification.

D. Current résumés of training and technical personnel who will be assigned to the classroom and/or on-the-job (hands-on) phases of training. Résumés shall be identified by course.

E. The ratio of training and technical personnel to trainees for both classroom and on-the-job phases of each course.

F. A narrative (verbal) and graphic (pictorial) explanation of the system's operational scheme and features that will be covered in the Operations-related courses.

G. A narrative (verbal) and graphic (pictorial) explanation of the system's operational and maintenance schemes that will be covered in the maintenance-related courses.

H. A comprehensive training manual for each course in the form and content that will be used in both classroom and on-the-job phases of training. In addition to other information, the manual shall contain an agenda, the hours devoted to each topic/subject, and the objectives for each lesson. Such manual shall be prepared in English.

I. A descriptive listing, by course, of all training aids to be used during both classroom and on-the-job phases of training.

J. A narrative description of the practices to be followed to measure, record, and report each trainee's progress and proficiency on completion of formal training. Also, a description of the method(s) to be used to certify trainees as being capable to operate and/or maintain the system/subsystem.

K. Samples of written examinations to be used for each course.

L. A narrative description of the methods proposed to provide make-up instruction for trainees who miss training sessions due to illness or approved absences.

M. A bibliography containing a listing of reference publications to be used in conjunction with classroom and on-the-job training.

1.4 EMPLOYER-NOMINATED STAFF

A. Employer will nominate staff to receive training. Such trainees may be expected to be familiar with the fundamental concepts of the operation and maintenance of similar systems and be basically qualified in both written and spoken English. Each trainee will be designated by the Employer in one of the three categories (trainee, supervisor-trainee, or instructor-trainee) for operations, maintenance, or computer software training.

1.5 ASSISTANCE TO KEY TECHNICAL PERSONNEL

A. All key technical personnel of the Contractor shall have a general responsibility to assist Employer-nominated trainees in conducting training for the operations and maintenance of the system/subsystem. Such assistance shall be a part of their duties and shall be included in their job descriptions.

1.6 QUALIFICATIONS OF TRAINING PERSONNEL

A. Personnel designated to conduct training shall be fully qualified and shall have a minimum of three years' experience with the system/subsystems as installed, or comparable successful experience as a trainer.

1.7 QUALITY ASSURANCE

A. Facilitator Qualifications: A firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for the Works, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: A factory-authorized service representative, complying with requirements in General Requirements Section "Quality Requirements," experienced in operation and maintenance procedures and training.

C. Review methods and procedures related to demonstration and training including, but not limited to, the following:

1.8 COORDINATION

A. Coordinate instruction schedule with the Employer's operations. Adjust schedule as required to minimize disrupting the Employer's operations.

B. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by the Engineer.

18.

PART 2 - PRODUCTS

2.1 TRAINING PROGRAM

A. Scope: Training shall be in sufficient scope to ensure that all trainees who complete the program will be certified as capable to operate and/or maintain the equipment, systems, and facilities furnished and installed under this contract, and to ensure a smooth transition between construction and operation activities.

B. Language: Training shall be conducted in English.

C. Program Structure: Develop an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, if any and as follows:

1. Landscape irrigation and sprinkler systems.
2. Specialties; including (if any), but not limited to, flagpoles, fire protection specialties, operable partitions, saunas, etc.
3. Fire-protection systems; including (if any), but not limited to, fire alarm, fire pumps, fire-extinguishing systems, etc.
4. Intrusion detection systems.
5. Medical equipment, including medical gas equipment and piping.
6. Laboratory equipment; including (if any), but not limited to, laboratory air and vacuum equipment and piping.
7. Water supply systems; including (if any), but not limited to, water supply wells, water softeners, pumps, storage and distribution, etc.
8. Compressed-air systems.
9. Refrigeration systems; including (if any), but not limited to, chillers, cooling towers, condensers, pumps, distribution piping, etc.
10 .HVAC systems; including (if any), but not limited to, air-handling equipment, air distribution systems,terminal equipment and devices, split system units, etc.
11. HVAC instrumentation and controls and/or building management and control system.
12. Electrical service and distribution; including (if any), but not limited to, transformers, switchboards, panelboards, uninterruptible power supplies, motor controls, etc.
13. Packaged engine generators, including transfer switches.
14. Lighting equipment and controls.
15. Communication systems; including (if any), but not limited to, intercommunication, surveillance, clocks and programming, voice and data, television equipment.

D. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instructions for the following:

1. Basis of System Design, Operational Requirements, and Criteria: Include the following:

a. System, subsystem, and equipment descriptions.
b. Performance and design criteria if the Contractor is delegated design responsibility.
c. Operating standards.
d. Regulatory requirements.
e. Equipment function.

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f. Operating characteristics.
g. Limiting conditions.
h. Performance curves.

2. Documentation: Review the following items in detail:

i. Emergency manuals.
j. Operations manuals.
k. Maintenance manuals.
l. Project Record Documents.
m. Identification systems.
n. Warranties and bonds.
o. Maintenance service agreements and similar continuing commitments.

3. Emergencies: Include the following, as applicable:

p. Instructions on meaning of warnings, trouble indications, and error messages.
q. Instructions on stopping.
r. Shutdown instructions for each type of emergency.
s. Operating instructions for conditions outside of normal operating limits.
t. Sequences for electric or electronic systems.
ii. Special operating instructions and procedures.

4. Operations: Include the following, as applicable:

jj. Startup procedures.
kk. Equipment or system break-in procedures.
ll. Routine and normal operating instructions.
mm. Regulation and control procedures.
nn. Control sequences.
oo. Safety procedures.
pp. Instructions on stopping.
qq. Normal shutdown instructions.
rr. Operating procedures for emergencies.
ss. Operating procedures for system, subsystem, or equipment failure.
tt. Seasonal and weekend operating instructions.
uu. Required sequences for electric or electronic systems.
vv. Special operating instructions and procedures.

5. Adjustments: Include the following:

ww. Alignments.
xx. Checking adjustments.
yy. Noise and vibration adjustments.
zz. Economy and efficiency adjustments.

6. Troubleshooting: Include the following:

aaa. Diagnostic instructions.
bbb. Test and inspection procedures.

7. Maintenance: Include the following:

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ccc. Inspection procedures.
ddd. Types of cleaning agents to be used and methods of cleaning.
eee. List of cleaning agents and methods of cleaning detrimental to product.
fff. Procedures for routine cleaning
ggg. Procedures for preventive maintenance.
hhh. Procedures for routine maintenance.
iii. Instruction on use of special tools.

8. Repairs: Include the following:

jjj. Diagnosis instructions.
kkk. Repair instructions.
lll. Disassembly; component removal, repair, and replacement; and reassembly instructions.
mmm. Instructions for identifying parts and components.
nnn. Review of spare parts needed for operation and maintenance.

2.2 TRAINING AIDS

A. Training aids shall include: approved operation and maintenance (O&M) manual(s) prepared by the Contractor as part of this Contract; training manuals, including course outline, basic text of instructions modules, and trainee workbook; films; slides; video tape(s); charts; models; hand-outs; catalogues; samples; and other visual and written aids to complement instruction. In addition, spare parts and other special hardware shall be provided to support " hands-on" familiarization with the equipment or systems.

2.3 TRAINING MANUALS

A. Training manuals in English shall be provided for each training course covering both the classroom and on-the-job phases. The organization of the manuals shall follow the same sequence as the course's scheduled presentation of material, providing such additional background and supplementary information as a trainee may need to understand the O&M manuals. The training manuals shall contain course agenda, schedule of hours devoted to each topic or subject, and objectives for each lesson.

2.4 DISPOSITION

A. Upon completion of the training program, all training aids, test equipment, and similar items provided by the Contractor for this training program shall become the property of the Employer.

PART 3 - EXECUTION

3.1 GENERAL

A. PREPARATION

1. Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a combined training manual.

2. Set up instructional equipment at instruction location.
3. Scheduling: Provide instruction at mutually agreed-on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.

   ooo. Schedule training with the Employer, through the Engineer, with at least seven (7) days' advance notice.

B. SCOPE:

1. Training shall be performed in accordance with an approved training program. All training aids shall be available and approved prior to start of training.
2. Each training course shall usually include classroom (theoretical) training and on-the-object (hands-on) training to provide practical experience. Where system/subsystem interfaces occur, such as with equipment or facilities furnished under other contracts, the Contractor shall cooperate with and assist the Employer/Engineer in coordinating timing and scheduling of training so that trainees can be made familiar with the operation and maintenance of the system subsystem as a complete, in-place, operating unit.
3. Laboratory equipment, spare parts, and mockup models may be used for theoretical orientation. However, practical "hands-on" familiarization shall be provided on equipment that is either installed and operable or ready to be installed and capable of being operated. System spares of special hardware items may be used to support "hands-on" familiarization as long as said spares are returned to inventory in satisfactory condition in accordance with approved procedures.

C. Proficiency: At designated intervals, each trainee's proficiency level shall be determined by written, oral, and practical performance tests.

D. Completion: Upon satisfactory completion of the training, it shall be certified, in writing, that each trainee, by name, is qualified to operate or maintain the system, supervise or instruct, as applicable.

E. Make-up Instruction: Make-up instruction shall be scheduled to accommodate shift work and authorized absences.

F. Training Categories: Training shall be provided using approved programs in the following categories:

1. Operations.
2. Maintenance.
3. Computer Software.
4. Operations – Supervisor.
5. Operations – Instructor.
6. Maintenance – Supervisor.
7. Maintenance – Instructor.
8. Other Categories as may be required by the Contractor's system design.

G. Operations Training: Operators shall be provided with a thorough training in all aspects of system/subsystem operation under both normal and abnormal conditions. This training shall include, but not be limited to:

1. Orientation to provide overview of system/subsystem purpose, configuration, and operations.

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2. Terminology.
3. Operations theory and interface.
4. Equipment appearance, functions, concepts and operation.
5. Operating modes, practices, and procedures under normal and emergency conditions.
6. Safety precautions.
7. On-the-job operating experience covering all system/subsystem operating functions, activities, and tasks including those associated with degraded operating modes; failure recognition, and recovery processes.
8. Familiarity with the content and use of O&M manuals and related reference publications.

H. Maintenance Training: Maintenance training shall cover all on-site routine preventive, and remedial maintenance of the system/subsystem. This training shall include, but not be limited to:

1. Orientation to provide overview of system/subsystem purpose, configuration, and operations.
2. Terminology.
3. Operations theory and interfaces.
4. Equipment appearance, layout, functions, concepts and operations.
5. Operating modes, practices, and procedures under normal and emergency conditions.
6. Safety precautions.
7. Use of tools and test equipment.
8. Preventive maintenance.
9. Troubleshooting, diagnostic, and testing.
10. Assembly and disassembly.
11. Repair and parts replacement.
12. Parts ordering practices and storage.
13. Failure and recovery procedures.
14. System/subsystem cabling.
15. Familiarity with and use of O&M manuals and other reference materials.

I. Computer Software Training: Computer software training shall cover the use, management, development, and maintenance of system/subsystem software. This training shall include, but not be limited to:

1. Orientation to provide overview of system/subsystem concepts, configurations, and operation.
2. Basis programming and software architecture.
3. System executive and utility program.
4. Application program.
5. Modification of command lines codes.
6. Interrupt logic.
7. Intra-and inter system interfaces.
8. Processor system operating procedures.
9. Diagnostics.
10. Communications software.
11. File management.
12. Hands-on training.
13. Safety precautions.
14. Familiarization with and use of O&M manuals and other reference materials.

J. Operations-Supervisor Training: Supplemental and specialized training shall be provided for supervisor level trainees. This training shall impart the knowledge and skills necessary to schedule, supervise, and improve the proficiency of operating and computer software personnel.

K. Operations - Instructor Training: Supplemental and specialized training shall be provided to prepare graduate trainees or nominated instructors successfully to instruct future trainees in system/subsystem operations and computer software.

L. Maintenance - Supervisor Training: Supplemental and specialized training shall be provided for supervisory level trainees. This training shall impart the knowledge and skills necessary to schedule, supervise, and improve the proficiency of maintenance personnel.

M. Maintenance - Instructor Training: Supplemental and specialized training shall be provided to prepare graduate trainees of nominated instructions successfully to instruct future trainees in all phases of system/subsystem maintenance.

N. Cleanup: Collect used and leftover educational materials and remove from instruction location. Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.

END OF SECTION 01820

DETAILED TABLE OF CONTENTS

PART 2A

CONCRETE SPECIFICATION

SECTION 03300 - CAST-IN-PLACE CONCRETE
SECTION 03331 - CAST-IN-PLACE ARCHITECTURAL CONCRETE
SECTION 03500 - GLASSFIBER REINFORCED CEMENT (GRC)
SECTION 03544 - CEMENT-BASED SCREED

SECTION 03300 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section specifies cast-in-place concrete, including formwork, reinforcement, concrete materials, mix design, placement procedures, and finishes.

B. Cast-in-place concrete includes, but is not limited to, the following:

1. Foundations and footings.
2. Piles.
3. Drilled piers.
4. Slabs-on-grade.
5. Fill for steel deck.
6. Walls.
7. Duct banks.
8. Structural framing members.
9. Equipment pads and bases.
10. Fill for steel pan stairs.
11. Landscape concrete paving for walkways and bands.
12. Suspended slabs.
13. Stairs.
14. Columns.
15. Others.

C. Related Sections: The following sections contain requirements that relate to this section:

1. Concrete Specification Section "Lightweight Concrete Roof Insulation" for cast-in-place lightweight concrete roof insulation.
2. Steel Specification Section, "Steel Deck" for steel deck construction.

1.3 DEFINITIONS

A. Cementitious Materials: Portland cement alone or in combination with one or more of blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume.

1.4 REFERENCES

European Standards (EN) shall take precedence over all other standards.

Performance requirements of materials and initial type testing and procedures for factory production control, including the on-going testing and traceability regimes, shall be certified by means of a Declaration of Performance and CE Marking, as required under the EU Construction Products Regulation No. 305/2011.

A. European Standards (EN):

EN 206	Concrete - Specification, Performance, Production and Conformity.
EN 1992	Design of Concrete Structures.
EN 13670	Execution
EN 13369	Precast Concrete
EN 197	Cement
EN 1008	Mixing Water
EN 12620	Aggregates for Concrete
EN 450	Fly Ash for Concrete
EN 13263	Silicia Fumefor Concrete
EN 15167	Ground Granulated Blast Furnaceslag for Concrete
EN 13055	Lightweight Aggregates
EN 934	Admixtures for Concrete
EN 14889	Fibres for Concrete
EN 12878	Pigments
EN 12350	Testing of Fresh Concrete
EN 12390	Testing of Hardened Concrete
EN 13791	Assessment of Concrete Strength in Structures
EN 12504	Testing Concrete in Structures

B. American Association of State Highway and Transportation Officials (AASHTO):

AASHTO M182	Standard Specification for Burlap Cloth Made from Jute or Kenaf
AASHTO T26	Standard Method of Test for Quality of Water to be used in Concrete

C. American Concrete Institute (ACI):

ACI 117	Specifications for Standard Tolerances for Concrete Construction and Materials
ACI 301	Specifications for Structural Concrete for Buildings
ACI 304	Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 305	Hot Weather Concreting
ACI 309	Guide for Consolidation of Concrete
ACI 315	Standard Practice for Detailing Reinforced Concrete Structures
ACI 318	Building Code Requirements for Reinforced Concrete
ACI 347	Formwork for Concrete
ACI 504R	Guide to Joint Sealants for Concrete Structures

D. American National Standards Institute (ANSI):

ANSI Q9001	Quality Systems – Model for Quality Assurance in Design, Development, Production, Installation, and Servicing
ANSI Q9002	Quality Systems – Model for Quality Assurance in Production, Installation, and Servicing

E. American Society for Testing and Materials (ASTM):

ASTM A 82	Standard Specification for Steel Wire Reinforcement, Plain, for Concrete
ASTM A 185	Standard Specification for Steel Welded Wire Fabric, Plain, for Concrete Reinforcement
ASTM A 496	Standard Specification for Steel Wire, Deformed, for Concrete Reinforcement
ASTM A 497	Standard Specification for Steel Welded Wire Reinforcement, Deformed, for Concrete
ASTM A 615M	Standard Specification for Deformed and Plain Billet- Steel Bars for Concrete Reinforcement
ASTM A 706M	Standard Specification for Low-Alloy Steel Deformed and Plain Bars
ASTM A 775/A775M	Standard Specification for Epoxy-Coated Reinforcing Steel Bars
ASTM A 780	Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM A 820	Standard Specification for Steel Fibers for Fiber-Reinforced Concrete
ASTM C 31	Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C 33	Standard Specification for Concrete Aggregates
ASTM C 39	Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C 40	Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C 42	Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C 88	Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C 94	Standard Specification for Ready-Mixed Concrete
ASTM C 114	Standard Test Methods for Chemical Analysis of Hydraulic Cement
ASTM C 117	Standard Test Method for Materials Finer Than 75-Micrometer (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C 127	Standard Test Method for Density, Relative Density (Specific Gravity) and Absorption of Coarse Aggregate
ASTM C 128	Standard Test Method for Density, Relative Density Specific Gravity) and Absorption of Fine Aggregate
ASTM C 131	Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C 142	Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTM C 143	Standard Test Method for Slump of Hydraulic Cement Concrete
ASTM C 150	Standard Specification for Portland Cement
ASTM C 171	Standard Specification for Sheet Materials for Curing Concrete
ASTM C 172	Standard Practice for Sampling Freshly Mixed Concrete
ASTM C 186	Standard Test Method for Heat of Hydration of Hydraulic Cement
ASTM C 191	Standard Test Method for Time of Setting of Hydraulic Cement by Vicat Needle
ASTM C 192	Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory

ASTM C 219	Standard Terminology Relating to Hydraulic Cement
ASTM C 227	Standard Test Method for Potential Alkali Reactivity of Cement-Aggregate Combinations (Mortar-Bar Method)
ASTM C 289	Standard Test Method for Potential Reactivity of Aggregates (Chemical Method)
ASTM C 295	Standard Guide for Petrographic Examination of Aggregates for Concrete
ASTM C 309	Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C 311	Standard Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete
ASTM C 330	Standard Specification for Lightweight Aggregates for Structural Concrete
ASTM C 494	Standard Specification for Chemical Admixtures for Concrete
ASTM C 586	Standard Test Method for Potential Alkali Reactivity of Carbonate Rocks for Concrete Aggregates (Rock Cylinder Method)
ASTM C 595	Standard Specification for Blended Hydraulic Cements
ASTM C 618	Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete
ASTM C 881	Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C 920	Standard Specification for Elastomeric Joint Sealants
ASTM C 989	Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars
ASTM C 1064	Standard Test Method for Temperature of Freshly Mixed Portland Cement Concrete
ASTM C 1077	Standard Practice for Laboratories Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Laboratory Evaluation
ASTM C 1105	Standard Test Method for Length Change of Concrete Due to Alkali-Carbonate Rock Reaction
ASTM C 1116	Standard Specification for Fiber-Reinforced Concrete and Shotcrete
ASTM D 209	Standard Specification for Lampblack Pigment

ASTM D 512	Standard Test Methods for Chloride Ion in Water
ASTM D 516	Standard Test Method for Sulfate Ion in Water
ASTM D 1751	Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types
ASTM D 3963/D 3963M	Standard Specification for Fabrication and Jobsite Handling of Epoxy-Coated Steel Reinforcing Bars
ASTM E 154	Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover
ASTM E 329	Standard Specification for Agencies Engaged in the Testing and/or Inspection of Materials used in Construction
ASTM E 548	Standard Guide for General Criteria Used for Evaluating Laboratory Competence
ASTM E 1155M	Standard Test Method for Determining FF Floor Flatness and FL Floor Levelness Numbers (Metric)
ASTM E 1745	Standard Specification for Water Vapor Retarders Used in Contact with Soil or Granular Fill under Concrete Slabs

F. British Standards:

BS 812	Testing Aggregates
BS 882	Specification for Aggregates from Natural Sources for Concrete
BS 1881	Methods of Testing Concrete
BS 1199 and 1200	Specification for Building Sands from Natural Sources
BS 4027	Specification for Sulfate-Resisting Portland Cement
BS 4449	Specification for Carbon Steel Bars for the Reinforcement of Concrete
BS 4483	Steel Fabric for the Reinforcement of Concrete
BS 5328	Methods for Specifying Concrete Including Ready-Mixed Concrete
BS 8007	Code of Practice for Design of Concrete Structures for Retaining Aqueous Liquids
BS 8110	Structural Use of Concrete Part 1 (1997): Code of Practice for Design and Construction, Part 2 (1985): Code of Practice for Special Circumstances, Part 3 (1985): Design Charts for Singly Reinforced Beams, Doubly Reinforced Beams and Rectangular Columns

BS 8666	Specification for Scheduling, Dimensioning, Bending and Cutting of Steel Reinforcement for Concrete
EN 197	Part 1: Cement. Composition, Specifications and Conformity Criteria for Common Cements

G. Commercial Standards (CS):

Concrete Plant Manufacturers Bureau - Concrete Plant Standards

National Ready-Mixed Concrete Association (NRMCA) - Check List

H. Concrete Reinforcing Steel Institute (CRSI):

CRSI	Manual of Standard Practice

I. Corps of Engineers (CE):

CE CRD-C119	Test for Flat and Elongated Particles
CE CRD-C300	Curing Compound
CE CRD-C513	Rubber Waterstops
CE CRD-C572	PVC Waterstops

J. National Standard of Canada:

CAN/CSA A23.5 - M86Supplementary Cementing Materials.

1.5 SUBMITTALS

A. Product Data: For proprietary materials and items, including reinforcement and forming accessories, admixtures, patching compounds, waterstops, joint systems, curing compounds, dry-shake finish materials, and others as requested by the Engineer.

B. Design Mixes: For each concrete mix. Include alternate mix designs when characteristics of materials, project conditions, weather, test results, or other circumstances warrant adjustments.

C. Steel Reinforcement Shop Drawings: Details of fabrication, bending, and placement, prepared according to BS 8666, "Specification for Scheduling, Dimensioning, Bending and Cutting of Steel Reinforcement for Concrete." Include material, grade, bar schedules, stirrup spacing, bent bar diagrams, arrangement, and supports of concrete reinforcement. Include special reinforcement required for openings through concrete structures.

D. Formwork Shop Drawings: Prepare shop drawings for formwork indicating fabrication and erection of forms for specified finish concrete surface. Show form construction including jointing, especial form joints or reveals, location and pattern of form tie placement. Prepare formwork drawings by or under the supervision of a qualified professional engineer detailing fabrication, assembly, and support of formwork. The Engineer’s review is for general architectural applications and features only. Design and engineering of formwork for structural stability and efficiency are the Contractor's responsibility.

1. Shoring and Reshoring: Indicate proposed schedule and sequence of stripping formwork, shoring removal, and installing and removing reshoring.

E. Welding Certificates: Copies of certificates for welding procedures and personnel.

F. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and any other information required by the Engineer.

G. Material Test Reports: From a qualified testing agency indicating and interpreting test results for compliance of the following with requirements indicated, based on comprehensive testing of current materials:

H. Material Certificates: Signed by manufacturers and contractor certifying that each of the following items complies with specified requirements:

1. Cementitious materials and aggregates.
2. Form materials and form-release agents.
3. Steel reinforcement and reinforcement accessories:
   A copy of the manufacturer's test certificate for ultimate strength, elongation and cold bending, together with the chemical analysis of the steel shall be submitted to the Engineer for each consignment of reinforcing steel delivered to the Project site.
4. Fiber reinforcement.
5. Admixtures:
   Material certificates in lieu of material laboratory test reports when permitted by the Engineer. Material certificates shall be signed by the manufacturer and the Contractor, certifying that each material item complies with specified requirements. Provide certification from admixture manufacturers that chloride content complies with specified requirements.
6. Waterstops.
7. Curing materials.
8. Floor and slab treatments.
9. Bonding agents.
10. Adhesives.
11. Vapor retarders.
12. Epoxy joint filler.
13. Joint-filler strips.
14. Repair materials.

I. Samples: Samples of materials as requested by the Engineer, with names, sources, and descriptions, including, but not limited to, the following:

1. Color finishes.
2. Normal-weight aggregates.
3. Reglets.
4. Waterstops, reinjectable hosing, water swelling gaskets.
5. Vapor retarder/barrier.
6. Form liners.
7. Joint fillers.
8. Sealants.

J. Minutes of preinstallation conference.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Codes and Standards: Comply with 2001 Manual of Concrete Practice Parts 1, 2, 3, 4 & 5, and CRSI "Manual of Standard Practice" except where more stringent requirements are shown or specified.

C. Concrete Quality Control Engineer: Appoint a full-time Concrete Quality Control Engineer (CQCE) to ensure that concrete is properly produced, placed, cured and protected. The CQCE shall be equivalent to a Member of the Institute of Concrete Technology (MICT) and shall be responsible for the maintenance and submission of all specified records. The CQCE shall not report to his own company's site construction management but to his own company's management and to the Engineer. The CQCE shall set standards of quality and insist that these standards be followed, prepare a Quality Control Program for the inspection and testing of concrete and the maintenance of all reports and records to meet the specified requirements and requirements of the Engineer.

1. The CQCE shall be authorized to:

    a. Postpone concreting operations until outstanding requirements are corrected.
    b. Reject materials or workmanship that do not conform to this Specification.
    c. Prevent the use of equipment that could cause improper construction relative to this Specification.
    d. Stop any work that is not being done in accordance with specified requirements.
    e. Report within 24 hours and provide records to and as required by the Engineer upon discovery of non-compliance.

D. The Contractor shall operate a Quality Assurance System in accordance with ANSI Q9002. This Quality Assurance Manager shall be responsible for the preparation of a Quality Plan for approval of the operations specified in this Section. The Quality Plan shall include, among other things, the list and schedule of the Quality Control audits that the Quality Assurance Manager or his designee shall make.

E. Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products complying with EN 206 requirements for production facilities and equipment.

F. Engage an independent testing agency acceptable to the Engineer to perform material evaluation tests and qualified according to ISO/IEC 17025 and ISO 9001 to conduct the testing indicated, as documented according to EN 12350 and EN 12390 and to design concrete mixes.

1. Personnel conducting field tests shall be qualified as ACI concrete field testing technician, Grade 1 according to ACI CP-1 or an equivalent certification program.

G. Materials and installed work may require testing and retesting at any time during progress of Work. Tests, including retesting of rejected materials for installed Work, shall be done at the Contractor's expense.

H. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, each aggregate from one source, and each admixture from the same manufacturer.

I. Welding: Qualify procedures and personnel according to BS 7123, "Specification for Metal Arc Welding of Steel for Concrete Reinforcement."

J. Mockups: Before casting concrete that is exposed to view on surfaces of the completed structure or building, cast a mockup for each exposed element, including slabs if applicable, to demonstrate typical joints, surface finish, texture, color, tolerances, quality of materials and standard of workmanship in the completed Work.

1. Build mockups in the location and of the size indicated or, if not indicated, as directed by the Engineer.
2. Notify the Engineer seven days in advance of dates and times when mockups will be constructed.
3. In the presence of the Engineer, damage parts of exposed surfaces as selected by the Engineer, and demonstrate materials and techniques proposed for repairs to match adjacent undamaged surfaces.
4. Obtain the Engineer's approval of mockups before starting cast-in-place concrete elements exposed to view.
5. If the Engineer determines that any mockup does not meet requirements, demolish and remove from the site and cast another until the mockup is approved.
6. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
7. Demolish and remove mockups when directed.

K. Preinstallation Conference: Conduct conference at Project site to comply with requirements as follows:

1. At least 35 days prior to submitting design mixes, conduct a meeting to review detailed requirements for preparing concrete design mixes and to determine procedures for satisfactory concrete operations. Review requirements for submittals, status of coordinating work, and availability of materials. Establish preliminary work progress schedule and procedures for materials inspection, testing, and certifications. Require representatives of each entity directly concerned with cast-in-place concrete to attend conference, including, but not limited to, the following:

a. Contractor’s superintendent.
b. Agency responsible for concrete design mixes.
c. Agency responsible for field quality control.
d. Ready-mix concrete producer.
e. Concrete subcontractor.
f. Primary admixture manufacturers.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle steel reinforcement to prevent bending and damage.

1. Avoid damaging coatings on steel reinforcement.
2. Repair damaged epoxy coatings on steel reinforcement according to ASTM D 3963/D 3963M.

PART 2 - PRODUCTS

2.1 FORM-FACING MATERIALS

A. Void Forms: Biodegradable paper surface, treated for moisture resistance, structurally sufficient to support weight of plastic concrete and other superimposed loads.

B. Chamfer Strips: Wood, metal, PVC, or rubber strips, 20 mm by 20 mm, unless otherwise indicated.

2.2 FORM MATERIALS

A. Forms for Exposed Finish Concrete: Plywood, metal, metal-framed plywood faced, or other acceptable panel-type materials to provide continuous, straight, smooth, exposed surfaces. Furnish in largest practicable sizes to minimize number of joints and to conform to joint system shown on Drawings.

B. Forms for Unexposed Finish Concrete: Plywood, lumber, metal, or another acceptable material. Provide lumber dressed on at least two edges and one side for tight fit.

C. Forms for Textured Finish Concrete: Units of face design, size, arrangement, and configuration to match the Engineer's control sample. Provide solid backing and form supports to ensure stability of textured form liners.

D. Forms for Cylindrical Columns and Supports: Metal, glass-fiber-reinforced plastic, or paper or fiber tubes that will produce smooth surfaces without joint indications. Provide units with sufficient wall thickness to resist wet concrete loads without deformation.

E. Pan-Type Forms: Glass-fiber-reinforced plastic or formed steel, stiffened to support weight of placed concrete without deformation.

F. Carton Forms: Biodegradable paper surface, treated for moisture-resistance, structurally sufficient to support weight of plastic concrete and other superimposed loads.

G. Form Release Agent: Provide commercial formulation form release agent with a maximum of 350 g/L volatile organic compounds (VOCs) that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.

   1. Formulate form release agent with rust inhibiters for steel facing materials.

H. Form Ties: Factory-fabricated, adjustable-length, removable or snap-off metal form ties designed to prevent form deflection and to prevent spalling of concrete upon removal. Provide units that shall leave no metal closer than 38 mm to the plane of the exposed concrete surface. No permanent metallic part shall have less concrete cover than the reinforcement. Provide ties that, when removed, will not leave holes larger than 25 mm in diameter in the concrete surface. Furnish ties with integral water-barrier plates to walls indicated to receive damp-proofing or waterproofing.

I. Void Forms: Biodegradable paper surface, treated for moisture resistance structural sufficient to support weight of plastic concrete and other superimposed loads.

J. Chamfer Strips: Wood, metal, PVC, or rubber strips 20 mm x 20 mm, size as indicated on drawing.

2.3 STEEL REINFORCEMENT

A. Reinforcing Bars: EN 10080, or BS 4449 grade 460 Type 2 deformed, uncoated. One test per 5000 m length delivered to site.

B. Plain-Steel Wire: BS 4482, Steel wire for the Reinforcement of Concrete Products

C. Welded Wire Fabric: BS 4483, welded steel wire fabric.

D. Only new material shall be furnished. On receipt and at time of installation, material shall be free of loose rust and loose mill scale, deleterious amounts of salts and coatings that reduce or destroy bond. Tight rust and mill scale or surface irregularities are acceptable if the weight and dimensions, including height of deformations and tensile properties, of a test specimen that has been wire-brushed by hand, are not less that those required by the applicable Standards.

E. Reinforcement shall be accurately bent, cut or formed to the dimensions and configuration shown on Drawings and within the tolerances specified in BS 8666. Reinforcement shall be bent cold using pin sizes in accordance with BS 8666. Bars may be preheated only if prior approval has been requested and received. Reinforcement shall not be rebent or straightened without prior approval.

1. Reinforcement having a reduced section, kinks, visible transverse cracks at bends, or otherwise damaged in any way shall not be used. Galvanized steel shall not be used for reinforcement.
2. Reinforcement shall not be welded unless specifically shown on Drawings or permitted as an exception and then only after approval of the welding method appropriate to the grade of steel and the type of welding rod to be used.

2.4 REINFORCEMENT ACCESSORIES

A. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire fabric in place. Manufacture bar supports according to CRSI's "Manual of Standard Practice" from steel wire, plastic, or precast concrete or fiber-reinforced concrete of greater compressive strength than concrete, and as follows:

1. For concrete surfaces exposed to view where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected or CRSI Class 2 stainless-steel bar supports.
2. Other reinforcement supports shall consist of concrete spacer blocks made of the same materials, to the same specified requirements and with the same inherent properties as the parent material with the exception that the maximum aggregate size shall be appropriate for the thickness of cover to the reinforcement.

B. Joint Dowel Bars: Plain-steel bars, ASTM A 615M, Grade 60 (420 MPa). Cut bars true to length with ends square and free of burrs.

C. Epoxy-Coated Joint Dowel Bars: ASTM A 775/A 775M; with ASTM A 615M, Grade 60 (420 MPa), plain-steel bars.

D. Epoxy Repair Coating: Liquid, two-part, epoxy repair coating; compatible with epoxy coating on joint dowel bars and complying with ASTM A 775/A 755M.

E. Zinc Repair Material: ASTM A 780, zinc-based solder, paint containing zinc dust, or sprayed zinc.

F. Mechanical Splices (Couplers) of deformed high yield steel bars are to consist of two seamless steel sleeves and interconnecting high tensile steel stud with plastic protection caps for threaded section of sleeve. To be tested and the test to exceed 135% of the specified yield strength of grade 60 bar.

2.5 CONCRETE MATERIALS

A. Portland Cement: Cement shall be low alkali with chemical composition in accordance with EN 197: Part 1. The magnesia content shall be limited to 4 percent by weight of cement. Use one brand of cement throughout Project unless otherwise approved by the Engineer. Manufacturer's test certification shall be supplied for each delivery of cement and shall confirm that the cement complies with the above requirements and shall be submitted by the Contractor not later than the day of delivery of the cement. The Engineer shall have the right to call for tests, the cost of which are to be borne by the Contractor, on each delivery of cement to confirm that the cement meets the following requirements.

1. Use Ordinary Portland Cement (OPC) conforming to EN 197, CEM 1. Cement meeting the requirements of rapid hardening Portland cement shall not be used and the heat of hydration shall not exceed 325 kj/kg. C_3A content shall be a maximum of 8 percent by weight.
2. Use sulfate-resisting Portland cement conforming to BS 4027, C_3A content shall not be more than 5% by weight, in all concrete at or below the finished ground floor level.
3. Sampling shall be carried out in accordance with EN 196: Part 7.
4. Test cement for fineness by air permeability apparatus in accordance with ASTM C 204 to meet the requirements of ASTM C 150.
5. Test cement for soundness, Autoclave expansion in accordance with ASTM C151.

B. Silica Fume: Silica fume (SF) used as a cement replacement shall be in accordance with Type U, specified in Canadian National Standard CAN/CSA-A23.5 - M86. There shall be a 5 percent replacement of the OPC. The silica fume shall be obtained from an approved supplier.

1. The approved supplier shall provide documentation to establish the following:

   a. That the silica fume complies with Type U of CAN/CSA-A23.5 - M86.
   b. The silica fume results from the production of silicon or ferro-silicon alloys containing at least 85 percent silicon.
   c. That source of supply shall remain the same for the construction period of the project.
   d. That the supply shall be adequate to meet the anticipated peak requirement.

   e. Chemical analyses to give the percentages of the following materials:

Si 02	C
Ca 0	Fe2 03
Al2 03	Na2 0
K2 0	Mg 0
S03	

   f. Loss on ignition.
   g. Percentage of particles greater than 44 mm.
   h. Specific surface area and method of test, together with corresponding particle size.

2. Delivery of silica fume to the concrete batching plant shall be in dry powder form, with a bulk density between 200 and 650 kg/cu. m.
3. Manufacturer's test certification shall be supplied for each delivery of silica fume and shall confirm that the silica fume complies with the above requirements and shall be submitted by the Contractor no later than the day of delivery of the silica fume.
4. The Engineer shall have the right to call for tests, the cost of which are to be borne by the Contractor, on each delivery of silica fume, if necessary to establish or confirm that the silica fume meets the above requirements.

C. Select supplementary cementing materials from subparagraphs below, if permitted. Blending of fly ash or slag with Portland cement is accomplished at mixing plant. Fly ash, slag, or pozzolanic materials may slow rate of concrete strengthening and affect color uniformity.

1. Fly Ash: EN 450 or ASTM C 618, Class C or F.
2. Ground Granulated Blast-Furnace Slag: EN 15167 or ASTM C 989, Grade 100 or 120.

D. Normal-Weight Aggregates: Aggregates shall be from approved sources and shall conform to the requirements of ASTM C 33 and BS 882. Petrographic analyses shall be made in accordance with ASTM C 295. Aggregates for exposed concrete shall be from a single source and shall not contain substances that cause spalling. Only aggregates not susceptible to alkali aggregate reaction shall be used. The Contractor shall supply samples of the materials for approval by the Engineer and each aggregate source shall be subject to monitoring by the Engineer. Grading of aggregate shall be to the completion of BS 882.

1. Coarse Aggregate: Coarse aggregate size shall be 20 mm nominal and those retained on a 5mm sieve and shall consist of crushed or uncrushed gravel or crushed stone and shall be selected, recrushed, finish screened and washed with water meeting the requirements of Paragraph 2.5.F as necessary to comply with the following:

Frequency of Tests	Test Description	Standard	Limit
Initial	Los Angeles Abrasion Loss (Grading A or B)	ASTM C 1 31	25% maximum
Initial	Ratio of Los Angeles Abrasion Loss at 100 & 500 Revolutions (100/500 Revolutions Value)	ASTM C 1 31 Note 6	0.25% maximum
1 per day	Clay Lumps and Friable Particles	ASTM C 1 42	1.0% maximum
1 per day	Material Finer than 75 Microns	ASTM C 1 17	1.0% maximum
1 per 7 days	Water Absorption	ASTM C 1 27	2.0% maximum
1 per 3 days	Chlorides as Cl	BS 812	0.03% maximum*
1 per 3 days	Sulfates as SO_3	BS 812	0.3% maximum*
1 per 30 days	Magnesium Sulfate Soundness Loss (5 cycles)	ASTM C 8 8	5.0% maximum
1 per 3 days	Flakiness Index	BS 812	25% maximum
1 per 3 days	Elongation Index	BS 812	25% maximum
	Reactive Silica	ASTM C 2 27	Per Appendix X1.3.7 of ASTM C 33
1 per 7 days	Specific gravity		Minimum 2.6
1 per 2 days	Moisture Content		

* Additionally, limits specified in Paragraph 2.14.H for the total salt content of concrete shall not be exceeded.

2. Fine Aggregate: Fine aggregate, those passing a 5mm sieve, shall consist of crushed gravel, crushed stone or natural sand with rounded or surrounded particles and shall be washed as necessary to comply with the following:

Test Description	Standard	Limit
Clay Lumps and Friable Particles	ASTM C 142	1.0% maximum
Material Finer than 75 Microns	ASTM C 117	2.0% maximum
Water Absorption	ASTM C 128	1.0% maximum
Chlorides as Cl	BS 812	0.06% maximum*
Sulfates as SO_3	BS 812	0.30% maximum*
Organic Impurities	ASTM C 40	Lighter than Standard

* Additionally, limits specified in Paragraph 2.14.H for the total salt content of concrete shall not be exceeded.

3. Certification: Obtain from each proposed source of supply Test Certification to confirm that the aggregates comply with the above requirements. The following information shall be provided:

   a. Quarry location.
   b. Aggregate type.
   c. Petrographic analysis report.
   d. Grading curve.
   e. Shape and surface texture.
   f. Flakiness index.
   g. 10 percent fines value.
   h. Impact test.
   i. Shell content.
   j. Chloride and sulfate content.
   k. Relative density.
   l. Water absorption value and moisture content.
   m. Silt, clay, and dust content.
   n. Results of reactive silica tests.
   o. Organic impurities (fine aggregate only).

4. Testing:

   a. When a source of supply for each aggregate type had been established, samples of materials delivered to Project site shall be taken for testing in accordance with BS 812 as follows:

      1) Tests for clay, silt and dust, and sieve analysis shall be carried out for every 20 tons of fine aggregate and every 40 tons of coarse aggregate.
      2) Chemical analyses shall be carried out on every 100 tons of aggregate.

   b. The Engineer shall have the right to call for additional samples at any time for testing of aggregates delivered to the Project site or of aggregates at the source of supply in order to confirm that the aggregates meet the above requirements.

5. Transportation: During transportation to the Project site, all aggregates shall be protected from wind-borne contaminants. If these contaminants are present at time of delivery to the Project site, then the aggregates shall be washed with water meeting the requirement

of Paragraph 2.5.F. Transport vehicles shall be cleaned of possible contamination due to previous use.

6. Storage: Aggregates shall be stored (under shade) on hard concrete floors or other approved materials having sufficient slope to ensure adequate drainage of aggregate before being used for concrete and each size and type shall be stored in separate heaps without intermixing. Storage shall prevent contamination of the aggregates by foreign material including wind blown dust. Fine and coarse aggregates shall be separated by permanent substantial partitions. Methods of storing, shading and cooling aggregates shall be approved by the Engineer.

E. Water: Water used for mixing concrete, ice production, washing and cooling aggregates, and curing concrete shall be free from impurities, oil, acid, salts, alkali, organic matter, and other potentially deleterious substances in accordance with EN 1008 and when tested in accordance with EN 1008. Additionally, the limits specified in Paragraph 2.14.H for the total salt content of the concrete shall not be exceeded.

1. Once a source of satisfactory supply has been established, further tests shall be made daily with a portable electrical conductivity probe calibrated against the satisfactory supply. If the conductivity exceeds that of the satisfactory supply, then further chemical tests shall be performed.

F. Lightweight Aggregate: EN 13055

1. Nominal Maximum Aggregate Size: 10 mm.

2.6 ADMIXTURES

A. Admixtures containing Chlorides shall not be used.

B. General: No admixture shall be used in the concrete without the Engineer's written approval and under no circumstances shall admixtures containing chlorides or other corrosive agents be allowed. Admixture compatibility with the type of cement used shall be proven.

C. The Contractor shall perform a trial batch and casting to substantiate the manufacturer’s claims of workability, retardation and air entrainment (0 to 1.0 percent maximum), as specified in Article 2.14. Admixtures shall comply with the following standards: ASTM C494/C494 M, EN 934 and EN 480. Also, admixture shall comply with EN 12878 for pigments of cement.

D. Air-Entraining Admixture: No air entraining agent shall be used.

E. Admixtures shall be incorporated into the mix design strictly in accordance with the manufacturer’s written instructions.

F. High-Range Water-Reducing Admixture (Superplasticizer): ASTM C 494, Type G.

1. If necessary, and only with the Engineer’s approval, a naphthalene sulphonate retarding superplasticizer shall be used to increase workability of the concrete and retard the initial set.
2. Products: To produce fluid concrete with a slump value at least 200 mm, easily flowing, but at the same time free from segregation and having the same water/cement ratio as that of a no slump concrete with admixture. The product shall result in concrete that

remains workable for a minimum of 3 hours at +20 deg.C (68 deg.F) and for a minimum of 1 hour at +40 deg. C (104 deg. F).

3. Obtain from the retarding superplasticiser supplier, details of the material for review by the Engineer and confirmation that it is in accordance with specified requirements. Confirmation shall be obtained that the retarding superplasticiser is compatible with any pozzolan that is used.

4. Glare-Reducing Agent: For landscape concrete paving, provide material for reducing glare. Comply with ASTM D 209.

G. Water-Reducing Admixture (Plasticizer): ASTM C 494, Type A.

H. Water-Reducing and Accelerating Admixture: ASTM C 494, Type E.

I. Water-Reducing and Retarding Admixture: ASTM C 494, Type D.

J. Corrosion-Inhibiting Admixture: Commercially formulated, mixed cathodic and anodic inhibitor based on Amines and Alcohol; capable of forming a protective barrier and absorbed on the reinforcement surface of concrete for protecting steel bars and minimizing chloride reactions with steel reinforcement in concrete.

2.7 FIBER REINFORCEMENT

To comply with EN 14889 and where not specified, to comply as follows:

A. Carbon-Steel Fiber: ASTM A 820, deformed, minimum 60 mm long, and of diameter or effective diameter indicated.

1. Fiber: Type 1, cold-drawn wire, or Type 2, cut sheet.

B. Synthetic Fiber: Fibrillated or monofilament polypropylene fibers engineered and designed for use in concrete, complying with ASTM C 1116, Type III, 15 to 40 mm long.

2.8 WATERSTOPS

A. Waterstops: Provide flat, dumbbell-type waterstops at construction joints and other joints as indicated. Waterstops shall be sized to suit joints.

1. Flexible Rubber Waterstops: CE CRD-C 513, for embedding in concrete to prevent passage of fluids through joints. Factory fabricate corners, intersections, and directional changes.

a. Select profile from five subparagraphs below.
b. Profile: Flat, dumbbell with center bulb.
c. Profile: Flat, dumbbell without center bulb.
d. Profile: Ribbed with center bulb.
e. Profile: Ribbed without center bulb.
f. Profile: As indicated.

2. Flexible PVC Waterstops: CE CRD-C 572, for embedding in concrete to prevent passage of fluids through joints. Factory fabricate corners, intersections, and directional chnges.

a. Select profile from five subparagraphs below.
b. Profile: Flat, dumbbell with center bulb.
c. Profile: Flat, dumbbell without center bulb.
d. Profile: Ribbed with center bulb.
e. Profile: Ribbed without center bulb.
f. Profile: As indicated.

B. Self-Expanding Strip Waterstops: Manufactured rectangular or trapezoidal strip, sodium bentonite or other hydrophylic material for adhesive bonding to concrete.

C. Re-injectable Hosing: Re-injectable hosing shall be used where shown on Drawings. Re-injectable hosing shall comply with the following requirements:

1. The hose shall consist of a hollow PVC core with lateral openings covered by neoprene valves. The hose shall be applied in 10 to 12 m lengths with entry port and vent ends terminating in junction boxes. The hose shall permit the option of re-injection. The hose shall be laid and injected by an approved applicator.
2. The joint shall be hydraulically tested to a pressure at least 2.0 bar higher than the expected local conditions. If the hose fails to hold the pressure for 10 minutes, the hose shall be injected with a vinyl ester based flexible resin as required to seal the joint and then retested.

2.9 VAPOR RETARDERS, CURING COMPOUNDS AND FINISH

To comply with EN 934-1 and 2 and where not specified, to comply as follows:

A. Vapor Retarder: ASTM E 1745, Class C, of one of the following materials; or polyethylene sheet, ASTM D 4397, not less than 0.25 mm thick:

1. Nonwoven, polyester-reinforced, polyethylene coated sheet; 0.25 mm thick.
2. Three-ply, nylon- or polyester-cord-reinforced, laminated, high-density polyethylene sheet; 0.18 mm thick.

B. Vapor Retarder: 2.8 mm thick, semiflexible, seven-ply sheet membrane consisting of reinforced core and carrier sheet with fortified asphalt layers, protective weathercoating, and removable plastic release liner. Furnish manufacturer's accessories including bonding asphalt, pointing mastics, and self-adhering joint tape.

1. Water-Vapor Permeance: 0.00 ng/Pa x s x sq. m; ASTM E 154.
2. Tensile Strength: 24.5 kN/m; ASTM E 154.
3. Puncture Resistance: 400 N; ASTM E 154.

C. Related Materials:

1. Polyethylene sheet not less than 8 units thick.
2. Water resistant barrier consisting of heavy Kraft papers laminated together with glass fiber reinforcement and over coated with black polyethylene on each side.

D. Fine-Graded Granular Material: Clean mixture of crushed stone, crushed gravel, and manufactured of natural sand; 100 percent passing a 4.75 mm sieve and 10 to 30 percent passing a 0.15 mm sieve; meeting deleterious substance limits of EN 12620 for aggregates.

E. Granular Fill: Clean mixture of crushed stone or crushed or uncrushed gravel; with 100 percent passing a 38 mm sieve and 0 to 5 percent passing a 2.36 mm sieve.

F. Sand Cushion: Clean, manufactured or natural sand.

G. Absorptive Cover: Burlap cloth made from jute or kenaf weighing approximately 0.29 kg/sq. m and complying with AASHTO M182, Class 2.

H. Moisture-Retaining Cover (Impervious Sheeting): One of the following, complying with ASTM C 171:

1. Waterproof paper.
2. Polyethylene film.
3. Polyethylene-coated burlap.

I. Curing Compound: Curing Compound shall be a non-degradable Pliolite Resin in accordance with EN 206 and ASTM C 309, Type I, Class B. It shall in addition to curing, act as both a sealer and primer coat for most subsequent coatings. The Contractor shall obtain confirmation from the supplier of the curing compound that it is compatible with the final applied finishes and meets the above requirements.

J. Evaporation Control: Monomolecular film-forming compound applied to exposed concrete slab surfaces for temporary protection from rapid moisture loss.

K. Underlayment Compound: Free-flowing, self-leveling, pumpable, cement-based compound for applications from 25 mm thick to feathered edges. The underlay shall be applied in passes not thicker than 6 mm at any interval.

L. Colored Wear-Resistant Finish: Packaged dry combination of materials consisting of Portland cement, graded aggregate, coloring pigments, and plasticizing admixture. Use coloring pigments that are finely ground nonfading mineral oxides interground with cement. Color shall be as selected by the Engineer from manufacturers' standards, unless otherwise indicated.

2.10 FLOOR AND SLAB TREATMENTS

A. Slip-Resistive Aggregate Finish: Factory-graded, packaged, rustproof, nonglazing, abrasive aggregate of fused aluminum-oxide granules or crushed emery with emery aggregate containing not less than 50 percent aluminum oxide and not less than 25 percent ferric oxide; unaffected by freezing, moisture, and cleaning materials.

B. Unpigmented Mineral Dry-Shake Floor Hardener: Factory-packaged dry combination of Portland cement, graded quartz aggregate, and plasticizing admixture.

C. Pigmented Mineral Dry-Shake Floor Hardener: Factory-packaged dry combination of Portland cement, graded quartz aggregate, coloring pigments, and plasticizing admixture. Use coloring pigments that are finely ground, nonfading mineral oxides interground with cement.

1. Colors: As selected by the Engineer from manufacturer's full range for these characteristics.

D. Penetrating Liquid Floor Treatment: Chemically reactive, waterborne solution of inorganic silicate or siliconate materials and proprietary components; odorless; colorless; that penetrates, hardens, and densifies concrete surfaces.

2.11 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 305 g/sq. m when dry.

C. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

D. Water: Potable.

E. Clear, Solvent-Borne, Membrane-Forming Curing Compound: EN 206 and ASTM C 309, Type 1, Class B.

F. Clear, Waterborne, Membrane-Forming Curing Compound: EN 206 and ASTM C 309, Type 1, Class B.

G. Clear, Solvent-Borne, Membrane-Forming Curing and Sealing Compound: EN 206 and ASTM C 1315, Type 1, Class A.

2.12 RELATED MATERIALS

A. Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber, or ASTM D 1752, cork or self-expanding cork.

B. Epoxy Joint Filler: Two-component, semirigid, 100 percent solids, epoxy resin with a Shore A hardness of 80 per ASTM D 2240.

C. Bonding Agent: ASTM C 1059, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

D. Epoxy-Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class and grade to suit requirements, and as follows:

1. Type II, non-load bearing, for bonding freshly mixed concrete to hardened concrete.
2. Types I and II, non-load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.
3. Types IV and V, load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

E. Reglets: Fabricate reglets of not less than 0.55 mm thick galvanized steel sheet. Temporarily fill or cover face opening of reglet to prevent intrusion of concrete or debris.

F. Dovetail Anchor Slots: Hot-dip galvanized steel sheet, not less than 0.85 mm thick, with bent tab anchors. Temporarily fill or cover face opening of slots to prevent intrusion of concrete or debris.

2.13 REPAIR MATERIALS

A. Repair Underlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 3 mm and that can be feathered at edges to match adjacent floor elevations.

1. Cement Binder: Portland cement or hydraulic or blended hydraulic cement as defined in EN 197.
2. Primer: Product of underlayment manufacturer recommended for substrate, conditions, and application.
3. Aggregate: Well-graded, washed gravel, 3 to 6 mm or coarse sand as recommended by underlayment manufacturer.
4. Compressive Strength: Not less than 30 MPa at 28 days when tested.

B. Repair Topping: Traffic-bearing, cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 6 mm.

1. Cement Binder: Portland cement or hydraulic or blended hydraulic cement as defined in EN 197.
2. Primer: Product of topping manufacturer recommended for substrate, conditions, and application.
3. Aggregate: Well-graded, washed gravel, 3 to 6 mm or coarse sand as recommended by topping manufacturer.
4. Compressive Strength: Not less than 40 MPa at 28 days when tested.

2.14 CONCRETE MIXES

A. Prepare design mixes for each type and strength of concrete determined by either laboratory trial mix or field test data bases, as follows:

1. Proportion normal-weight concrete according to EN 206
2. Proportion lightweight structural concrete according to EN 206 and EN 13055

B. Use a qualified independent testing agency acceptable to the Engineer for preparing and reporting proposed mix designs for the laboratory trial mix basis.

1. Do not use the same testing agency for field quality control.

C. Submit written reports to the Engineer of each proposed mix for each class of concrete at least 15 days prior to start of Work. Do not begin concrete production until proposed mix designs have been reviewed and approved by the Engineer.

D. Design mixes to provide normal-weight concrete with the following properties unless otherwise indicated on Drawings:

1. Blinding Concrete: Proportion normal-weight concrete mix as follows:

a. Compressive Strength (28 Days): 20 MPa.

b. Maximum Slump: 100 mm.

2. Footings and Foundation Walls: Proportion normal-weight concrete mix as follows:

a. Compressive Strength (28 Days): 40 MPa.
b. Select slump limits from subparagraphs below or revise to suit Project.
c. Maximum Slump: 100 mm.
d. Delete subparagraph below if superplasticizers are not permitted.
e. Maximum Slump for Concrete Containing High-Range Water-Reducing Admixture: 200 mm after admixture is added to concrete with 50 to 100 mm slump.

3. Slab-on-Grade: Proportion normal-weight concrete mix as follows:

a. Compressive Strength (28 Days): 40 MPa.
b. Maximum Slump: 100 mm.

4. Suspended Slabs: Proportion normal-weight concrete mix as follows:

a. Compressive Strength (28 Days): 40 MPa.
b. Maximum Slump: 100 mm.

5. Suspended Slabs: Proportion lightweight structural concrete mix as follows:

a. Compressive Strength (28 Days): 40 MPa.
b. Maximum Slump: 100 mm.

6. Building Frame Members: Proportion normal-weight concrete mix as follows:

a. Compressive Strength (28 Days): 40 MPa.
b. Maximum Slump: 100 mm.
c. Maximum Slump for Concrete Containing High-Range Water-Reducing Admixture: 200 mm after admixture is added to concrete with 50 to 100 mm slump.

E. Cement and Pozzolan Content:

1. Structural concrete shall contain a minimum of 355 kg/cu. m of Ordinary Portland Cement (OPC) plus 25 kg/cu. m of silica fume (SF). All concrete below ground shall have protective coating as specified in Articles 3.17 and 3.18.
2. Blinding/mudmat concrete shall contain sufficient OPC to obtain the specified design strength.

F. Water-Cement Ratio: The free water-cement ratio shall not exceed 0.40. Where pozzolans are used, the water-cement ratio shall be the water content divided by the cement plus pozzolan content. The water-cement ratio shall be continuously checked at the mixer with due allowance made for water contained in the aggregates. Under no circumstance shall water be added between the mixer and the place of concrete placement. The Engineer may require that the water-cement ratio be checked during tests performed on fresh concrete samples taken at the time of placement as specified in Article 3.21.

G. Slump Limits: The slump of concrete mixes shall be such that the concrete can be transported, placed into the forms, and compacted without segregation in accordance with Article 3.8. If no superplasticizer is required, the slump at time of placement shall be 50-75 mm as measured in

accordance with EN 206. In accordance with Article 3.8, the concrete shall at all times exhibit a mobility that enables this placement to be affected with the minimum amount of difficulty.

H. Total Salt Content:

1. Chlorides: The total chloride content (sum of both acid soluble and water soluble chlorides) of the concrete from all sources, expressed as chloride ion, shall not exceed 0.15 percent by weight of dry cement, inclusive of pozzolan, when tested in accordance with BS 1881.
2. Sulfates: The total sulfate content of the concrete from all sources, expressed as SO_3, when tested in accordance with BS 1881, shall not exceed 3 percent by weight of dry cement, inclusive of pozzolan.

I. Initial Setting Time:

1. The initial setting time shall be not less than one hour after the production concrete is discharged into the form. With a maximum time between mixing and placing concrete of one hour, the total time between mixing and initial set shall be a minimum of 2 hours. There shall be a maximum setting time of 6 hours.
2. When trial mixes are made to determine the workability of the concrete, the initial setting time of the cement paste shall be determined using the method defined in ASTM C 191 but at the maximum allowable temperature and with same proportions of pozzolan and retarding superplasticizer as specified in this Specification.

J. Test Mixes - Structural Grade Concrete: When the proposed workability, proportions of aggregates and superplasticizer, and strength for each grade of concrete have been established, test concrete shall be produced for approval by the Engineer. The following shall be performed for each grade of concrete:

1. Six separate test mixes shall be made and cured in accordance with EN 206 and tested in accordance with EN 12390.
2. The 36 cylinders for each grade of concrete shall be tested for compressive strength at 28 days and the mean strength and standard deviation established for each grade.
3. The test mixes shall be accepted provided that:

    a. The mix proportions and workability are in accordance with this Specification.
    b. The standard deviation for each grade is 3.5 MPa or less.
    c. The mean strength for structural concrete exceeds the specified design strength by 6 MPa.

K. Test Mixes - Blinding or Mudmat Concrete:

1. Concrete for blinding or mudmat shall be a designed mix with the specified design strength of 20 MPa. The trial mix proportions shall be left to the discretion of the Contractor. Once they have been chosen, three separate test mixes shall be made with those proportions and three 150 mm test cylinders shall be made from each mix for testing at 28 days. The trial mix proportions and water/cement ratio shall be approved if the average strength of the nine cylinders is not less than 10 N/sq. mm.
2. No production test shall be made for this grade of concrete but the Engineer will monitor the mix proportions and water/cement ratio.

L. Adjustment to Concrete Mixes: Mix design adjustments may be requested when characteristics of materials, job conditions, weather, test results, or other circumstances warrant. Laboratory

test data for revised mix design and strength results shall be submitted to and approved by the Engineer before using in the Work.

M. Absorption Test: The absorption of the hardened concrete from the trial mixes for structural grades shall be tested as a measure of the concrete's ability to resist the ingress of aggressive salts. No absorption tests shall be required for blinding or mudmat concrete. Absorption tests shall be in accordance with modified BS 1881: Part 122 as follows:

1. After trial mixes have been accepted, three 150 mm cylinders shall be cast from each grade of concrete and immersed in water at 20 deg. C (68 deg. F) for 7 days.
2. At the end of the curing period, 75 mm diameter core specimens shall be cut along the longitudinal axis of each cylinder to a depth of 75 mm.
3. The specimens shall be dried in an oven at 105 deg. C (221 deg. F) for 72 hours.
4. The specimens shall be cooled in a dry airtight vessel for 24 hours, weighed, and than immediately immersed in a tank containing water at 20 deg. C (68 deg.F) with the longitudinal axis of the cores horizontal, and with 25 mm depth of water over the specimens.
5. The specimens shall be immersed for 24 hours, then removed, shaken, surface dried, and reweighed. The water absorption shall be calculated as the increase in mass resulting from immersion, expressed as a percentage of the dry mass.
6. If the cores lengths differ from 75 mm, a correction factor (graph given in BS 1881) shall be applied.
6. The mean of the corrected absorption figure for each concrete grade shall be calculated and the absorption of the concrete mixes shall be acceptable if the mean absorption is less than 2 percent at 30 minutes. The mean absorption figure and the lowest absorption figure shall be recorded for each grade and used for comparison purposes with absorption tests carried out on cores cut from in-situ concrete, as scheduled in Paragraph 3.21.C.4.

N. Permeability Tests:

1. DIN 1048 Depth of Penetration shall be carried out as a measure of water permeability on prototype samples of different components of the structure before the mix design or manufacturing method are approved. The prototype samples shall be short sections of not less than 1.0m x 1.0m x full thickness. The samples shall be fully reinforced and cast in similar environment as the actual structure. The tests shall be carried out in accordance with DIN 1048 at age 28 days and the maximum value of penetration shall not exceed the following limits:

In-situ construction:	Less than 25 mm.
Precast units:	Less than 20 mm.

2. The chloride permeability of the concrete shall be determined using cores taken from the prototype samples, in accordance with AASHTO T277 "Rapid Determination of the Chloride Permeability of Concrete". The following limits shall not be exceeded:

In-situ construction:	1500 coulombs.
Precast units:	1000 coulombs.

3. The concrete will be deemed to comply with the permeability requirements of this specification if all test results comply with the limits indicated in paragraphs 1 and 2.

4. If any of the DIN 1048 and the rapid chloride permeability test results of unprotected surfaces of a sample failed to meet the above requirements, then all the concrete work represented by such sample shall be deemed not to comply with the permeability requirements.
5. If, in the opinion of the Engineer, the concrete that has failed to meet the permeability requirements of this Specification is likely to cause durability problems, three cores shall be cut from the area represented by failed samples for additional DIN 1048 and rapid chloride permeability testing. The actual location shall be decided by the Engineer.

O. Test Construction:

1. Test Foundation: A test foundation footing and plinth shall be cast on grade to the details provided by the Engineer in accordance with specified requirements. This shall be performed before any permanent works are constructed. The concrete shall be cured for the period required in Article 3.13, after which an epoxy coating shall be applied to that part of the plinth which would normally be above ground and a bitumen coating applied to the remainder of the plinth and the footing, all in accordance with Article 3.17.
2. Test Floor Slab: A 3 m by 4 m test area of 150 mm thick floor slab constructed of above ground reinforced concrete in accordance with this Specification shall be cast on grade and given a Class U4 finish as specified in Paragraph 3.11.E. The concrete shall be cured for the period specified in Article 3.13.

P. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than Portland cement in concrete as follows:

1. Fly Ash: 25 percent.
2. Combined Fly Ash and Pozzolan: 25 percent.
3. Ground Granulated Blast-Furnace Slag: 50 percent.
4. Combined Fly Ash or Pozzolan and Ground Granulated Blast-Furnace Slag: 50 percent Portland cement minimum, with fly ash or pozzolan not exceeding 25 percent.
5. Silica Fume: 7 percent.
6. Combined Fly Ash, Pozzolans, and Silica Fume: 35 percent with fly ash or pozzolans not exceeding 25 percent and silica fume not exceeding 7 percent.
7. Combined Fly Ash or Pozzolans, Ground Granulated Blast-Furnace Slag, and Silica Fume: 50 percent Portland cement minimum, with fly ash or pozzolans not exceeding 25 percent and silica fume not exceeding 7 percent.

Q. Limit water-soluble, chloride-ion content in hardened concrete to 0.15 percent by weight of cement.

R. Synthetic Fiber: Uniformly disperse in concrete mix at manufacturer's recommended rate, but not less than 0.60 kg/cu. m.

S. Admixtures: Use admixtures according to the manufacturer's written instructions.

1. Use water-reducing admixture or high-range water-reducing admixture (superplasticizer) in concrete, as required, for placement and workability.
2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
3. Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.50.
4. Use corrosion-inhibiting admixture in concrete mixes where indicated.

a. All to be as required and approved by the Engineer, for placement and workability. Use admixtures in accordance with the manufacturer's instructions. Ensure that the correct quantity of admixture is used at all times. The equipment to be used for dispensing and the method of incorporating the admixture into the concrete shall be subject to approval. The dispensing unit shall be translucent so that the operator can see the discharge of the admixture.

2.15 FABRICATING REINFORCEMENT

A. Fabricate steel reinforcement according to EN 10080.

2.16 CONCRETE MIXING

A. General: Concrete production shall be in accordance with EN 206. A checklist for concrete production shall be produced, such as that used by the NRMCA or approved equal.

1. Batching of materials shall be by weight. All weighing equipment shall be calibrated and documentation shall be provided to establish that the accuracy is continuously maintained in accordance with the requirements of ISO/IEC 17025. Batching scale accuracy shall be in accordance with the Concrete Plant Standards of the Concrete Plant Manufacturers Bureau, or approved equal.
2. Furnish the necessary equipment and establish accurate procedures for determining the quantities of free moisture in the aggregates. Moisture determinations shall be made daily and whenever there is an apparent change in the moisture content. The moisture content shall be recorded. The moisture of aggregates shall be utilized in adjusting the weight of aggregate added to the mix. The water added to the mix shall be similarly adjusted.

B. Job-Site Mixing: All concrete mixed on Project site shall be in a batch mixer of approved size and design complying with EN 206 and producing a uniform distribution of the materials throughout the mixed concrete in accordance with EN 12350 uniformity test. The contents of the drum shall be completely discharged before re-charging. After all the materials are in the mixer, mixing shall continue until the whole of the materials are uniformly distributed and the mass is of uniform color and consistency. In the case of concrete that contains silica fume with a density between 400-650 kg/cu. m, the mixing time shall be 50 percent greater than the requirement for concrete without silica fume.

1. Whenever mixing is to be suspended for half an hour or longer, the drum of the mixer shall be thoroughly washed out with clean water. Provide a competent operator who shall be in continuous control of the mixer. No retempering of concrete, which has partially hardened, by the addition of cement, aggregate, or water shall be allowed.
2. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mix type, mix time, quantity, and amount of water introduced.

C. Ready-Mixed Concrete:

1. Ready-Mixed concrete shall comply with the requirements of ASTM C 94 or BS 5328 and as follows:

a. Concrete shall be centrally mixed off site and transported in an agitator truck. Truck mixing shall not be permitted.

b. The plant and trucks shall be certified as meeting the requirements of the NRMCA Check List, or approved equal.
c. Details and information regarding the supplier proposed by the contractor shall be submitted to the Engineer for approval.
d. The sewer of ready mixed concrete shall not subsequently be charged without further approval of the Engineer.

2. When air temperature is between 30 deg.C (82 deg.F) and 32 deg.C (90 deg. F), delivery time from the time water is added to the mix until it is placed in its final position in the form shall not exceed 60 minutes. When air temperature is above 32 deg.C (90 deg. F), delivery time shall not exceed 45 minutes.

3. Before discharging concrete at the point of delivery, provide the Engineer with a delivery ticket for each batch of concrete containing the following information as a minimum:

a. Name or number of off-site concrete depot.
b. Serial number for ticket.
c. Date.
d. Time of dispatch.
e. Truck number.
f. Name of Supplier.
g. Grade or mix description of concrete.
h. Type of cement.
i. Cement content.
j. Water/cement ratio.
k. Nominal maximum size of aggregate.
l. Source of aggregate, maximum size, weight of fine and coarse aggregate.
m. Type or name of admixture, if included.
n. Quantity of concrete in cubic meters.
o. Certifying that chlorides and sulfate contents are within specified limits and stating their values.
p. Amount of concrete in cubic meter.

PART 3 - EXECUTION

3.1 FORMWORK

A. Design, erect, shore, brace, and maintain formwork, according to EN 12812, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until concrete structure can support such loads. Design of formwork shall be the sole responsibility of the Contractor.

B. Construct formwork so that concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of EN 206.

C. Limit concrete surface irregularities, designated by EN 206 as abrupt or gradual, as follows:

1. Class A, 4 mm.
2. Class B, 6 mm.
3. Class C, 10 mm.
4. Class D, 25 mm.

D. Construct forms tight enough to prevent loss of concrete mortar.

E. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical. Kerf wood inserts for forming keyways, reglets, recesses, and the like, for easy removal.

1. Do not use rust-stained steel form-facing material.

F. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.

G. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.

H. Chamfer exterior corners and edges of concrete receiving applied waterproofing membranes.

I. Do not chamfer corners or edges of concrete unless otherwise indicated.

J. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.

K. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

L. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

M. Coat contact surfaces of forms with form release agent, according to the manufacturer's written instructions, before placing reinforcement.

N. Where it is required to use internal ties and spacers, their type, spacing and use shall be to the, approved of the Engineer. In no circumstances shall these ties protrude out of the finished concrete, all ties must be cut back into the structural concrete and the surface made good to satisfy the requirements of the minimum spacing and cover.

3.2 EMBEDDED ITEMS

A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use Setting Drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

1. Install anchor bolts, accurately located, to elevations required.

2. Install reglets to receive top edge of foundation sheet waterproofing and to receive through-wall flashings in outer face of concrete frame at exterior walls, where flashing is shown at lintels, shelf angles, and other conditions.
3. Install dovetail anchor slots in concrete structures as indicated.

3.3 REMOVING AND REUSING FORMS

A. General: Formwork, for sides of beams, walls, columns, and similar parts of the Work, that does not support weight of concrete may be removed after cumulatively curing at not less than 10 deg.C (50 deg. F) for 24 hours after placing concrete provided concrete is hard enough to not be damaged by form-removal operations and provided curing and protection operations are maintained.

B. The Engineer shall be notified when the Contractor intends to remove any formwork at least 6 hours prior to starting the process.

C. Leave formwork, for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved the following:

1. 28-day design compressive strength.
2. Determine compressive strength of in-place concrete by testing representative field- or laboratory-cured test specimens according to ACI 301.
3. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

D. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form release agent.

E. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by the Engineer.

3.4 SHORES AND RESHORES

A. Comply with ACI 318M, ACI 301, and recommendations in ACI 347R for design, installation, and removal of shoring and reshoring.

B. In multistory construction, extend shoring or reshoring over a sufficient number of stories to distribute loads in such a manner that no floor or member will be excessively loaded or will induce tensile stress in concrete members without sufficient steel reinforcement.

C. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

3.5 VAPOR RETARDERS

A. Vapor Retarder: Place, protect, and repair vapor retarder sheets according to EN 934 and manufacturer's written instructions.

B. Fine-Graded Granular Material: Cover vapor retarder with fine-graded granular material, moisten, and compact with mechanical equipment to elevation tolerances of plus 0 mm or minus 20 mm.

C. Granular Fill: Cover vapor retarder with granular fill, moisten, and compact with mechanical equipment to elevation tolerances of plus 0 mm or minus 20 mm.

1. Place and compact a 15 mm thick layer of fine-graded granular material over granular fill.

D. General: Place vapor retarder/barrier sheeting in position with longest dimension parallel with direction of pour.

E. Lap joints 150 mm and seal with manufacturer's recommended mastic or pressure-sensitive tape. Cover vapor retarder/barrier with sand cushion and compact to depth indicated.

3.6 STEEL REINFORCEMENT

A. General: Comply with EN 10080 for placing reinforcement.

1. Avoid cutting or puncturing vapor retarder/barrier and waterproofing membranes during reinforcement placement and concreting operations. Repair damages before placing concrete.

B. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars.

1. Shop- or field-weld reinforcement according to BS 7123, where indicated.

C. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

D. Install welded wire fabric in longest practicable lengths on bar supports spaced to minimize sagging. Lap edges and ends of adjoining sheets at least one mesh spacing. Offset laps of adjoining sheet widths to prevent continuous laps in either direction. Lace overlaps with wire.

E. Shipping and Storage:

1. Reinforcement shall be handled and shipped in a manner to avoid bending or other damage to the bars. Bars shall be bundled, separated in sizes and clearly marked by diameter size preferably for one placement, in accordance with the placement schedule and as follows:

a. Bars for separate buildings or large structures shall not be bundled together. Bars for small structures may be bundled together but each bar or group of bars that have the same piece mark shall be tagged and coded.
b. Metal tags or approved equal shall be provided and labeled with legible markings.
c. All bundles shall be tagged at each end. Tags shall show piece marks corresponding to the mark numbers on the placement drawings and on the bar list.
d. Bars shall be bundled in the largest size practical for handling and shipping.

2. Reinforcement shall be stored 1m above ground on platforms, skids or other approved supports and suitably spaced. Contact with the soil shall be avoided. Proper drainage and protection from the elements shall be provided to minimize corrosion.

F. Clean reinforcement of loose rust and mill scale, earth, ice, and other materials that reduce or destroy bond with concrete.

G. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement on concrete blocks of a size to give the correct cover to the reinforcement. Concrete spacer blocks shall be made of the same materials, to the same specified requirements and have the same inherent properties as the parent material, but with the exception that the maximum aggregate size shall be appropriate for the thickness of cover to the reinforcement.

1. Where cathodic protection is utilized, reinforcement shall be isolated from the electrical grounding system.
2. Chairs made of reinforcement shall be used to support the top mats of slab reinforcement and they shall be so dimensioned as to be stable during concreting operations. The chairs shall themselves be supported on concrete blocks as specified above.

H. Place reinforcement to maintain minimum coverages as indicated for concrete protection. Arrange, space, and securely tie bars and bar supports to hold reinforcement in position during concrete placement operations. Ties at intersections shall be made with 1.5 mm diameter annealed wire with wire ends directed into concrete, not toward exposed concrete surfaces.

I. Concrete Cover:

1. Concrete cover to reinforcement shall be as indicated on Drawings but shall not be less than the following:

a. Cover for all concrete in contact with soil and in foundations up to the structural steel or equipment base plate level shall be 75 mm.
b. Cover for concrete exposed to marine conditions shall be 75 mm.
c. Cover for all concrete below grade and waterproofed shall be 50 mm.
d. Cover for all other exterior exposed concrete faces shall be 50 mm.
e. Cover for all other interior protected faces shall be 40 mm, except slabs which shall be 25 mm.

2. Cover to reinforcement shall be checked before any concrete is cast. The bending of reinforcement at a cold joint is not permitted. Concrete cover shall be checked with a cover meter as soon as formwork is removed.

J. All lap splices shall be in accordance with EN 1992 tension lap splice unless otherwise shown on Drawings. All reinforcement bars shall be developed in accordance with BS 4483 unless otherwise shown on Drawings. Welded wire fabric shall be lapped 1.5 mesh plus the extension on the wires unless otherwise shown on Drawings.

3.7 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints:

1. Locate and install construction joints so that they do not impair strength or appearance of the structure, and are acceptable to the Engineer. Unless otherwise shown or approved, provide and locate construction joints in accordance with ACI 301. Where construction joints are indicated in construction documents, no deviation shall be allowed without the approval of the Engineer. Additional joints shall be kept to a minimum and must be approved by the Engineer. The joint surface shall be roughened to remove laitance without disturbing the coarse aggregate by pressure jetting with air and water or by wire brushing. The joint shall be clean prior to placing fresh concrete. The new concrete shall be well worked against the old concrete to ensure a good joint.

2. The use of expanded metal or other perforated material is prohibited in construction joints.
3. Place construction joints perpendicular to main reinforcement. Continue reinforcement across construction joints except as indicated otherwise. Do not continue reinforcement through sides of strip placements.
4. Locate joints for beams, slabs, joists, and girders in the middle third of spans. Offset joints in girders a minimum distance of twice the beam width from a beam-girder intersection.
5. Locate horizontal joints in walls and columns at underside of floors, slabs, beams, and girders.
6. Space vertical joints in walls as indicated, or as required by the Engineer.
7. Use a bonding agent at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.
8. Use epoxy-bonding adhesive at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.
9. Waterstops/Re-injectable Hoses: Provide waterstops/re-injectable hoses in construction joints as indicated. Install waterstops to form a continuous diaphragm in each joint. Support and protect exposed waterstops during progress of Work. Field-fabricate joints in waterstops according to the manufacturer's printed instructions.

C. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness, as follows:

1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint to a radius of 3 mm. Repeat grooving of contraction joints after applying surface finishes. Eliminate groover tool marks on concrete surfaces.
2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 3 mm wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.

D. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

1. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface, unless otherwise indicated.
2. Terminate full-width joint-filler strips not less than 12 mm or more than 25 mm below finished concrete surface where joint sealants.
3. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.

E. Dowel Joints: Install dowel sleeves and dowels or dowel bar and support assemblies at joints where indicated.

1. Use dowel sleeves or lubricate or asphalt-coat one-half of dowel length to prevent concrete bonding to one side of joint.

F. Score Joints for Landscape Concrete Paving: As shown on Drawings.

G. Unless otherwise indicated on design drawings, joint sealing shall be in accordance with EN 15651 and ISO 11600.

3.8 WATERSTOPS

A. Flexible Waterstops: Install in construction joints as indicated to form a continuous diaphragm. Install in longest lengths practicable. Support and protect exposed waterstops during progress of Work. Field-fabricate joints in waterstops according to manufacturer's written instructions.

B. Self-Expanding Strip Waterstops: Install in construction joints and at other locations indicated, according to manufacturer's written instructions, bonding or mechanically fastening and firmly pressing into place. Install in longest lengths practicable.

3.9 CONCRETE PLACEMENT

A. General: Comply with EN 206.

B. Inspection: Before placing concrete, inspect and complete formwork installation, reinforcing steel, and items to be embedded or cast in. Notify other trades to permit installation of their work. Concrete shall not be placed until the condition of the reinforcement, other embedded items, and the formwork has been inspected and approved by the Engineer.

C. Transportation: Concrete, after being discharged from the mixer, shall be transported as rapidly as possible to its final position in the Work by agitator trucks, which shall prevent adulteration, segregation, loss of workability or contamination of the ingredients. The containers that convey the concrete shall be kept clean and free from hardened or partially hardened concrete.

1. The addition of water at the point of discharge is prohibited and trucks shall have the water tank completely disconnected from the drum.
2. The use of chutes, spouts, skips and pumps shall be permitted if approval is obtained. Under no circumstances shall any aluminum pipe or other conveying equipment containing aluminum be allowed to contact fresh concrete when it is conveyed to its point of placement.
3. Method of pouring and pouring sequence shall be submitted by the Contractor to the Engineer’s approval.

D. Placing Concrete in Forms: Deposit concrete in forms continuously or in horizontal layers no deeper than 450 mm and in a manner to avoid inclined construction joints. Where placement consists of several layers, place each layer while the preceding layer is still plastic to avoid cold joints. If a section cannot be placed continuously, provide construction joints as specified. Deposit concrete to avoid segregation at its final location.

1. Concrete shall not be dropped into place from a height exceeding 1.5 m nor through dense reinforcing steel, which could cause segregation of the coarse aggregate. Structural concreting against open excavation will not be permitted as the concrete cannot be coated afterwards.
2. When vertical lifts of concrete are interrupted or delayed for more than one hour, the surface of the unfinished concrete shall be thoroughly cleaned and washed with cement grout immediately before fresh concrete is added and the first layer of new concrete placed shall not exceed 150mm depth and particular care shall be taken with compaction of this new layer to ensure good bond.
3. Method of pouring and pouring sequence shall be submitted by the Contractor to the Engineer's approval.

E. Do not add water to concrete during delivery, at Project site, or during placement, unless approved by the Engineer.

F. Deposit concrete continuously or in layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as specified. Deposit concrete to avoid segregation.

G. Deposit concrete in forms in horizontal layers no deeper than 600 mm and in a manner to avoid inclined construction joints. Place each layer while preceding layer is still plastic, to avoid cold joints.

1. Consolidate placed concrete with mechanical vibrating equipment. Use equipment and procedures for consolidating concrete recommended by EN 206.
2. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations no farther than the visible effectiveness of the vibrator. Place vibrators to rapidly penetrate placed layer and at least 150 mm into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mix constituents to segregate.

H. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.

1. Consolidate concrete during placement operations so that concrete is thoroughly worked around reinforcement and other embedded items and into corners.
2. Maintain reinforcement in position on chairs during concrete placement.
3. Screed slab surfaces with a straightedge and strike off to correct elevations.
4. Slope surfaces uniformly to drains where required.
5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, free of humps or hollows, before excess moisture or bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.

I. Cold-Weather Placement: Comply with EN 206 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.

1. When air temperature has fallen to or is expected to fall below 5 deg.C (41 deg. F), uniformly heat water and aggregates before mixing to obtain a concrete mixture temperature of not less than 10 deg.C (50 deg.F) and not more than 27 deg.C (81 deg.F) at point of placement.

2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators, unless otherwise specified and approved in mix designs.

J. Compaction and Vibration: Full compaction of the concrete shall be achieved throughout the entire depth of the layer. It shall be thoroughly worked against the formwork and around the reinforcement and successive layers shall be thoroughly bonded together. Air bubbles formed during the mixing and casting shall be expelled particular care shall be taken where sloping formwork is used.

1. Consolidate placed concrete by mechanical vibrating equipment supplemented by hand-spading, rodding, or tamping. Use equipment and procedures for consolidation of concrete complying with EN 206.
2. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations no farther than the visible effectiveness of the machine. Place vibrators to rapidly penetrate placed layer and at least 150 mm into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to set. At each insertion, avoid over vibration and limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mix to segregate.

K. Placing Concrete Slabs: Deposit and consolidate concrete slabs in a continuous operation, within limits of construction joints, until completing placement of a panel or section. Consolidate concrete during placement operations so that concrete is thoroughly worked around reinforcement, other embedded items and into corners. Bring slab surfaces to correct level with a straightedge and strike off. Use bull floats or darbies to smooth surface free of humps or hollows. Do not disturb slab surfaces prior to beginning finishing operations. Maintain reinforcing in proper position during concrete placement.

L. Hot-Weather Placement: When hot weather conditions exist that would impair quality and strength of concrete, place concrete complying with ACI 305 and as specified.

1. Cool ingredients before mixing to maintain concrete temperature at time of placement to below 25 deg.C (68 deg. F). Mixing water may be chilled or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is the Contractor's option.
2. Concrete temperature shall not exceed 32 deg. C (90 deg.F) and the temperature differential shall not exceed 25 deg. C (68 deg. F).
3. No concreting operation shall be carried out at ambient temperature of 40 deg. C (104 deg. F) or more.
4. Cover reinforcing steel with water-soaked burlap if it becomes too hot, so that steel temperature will not exceed the ambient air temperature immediately before embedding in concrete.
5. Fog spray forms, reinforcing steel, and subgrade just before placing concrete. Keep subgrade moisture uniform without puddles or dry areas.
6. Use water-reducing retarding admixture when required by high temperatures, low humidity, or other adverse placing conditions, as acceptable to the Engineer.
7. Shade mixing plant and trucks, aggregates, water tank, and cement silo.
8. Paint white the mixing plant, trucks, water tank, and cement silo.
9. Insulate the water tank and supply piping.

10. Provide necessary shades over and around the concrete being poured to prevent sun rays from coming into direct contact with the surface of the concrete and the formwork for a period of about 7 days (minimum from the time of pouring concrete).
11. Concrete placing shall be completed as quickly as possible to reduce transit time.
12. Curing of exported concrete shall be immediately carried out.

3.10 FINISHING FORMED SURFACES

A. Rough-Formed Finish (Class F1): Provide a rough-formed finish on formed concrete surfaces not exposed to view in the finished Work or concealed by other construction. Concrete surface texture is that imparted by form-facing material used, with tie holes and defective areas repaired and patched, and fins and other projections exceeding 7 mm in height rubbed down or chipped off. This finish class is not applicable to elements where backfill is to be placed against the concrete.

B. Smooth-Formed Finish (Class F2): Provide a smooth-formed finish on formed concrete surfaces exposed to view or to be covered with a coating material applied directly to concrete, such as waterproofing, damp-proofing, veneer plaster, painting, or another similar system. This is an as-cast concrete surface obtained with selected form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch defective areas with fins and other projections completely removed and smoothed. No ledges shall be permitted at the position of joints in the formwork.

C. Smooth-Rubbed Finish (Class F3): Provide smooth-rubbed finish not later than 1 day after form removal on scheduled concrete surfaces that have received smooth-formed finish treatment. Moisten concrete surfaces and rub with carborundum brick or another abrasive until producing a uniform color and texture. Do not apply cement grout other than that created by the rubbing process.

D. Grout-Cleaned Finish (Class F4): Provide grout-cleaned finish on scheduled concrete surfaces that have received smooth-formed finish treatment.

1. Combine one part Portland cement to one and one-half parts fine sand by volume, and a 50:50 mixture of acrylic or styrene butadiene-based bonding admixture and water to form the consistency of thick paint. Blend standard Portland cement and white Portland cement in amounts determined by trial patches so that final color of dry grout shall match adjacent surfaces.
2. Thoroughly wet concrete surfaces, apply grout to coat surfaces, and fill small holes. Remove excess grout by scraping and rubbing with clean burlap. Keep damp by fog spray for at least 36 hours after rubbing.

E. Cork-Floated Finish: Wet concrete surfaces and apply a stiff grout. Mix one part Portland cement and one part fine sand with a 1:1 mixture of bonding agent and water. Add white Portland cement in amounts determined by trial patches so color of dry grout will match adjacent surfaces. Compress grout into voids by grinding surface. In a swirling motion, finish surface with a cork float.

F. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.

3.11 FINISHING FLOORS AND SLABS

A. General: Comply with recommendations in EN 197 for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Scratch Finish (U1): Apply scratch finish to monolithic slab surfaces receiving concrete floor topping or mortar setting beds for tile, Portland cement terrazzo, and other bonded applied cementitious finish flooring material, and where indicated. After placing slabs, finish surface to tolerances of F(F) 15 (floor flatness) and F(L) 13 (floor levelness) measured in accordance with EN 13670. Slope surfaces uniformly to drains where required. After leveling, roughen surface before final set with stiff brushes, brooms, or rakes.

C. Nonslip Broom Finish (U2): Apply a nonslip broom finish to exterior concrete platforms, steps, ramps, and elsewhere as indicated. Immediately after float finishing, slightly roughen concrete surface by brooming with fiber-bristle broom perpendicular to main traffic route. Coordinate required final finish with the Engineer before application.

D. Float Finish (U3): Apply float finish to monolithic slab surfaces receiving trowel finish and other finishes as specified; slab surfaces to be covered with membrane or elastic waterproofing, membrane or elastic roofing, or sand-bed terrazzo; and where indicated. After screeding, consolidating, and leveling concrete slabs, do not work surface until ready for floating. Begin floating, using float blades or float shoes only, when surface water has disappeared, or when concrete has stiffened sufficiently to permit operation of power-driven floats, or both. Consolidate surface with power-driven floats or by hand-floating if area is small or inaccessible to power units. Finish surfaces to tolerances of F(F) 18 (floor flatness) and F(L) 15 (floor levelness) measured in accordance with EN 13670. Cut down high spots and fill low spots. Uniformly slope surfaces to drains. Immediately after leveling, refloat surface to a uniform, smooth, granular texture.

E. Trowel Finish (U4): Apply a trowel finish to monolithic slab surfaces exposed to view and slab surfaces to be covered with resilient flooring, carpet, ceramic or quarry tile, paint, or another thin film finish coating system. This finish is also applicable to tops of buried foundations since they have to be subsequently coated.

1. After floating, begin first trowel-finish operation using a power-driven trowel. Begin final troweling when surface produces a ringing sound as trowel is moved over surface. Consolidate concrete surface by final hand-troweling operation, free of trowel marks, uniform in texture and appearance, and finish surfaces to tolerances in accordance with EN 13670 of the following:

    a. Specified overall values of flatness, F(F) 25; and levelness, F(L) 20; with minimum local values of flatness, F(F) 17; and levelness, F(L) 15.
    b. Specified overall values of flatness, F(F) 35; and levelness, F(L) 25; with minimum local values of flatness, F(F) 24; and levelness, F(L) 17; for slabs-on-grade.
    c. Specified overall values of flatness, F(F) 30; and levelness, F(L) 20; with minimum local values of flatness, F(F) 24; and levelness, F(L) 15; for suspended slabs.
    d. Specified overall values of flatness, F(F) 45; and levelness, F(L) 35; with minimum local values of flatness, F(F) 30; and levelness, F(L) 24.
2. Grind smooth any surface defects that would telegraph through applied floor covering system.

3. Finish and measure surface so gap at any point between concrete surface and an unleveled freestanding 3 m long straightedge, resting on two high spots and placed anywhere on the surface, does not exceed the following:

   a. 3 mm.

F. Trowel and Fine Broom Finish (U5): Where ceramic or quarry tile is to be installed with thin-set mortar, apply a trowel finish as specified, then immediately follow by slightly scarifying the surface with a fine broom.

G. Colored Wear-Resistant Finish (U6): Apply a colored wear-resistant finish to monolithic slab surface indicated.

1. Apply dry shake materials for the colored wear-resistant finish at a rate of 5 kg/sq. m, unless a greater amount is recommended by material manufacturer.
2. Cast a trial slab approximately 3 m square to determine actual application rate, color, and finish, as acceptable to the Engineer.
3. Following placement, vibrating and leveling, float the concrete with a wooden float to "open" the surface and allow the excess moisture and air to escape.
4. Once the sheen has disappeared, apply floor hardener as a dry shake onto the wet surface. Apply approximately one-half of the material and then float into the surface with a wooden float.
5. Following the first float, apply the balance of the material and float in the same fashion.
6. Once the surface is firm enough to take foot traffic, use a power float to finish the surface to a smooth and non-slip finish.
7. After floating, apply a trowel finish as specified. Cure slab surface with a curing compound recommended by the dry shake material manufacturer. Apply the curing compound immediately after the final finishing.

H. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, and ramps, and elsewhere as indicated.

1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route. Coordinate required final finish with the Engineer before application.

I. Slip-Resistive Aggregate Finish: Before final floating, apply slip-resistive aggregate finish where indicated and to concrete stair treads, platforms, and ramps. Apply according to manufacturer's written instructions and as follows:

1. Uniformly spread 12 kg/10 sq. m of dampened slip-resistive aggregate over surface in one or two applications. Tamp aggregate flush with surface, but do not force below surface.
2. After broadcasting and tamping, apply float finish.
3. After curing, lightly work surface with a steel wire brush or an abrasive stone, and water to expose slip-resistive aggregate.

3.12 MISCELLANEOUS CONCRETE ITEMS

A. Filling In: Fill in holes and openings left in concrete structures, unless otherwise indicated, after work of other trades is in place. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete Work.

B. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and by steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.

C. Equipment Bases and Foundations: Provide machine and equipment bases and foundations as shown on Drawings. Set anchor bolts for machines and equipment at correct elevations, complying with diagrams or templates of manufacturer furnishing machines and equipment.

D. Steel Pan Stairs: Provide concrete fill for steel pan stair treads, landings, and associated items. Cast-in inserts and accessories as shown on Drawings. Screed, tamp, and trowel-finish concrete surfaces.

E. Tolerance and concrete dimensions for in-situ concrete members shall, under no circumstances, exceed the permissible ones as indicated in the EN 13670.

3.13 CONCRETE PROTECTION AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with EN 206 for cold-weather protection and with recommendations in EN 206 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 1 kg/sq. m x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing by one or a combination of the following methods:

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces, by one or a combination of the following methods:

1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:

    a. Water.
    b. Continuous water-fog spray.
    c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 300 mm lap over adjacent absorptive covers.

2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 300 mm, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer recommends for use with floor coverings.

3. Curing Compound: Shall be to the Engineer's approval applied uniformly in continuous operation by power spray or roller according to the manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
4. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

E. Curing Methods: Cure concrete by curing compound, by moist curing, by moisture-retaining cover curing, or by combining these methods, as specified below.

1. Horizontal Surfaces: Horizontal surfaces shall be saturated with water and then treated with curing compound. Apply curing compound as soon as final finishing operations are complete (within 2 hours and after surface water sheen has disappeared). Apply uniformly in continuous operation by power spray or roller according to manufacturer's directions. Recoat areas subjected to heavy rainfall within 3 hours after initial application. The surfaces shall then be bonded and flooded with water or draped with wet burlap together with a perforated soaker hosepipe, covered with white impervious sheeting held firmly in place along all edges and kept continuously wet for the duration of the curing period.
2. Vertical Surfaces: Vertical timber formwork shall be draped with wet burlap as soon as concrete is placed. Vertical surfaces shall be treated with curing compound as soon as formwork is removed, draped with wet burlap, covered with white impervious sheeting held firmly in place along all edges and kept continuously wet for the duration of the curing period. Care shall be taken to avoid drying winds.
3. Impervious Sheeting: Impervious sheeting shall be in accordance with ASTM C 171.
4. Curing Compound: Curing compound shall be white and in accordance with CE CRD-C300. It shall be a water-based wax emulsion for concrete above grade, which will not be subsequently coated. Otherwise the curing compound shall be compatible with the coating. Obtain confirmation from the supplier of the curing compound that it meets the above requirements.

3.14 LIQUID FLOOR TREATMENTS

A. Penetrating Liquid Floor Treatment: Prepare, apply, and finish penetrating liquid floor treatment according to manufacturer's written instructions.

1. Remove curing compounds, sealers, oil, dirt, laitance, and other contaminants and complete surface repairs.
2. Do not apply to concrete that is less than seven days old.

3. Apply liquid until surface is saturated, scrubbing into surface until a gel forms; rewet; and repeat brooming or scrubbing. Rinse with water; remove excess material until surface is dry. Apply a second coat in a similar manner if surface is rough or porous.

B. Sealing Coat: Uniformly apply a continuous sealing coat of curing and sealing compound to hardened concrete by power spray or roller according to manufacturer's written instructions.

3.15 JOINT FILLING

A. Prepare, clean, and install joint filler according to manufacturer's written instructions.

1. Defer joint filling until concrete has aged at least six months. Do not fill joints until construction traffic has permanently ceased.

B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.

C. Install semi-rigid epoxy joint filler full depth in saw-cut joints and at least 50 mm deep in formed joints. Overfill joint and trim joint filler flush with top of joint after hardening.

3.16 CONCRETE SURFACE REPAIRS

A. Concrete exposed by the removal of formwork shall be inspected by the Engineer before any remedial work, subsequent coating or other treatment that would hinder the proper inspection of the concrete is carried out. Any concrete not complying with this requirement shall be liable for rejection.

B. Concrete not meeting the specified requirements shall be removed and rebuilt without delay unless the Engineer approves that a repair may be satisfactorily effected. This agreement shall not preclude the subsequent rejection of the repaired work by the Engineer. The proposed method for removal and replacement of defective work shall be submitted to the Engineer for approval for each concrete placement before the removal commences.

C. All repairs approved by the Engineer shall be performed by a subcontractor specialized in the repair of concrete in the Middle East and prepared to guarantee the work. Any repair method submitted for approval shall produce a result that is as impermeable as the original concrete. Subsequent tests on the repaired concrete shall be carried out at the discretion of the Engineer in order to establish the quality of the repair, particularly at the joint between the original and the repaired concrete.

D. Defective Concrete: Repair and patch defective areas when approved by the Engineer. Remove and replace concrete that cannot be repaired and patched to the Engineer's approval.

E. Patching Mortar: Mix dry-pack patching mortar, consisting of one part Portland cement to two and one-half parts fine aggregate passing a 1.2 mm sieve, using only enough water for handling and placing.

F. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and stains and other discolorations that cannot be removed by cleaning.

1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 13 mm in any dimension in solid concrete but not less than 25 mm in depth. Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent. Fill and compact with patching mortar before bonding agent has dried. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
2. Repair defects on surfaces exposed to view by blending white Portland cement and standard Portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.
3. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by the Engineer.

G. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template.

1. Repair finished surfaces containing defects. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.25 mm wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
2. After concrete has cured at least 14 days, correct high areas by grinding.
3. Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
4. Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
5. Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 6 mm to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
6. Repair defective areas, except random cracks and single holes 25 mm or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least 20 mm clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mix as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
7. Repair random cracks and single holes 25 mm or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hours.

H. Perform structural repairs of concrete, subject to the Engineer's approval, using epoxy adhesive and patching mortar.

I. Repair materials and installation not specified above may be used, subject to the Engineer's approval.

3.17 CONCRETE PROTECTION

A. The following are typical concrete protection and waterproofing requirements except at basements, tunnels, and other buried structures. Concrete protection and waterproofing requirements for basements, tunnels, and other buried structures shall be as specified in Article 3.18.

B. General: All concrete shall have surface holes investigated and filled with a filler compatible with the coating to be applied (if that coating alone cannot fill the hole) provided that they are small and formed by air bubbles caught against the formwork. For surface holes having any dimension greater than 10 mm, refer to Article 3.16.

C. Blinding or Mudmat Concrete: Unless otherwise shown on Drawings, all foundations shall be placed on 50 mm thick blinding or mudmat concrete. Prior to placing this concrete, the subsoil shall be well consolidated and free from loose material.

D. Impervious Membrane: Concrete shall be cast onto impervious sheeting complying with ASTM E 154 laid over the concrete blinding or mudmat. Overlap shall be 150 mm and the sheeting shall extend 150 mm beyond the edge of all foundations.

E. Where piling or drilled piers are involved, the impervious membrane shall consist of bituminous roofing felt impregnated with small stone chippings. The stone chipping covered surface shall be laid uppermost so that when concrete is cast onto it, it shall be firmly adhered to the pilecap.

F. Surfaces Below Ground Level: Surfaces below ground level shall be coated with 3 coats of modified bitumen giving a total coating thickness of 1 mm. A square scrape master sample shall be taken to determine the weight of specified thickness of coating from a 300 mm by 300 mm patch. The weight can then be compared with test samples. With surfaces at grade coated as specified, the bitumen shall overlap the epoxy coating by 75 mm. Before soil is backfilled around the foundations, the coated surfaces shall be permanently covered with impervious sheeting as specified in Paragraph 3.17.D and overlain with protection board, suitably banded to retain it in position. Overlap of the sheeting shall be 150 mm.

G. Exposed External Surfaces at Ground Level:

1. Exposed external concrete surfaces for a depth of 150 mm below and 300 mm above ground level, or to the underside of base plates (whichever is lowest), shall be primed with a low viscosity primer and coated with 2 coats of a light gray colored epoxy paint with a minimum thickness of 125 microns per coat.
2. Application trials shall be carried out on the chosen coating. Concrete surface preparation and application shall be in accordance with the manufacturer’s instructions.

H. Exposed External Surfaces Above Ground Level: General exposed exterior surfaces above grade shall be coated as specified in Paragraph 3.17.F. The coating shall extend to the underside of the baseplates; i.e., cover the grout.

I. External Surfaces Exposed to Saline Waters: Exterior surfaces in the sea splash zone or in the tidal zone, or the interior faces of structures containing saline water, shall be coated as specified in Paragraph 3.17.F.

3.18 CONCRETE PROTECTION - BASEMENTS, TUNNELS, AND OTHER BURIED STRUCTURES

A. Install a waterproofing system on the exterior surfaces of the basement and adjoining tunnel structure consisting of a sheet membrane system such as the following:

1. Underside of Basement and Tunnel Floor Slab.
2. Vertical and Buried Roof Elements.
3. Pile Caps.

B. Alternative waterproofing systems may be submitted to the Engineer for review and approval. Submittal shall include technical data and case histories sufficient for the Engineer to do a technical evaluation of the proposed systems.

C. Install membrane system in accordance with the membrane manufacturer's instructions.

D. Submit the following to the Engineer for approval before beginning the work:

1. List of all membrane materials, joint compounds, and concrete surfacers to be used in the Work. List shall identify the specific products by manufacturer and catalog number.
2. Procedures for material storage and handling, surface preparation, environmental control, application sequence, overlap dimensions, touch-up and repair, curing, and inspection of the membrane system. The membrane manufacturer's published instructions and installation details shall be attached as part of submitted procedures. Conflicts between the material manufacturer's recommendations and Contract Documents shall be noted in writing to the Engineer for resolution.
3. Cleaning and installation verification forms for daily inspection records. A detailed cleaning and installation verification report in accepted form shall be completed each day during the work and submitted for record. The final verification report shall include a statement of completion conformance verifying that the required materials were used and that the accepted application procedures and specified requirements were followed.
4. Details concerning corners, bottom slab to wall tie-ins, slab/wall/roof penetrations, terminations, control joints, expansion joint, and crack control.
5. Manufacturer material safety data sheets for all materials used in the execution of the work.

E. Arrange for the material manufacturer's technical representative to be present at the beginning of work and to qualify installation personnel in the installation of the manufacturer's waterproofing products.

F. Joints: Unless otherwise shown on design drawings, joints shall be designed and constructed in accordance with EN 1992 and EN 206. Details and positioning of joints, together with the materials to be used, shall be submitted for the Engineer's approval.

1. Waterstops shall be in accordance with Paragraph 2.8A. Jointing of waterstops shall be made by welding in an approved fashion. Lapping of waterstops at joints and the use of adhesives for jointing purposes will not be permitted unless approved by the Engineer. Waterstops shall not be perforated or damaged. Concrete shall be carefully placed and compacted to ensure dense impervious concrete, particularly around the ribs of waterstops. At all joints except for expansion joints as indicated on Drawings, the concrete shall be placed up to the centerline of the waterstop. All starters to walls of watertight construction shall be cast using hung formwork so that the concrete in the starters is placed simultaneously with the concrete in the slab.

3.19 GROUTING

A. Preparation: Concrete foundation top shall be cleaned of dirt, laitance, oil and grease. Anchor bolt boxes and sleeves shall be cleaned of all polystyrene and other deleterious material. The surface of the concrete shall be thoroughly wetted just prior to grouting but shall contain no excess water, particularly in the bolt boxes and sleeves.

B. Materials:

1. Type G1: For interior bases protected from weather and saline bearing waters and not subject to heavy or vibratory loads, grout shall consist of one part Portland cement to two parts well graded sand by volume. Sand shall comply with Paragraph 2.5.C.2. A retarding superplasticizer may be used if necessary to obtain the correct fluidity in high ambient temperatures.
2. Type G2: For all heavily loaded structural column bases and equipment bases subjected to vibratory loads, a proprietary non-shrink, non-metallic high strength grout especially formulated for high-temperature work.
3. Type G3: For all other exterior work exposed to potential saline ingress, a proprietary general purpose non-shrink grout, especially formulated for high temperature work.

C. Placement:

1. The manufacturer's recommendations shall be followed for proprietary grouts. The temperature of the grout at time of placement shall not exceed 25 deg.C (77 deg. F) and the temperature of the elements in contact with the grout shall not exceed 40 deg.C (104 deg. F) . To obtain the required temperatures, it may be necessary to do the following:

    a. Shield the materials from the direct rays of the sun.
    b. Mix materials with flaked ice.
    c. Cool base plates with water but ensure that anchor bolt pockets are free from water.
    d. Require certification of plant and trucks to meet requirements of the NRMCA Check List or approved equal.

2. Grout strength shall not be less than 30 N/sq. mm at 28 days. Document to Engineer that this strength is being achieved. Grouting shall not proceed until the steel work or equipment has been leveled and plumbed with the bases being supported in the meantime by steel packers and shims.
3. Completely fill anchor bolt sleeves with grout before placing grout under base plates. The gravity grouting method shall be used wherein the flowable self-leveling grout is poured on one side of a base until it flows out at the opposite side. Packers and shims used to level bases shall be removed after the grout has set and the resulting pocket repaired with similar grout.

3.20 WATER-RETAINING CONSTRUCTION

A. General: Water-retaining construction shall comply with this Specification.

B. Joints: Joints shall be designed and constructed in accordance with EN 1992. Details and positioning of joints, together with the materials to be used, shall be shown on the Drawings.

1. Waterstops shall be in accordance with Paragraph 2.8.A. Jointing of waterstops shall be made by welding in an approved fashion. Lapping of waterstops at joints and the use of adhesives for jointing purposes shall not be permitted unless specifically authorized. Waterstops shall not be perforated or damaged. Concrete shall be carefully placed and compacted to ensure dense impervious concrete, particularly around the ribs of

waterstops. At all joints the concrete shall be placed up to the centerline of the waterstop. All starters to walls of watertight construction shall be cast using hung formwork so that the concrete in the starters is placed simultaneously with the concrete in the slab.

C. Coating: Coat inside faces of structures containing saline waters as specified in Paragraph 3.17.F.

D. Testing: In addition to the testing required in Paragraph 3.21.C, further tests to determine the watertightness of the structure shall be performed in accordance with EN 1992. The structure shall be filled with fresh water to the designed level and after a period to allow for absorption of water, the faces remote from the liquid shall be inspected for leaks over a 7-day period. Any defects shall be repaired by an approved method, which could involve demolition and rebuilding, or lining of the structure.

3.21 QUALITY CONTROL AND TESTING

A. General:

1. Testing Laboratory:

a. Employ an independent testing agency to perform tests and to submit test reports.
b. Be responsible for taking, identifying and delivering to the test laboratory all test samples called for in this Specification. The testing laboratory shall be responsible for the testing. Collect all test results and deliver them to the Engineer in the format and detail as specified.

2. Testing Laboratory Qualifications: The testing laboratory shall be accredited to ISO/IEC 17025 or an equivalent National Standard and shall have a Quality System in accordance with ISO 9001.

B. Quality Control - Testing on Fresh Concrete:

1. Compressive Strength Test for Structural Concrete:

a. Sampling, curing and testing shall be performed using the relevant procedures in EN 206, EN 12350 and EN 12390.

1) Samples for production concrete cylinders shall be taken at the point of placement at the average rate of one per 25 cu. m of concrete placed, with a minimum of one sample taken every day that the mix is used. A sample shall consist of four 150 mm cylinders molded and stored for laboratory-cured test specimens except when field-cured test specimens are required. One cylinder is for testing at 7 days after casting, two for testing at 28 days after casting, and one reserved for later testing if required.
2) If frequency of testing provides fewer than 5 strength tests for a given class of concrete, conduct testing from at least 5 randomly selected batches or from each batch if fewer than 5 are used.
3) When total quantity of a given class of concrete is less than 25 cu. m, the Engineer may waive strength testing if adequate evidence of satisfactory strength is provided.
4) When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, evaluate current operations and provide corrective procedures for protecting and curing the in-place concrete.

b. Records: Records shall be kept of the mix details and position in the works of all batches of concrete and of all samples taken for cylinders and other specimens and of their test results. A copy shall be supplied to the Engineer within 24 hours after recording/testing. Records shall contain, but not be limited to, the following information:

1) Date, time, location, and volume of pour.
2) Ambient temperature and humidity.
3) Concrete temperature (at time of placement).
4) Cement type and manufacture.
5) Concrete type and class.
6) Aggregate type and source.
7) Admixture details.
8) Water/cement ratio.
9) Identification of test cylinder.
10) Name of concrete testing service.
11) Date and time of sampling.
12) Method of compaction.
13) Date of testing and results of test.
14) Age of sample in days, weight in grams, density in kg/cu. m.
15) Crushing load in newtons and crushing strength in N/sq. mm.
16) Signatures of person preparing cylinder and of person crushing cylinder.
17) Results of testing.

c. For the 28-day tests, the concrete will be deemed to comply with the specified design strength if the average strength determined from all sets of 3 consecutive tests is at least equal to the specified design strength and no individual strength test falls below the specified design strength by more than 3.5 N/sq. mm. Any concrete not complying with the specified design strength shall be at risk for removal and replacement at the Contractor's expense.

d. The 28-day cylinder crushing results shall be grouped consecutively in groups of 40 and each group shall have a standard deviation less than 3.5 N/sq. mm. If the standard deviation is greater than or equal to 3.5 N/sq. mm, then concrete production shall be investigated by the Engineer and further tests on trial mixes may be required.

e. Tests shall be carried out at 7 days to establish a relationship between the 7-day and 28-day strengths. This relationship shall be used to interpret future test results in order to predict the corresponding 28-day strength. The Engineer shall be advised without delay of any 7-day test result indicating that the corresponding 28-day strength will likely fail to meet the specified strength so that any necessary action can be taken to minimize the effect of such possible failure.

2. Compressive Strength Test for Blinding or Mudmat Concrete: There shall be no production tests on blinding or mudmat concrete. The Engineer will require compressive strength tests if it is believed that the characteristic strength is below 10 N/sq. mm. Characteristic strength is defined as that value of the cube strength below which 5 percent of all possible test results would be expected to fall. If the tests confirm that the characteristic strength is less than 10 N/sq. mm, then the Engineer will require revisions to the mix design to ensure that the concrete meets the specified requirements.

3. Measurement of Concrete Temperature: Temperature measurements shall be in accordance with EN 206. Concrete temperature shall be measured 50 mm below the surface prior to and at the point of placement and recorded on the pour card for each pour.

There shall be one test hourly when air temperature is 4 deg.C (39 deg. F) and below and when air temperature is 27 deg.C (81 deg. F) and above, and one test for each set of compressive-strength specimens. Concreting shall stop if the temperature of the concrete does not meet the requirements of Paragraph 3.9.L

4. Cement Content and Water/Cement Ratio: Samples of freshly mixed concrete shall be chosen by the Engineer at least once per month from each structural grade and determination made of cement content and water/cement ratio in accordance with EN 206 and BS 1881. If the cement content is less than 90 percent of the requirement in Paragraph 2.14.F or the water/cement ratio greater than 110 percent of the requirement in Paragraph 2.14.H then there shall be an investigation to establish the cause and the Engineer may reject the casted concrete for the pour from which the samples were taken. The Engineer may request an analysis of fresh concrete from any pour. No analysis of freshly mixed concrete is required for blinding or mudmat concrete.
5. Salt Content: The total concentration of sulfates and chlorides in fresh concrete shall be measured at least once a week for all structural grades of concrete. Tests shall be in accordance with EN 206, BS 1881 and EN 12390. Concentrations of each ion shall not exceed the limits specified in Paragraph 2.5.C. If these limits are exceeded, the concrete pour from which the samples were taken shall be rejected and further tests performed on the casted concrete in accordance with Paragraph 3.21.C to determine the total extent of the problem.
6. Slump: Slump tests shall be performed in accordance with EN 206 and EN 12350. There shall be a minimum of one test at the point of discharge for each day's pour for each type of concrete. Additional tests shall be performed when concrete consistency appears to have changed

C. Quality Control - Testing on Hardened Concrete:

1. General: The Engineer may request samples to be taken and tests carried out on any hardened structural grade concrete as specified below if he suspects that the concrete does not meet the specified requirements. If the tests confirm that the concrete does not meet the requirements of this Specification, then the Engineer may require the concrete to be removed at the Contractor’s expense. If the tests confirm that the concrete meets the requirements of this Specification, then the cost of taking the samples shall not be at the Contractor’s expense.
2. Compressive Strength Tests: The Engineer may request cores to be drilled from a particular pour. 100 mm diameter cores shall be drilled as requested, in accordance with EN 12390, and sent for crushing. If the cores from that pour have an average compressive strength less than 85 percent of the characteristic strength or any individual core has a compressive strength less than 75 percent of the characteristic strength, it shall be evidence that the concrete from which it was taken is not in accordance with the specified requirements.
3. Concrete Cover: The Engineer may check the concrete cover over the reinforcement with a cover meter. Any indication that the cover is generally less than the requirements specified in Paragraph 3..6.I shall be checked by limited surface concrete removal. If it is confirmed that the actual cover is generally less than specified, then the concrete shall be removed at the Contractor’s expense. In the case of localized lack of cover and where appearance is not important, a repair shall be effected by removal of the inadequate cover and the cutting back of concrete for 50 mm behind the reinforcement. Resurfacing of the concrete with the specified cover shall be carried out as a repair by a specialist subcontractor as specified in Article 3.16.
4. Absorption: A sample of three 75 mm diameter cores, 75 mm long, shall be taken from hardened concrete if directed by the Engineer and tested in accordance with Paragraph 2.14.M. Should the absorption of any core exceed by more than 1 percent the highest

approved test result, then the concrete from which it was cut shall be removed. No absorption test shall be required for blinding or mudmat concrete.

5. Salt Content: Engineer may request samples to be taken from two 20 mm diameter drillings into the concrete surface, spaced 75 mm apart. Each drilling shall proceed in 25 mm increments for a total depth of 100 mm and the dust from the 4 samples in each hole shall be sent for sulfate and chloride content analysis in accordance with BS 1881. Should the tests show that the limits specified in Paragraph 2.14.H are exceeded, the concrete shall be removed.
6. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be used but shall not be used as the sole basis for acceptance or rejection.
7. Additional Tests: The testing agency shall make additional tests of in-place concrete when test results indicate specified concrete strengths and other characteristics have not been attained in the structure, as directed by the Engineer. Testing agency may conduct tests to determine adequacy of concrete by cored cylinders complying with EN 12390, or by other methods as directed.

END OF SECTION 03300

19.

SECTION 03331 - CAST-IN-PLACE ARCHITECTURAL CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section specifies cast-in-place architectural concrete, including formwork, reinforcement accessories, concrete materials, concrete mix design, placement procedures, and finishes.

B. Related Sections include the following:

1. Concrete Specification Section "Cast-in-Place Concrete" for structural and general cast-in-place concrete construction, including formed and unformed finishes.

1.3 DEFINITION

A. Cast-in-Place Architectural Concrete: Concrete that is exposed to view on surfaces of the completed structure or building and that requires special concrete materials, formwork, placement, or finishes to obtain specified architectural appearance.

B. Design Reference Sample: Sample designated by the Engineer in the Contract Documents that reflects acceptable surface quality and appearance of cast-in-place architectural concrete.

C. Reveal: Projection of the coarse aggregate from the matrix after exposure.

1.4 SUBMITTALS

A. Product Data: For each type of manufactured material and product indicated.

B. Design Mixes: For each concrete mix. Include alternate mix designs when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

C. Shop Drawings: Show formwork construction including form-facing joints, rustications, construction and contraction joints, form joint-sealant details, form tie location and patterns, inserts and embedments, cutouts, cleanout panels, and other items that visually affect cast-in-place architectural concrete.

D. Samples: For each of the following materials:

1. Form-facing panel.
2. Form release agent.
3. Form ties.

4. Form liners.
5. Cement.
6. Coarse- and fine-aggregate gradations.
7. Chamfers and rustications.
8. Curing compound.
9. Coloring admixtures.

E. Samples for Verification: Architectural concrete samples, cast vertically, approximately 450 by 450 by 50 mm, of finishes, colors, and textures to match the design reference sample. Include Sample sets showing the full range of variations expected in these characteristics.

F. Material Test Reports: From a qualified testing agency indicating and interpreting test results of the following for compliance with requirements indicated, based on comprehensive testing of current materials:

G. Material Certificates: Signed by manufacturers certifying that each of the following materials complies with requirements:

1. Cementitious materials and aggregates.
2. Admixtures.
3. Curing compounds.

H. Placement Schedule: Submit concrete placement schedule before start of architectural concrete placement operations. Include location of all joints including construction joints.

I. Minutes of preinstallation conference.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Installer Qualifications: An experienced cast-in-place architectural concrete contractor who has specialized in installing cast-in-place architectural concrete similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

1. Installer shall retain a quality-control inspector, experienced in inspecting cast-in-place architectural concrete. Inspectors should be certified as a reinforced concrete especial inspector by local authorities.

C. Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products complying with EN 206 requirements for production facilities and equipment.

D. Testing Agency Qualifications: An independent testing agency, acceptable to authorities having jurisdiction, qualified according to ISO/IEC 17025 and ISO 9001 to conduct the testing indicated, as documented according to EN 206 .

19.

1. Personnel conducting field tests shall be qualified as ACI Concrete Field Testing Technician, Grade 1, according to ACI CP-1 or an equivalent recognized certification program.

E. Source Limitations for Cast-in-Place Architectural Concrete: Obtain each color, size, type, and variety of concrete material and concrete mix from one manufacturer with resources to provide cast-in-place architectural concrete of consistent quality in appearance and physical properties.

F. Comply with ACI 303.1, "Specification for Cast-in-Place Architectural Concrete"; EN 206, "Concrete - Specification, Performance, Production and Conformity"; and BS EN 13670 ACI 117, "Specifications for Tolerances" unless more stringent provisions are indicated.

G. Sample Panels: Before casting architectural concrete, produce sample panels to demonstrate the approved range of selections made under sample Submittals. Produce a minimum of 3 sets of full-scale sample panels, cast vertically, approximately 1200 by 1200 by 150 mm minimum, to demonstrate the expected range of finish, color, and texture variations.

1. Locate panels as indicated or, if not indicated, as directed by the Engineer.
2. Demonstrate methods of curing aggregate exposure, sealers, and coatings, as applicable.
3. In the presence of the Engineer, damage part of an exposed-face surface for each finish, color, and texture, and demonstrate materials and techniques proposed for repair of tie holes and surface blemishes to match adjacent undamaged surfaces.
4. Maintain sample panels during construction in an undisturbed condition as a standard for judging the completed Work.
5. Demolish and remove sample panels when directed.

H. Mockups: Before casting architectural concrete, build mockups to verify selections made under sample Submittals and to demonstrate aesthetic effects and qualities of materials and execution. Build mockups to comply with the following requirements, using materials indicated for the completed Work:

1. Build mockups in the location and of the size indicated or, if not indicated, as directed by the Engineer.
2. Notify the Engineer seven days in advance of dates and times when mockups will be constructed.
3. Demonstrate curing, cleaning, and protecting of cast-in-place architectural concrete, finishes, and contraction joints, as applicable.
4. In the presence of the Engineer, damage part of the exposed surface of cast-in-place architectural concrete for each finish, color, and texture required, and demonstrate materials and techniques proposed for repairs to match adjacent undamaged surfaces.
5. Obtain the Engineer's approval of mockups before casting architectural concrete.
6. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
7. Demolish and remove mockups when directed.
8. Approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

I. Preinstallation Conference: Conduct conference at Project site to comply with requirements in General Requirements Section "Project Management and Coordination."

19.

PART 2 - PRODUCTS

2.1 FORM-FACING MATERIALS

A. General: Comply with Concrete Specification Section "Cast-in-Place Concrete" for formwork and other form-facing material requirements.

B. Form-Facing Panels for As-Cast and Exposed-Aggregate Finishes: Steel, glass-fiber-reinforced plastic, or other approved nonabsorptive panel materials that will provide continuous, true, and smooth architectural concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

C. Forms for Cylindrical Columns, Pedestals, and Supports: Metal, glass-fiber-reinforced plastic, paper, or fiber tubes that will provide surfaces with gradual or abrupt irregularities not exceeding specified formwork surface class. Provide units with sufficient wall thickness to resist plastic concrete loads without detrimental deformation.

D. Pan-Type Forms: Glass-fiber-reinforced plastic or formed steel, stiffened to resist plastic concrete loads without detrimental deformation.

E. Form Liners: Units of face design, texture, arrangement, and configuration indicated to match design reference sample. Furnish with manufacturer's recommended liquid-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent surface treatments of concrete.

F. Rustication Strips: Metal, rigid plastic, or dressed wood with sides beveled and back kerfed; nonstaining.

G. Chamfer Strips: Metal, rigid plastic, elastomeric rubber, or dressed wood, 19 by 19 mm, minimum; nonstaining.

H. Form Joint Tape: Compressible foam tape, pressure sensitive, BS 10948, minimum 6 mm thick.

I. Form Joint Sealant: Elastomeric sealant complying with BS 10948, that adheres to form joint substrates.

J. Sealer: Penetrating, clear, polyurethane wood form sealer formulated to reduce absorption of bleedwater and prevent migration from wood of set-retarding chemicals.

K. Form Release Agent: Commercially formulated form release agent that will not bond with, stain, or adversely affect architectural concrete surfaces and will not impair subsequent treatments of those surfaces.

   1. Formulate form release agent with rust inhibitor for steel form-facing materials.

L. Surface Retarder: Chemical liquid set retarder, for application on form-facing materials, capable of temporarily delaying final hardening of newly placed concrete surface to depth of reveal specified.

M. Form Ties: Factory-fabricated, glass-fiber-reinforced plastic, internally disconnecting, or removable ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.

1. Furnish ties that, when removed, will leave holes not larger than 19 mm in diameter on concrete surface.
2. Furnish internally disconnecting ties that will leave no corrodible metal closer than 38 mm, plus reveal projection of exposed aggregate, from the plane of architectural concrete surface.
3. Furnish glass-fiber-reinforced plastic ties, not less than 13 mm in diameter, of color selected by the Engineer from manufacturer's full range.
4. Furnish ties with integral water-barrier plates to walls indicated to receive damp-proofing or waterproofing.

2.2 REINFORCEMENT ACCESSORIES

A. General: Comply with Concrete Specification Section "Cast-in-Place Concrete" for steel reinforcement and other requirements for reinforcement accessories.

B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire fabric in place. Where legs of wire bar supports contact forms, use all-plastic, plastic-protected, or stainless-steel bar supports.

2.3 CONCRETE MATERIALS

A. Portland Cement: EN 197, CEM 1, gray or white color, of same type, brand, and source for entire Project.

1. Fly Ash: EN 450
2. Ground Granulated Blast-Furnace Slag: EN 15167

B. Blended Hydraulic Cement: EN 197.

C. Normal-Weight Coarse Aggregate: EN 206, from the same source for entire Project, and as follows:

1. Weathering Region and Class: Severe, 5S or Moderate, 5M or Negligible, 1N.
2. Nominal Maximum Aggregate Size: 19 mm.
3. Gradation: Uniformly graded.

D. Normal-Weight Fine Aggregate: EN 206, manufactured or natural sand, from the same source for entire Project.

E. Water: Potable, complying with except free of wash water from mixer washout operations.

F. Chemical Admixtures: Certified by manufacturer to contain not more than 0.1 percent water-soluble chloride ions by mass of cementitious material and to be compatible with other admixtures and cementitious materials. Do not use admixtures containing calcium chloride.

1. Air-Entraining Admixture: EN 934 and ASTM C 260.
2. Water-Reducing Admixture: EN 934 and ASTM C 494, Type A.
3. High-Range, Water-Reducing Admixture: EN 934 and ASTM C 494, Type F.
4. Water-Reducing and Accelerating Admixture: EN 934 and ASTM C 494, Type E.
5. Water-Reducing and Retarding Admixture: EN 934 and ASTM C 494, Type D.

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G. Coloring Admixture: EN 12878, synthetic mineral-oxide pigments or colored water-reducing admixtures, free of carbon black; color stable, nonfading, and resistant to lime and other alkalis.

1. Color: As selected by the Engineer from manufacturer's full range.

2.4 CURING MATERIALS

A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, washed to prevent concrete staining, weighing approximately 305 g/sq. m when dry.

B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

C. Clear, Solvent-Borne and Waterborne, Liquid Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B.

1. For integrally colored concrete, curing compound shall be pigmented type approved by coloring admixture manufacturer.
2. For concrete indicated to be sealed, curing compound shall be compatible with sealer.

2.5 REPAIR MATERIALS

A. Bonding Agent: ASTM C 1059, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

B. Epoxy-Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class and grade to suit requirements, and as follows:

1. Type II, non-load bearing, for bonding freshly mixed concrete to hardened concrete.
2. Types I and II, non-load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

2.6 CONCRETE MIXES

A. Prepare design mixes for each type and strength of cast-in-place architectural concrete determined by either laboratory trial mix or field test databases. Proportion concrete according to EN 206.

B. Use a qualified independent testing agency for preparing and reporting proposed concrete mix designs for the laboratory trial mix basis.

C. Proportion concrete mix as follows:

1. Compressive Strength (28 Days): 40 MPa.
2. Maximum Water-Cementitious Materials Ratio: 0.46.
3. Maximum Slump: 100 mm.
4. Maximum Slump for Concrete Containing High-Range, Water-Reducing Admixture: 200 mm after admixture is added to concrete with 50 to 100 mm slump.

D. Air Content: Air content is not allowed to used.

E. Coloring Admixture: Add coloring admixture to architectural concrete mix according to manufacturer's written instructions.

2.7 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver cast-in-place architectural concrete according to EN 206, and furnish batch ticket information.

1. When air temperature is between 30 and 32 deg. C (86 and 90 deg. F), reduce mixing and delivery time from 90 to 75 minutes; when air temperature is above 32 deg. C (90 deg. F), reduce mixing and delivery time to 60 minutes.

PART 3 - EXECUTION

3.1 FORMWORK

A. General: Comply with Concrete Specification Section "Cast-in-Place Concrete" for formwork, embedded items, and shoring and reshoring.

B. In addition to ACI 303.1 limits on form-facing panel deflection, limit concrete surface irregularities, designated by ACI 347R as abrupt or gradual, as follows:

1. Class A, 3 mm.

C. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast-in-place surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical. Kerf wood rustications, keyways, reglets, recesses, and the like, for easy removal.

1. Do not use rust-stained, steel, form-facing material.

D. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.

E. Chamfer exterior corners and edges of cast-in-place architectural concrete.

F. Coat contact surfaces of wood rustications and chamfer strips with sealer before placing reinforcement.

G. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.

H. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

I. Seal form joints and penetrations at form ties with form joint tape or form joint sealant to prevent mortar leaks.

J. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

K. Coat contact surfaces of forms with form release agent, according to manufacturer's written instructions, before placing reinforcement.

L. Coat contact surfaces of forms with surface retarder, according to manufacturer's written instructions, before placing reinforcement.

M. Place form liners accurately to provide finished surface texture indicated. Provide solid backing and attach securely to prevent deflection and maintain stability of liners during concreting. Prevent form liners from sagging and stretching in hot weather. Seal joints of form liners and form liner accessories to prevent mortar leaks. Coat form liner with form release agent.

3.2 REINFORCEMENT AND INSERTS

A. General: Comply with Concrete Specification Section "Cast-in-Place Concrete" for fabricating and installing steel reinforcement.

B. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

3.3 REMOVING AND REUSING FORMS

A. Formwork, for sides of beams, walls, columns, and similar parts of the Work, that does not support weight of concrete may be removed after cumulatively curing at not less than 10 deg. C (50 deg. F) for 24 hours after placing concrete, provided concrete is hard enough to not be damaged by form-removal operations and provided curing and protection operations are maintained.

1. Schedule form removal to maintain surface appearance that matches approved mockups.
2. Cut off and grind glass-fiber-reinforced plastic form ties flush with surface of concrete.

B. Leave formwork, for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved 28-day design compressive strength. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

C. Clean and repair surfaces of forms to be reused in the Work. Do not use split, frayed, delaminated, or otherwise damaged form-facing material. Apply new form release agent.

D. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for architectural concrete surfaces.

3.4 JOINTS

A. Construction Joints: Install construction joints true to line with faces perpendicular to surface plane of cast-in-place architectural concrete so strength and appearance of concrete are not impaired, at locations indicated or as approved by the Engineer.

1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints, unless otherwise indicated.
2. Use bulkhead forms with keys of plywood, wood, or expanded galvanized steel sheet, unless otherwise indicated. Embed keys at least 38 mm into concrete. Align construction joint within rustications attached to form-facing material.
3. Locate joints for beams, slabs, joists, and girders in the middle third of spans. Offset joints in girders a minimum distance of twice the beam width from a beam-girder intersection.
4. Locate horizontal joints in walls and columns at underside of floors, slabs, beams, and girders and at the top of footings or floor slabs.
5. Space vertical joints in walls as indicated. Locate joints beside piers integral with walls, near corners, and in concealed locations where possible.
6. Use bonding agent or epoxy-bonding adhesive at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.

B. Contraction Joints: Form weakened-plane contraction joints true to line with faces perpendicular to surface plane of cast-in-place architectural concrete so that strength and appearance of concrete are not impaired, at locations indicated or as approved by the Engineer.

3.5 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Do not add water to concrete during delivery, at Project site, or during placement, unless approved by the Engineer.

C. Deposit concrete continuously between construction joints. Deposit concrete to avoid segregation.

D. Deposit concrete in forms in horizontal layers no deeper than 600 mm and in a manner so as to avoid inclined construction joints. Place each layer while preceding layer is still plastic, to avoid cold joints.

1. Consolidate placed concrete with mechanical vibrating equipment. Use equipment and procedures for consolidating concrete recommended by ACI 309R.
2. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations no farther than the visible effectiveness of the vibrator. Place vibrators to rapidly penetrate placed layer and at least 150 mm into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing concrete mix constituents to segregate.

E. Hot-Weather Placement: Place concrete according to recommendations in EN 206 and as follows, when hot-weather conditions exist:

1. Cool ingredients before mixing to maintain concrete temperature below 32 deg. C (90 deg. F) at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is the Contractor's option.
2. Cover steel reinforcement with water-soaked burlap so that steel temperature will not exceed ambient air temperature immediately before embedding in concrete.

3. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade moisture uniform without standing water, soft spots, or dry areas.

3.6 FINISHES, GENERAL

A. Architectural Concrete Finish: Match the Engineer's design reference sample, identified and described as indicated.

B. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces.

1. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces, unless otherwise indicated.

C. Maintain uniformity of special finishes over construction joints, unless otherwise indicated.

3.7 AS-CAST FORMED FINISHES

A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defective areas repaired and patched. Remove fins and other projections exceeding ACI 347R limits for class of surface specified.

B. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defective areas. Remove fins and other projections exceeding 3 mm in height.

C. Rubbed Finish: Apply the following to smooth-formed finished concrete:

1. Smooth-Rubbed Finish: Not later than one day after form removal, moisten concrete surfaces and rub with carborundum brick or another abrasive until producing a uniform color and texture. Do not apply cement grout other than that created by the rubbing process.
2. Grout-Cleaned Finish: Wet concrete surfaces and apply grout of a consistency of thick paint to coat surfaces and fill small holes. Mix 1 part Portland cement to 1-1/2 parts fine sand with a 1:1 mixture of bonding admixture and water. Add Portland cement in amounts determined by trial patches so color of dry grout matches adjacent surfaces. Scrub grout into voids and remove excess grout. When grout whitens, rub surface with clean burlap and keep surface damp by fog spray for at least 36 hours.

3. Cork-Floated Finish: Wet concrete surfaces and apply a stiff grout. Mix 1 part Portland cement and 1 part fine sand with a 1:1 mixture of bonding agent and water. Add Portland cement in amounts determined by trial patches so color of dry grout matches adjacent surfaces. Compress grout into voids by grinding surface. In a swirling motion, finish surface with a cork float.

D. Form Liner Finish: Cast panel against form liners placed, secured, and sealed over formwork panels to produce a textured surface free of pockets, streaks, and honeycombs. Produce a surface appearance of uniform color and texture.

3.8 EXPOSED-AGGREGATE FINISHES

A. Scrubbed Finish: After concrete has achieved a compressive strength from 6.9 to 10.3 MPa, apply scrubbed finish. Wet concrete surfaces thoroughly and scrub with stiff fiber or wire brushes, using water freely, until top mortar surface is removed and aggregate is uniformly exposed. Rinse scrubbed surfaces with clean water. Maintain continuity of finish on each surface or area of Work. Remove only enough concrete mortar from surfaces to match design reference sample or mockup.

B. High-Pressure Water-Jet Finish: Perform high-pressure water jetting on concrete that has achieved a minimum compressive strength of 31 MPa. Coordinate with formwork removal to ensure that surfaces to be high-pressure water-jet finished are treated at the same age for uniform results.

1. Surface Continuity: Perform high-pressure water-jet finishing in as continuous an operation as possible, maintaining continuity of finish on each surface or area of Work. Maintain required patterns or variances in reveal projection to match design reference sample or mockup.

C. Abrasive-Blast Finish: Perform abrasive blasting after the compressive strength of concrete exceeds 13.8 MPa. Coordinate with formwork removal to ensure that surfaces to be abrasive blasted are treated at the same age for uniform results.

1. Surface Continuity: Perform abrasive-blast finishing in as continuous an operation as possible, maintaining continuity of finish on each surface or area of Work. Maintain required patterns or variances in depths of blast to match design reference sample or mockup.
2. Abrasive Blasting: Abrasive blast corners and edges of patterns carefully, using backup boards, to maintain uniform corner or edge line. Determine type of nozzle, nozzle pressure, and blasting techniques required to match design reference sample or mockup.
3. Depth of Cut: Use an abrasive grit of proper type and gradation to expose aggregate and surrounding matrix surfaces to match design reference sample or mockup, as follows:

    a. Brush: Remove cement matrix to dull surface sheen and expose face of fine aggregate. No significant reveal.
    b. Light: Expose fine aggregate with occasional exposure of coarse aggregate and uniform color. Maximum reveal 1.5 mm.
    c. Medium: Generally expose coarse aggregate with slight reveal. Maximum reveal 6 mm.
    d. Heavy: Expose and reveal coarse aggregate to a maximum projection of one-third its diameter. Reveal range 6 to 13 mm.

D. Bushhammer Finish: Allow concrete to cure at least 14 days before starting bushhammer surface finish operations.

1. Surface Continuity: Perform bushhammer finishing in as continuous an operation as possible, maintaining continuity of finish on each surface or area of Work. Maintain required patterns or variances of cut as shown on Drawings or to match design reference sample or mockup.
2. Surface Cut: Maintain required depth of cut and general aggregate exposure. Use power hammerheads for large, flat surfaces and hand hammers for small areas, at corners and edges, and for restricted locations where power tools cannot reach.

3.9 CONCRETE CURING

A. Protect freshly placed concrete from premature drying and excessive cold or hot temperatures according to EN 206.

B. Begin curing immediately after removing forms from concrete. Cure by one or a combination of the following methods that will not mottle, discolor, or stain concrete:

1. Moisture Curing: Keep exposed surfaces of cast-in-place architectural concrete continuously moist for not less than seven days with the following materials:

a. Water.
b. Continuous water-fog spray.
c. Absorptive cover, water saturated and kept continuously wet. Cover concrete surfaces and edges with 300 mm lap over adjacent absorptive covers.

2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 300 mm, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period; use cover material and waterproof tape.
3. Curing Compound: Mist concrete surfaces with water. Apply curing compound uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.

3.10 FIELD QUALITY CONTROL

A. General: Comply with Concrete Specification Section "Cast-in-Place Concrete" for field quality-control requirements.

3.11 REPAIRS, PROTECTION, AND CLEANING

A. Repair and cure damaged finished surfaces of cast-in-place architectural concrete when approved by the Engineer. Match repairs to color, texture, and uniformity of surrounding surfaces and to repairs on approved mockups.

1. Remove and replace cast-in-place architectural concrete that cannot be repaired and cured to the Engineer's approval.

B. Protect corners, edges, and surfaces of cast-in-place architectural concrete from damage; use guards and barricades.

C. Protect cast-in-place architectural concrete from staining, laitance, and contamination during remainder of construction period.

D. Clean cast-in-place architectural concrete surfaces after finish treatment to remove stains, markings, dust, and debris.

E. Wash and rinse surfaces according to concrete finish applicator's written recommendations. Protect other Work from staining or damage due to cleaning operations.

19.

1. Do not use cleaning materials or processes that could change the appearance of cast-in-place architectural concrete finishes.

END OF SECTION 03331

20.

SECTION 03500 - GLASSFIBER REINFORCED CEMENT (GRC)

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Provide special glassfiber reinforced cement (GRC) shapes in accordance with the requirements of the contract documents.

2. The installing contractor shall be responsible for verifying that all required blocking is provided and installed in the correct locations for GRC component's proper installation.

3. Works including the following:

   a. Supply of GRC units.
   b. Erection/installation.
   c. Joint treatment
   d. Supply and installation of backup supports, shimming labor and materials, etc.

B. Related Sections include the following:

1. Steel Specification Section "Architectural Joint Systems".
2. Steel Specification Section "Metal Fabrications"
3. Steel Specification Section "Cold Formed Metal Framing"
4. Woods & Plastics Specification Section "Interior Plastic Ornamentation".

1.3 DESCRIPTION OF WORK

A. This specification is intended to outline the general requirements of the (GRC) units as they pertain to the overall design of the project. The manufacturer's recommendations shall not govern the work in this section.

B. The installing contractor shall perform all work in this section, including installation, caulking, filling and patching and will assume responsibility for coordinating installation with associated trades.

C. GRC is a decorative non-load bearing material and as such the manufacturer cannot be held responsible for structural, load, or seismic considerations.

20.

1.4 SUBMITTALS

A. Product Data: For each type of product and process specified and incorporated into items of GRC during fabrication, finishing, and installation.

1. Include data for GRC-preservative treatment from chemical treatment manufacturer and certification by treating plant that treated materials comply with requirements.

B. Shop Drawings: Show location of each item, dimensioned plans and elevations, large-scale details, and other components.

1. Show details full size and supports.
2. Show locations and sizes of blocking, including concealed blocking and reinforcement specified in other Sections.

C. Samples for Verification: For the following:

1. Submit sample of each type.

D. Product Certificates: Signed by manufacturers of GRC certifying that products furnished comply with requirements.

E. Qualification Data:

1. The manufacturer of glassfiber reinforced cement work shall submit evidence of satisfactory projects completed over the last 10 years.
2. The installer of the work specified herein shall be approved by the manufacturer of the specified material.
3. The installer of the work, with more than 5 years experience in the installation of GRC units, shall carry out the installation efficiently and co-operate fully with other trades.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Installer Qualifications: An experienced installer who has completed GRC work similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

C. Fabricator Qualifications: A firm experienced in producing GRC work similar to that indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Source Limitations for Fabrication and Installation: Engage a qualified GRC working firm to assume undivided responsibility for fabricating and installing GRC work specified in this Section.

20.

E. Quality Standard: Except as otherwise additionally indicated on the drawings or specified herein, the standards referred to below, shall apply to work under this section:

1. ASTM C 473 Nail Pull Resistance
2. ASTM C 473 Humidified Deflection
3. ASTM C 947 Elastic Limits
4. ASTM D 256 Impact Resistance
5. ASTM D 638 Ultimate Tensile Strength
6. ASTM D 638 Young's Modulus
7. ASTM D 696 Coefficient of Linear Thermal Expansion
8. ASTM D 790 Flexural Strength
9. ASTM D 790 Flexural Modulus
10. ASTM D 2583 Barcol Hardness
11. ASTM E 84 Flame Spread Index
12. ASTM E 84 Smoke Developed Index

1.6 DESIGN CRITERIA

A. Unless stated on the drawings, fabrication tolerances are as indicated below.

1.	Dimensional - all directions (0' - 10')	± 1/8" (3.2mm)
2.	Dimensional - all directions (11' - 20')	± 3/16" (4.8mm)
3.	Straightness along an edge or surface	± 1/8"/linear ft.
4.	All reveals, setbacks or returns	5E draft
5.	All corners	1/16" - 1/8" radius

1.7 PROJECT CONDITIONS

A. Weather Limitations: Proceed with installation of exterior GRC work only when existing and forecasted weather conditions permit work to be performed and at least one coat of specified finish to be applied without exposure to rain, snow, or dampness.

B. Field Measurements: Where GRC work is indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1. Locate concealed framing, blocking, and reinforcements that support GRC work by field measurements before being enclosed and indicate measurements on Shop Drawings.

1.8 COORDINATION

A. Coordinate sizes and locations of framing, blocking, reinforcements, and other related units of Work specified in other Sections to ensure that GRC work can be supported and installed as indicated.

1.9 DELIVERY, STORAGE, AND HANDLING

20.

A. Units shall be handled and transported in a manner so as not to create damage or excessive stresses.

B. GRC units shall be stored level on a clean dry surface in an area protected from weather, moisture and damage. The units shall not be stacked or leaned unless instructed otherwise by the manufacturer.

1.10 WARRANTY

A. The manufacturer warrants that the delivered material supplied will conform to sample and to specifications and will be free from defects in workmanship or material under normal use and conditions for a period of one (1) year from date of substantial completion. Should defects, attributable to the manufacturer, appear within one (1) year of the date of substantial completion, the manufacturer has the option of replacing or repairing the defective material.

B. Limitations: The aforementioned general warranty is exclusive. All other warranties whether expressed or implied or arising by operation of Alat FEZ law, usage of trade, course of dealings or otherwise, are excluded. The only warranties are those expressed above. The manufacturer shall not be liable for any penalty or for any loss or damages associated with the removal or installation of its product or claims of third parties against the Purchaser.

PART 2 - PRODUCTS

2.1 MATERIALS

A. GRC units shall be prefabricated with glassfiber reinforced cement.

B. Units will be suitably reinforced with additional materials as required.

C. GRC units are supplied primer ready. Finishing is specified elsewhere and shall be applied after the GRC units are installed.

D. Exposed fasteners shall be stainless steel. All other fastening or attachment devices shall be appropriately plated or galvanized.

E. Where cast-in texture is specified, the texture shall be approved at time of sample submission.

F. Related product sections include:
1. Columns
2. Trim
3. Cementitious Panels

2.2 PHYSICAL PROPERTIES

A.	Glass content		5 - 6% by weight,
B.	Weight (depending on reinforcing)		2.9 lbs/sq.ft.
C.	Shell Thickness		3/8” nominal (1/4" min.)
D.	Flame Spread Index	(ASTM E 84)	0 (Class 1) (Class A)
E.	Smoke Developed Index	(ASTM E 84)	0 (Class 1) (Class A)
F.	Flexural Strength	(ASTM D 790)	4,650 psi (32.1 MPa)

G.	Flexural Modulus	(ASTM D 790)	1.43 x 10^6 (9,860 MPa)
H.	Elastic Limit	(ASTM C 947)	981 psi (6.70 MPa) avg.
I.	Ultimate Tensile Strength	(ASTM D 638)	1,480 psi (10.2 MPa)
J.	Impact Resistance	(ASTM D 256)	7.83 ft.lb./in (mean value)
K.	Barcol Hardness	(ASTM D 2583)	81 Avg.
L.	Coef. of Thermal Expansion	(ASTM D 696)	5.50x10^{-6} in/in EF/9.95 x 10^{-6} mm/mm
M.	Nail Pull (wood embedment)	(ASTM C 473)	466 lbf. Avg.
N.	Nail Pull (metal embedment)	(ASTM A 473)	139 lbf. Avg.
O.	Humidified Deflection	(ASTM C 473)	0.2 mm. mean deflection
P.	Compressive Strength	(ASTM C 472)	10,364 psi.
Q.	Young's Modulus	(ASTM D 638)	1.48 x 10^6

PART 3 - EXECUTION

3.1 EXAMINATION

A. Prior to the manufacture of components, the installer shall check all pertinent site dimensions and conditions against the manufacture's drawings and relay discrepancies to the manufacturer for inclusion in the drawings.

B. Prior to installation, the installer shall compare job site dimensions and conditions against the Architect's drawings and shall report any discrepancies to the General Contractor and the Architect. Work shall not proceed until discrepancies are corrected

C. Prior to installation, the installer shall examine pertinent job-site conditions to insure proper arrangement and fit of the work. Start of work implies acceptance of job-site condition.

D. All surfaces or framing structures shall be plumb and true as required.

3.2 ERECTION

A. Install work as indicated on drawings, as specified herein and in accordance with approved shop drawings and manufacturer's recommendations.

B. Provide all support framing/reinforcing/support brackets required for work of this section and to ensure solid and secure installation.

C. Provide temporary supports to maintain position as units are being installed.

D. Fasten units with screws through the face of the GRC unit(s), or from the rear.

E. Adequate control and/or expansion joints shall be employed.

F. Pre-drill fastener holes in GRC components. Clean fastener holes to remove dirt and oil.

G. Attach GRC components to framing and substrate with steel drill screws. Do not use pneumatic staple guns. Countersink screw heads below adjoining finished surface.

H. Fasten not less than 5/16" from edge or end.

I. Cover screw heads with joint compound to produce, smooth, flush and level surfaces.

J. GRC units shall be handled with care and lifted with appropriate equipment.

K. GRC units shall be installed true and plumb, shimmed where necessary.

L. Caulk joints where required with a one-compound elastomeric low modulus urethane sealant equivalent to Sonolastic Ultra. Control joints shall be provided where required as specified by Architect. Color of caulk to be selected by Architect.

M. Expansion joints shall be installed as per Architect's recommendation.

N. Anchors and fasteners: Type 304 stainless steel where exposed; hot dip galvanized steel where unexposed.

3.3 PATCHING AND CLEANING

A. Repair any defects found after the work of all trades has been completed, regardless of how, or by whom, the damage was caused. Patching shall match the original work. Use Auto Body filler or a polymer modified cementitious product with migrating corrosion inhibitors equivalent to Sika Top to repair any defects and areas requiring patching. Select the correct product by application.Repair any defects found after the work of all trades has been completed, regardless of how, or by whom, the damage was caused. Patching shall match the original work. Use Auto Body filler or a polymer modified cementitious product with migrating corrosion inhibitors equivalent to Sika Top to repair any defects and areas requiring patching. Select the correct product by application.

B. Patch all countersunk fasteners and damages to match unit's texture, finished flush with face of unit.

C. Soiled units may be cleaned using detergent and water.

3.4 FINISHING

A. All GRC requires site prime and paint. Use only breathable primers and paints/coatings equivalent to Sherwin-Williams Loxon A-100 breathable primers and paints.

END OF SECTION 03500

SECTION 03544 - CEMENT-BASED SCREED

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes:

1. Standard aggregate concrete floor screed underlayment for interior finish flooring.
2. Lightweight aggregate concrete floor screed for interior application only.
3. Standard aerated concrete floor screed for interior application only.

B. Related Sections include the following:

1. Concrete Specification Section "Concrete Floor Topping" for cementitious wear toppings applied over base slabs.
2. Concrete Specification Section "Cement-Based Underlayment" for cementitious self-leveling underlayment applied under floor finishes.
3. Concrete Specification Section " Lightweight Concrete Roof Insulation".

1.3 DEFINITIONS

A. Screed: A layer of cement-sand or concrete mix, with or without reinforcement, applied over structural floor or roof deck slab to achieve correct level and receive another finish material.

1.4 SUBMITTALS

A. Product Data: For each type of product indicated.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating brand name and directions for storage, mixing with other components, and application.

B. Store materials to comply with manufacturer's written instructions to prevent deterioration from moisture or other detrimental effects.

1.7 PROJECT CONDITIONS

A. Environmental Limitations: For proprietary products, comply with manufacturer's written recommendations for substrate temperature and moisture content, ambient temperature and humidity, ventilation, and other conditions affecting screed performance.

B. Close areas to traffic during screed application and for appropriate time period after application as recommended in writing by the manufacturer.

1.8 COORDINATION

A. Coordinate cement-based screed with requirements of finish flooring products, including adhesives

PART 2 - PRODUCTS

1.8 SCREED MATERIALS

A. Screed: Cement-sand mix with admixtures applied over structural deck to receive another finishing material.

1. Portland Cement: EN 197, CEM 1
2. Compressive Strength: Not less than 20 MPa at 28 days when tested according to EN 12390.
3. Maximum Slump: 125 mm.
4. Screed Additive: Plasticisers and water-retaining agents formulated for use with screed when applied to substrate and conditions indicated, and to approval.

B. Normal-Weight Aggregate: EN 206 and as follows:

1. Fine aggregate, consisting of sand or crushed stone screenings, clean, hard, free of deleterious matter. Grade by weight to pass sieves as follows:

a. 9.5 mm: 100 percent.
b. 4.8 mm: 95-100 percent.
c. 2.4 mm: 80-90 percent.
d. 1.2 mm: 50-75 percent.
e. 600 µm: 30-50 percent.
f. 300 µm: 10-20 percent.

g. 150 µm: 2-5 percent.

2. Coarse aggregate, consisting of gravel or crushed stone, clean, hard, free of deleterious matter. Grade by weight to pass sieves as follows:

a. 12.5 mm: 100 percent.
b. 9.5 mm: 30-50 percent.
c. 4.8 mm: 0-15 percent.
d. 2.4 mm: 0-5 percent.

C. Lightweight Aerated Screed: Not less than 75 mm thick and having the following properties:

1. Dry Density: 400 kg/m³.
2. Sand Content: Nil.
3. Compressive Strength (28 days): 1.0 - 1.2 N/mm².
4. Thermal Conductivity: 0.08 W/m K.
5. Cement and Foaming Agent Content: Adjust as necessary to obtain the above properties.

D. Lightweight Aerated Screed: Not less than 75 mm thick, and having the following properties:

1. Dry Density: 800 kg/m³.
2. Sand Content: 420 kg/m³.
3. Compressive Strength (28 days): 1.5 - 2.0 N/mm².
4. Thermal Conductivity: 0.17 - 0.23 W/m K.
5. Cement and Foaming Agent Content: Adjust as necessary to obtain the above properties.

E. Lightweight Aggregate Screed: Comprising ordinary Portland cement and expanded clay. Not less than 80 mm thick, and having the following properties:

1. Nominal Maximum Aggregate Size: 8 - 16 mm.
2. Bulk Density in Dry: 300-400 kg/m³.
3. Cement: 125 kg.
4. Lightweight Aggregate: 1000 lt. or as directed by the manufacturer to achieve specified density, compressive strength and thermal conductivity.
5. Concrete Density: 500 - 600 kg/m³.
6. Compressive Strength: > 2.5 MPa.
7. Thermal Conductivity: 0.12 W/m K.

F. Lightweight Aggregate Screed: Comprising ordinary Portland cement and pumice. Not less than 80 mm thick, and having the following properties:

1. Nominal Maximum Aggregate Size: 0 - 8 mm.
2. Bulk Density in Dry: 700 kg/m³.
3. Cement: 200 kg.
4. Sand Content: Nil.
5. Lightweight Aggregate: 1000 lt. or as directed by the manufacturer to achieve specified density, compressive strength and thermal conductivity.
6. Concrete Density: 900 kg/m³.
7. Compressive Strength: > 5 MPa.
8. Thermal Conductivity: 0.226 W/m K.

G. Lightweight Aggregate Screed: Comprising ordinary Portland cement and pumice. Not less than 80 mm, and having the following properties:

1. Nominal Maximum Aggregate Size: 0 - 8 mm.
2. Bulk Density in Dry: 700 kg/m³.
3. Cement: 350 kg.
4. Sand Content: 425 kg.
5. Lightweight Aggregate: 1000 lt. or as directed by the manufacturer to achieve specified density, compressive strength and thermal conductivity.
6. Concrete Density: 1600 kg/m³.
7. Compressive Strength: > 14 MPa.
8. Thermal Conductivity: 0.50 W/m K.

H. Reinforcement: ASTM A 185, welded steel wire fabric.

I. Water: Potable and at a temperature of not more than 21 deg. C (70 deg. F).

J. Fibrous Reinforcement: 100% virgin polypropylene, fibrillated fibers containing no reprocessed olefin materials and specifically manufactured to an optimum gradation for use as concrete secondary reinforcement. Proportion shall be 0.9 kg per m³ of concrete:

1. Specific Gravity: 0.91.
2. Tensile strength: 345 - 758 N/mm².

1.9 CURING MATERIALS

A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 305 g/sq. m when dry.

B. Water: Potable EN 1008

1.10 DESIGN MIXES

A. Prepare design mixes for each type and strength of lightweight concrete screed by either laboratory trial batch or field-test data methods. For trial batch method, use an approved qualified independent testing agency for preparing and reporting proposed mix designs.

B. Limit water-soluble chloride ions to the maximum percentage by weight of cement or cementitious material permitted by EN 206.

1.11 RELATED MATERIALS

A. Epoxy Joint Filler: Two-component, semi-rigid, 100 percent solids, epoxy resin with a Shore A hardness of 80

B. Joint-Filler Strips: asphalt-saturated cellulosic fiber or cork or self-expanding cork.

C. Water: Potable to EN 1008.

D. Acrylic-Bonding Agent: non-redispersible, acrylic emulsion or styrene butadiene.

E. Admixtures: Admixtures certified by manufacturer to contain not more than 0.1 % water-soluble chloride ions by mass of cementitious material and compatible with other admixture and cementitious materials. Do not use admixtures containing calcium chloride.

F. Epoxy Adhesive: two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class and grade to suit requirements.

1.12 MIXING

A. Bonding Slurry: Mix 1 part Portland cement and 1.5 parts sand with water and an acrylic-bonding agent according to manufacturer's written instructions to a thick paint consistency.

B. Screed Mix: Design mix, with or without admixture, to produce concrete floor topping material with the following characteristics:

1. Compressive Strength (28 days): Not less than 20 Mpa.
2. Maximum Slump: 125 mm.
3. Mix proportions: Submit proposed mix design for approval.
4. Minimum Cement-Sand Ratio: Not less than 1:4 by volume.
5. Maximum Water-Cement Ratio: 0.51.
6. Mix screed materials, admixtures and water in appropriate drum-type batch machine mixer or truck mixer according to manufacturer's written instructions.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates for conditions affecting performance of screed. Proceed with application only after unsatisfactory conditions have been corrected.

B. Verify that base slabs meet finish and surface profile requirements in Concrete Specification Section "Cast-in-Place Concrete."

3.2 PREPARATION

A. Existing Concrete: Remove existing surface treatments and deteriorated and unsound concrete. Mechanically abrade base slabs to produce a heavily scarified surface profile with an amplitude of 6 mm.

1. Prepare and clean existing base slabs according to topping manufacturer's written instructions. Fill voids, cracks, and cavities in base slabs.
2. Mechanically remove contaminants from existing concrete that might impair bond of topping.
3. Saw cut existing contraction and construction joints to a depth of 13 mm and fill with epoxy joint filler.

B. Install joint-filler strips where screed abuts vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

1. Extend joint-filler strips full width and depth of joint, terminating flush with screed surface, unless otherwise indicated.
2. Terminate full-width joint-filler strips 13 mm below screed surface where joint sealants
3. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.

3.3 SCREED APPLICATION

A. Start screed application in presence of the manufacturer's technical representative.

B. Fibrous Reinforcement: For reinforced screeds, apply fibrous reinforcing as per manufacturer's written instructions.

C. Deferred Screed: Within 72 hours of placing base slabs, mix and scrub bonding slurry into dampened concrete to a thickness of 1.5 to 3 mm, without puddling. Place screed while slurry is still tacky.

D. Place screed continuously in a single layer, tamping and consolidating to achieve tight contact with bonding surface. Do not permit cold joints or seams to develop within pour strip.

1. Screed surface with a straightedge and strike off to correct elevations.
2. Slope surfaces uniformly where indicated.
3. Begin initial floating using bull floats to form a uniform and open-textured surface plane free of humps or hollows.

E. Finishing: Consolidate surface with power-driven floats as soon as screed can support equipment and operator. Restraighten, cut down high spots, and fill low spots. Repeat float passes and restraightening until surface is left with a uniform, smooth, granular texture.

1. Hard Trowel Finish: After floating surface, apply first trowel finish and consolidate screed by power-driven trowel so no blisters develop. Continue troweling passes and restraighten until surface is smooth and uniform in texture.

a. Finish surfaces to specified overall values of flatness, F(F) 25; and levelness, F(L) 20; with minimum local values of flatness, F(F) 17; and levelness, F(L) 15, measured within 24 hours for a randomly trafficked floor surface.

b. Finish and measure surface so gap at any point between screed surface and an unleveled freestanding 3 m long straightedge, resting on two high spots and placed anywhere on the surface, does not exceed 6 mm.

F. Construction Joints: Construct joints true to line with faces perpendicular to surface plane of screed, at locations indicated or as approved by the Engineer.

1. Coat face of construction joint with epoxy adhesive at locations where screed is placed against hardened or partially hardened screed.

G. Contraction Joints: Form weakened-plane contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 3 mm wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before screed develops random contraction cracks.

1. Form joints in screed over contraction joints in base slabs, unless otherwise indicated.
2. Construct contraction joints for a depth equal to one-half of screed thickness, but not less than 13 mm deep.

3.4 PROTECTION AND CURING

A. General: Protect freshly placed screed from premature drying and excessive cold or hot temperatures.

B. Begin curing immediately after finishing screed. Cure by one or a combination of the following methods, according to screed manufacturer's written instructions:

1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with water or absorptive cover, water saturated and kept continuously wet. Cover screed surfaces and edges with 300 mm lap over adjacent absorptive covers.

3.5 JOINT FILLING

A. Prepare and clean contraction joints and install epoxy joint filler, according to manufacturer's written instructions, once screed has fully cured.

B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.

C. Install epoxy joint filler full depth of contraction joints. Overfill joint and trim joint filler flush with top of joint after hardening.

3.6 REPAIRS

A. Defective Screed: Repair and patch defective screed areas, including areas that have not bonded to concrete substrate.

3.7 FIELD QUALITY CONTROL

A. Sample Sets: At point of placement, testing and inspecting agency shall take a set of 3 molded-cube samples from the screed mix for the first 90 sq. m plus 1 set of samples for each subsequent 450 sq. m of screed, or fraction thereof, but not less than 6 samples for each day's placement. Samples shall be tested according to EN 12390 for compliance with compressive strength requirements.

3.8 CLEANING AND PROTECTING

A. Protect screeds from damage and wear during the remainder of construction period. Use protective methods and materials approved by the Engineer, including temporary covering.

B. Clean screeds not more than 4 days before dates scheduled for inspections intended to install final finishes.

END OF SECTION 03544

26.

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PART 2B

STEEL SPECIFICATION - METALS

SECTION 05120 - STRUCTURAL STEEL
SECTION 05210 - STEEL JOISTS
SECTION 05310 - STEEL DECK
SECTION 05400 - COLD-FORMED METAL FRAMING
SECTION 05500 - METAL FABRICATIONS
SECTION 05521 - PIPE AND TUBE RAILINGS
SECTION 05530 – GRATINGS
SECTION 05580 - FORMED-METAL FABRICATIONS
SECTION 05700 - ORNAMENTAL METAL
SECTION 05811 - ARCHITECTURAL JOINT SYSTEMS

SECTION 05120 - STRUCTURAL STEEL

PART 1 - GENERAL

1.1 STANDARDS

A. The requirements of EN 1993 and EN 1999 shall take precedence over all other standards. Other specifications outlined in this document may be used, where there isn't an equivalent hEN or EN / BS Standards.

1.2 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and General Requirements Specification Sections, apply to this Section.

1.3 SUMMARY

A. This Section includes structural steel.

B. This Section includes structural steel and architecturally exposed structural steel.

C. Related Sections: The following Sections contain requirements that relate to this Section:

1. General Requirements Section "Quality Control" for independent testing agency procedures and administrative requirements.
2. Steel Specification Section "Steel Deck" for field installation of shear connectors.
3. Steel Specification Section "Metal Fabrications" for loose steel bearing plates and miscellaneous steel framing.

1.4 PERFORMANCE REQUIREMENTS

A. Structural Performance: Engineer structural steel connections required by the Contract Documents to be selected or completed by the fabricator to withstand design loadings indicated.

B. Engineering Responsibility: Engage a fabricator who utilizes a qualified professional engineer to prepare calculations, Shop Drawings, and other structural data for structural steel connections.

1.5 SUBMITTALS

A. General: Submit each item in this Article according to the Conditions of the Contract and General Requirements Specification Sections.

B. Product Data for each type of product specified.

C. Shop Drawings detailing fabrication of structural steel components.

1. Include details of cuts, connections, splices, camber, holes, and other pertinent data.
2. Indicate welds by standard AWS symbols, distinguishing between shop and field welds, and show size, length, and type of each weld.
3. Indicate type, size, and length of bolts, distinguishing between shop and field bolts. Identify high-strength bolted slip-critical, direct-tension, or tensioned shear/bearing connections.
4. Include Shop Drawings signed and sealed by a qualified professional engineer responsible for their preparation.

D. Qualification data for firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

E. Mill test reports signed by manufacturers certifying that their products, including the following, comply with requirements.

1. Structural steel, including chemical and physical properties.
2. Bolts, nuts, and washers, including mechanical properties and chemical analysis.
3. Direct-tension indicators.
4. Shear stud connectors.
5. Shop primers.
6. Nonshrink grout.

F. As-Built Drawings: At project closeout, submit Record (As-Built) Drawings of completed work products, in accordance with requirements of the Specification as indicated in General Requirements.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Installer Qualifications: Engage an experienced Installer who has completed structural steel work similar in material, design, and extent to that indicated for this Project and with a record of successful in-service performance.

C. Fabricator Qualifications: Engage a firm experienced in fabricating structural steel similar to that indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to fabricate structural steel without delaying the Work.

D. Unless noted otherwise on drawings or in this specification all materials and workmanship shall comply with the National Structural Steelwork Specification for Building Construction 5th Edition.

E. Professional Engineer Qualifications: A professional engineer who is legally authorized to practice in the jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed

for projects with structural steel framing that are similar to that indicated for this Project in material, design, and extent.

F. Welding Standards: Comply with applicable provisions of BS 499, EN 756 and EN 760.

1. Present evidence of a Welding Quality Management System (WQMS) to EN ISO 3834 and a Responsible Welding Coordinator. Evidence of CE Certification to Execution Class 3 and 4 shall also be presented.

G. Mockups: Prior to installing architecturally exposed structural steel, construct mockups for each form of construction and finish required to demonstrate aesthetic effects as well as qualities of materials and execution. Build mockups to comply with the following requirements, using materials indicated for final unit of Work.

1. Locate mockups on-site in the location and of the size indicated or, if not indicated, as directed by the Engineer.
2. Notify the Engineer one week in advance of the dates and times when mockups will be constructed.
3. Demonstrate the proposed range of aesthetic effects and workmanship of steel surfaces and welded and bolted connections.

    a. Coordinate finish painting requirements of mockups

4. Obtain the Engineer's approval of mockups before start of final unit of Work.
5. Retain and maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.

    a. When directed, demolish and remove mockups from Project site.
    b. Approved mockups in an undisturbed condition at the time of Substantial Completion may become part of the completed Work.

H. Preinstallation Conference: Conduct conference at Project site to comply with requirements of General Requirements Section "Project Management and Coordination."

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver structural steel to Project site in such quantities and at such times to ensure continuity of installation.

B. Store materials to permit easy access for inspection and identification. Keep steel members off ground by using pallets, platforms, or other supports. Protect steel members and packaged materials from erosion and deterioration.

1. Store fasteners in a protected place. Clean and relubricate bolts and nuts that become dry or rusty before use.

2. Do not store materials on structure in a manner that might cause distortion or damage to members or supporting structures. Repair or replace damaged materials or structures as directed.

1.8 SEQUENCING

A. Supply anchorage items to be embedded in or attached to other construction without delaying the Work. Provide setting diagrams, templates, instructions, and directions, as required, for installation.

PART 2 - PRODUCTS

2.1 MATERIALS

Standard Number	**Description**	**Status**
EN 10016-1	Non-alloy steel rod for drawing and/or cold rolling - Part 1: General requirements	
EN 10016-2	Non-alloy steel rod for drawing and/or cold rolling - Part 2: Specific requirements for general purposes rod	
EN 10016-3	Non-alloy steel rod for drawing and/or cold rolling - Part 3: Specific requirements for rimmed and rimmed substitute low carbon steel rod	
EN 10016-4	Non-alloy steel rod for drawing and/or cold rolling - Part 4: Specific requirements for rod for special applications	
EN 10017	Steel rod for drawing and/or cold rolling - Dimensions and tolerances	
EN 10020	Definition and classification of grades of steel	
EN 10021	General technical delivery conditions for steel products	
EN 10024	Hot rolled taper flange I sections - Tolerances on shape and dimensions	
EN 10025	Hot rolled products of non-alloy structural steels; technical delivery conditions	Replaced by EN 10025-1/2004; 10025-2/2004
EN 10025-1	Hot rolled products of structural steels - Part 1: General technical delivery conditions	
EN 10025-2	Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels	
EN 10025-3	Hot rolled products of structural steels - Part 3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels	
EN 10025-4	Hot rolled products of structural steels - Part 4: Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels	

EN 10025-5	Hot rolled products of structural steels - Part 5: Technical delivery conditions for structural steels with improved atmospheric corrosion resistance	
EN 10025-6	Hot rolled products of structural steels - Part 6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition	
EN 10028-1	Flat products made of steels for pressure purposes - Part 1: General requirements	
EN 10028-2	Flat products made of steels for pressure purposes - Part 2: Non-alloy and alloy steels with specified elevated temperature properties	
EN 10028-3	Flat products made of steels for pressure purposes - Part 3: Weldable fine grain steels, normalized	
EN 10028-4	Flat products made of steels for pressure purposes - Part 4: Nickel alloy steels with specified low temperature properties	
EN 10028-5	Flat products made of steels for pressure purposes - Part 5: Weldable fine grain steels, thermomechanically rolled	
EN 10028-6	Flat products made of steels for pressure purposes - Part 6: Weldable fine grain steels, quenched and tempered	
EN 10028-7	Flat products made of steels for pressure purposes - Part 7: Stainless steels	
EN 10029	Hot rolled steel plates 3 mm thick or above - Tolerances on dimensions, shape and mass	
EN 10031	Semi-finished products for forging - Tolerances on dimensions shape and mass	
EN 10034	Structural steel I and H sections - Tolerances on shape and dimensions	
EN 10048	Hot rolled narrow steel strip - Tolerances on dimensions and shape	
EN 10051	Continuously hot-rolled uncoated plate, sheet and strip of non-alloy and alloy steels - Tolerances on dimensions and shape	
EN 10055	Hot rolled steel equal flange tees with radiused root and toes - Dimensions and tolerances on shape and dimensions	
EN 10056-1	Structural steel equal and unequal leg angles - Part 1: Dimensions	
EN 10056-2	Structural steel equal and unequal leg angles - Part 2: Tolerances on shape and dimensions	
EN 10058	Hot rolled flat steel bars for general purposes - Dimensions and tolerances on shape and dimensions	
EN 10059	Hot rolled square steel bars for general purposes - Dimensions and tolerances on shape and dimensions	
EN 10060	Hot rolled round steel bars for general purposes - Dimensions and tolerances on shape and dimensions	
EN 10061	Hot rolled hexagon steel bars for general purposes - Dimensions and tolerances on shape and dimensions	
EN 10067	Hot rolled bulb flats - Dimensions and tolerances on shape, dimensions and mass	
EN 10080	Steel for the reinforcement of concrete - Weldable reinforcing steel – General	
EN 10083-1	Steels for quenching and tempering - Part 1: General technical delivery conditions	

EN 10083-2	Steels for quenching and tempering - Part 2: Technical delivery conditions for non alloy steels	
EN 10083-3	Steels for quenching and tempering - Part 3: Technical delivery conditions for alloy steels	
EN 10084	Case hardening steels - Technical delivery conditions	
EN 10085	Nitriding steels - Technical delivery conditions	
EN 10087	Free-cutting steels - Technical delivery conditions for semi-finished products, hot-rolled bars and rods	
EN 10089	Hot-rolled steels for quenched and tempered springs - Technical delivery conditions	
EN 10090	Valve steels and alloys for internal combustion engines	
EN 10092-1	Hot rolled spring steel flat bars - Part 1: Flat bars - Dimensions and tolerances on shape and dimensions	
EN 10092-2	Hot rolled spring steel flat bars - Part 2: Ribbed and grooved spring leaves - Dimensions and tolerances on shape and dimensions	
EN 10108	Round steel rod for cold heading and cold extrusion - Dimensions and tolerances	
EN 10111	Continuously hot-rolled low carbon steel sheet and strip for cold forming - Technical delivery conditions	
EN 10113-2	Hot-rolled products in weldable fine grain structural steels - Part 2: delivery conditions for normalized rolled steels	Replaced by EN 10025-1:2004; EN 10025-3:2004
EN 10113-3	Hot-rolled products in weldable fine grain structural steels - Part 3: delivery conditions for thermomechanically rolled steels	Replaced by EN 10025-1:2004; EN 10025-4:2004
EN 10120	Steel sheet and strip for welded gas cylinders	
EN 10130	Cold rolled low carbon steel flat products for cold forming - Technical delivery conditions	
EN 10131	Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming - Tolerances on dimensions and shape	
EN 10132-1	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 1: General	
EN 10132-2	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 2: Case hardening steels	
EN 10132-3	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 3: Steels for quenching and tempering	
EN 10132-4	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 4: Spring steels and other applications	
EN 10137-2	Plates and wide flats made of high yield strength structural steels in the quenched and tempered or precipitation hardened conditions - Part 2: Delivery conditions for quenched and tempered steels	Replaced by EN 10025-1:2004; EN 10025-6:2004
EN 10137-3	Plates and wide flats made of high yield strength structural steels in the quenched and tempered or precipitation hardened conditions - Part 3: Delivery conditions for precipitation hardened steels	Withdrawn 2000
EN 10139	Cold rolled uncoated mild steel narrow strip for cold forming - Technical delivery conditions	
EN 10140	Cold rolled narrow steel strip - Tolerances on dimensions and shape	

EN 10149-1	Hot-rolled flat products made of high yield strength steels for cold forming - Part 1: General delivery conditions	
EN 10149-2	Hot-rolled flat products made of high yield strength steels for cold forming - Part 2: Delivery conditions for thermomechanically rolled steels	
EN 10149-3	Hot-rolled flat products made of high yield strength steels for cold forming - Part 3: Delivery conditions for normalized or normalized rolled steels	
EN 10155	Structural steels with improved atmospheric corrosion resistance - Technical delivery conditions	Replaced by EN 10025-5:2004
EN 10162	Cold rolled steel sections - Technical delivery conditions - Dimensional and cross-sectional tolerances	
EN 10163-1	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 1: General requirements	
EN 10163-2	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 2: Plate and wide flats	
EN 10163-3	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 3: Sections	
EN 10164	Steel products with improved deformation properties perpendicular to the surface of the product - Technical delivery conditions	
EN 10168	Steel products - Inspection documents - List of information and description	
EN 10207	Steels for simple pressure vessels - Technical delivery requirements for plates, strips and bars	
EN 10208-1	Steel pipes for pipelines for combustible fluids - Technical delivery conditions - Part 1: Pipes of requirement class A	
EN 10208-2	Steel pipes for pipelines for combustible fluids - Technical delivery conditions - Part 2: Pipes of requirement class B	
EN 10209	Cold rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions	
EN 10210-1	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions	
EN 10210-2	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 2: Tolerances, dimensions and sectional properties	

A. Shear Connectors: ASTM A 108, Grade 1015 through 1020, headed-stud type, cold-finished carbon steel, AWS D1.1, Type B.

B. Anchor Rods, Bolts, Nuts, and Washers: As follows:

1. Unheaded Rods: ASTM A 36M.
2. Headed Bolts: ASTM F 568, Property Class 4.6; carbon-steel, hex-head bolts; and carbon-steel nuts.
3. Washers: ASTM A 36M.

C. Nonhigh-Strength Bolts, Nuts, and Washers: ASTM F 568, Property Class 4.6; carbon-steel, hex-head bolts; carbon-steel nuts; and flat, unhardened steel washers.

1. Finish: Hot-dip zinc-coating, ASTM A 153, Class C.

D. High-Strength Bolts, Nuts, and Washers: ASTM A 325M, Type 1, heavy hex steel structural bolts, heavy hex carbon-steel nuts, and hardened carbon-steel washers.

1. Finish: Hot-dip zinc-coating, ASTM A 153, Class C.
2. Direct-Tension Indicators: ASTM F 959, Type 325.

a. Finish: Mechanically deposited zinc-coating, ASTM B 695, Class 50, epoxy coated.

E. High-Strength Bolts, Nuts, and Washers: ASTM A 490M, Type 1, heavy hex steel structural bolts, heavy hex carbon-steel nuts, and hardened carbon-steel washers, uncoated.

1. Direct-Tension Indicators: ASTM F 959, Type 490, uncoated.

F. Welding Electrodes: Comply with AWS requirements.

2.2 PRIMER

A. Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer with good resistance to normal atmospheric corrosion, complying with performance requirements of FS TT-P-664.

B. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds and repair painting galvanized steel, with dry film containing not less than 93 percent zinc dust by weight, and complying with DOD-P-21035A or SSPC-Paint 20.

C. Primer: In organic zinc silicate primer.

2.3 GROUT

A. Cement Grout: Portland cement, EN 197, CEM I; and clean, natural sand. Mix at ratio of 1-part cement to 2-1/2 parts sand, by volume, with minimum water required for placement and hydration.

B. Metallic, Shrinkage-Resistant Grout: Premixed, factory-packaged, ferrous aggregate grout, complying with EN 934, of consistency suitable for application, and a 30-minute working time.

C. Nonmetallic, Shrinkage-Resistant Grout: Premixed, nonmetallic, noncorrosive, nonstaining grout containing selected silica sands, Portland cement, shrinkage compensating agents, plasticizing and water-reducing agents, complying with EN 934, of consistency suitable for application, and a 30-minute working time.

D. As specified in Section 03300 “Cast-in-Place Concrete”.

2.4 FABRICATION

A. Fabricate and assemble structural steel in shop to greatest extent possible. Fabricate structural steel according to EN 1090.

1. Camber structural steel members where indicated.
2. Identify high-strength structural steel and maintain markings until steel has been erected.
3. Mark and match-mark materials for field assembly.
4. Fabricate for delivery a sequence that will expedite erection and minimize field handling of structural steel.
5. Complete structural steel assemblies, including welding of units, before starting shop-priming operations.
6. Comply with fabrication tolerance limits of EN 1090 for structural steel.

B. Fabricate architecturally exposed structural steel with exposed surfaces smooth, square, and free of surface blemishes, including pitting, rust and scale seam marks, roller marks, rolled trade names, and roughness.

1. Remove blemishes by filling, grinding, or by welding and grinding, prior to cleaning, treating, and shop priming.
2. Comply with fabrication requirements, including tolerance limits, of EN 1090 for architecturally exposed structural steel.

C. Thermal Cutting: Perform thermal cutting by machine to greatest extent possible.

1. Plane thermally cut edges to be welded.

D. Finishing: Accurately mill ends of columns and other members transmitting loads in bearing.

E. Shear Connectors: Prepare steel surfaces as recommended by the manufacturer of shear connectors. Use automatic end welding of headed-stud shear connectors according to AWS D1.1 and manufacturer's printed instructions.

F. Steel Wall Framing: Select true and straight members for fabricating steel wall framing to be attached to structural steel framing. Straighten as required to provide uniform, square, and true members in completed wall framing.

G. Welded Door Frames: Build up welded door frames attached to structural steel framing. Weld exposed joints continuously and grind smooth. Plug-weld fixed steel bar stops to frames. Secure removable stops to frames with countersunk, cross-recessed head machine screws, uniformly spaced not more than 250 mm o.c., unless otherwise indicated.

H. Holes: Provide holes required for securing other work to structural steel framing and for passage of other work through steel framing members, as shown on Shop Drawings.

1. Cut, drill, or punch holes perpendicular to metal surfaces. Do not flame-cut holes or enlarge holes by burning. Drill holes in bearing plates.
2. Weld threaded nuts to framing and other specialty items as indicated to receive other work.

2.5 SHOP CONNECTIONS

A. Shop install and tighten nonhigh-strength bolts, except where high-strength bolts are indicated.

B. Shop install and tighten high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

C. Shop install and tighten high-strength bolts according to RCSC's "Load and Resistance Factor Design Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1. Bolts: ASTM A 325M high-strength bolts, unless otherwise indicated.
2. Connection Type: Slip-critical, direct-tension, or tensioned shear/bearing connections as indicated.

D. Weld Connections: Comply with BS 499 for procedures, appearance and quality of welds, and methods used in correcting welding work.

1. Assemble and weld built-up sections by methods that will maintain true alignment of axes without warp.
2. Verify that weld sizes, fabrication sequence, and equipment used for architecturally exposed structural steel will limit distortions to allowable tolerances. Prevent surface bleeding of back-side welding on exposed steel surfaces. Grind smooth exposed fillet welds 13 mm and larger. Grind flush butt welds. Dress exposed welds.

2.6 PREFABRICATED BUILDING COLUMNS

A. Definition: Prefabricated building columns consist of assemblies composed of load-bearing structural steel members encased in manufacturer's standard insulating material for fire protection and wrapped in outer nonload-bearing steel sheet enclosures.

B. Fire-Test-Response Characteristics: Provide prefabricated building column assemblies identical to those of assemblies tested for the following fire-resistance ratings per EN 13501 and EN 1993 or another testing and inspecting agency acceptable to authorities having jurisdiction. Identify columns with appropriate markings of applicable testing and inspecting agency.

1. Fire-Resistance Rating: As indicated.

C. Column Configuration: Provide columns of sizes and shapes indicated. Fabricate connections to comply with details shown or required to suit type of structure indicated.

1. Concrete Fill: Structural concrete, manufacturer's standard mix, with a minimum 28-day compressive strength of 40 MPa, machine mixed and mechanically vibrated during placement to produce a concrete core free of voids.

D. Available Manufacturers: Subject to compliance with requirements, manufacturers offering prefabricated building columns that may be incorporated in the Work include, but are not limited to, the following:

E. Manufacturers: Subject to compliance with requirements, provide prefabricated building columns by one of the following:

1. Black Rock Column, Inc.
2. Dean Lally L.P.; Firetrol Division**.**

2.7 SHOP PRIMING

A. Shop prime steel surfaces, except the following:

1. Surfaces embedded in concrete or mortar. Extend priming of partially embedded members to a depth of 50 mm.
2. Surfaces to be field welded.
3. Surfaces to be high-strength bolted with slip-critical connections.
4. Surfaces to receive sprayed-on fireproofing.
5. Galvanized surfaces.

B. Surface Preparation: Clean surfaces to be painted. Remove loose rust, loose mill scale, and spatter, slag, or flux deposits. Prepare surfaces according to SSPC specifications as follows:

1. SSPC-SP 2 "Hand Tool Cleaning."
2. SSPC-SP 3 "Power Tool Cleaning."
3. SSPC-SP 5 "White Metal Blast Cleaning."
4. SSPC-SP 6 "Commercial Blast Cleaning."
5. SSPC-SP 7 "Brush-Off Blast Cleaning."
6. SSPC-SP 8 "Pickling."
7. SSPC-SP 10 "Near-White Blast Cleaning."
8. SSPC-SP 11 "Power Tool Cleaning to Bare Metal."

C. Priming: Immediately after surface preparation, apply primer according to manufacturer's instructions and at rate recommended by SSPC to provide a dry film thickness of not less than 1.5 mils (0.038 mm). Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

1. Stripe paint corners, crevices, bolts, welds, and sharp edges.
2. Apply 2 coats of shop paint to inaccessible surfaces after assembly or erection. Change color of second coat to distinguish it from first.

D. Painting: Apply a 1-coat, nonasphaltic primer complying with SSPC's "Painting System Guide No. 7.00" to provide a dry film thickness of not less than 1.5 mils (0.038 mm).

2.8 GALVANIZING

A. Hot-Dip Galvanized Finish: Apply zinc coating by the hot-dip process to structural steel indicated for galvanizing according to ISO EN 1461.

2.9 SOURCE QUALITY CONTROL

A. The Employer will engage an independent testing and inspecting agency to perform shop inspections and tests and to prepare test reports.

1. Testing agency will conduct and interpret tests and state in each report whether test specimens comply with or deviate from requirements.
2. Provide the testing agency with access to places where structural steel Work is being fabricated or produced so that required inspection and testing can be accomplished.

B. Correct deficiencies in or remove and replace structural steel that inspections and test reports indicate do not comply with specified requirements.

C. Additional testing, at the Contractor's expense, will be performed to determine compliance of corrected Work with specified requirements.

D. Shop-bolted connections will be tested and inspected according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

E. Shop-bolted connections will be tested and inspected according to RCSC's "Load and Resistance Factor Design Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1. Direct-tension indicator gaps will be verified to comply with ASTM F 959, Table 2.

F. In addition to visual inspection, shop-welded connections will be inspected and tested according to BS 499 and the inspection procedures listed below, at testing agency's option.

1. Liquid Penetrant Inspection
2. Magnetic Particle Inspection: performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration will not be accepted.
3. Radiographic Inspection: minimum quality level "2-2T."
4. Ultrasonic Inspection

G. In addition to visual inspection, shop-welded shear connectors will be inspected and tested according to requirements of BS 499 for stud welding and as follows:

1. Bend tests will be performed when visual inspections reveal either less than a continuous 360-degree flash or welding repairs to any shear connector.
2. Tests will be conducted on additional shear connectors when weld fracture occurs on shear connectors already tested.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Before erection proceeds, and with the steel erector present, verify elevations of concrete and masonry bearing surfaces and locations of anchorages for compliance with requirements.

B. Do not proceed with erection until unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Provide temporary shores, guys, braces, and other supports during erection to keep structural steel secure, plumb, and in alignment against temporary construction loads and loads equal in intensity to design loads. Remove temporary supports when permanent structural steel, connections, and bracing are in place, unless otherwise indicated.

1. Do not remove temporary shoring supporting composite deck construction until cast-in-place concrete has attained its design compressive strength.

3.3 ERECTION

A. Set structural steel accurately in locations and to elevations indicated and according to AISC specifications referenced in this Section.

B. Base and Bearing Plates: Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen surfaces prior to setting base and bearing plates. Clean bottom surface of base and bearing plates.

1. Set base and bearing plates for structural members on wedges, shims, or setting nuts as required.
2. Tighten anchor bolts after supported members have been positioned and plumbed. Do not remove wedges or shims but, if protruding, cut off flush with edge of base or bearing plate prior to packing with grout.
3. Pack grout solidly between bearing surfaces and plates so no voids remain. Finish exposed surfaces, protect installed materials, and allow to cure.

a. Comply with manufacturer's instructions for proprietary grout materials.

C. Maintain erection tolerances of structural steel within EN 1090.

1. Maintain erection tolerances of architecturally exposed structural steel within EN 1090.

D. Align and adjust various members forming part of complete frame or structure before permanently fastening. Before assembly, clean bearing surfaces and other surfaces that will be in permanent contact. Perform necessary adjustments to compensate for discrepancies in elevations and alignment.

1. Level and plumb individual members of structure.
2. Establish required leveling and plumbing measurements on mean operating temperature of structure. Make allowances for difference between temperature at time of erection and mean temperature at which structure will be when completed and in service.

E. Splice members only where indicated.

F. Remove erection bolts on welded, architecturally exposed structural steel; fill holes with plug welds; and grind smooth at exposed surfaces.

G. Do not use thermal cutting during erection.

H. Finish sections thermally cut during erection equal to a sheared appearance.

I. Do not enlarge unfair holes in members by burning or by using drift pins. Ream holes that must be enlarged to admit bolts.

3.4 FIELD CONNECTIONS

A. Install and tighten nonhigh-strength bolts, except where high-strength bolts are indicated.

B. Install and tighten high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

C. Install and tighten high-strength bolts according to RCSC's "Load and Resistance Factor Design Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1. Bolts: ASTM A 325M high-strength bolts, unless otherwise indicated.
2. Connection Type: Slip-critical, direct-tension, or tensioned shear/bearing connections as indicated.

D. Weld Connections: Comply with BS 499 for procedures, appearance and quality of welds, and methods used in correcting welding work.

1. Comply with AISC specifications referenced in this Section for bearing, adequacy of temporary connections, alignment, and removal of paint on surfaces adjacent to field welds.
2. Assemble and weld built-up sections by methods that will maintain true alignment of axes without warp.
3. Verify that weld sizes, fabrication sequence, and equipment used for architecturally exposed structural steel will limit distortions to allowable tolerances. Prevent surface bleeding of back-side welding on exposed steel surfaces. Grind smooth exposed fillet welds 13 mm and larger. Grind flush butt welds. Dress exposed welds.

3.5 FIELD QUALITY CONTROL

A. The Employer will engage an independent testing and inspecting agency to perform field inspections and tests and to prepare test reports.

1. Testing agency will conduct and interpret tests and state in each report whether tested Work complies with or deviates from requirements.

B. Correct deficiencies in or remove and replace structural steel that inspections and test reports indicate do not comply with specified requirements.

C. Additional testing, at the Contractor's expense, will be performed to determine compliance of corrected Work with specified requirements.

D. Field-bolted connections will be tested and inspected according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1. Direct-tension indicator gaps will be verified to comply with ASTM F 959, Table 2.

E. In addition to visual inspection, field-welded connections will be inspected and tested according to BS 499 and the inspection procedures listed below, at the testing agency's option.

1. Liquid Penetrant Inspection
2. Magnetic Particle Inspection: performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration will not be accepted.
3. Radiographic Inspection: minimum quality level "2-2T."
4. Ultrasonic Inspection

F. In addition to visual inspection, field-welded shear connectors will be inspected and tested according to requirements of BS 499 for stud welding and as follows:

1. Bend tests will be performed when visual inspections reveal either less than a continuous 360-degree flash or welding repairs to any shear connector.
2. Tests will be conducted on additional shear connectors when weld fracture occurs on shear connectors already tested

3.6 CLEANING

A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint. Apply paint to exposed areas using same material as used for shop painting.

1. Apply by brush or spray to provide a minimum dry film thickness of 1.5 mils (0.038 mm).

B. Touchup Painting: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on structural steel

C. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and apply galvanizing repair paint according to EN ISO 1461.

END OF SECTION 05120

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SECTION 05210 - STEEL JOISTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Open-web K-series steel joists.
2. KCS-type, open-web K-series steel joists.
3. LH-series long-span steel joists.
4. DLH-series deep long-span steel joists.
5. Joist girders.
6. Joist accessories.

B. Related Sections include the following:

1. Concrete Specification Section "Cast-in-Place Concrete" for installing bearing plates in concrete.
2. Masonry Specification Section "Unit Masonry Assemblies" for installing bearing plates in unit masonry.
3. Steel Specification Section "Metal Fabrications" for furnishing steel bearing plates.

1.3 DEFINITIONS

A. Special Joists: Joists requiring modification by the manufacturer to support nonuniform, unequal, or special loading conditions that invalidate SJI's "Standard Specifications Load Tables and Weight Tables for Steel Joists and Joist Girders."

1.4 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide special joists and connections capable of withstanding design loads within limits and under conditions indicated.

B. Design joists to withstand design loads with total load deflections no greater than the following:

1. Floor Joists: Vertical deflection of 1/360 of the span.
2. Roof Joists: Vertical deflection of 1/360 of the span.

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1.5 SUBMITTALS

A. Product Data: For each type of joist, accessory, and product indicated.

B. Shop Drawings: Show layout, mark, number, type, location, and spacings of joists. Include joining and anchorage details, bracing, bridging, accessories; splice and connection locations and details; and attachments to other construction.

1. Indicate locations and details of anchorage devices and bearing plates to be embedded in other construction.
2. Comprehensive engineering analysis certified by the qualified professional engineer responsible for its preparation.

C. Welding Certificates: Copies of certificates for welding procedures and personnel.

D. Mill certificates signed by manufacturers of bolts certifying that their products comply with specified requirements.

E. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

F. Research/Evaluation Reports: Evidence of steel joists' compliance with building code in effect for Project, from a model code organization acceptable to authorities having jurisdiction.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Manufacturer Qualifications: A firm experienced in manufacturing joists similar to those indicated for this Project and with a record of successful in-service performance.

1. Manufacturer must be certified by SJI to manufacture joists complying with SJI standard specifications and load tables.
2. Assumes responsibility for engineering special joists to comply with performance requirements. This responsibility includes preparation of Shop Drawings and comprehensive engineering analysis by a qualified professional engineer.
3. Professional Engineer Qualifications: A professional engineer who is legally authorized to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of joists that are similar to those indicated for this Project in material, design, and extent.

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C. SJI Specifications: Comply with SJI's "Standard Specifications Load Tables and Weight Tables for Steel Joists and Joist Girders" (hereafter, "Specifications"), applicable to types of joists indicated.

D. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code--Steel"; and AWS D1.3 "Structural Welding Code--Sheet Steel."

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle joists as recommended in SJI's "Specifications."

B. Protect joists from corrosion, deformation, and other damage during delivery, storage, and handling.

1.8 SEQUENCING

A. Deliver steel bearing plates and other devices to be built into concrete and masonry construction.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Steel: Comply with SJI's "Specifications" for chord and web members.

B. Steel Bearing Plates: ASTM A 36/A 36M.

C. Carbon-Steel Bolts and Threaded Fasteners: ASTM F 568M, Property Class 4.6, carbon-steel, hex-head bolts and threaded fasteners; carbon-steel nuts; and flat, unhardened steel washers.

1. Finish: Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

D. High-Strength Bolts and Nuts: ASTM A 325M, Type 1, heavy hex steel structural bolts, heavy hex carbon-steel nuts, and hardened carbon-steel washers.

1. Finish: Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

E. Welding Electrodes: Comply with AWS standards.

F. Galvanizing Repair Paint: ASTM A 780.

2.2 PRIMERS

A. Primer: SSPC-Paint 15, Type I, red oxide; FS TT-P-636, red oxide; or manufacturer's standard shop primer complying with performance requirements of either of these red-oxide primers.

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B. Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer with good resistance to normal atmospheric corrosion, complying with performance requirements in FS TT-P-664.

2.3 OPEN-WEB K-SERIES STEEL JOISTS

A. Manufacture steel joists according to "Standard Specifications for Open Web Steel Joists, K-Series," in SJI's "Specifications," with steel-angle top- and bottom-chord members, underslung ends, and parallel top chord; of joist type indicated.

1. Joist Type: K-series steel joists and KCS-type K-series steel joists.

B. Comply with AWS requirements and procedures for shop welding, appearance, quality of welds, and methods used in correcting welding work.

C. Provide holes in chord members for connecting and securing other construction to joists.

D. Top-Chord Extensions: Extend top chords of joists with SJI's Type S top-chord extensions where indicated, complying with SJI's "Specifications."

E. Extended Ends: Extend bearing ends of joists with SJI's Type R extended ends where indicated, complying with SJI's "Specifications."

F. Do not camber joists.

G. Camber joists as indicated**.**

H. Equip bearing ends of joists with manufacturer's standard beveled ends or sloped shoes if joist slope exceeds 1:48.

2.4 LONG-SPAN STEEL JOISTS

A. Manufacture steel joists according to "Standard Specifications for Longspan Steel Joists, LH-Series and Deep Longspan Steel Joists, DLH-Series," in SJI's "Specifications," with steel-angle top- and bottom-chord members; of joist type and end and top-chord arrangements as indicated.

B. Comply with AWS requirements and procedures for shop welding, appearance, quality of welds, and methods used in correcting welding work.

C. Provide holes in chord members for connecting and securing other construction to joists.

D. Camber long-span steel joists as indicated.

E. Equip bearing ends of joists with manufacturer's standard beveled ends or sloped shoes if joist slope exceeds 1:48.

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2.5 JOIST GIRDERS

A. Manufacture joist girders according to "Standard Specifications for Joist Girders," in SJI's "Specifications," with steel-angle top- and bottom-chord members; with end and top-chord arrangements as indicated.

B. Comply with AWS requirements and procedures for shop welding, appearance, quality of welds, and methods used in correcting welding work.

C. Provide holes in chord members for connecting and securing other construction to joist girders.

D. Camber joist girders as indicated.

E. Equip bearing ends of joists with manufacturer's standard beveled ends or sloped shoes if joist slope exceeds 1:48.

2.6 JOIST ACCESSORIES

A. Bridging: Provide bridging anchors and number of rows of horizontal or diagonal bridging of material, size, and type required by SJI's "Specifications" for type of joist, chord size, spacing, and span.

B. Fabricate steel bearing plates with integral anchorages of sizes and thicknesses indicated. Hot-dip zinc coat according to ASTM A 123/A 123M.

C. Steel bearing plates with integral anchorages are specified in Steel Specification Section "Metal Fabrications."

D. Supply ceiling extensions, either extended bottom-chord elements or a separate extension unit of enough strength to support ceiling construction. Extend ends to within 13 mm of finished wall surface, unless otherwise indicated.

E. Supply miscellaneous accessories, including splice plates and bolts required by joist manufacturer to complete joist installation.

2.7 CLEANING AND SHOP PAINTING

A. Clean and remove loose scale, heavy rust, and other foreign materials from fabricated joists and accessories to be primed by hand-tool cleaning, SSPC-SP 2 or power-tool cleaning, SSPC-SP 3.

B. Do not prime paint joists and accessories to receive sprayed fire-resistive materials.

C. Apply one shop coat of primer to joists and joist accessories to be primed to provide a continuous, dry paint film not less than 0.025 mm thick.

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PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine supporting substrates, embedded bearing plates, and abutting structural framing, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Do not install joists until supporting construction is in place and secured.

B. Install joists and accessories plumb, square, and true to line; securely fasten to supporting construction according to SJI's "Specifications," joist manufacturer's written recommendations, and requirements in this Section.

1. Before installation, splice joists delivered to Project site in more than one piece.
2. Space, adjust, and align joists accurately in location before permanently fastening.
3. Install temporary bracing and erection bridging, connections, and anchors to ensure that joists are stabilized during construction.
4. Delay rigidly connecting bottom-chord extensions to columns or supports until dead loads have been applied.

C. Field weld joists to supporting steel bearing plates. Coordinate welding sequence and procedure with placement of joists. Comply with AWS requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.

D. Bolt joists to supporting steel framework using carbon-steel bolts, unless otherwise indicated.

E. Bolt joists to supporting steel framework using high-strength structural bolts, unless otherwise indicated. Comply with RCSC's "Allowable Stress Design Specification for Structural Joints Using ASTM A 325 or ASTM A 490 Bolts" for high-strength structural bolt installation and tightening requirements.

F. Install and connect bridging concurrently with joist erection, before construction loads are applied. Anchor ends of bridging lines at top and bottom chords if terminating at walls or beams.

3.3 FIELD QUALITY CONTROL

A. Testing Agency: The Employer will engage a qualified independent testing and inspecting agency to inspect field welds and high-strength bolted connections.

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B. Field welds will be visually inspected according to AWS D1.1.

C. In addition to visual inspection, field welds will be tested according to AWS D1.1 and the following procedures, as applicable:

1. Radiographic Testing: ASTM E 94 and ASTM E 142.
2. Magnetic Particle Inspection: ASTM E 709.
3. Ultrasonic Testing: ASTM E 164.
4. Liquid Penetrant Inspection: ASTM E 165.

D. Bolted connections will be visually inspected.

1. High-strength, field-bolted connections will be tested and verified according to procedures in RCSC's "Allowable Stress Design Specification for Structural Joints Using ASTM A 325 or ASTM A 490 Bolts."

E. Correct deficiencies in Work that inspections and test reports have indicated are not in compliance with specified requirements.

F. Additional testing will be performed to determine compliance of corrected Work with specified requirements.

3.4 REPAIRS AND PROTECTION

A. Repair damaged galvanized coatings on galvanized items with galvanized repair paint according to ASTM A 780 and manufacturer's written instructions.

B. Touchup Painting: After installation, promptly clean, prepare, and prime or reprime field connections, rust spots, and abraded surfaces of prime-painted joists and accessories**,** bearing plates and abutting structural steel.

1. Clean and prepare surfaces by hand-tool cleaning, SSPC-SP 2, or power-tool cleaning, SSPC-SP 3.
2. Apply a compatible primer of the same type as the shop primer used on adjacent surfaces.

C. Touchup Painting

D. Provide final protection and maintain conditions, in a manner acceptable to the manufacturer and Installer, which ensure that joists and accessories are without damage or deterioration at time of Substantial Completion.

END OF SECTION 05210

SECTION 05310 - STEEL DECK

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Roof deck.
2. Acoustical roof deck.
3. Cellular roof deck.
4. Acoustical cellular roof deck.
5. Composite floor deck.
6. Cellular deck floor systems with electrical distribution.
7. Noncomposite form deck.
8. Noncomposite vented form deck.

B. Related Sections include the following:

1. Concrete Specification Section "Cast-in-Place Concrete" for concrete fill and reinforcing steel.
2. Concrete Specification Section "Lightweight Concrete Roof Insulation" for lightweight insulating concrete fill.
3. Steel Specification Section "Structural Steel" for shop-welded shear connectors.
4. Steel Specification Section "Metal Fabrications" for framing deck openings with miscellaneous steel shapes.

1.3 SUBMITTALS

A. Product Data: For each type of deck, accessory, and product indicated.

B. Shop Drawings: Show layout and types of deck panels, anchorage details, reinforcing channels, pans, deck openings, special jointing, accessories, and attachments to other construction.

1. For steel deck indicated to comply with certain design loadings, include structural analysis data sealed and signed by the qualified professional engineer who was responsible for its preparation.

C. Product Certificates: Signed by steel deck manufacturers certifying that products furnished comply with requirements.

D. Welding Certificates signed by Contractor certifying that welders comply with requirements specified under the "Quality Assurance" Article.

E. Product Test Reports: From a qualified testing agency indicating that each of the following complies with requirements, based on comprehensive testing of current products:

1. Mechanical fasteners.
2. Acoustical roof deck.

F. Research/Evaluation Reports: Evidence of steel deck's compliance with building code in effect for Project, from a model code organization acceptable to authorities having jurisdiction.

G. As-Built Drawings: At project close out, submit Record (As- Built) Drawings of completed work products, in accordance with requirements of the Specification as indicated in General Requirements.

1.4 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Engineer Qualifications: A professional engineer who is legally authorized to practice in the jurisdiction where project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for projects with structural steel framing that are similar to that indicated for this Project in material, design, and extent.

C. Testing Agency Qualifications: To qualify for acceptance, an independent testing agency shall demonstrate to the Engineer's satisfaction, based on evaluation of agency-submitted criteria conforming to ASTM E 699, that it has the experience and capability to satisfactorily conduct the testing indicated without delaying the Work.

D. Source Limitations for Cellular Deck Floor Systems with Electrical Distribution: Obtain cellular floor deck units and compatible electrical components, such as preset inserts, activation kits, afterset inserts, service fittings, header ducts, and trench header ducts, from the same manufacturer.

E. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code--Steel," and AWS D1.3, "Structural Welding Code--Sheet Steel."

1. Certify that each welder has satisfactorily passed AWS qualification tests for welding processes involved and, if pertinent, has undergone recertification.

F. Fire-Test-Response Characteristics: Where indicated, provide steel deck units identical to those steel deck units tested for fire resistance per ASTM E 119 by a testing and inspection agency acceptable to authorities having jurisdiction.

1. Fire-Resistance Ratings: Indicated by design designations from UL's "Fire Resistance Directory" or by Warnock Hersey or another testing and inspecting agency.
2. Steel deck units shall be identified with appropriate markings of applicable testing and inspecting agency.

G. AISI Specifications: Calculate structural characteristics of steel deck according to AISI's "Specification for the Design of Cold-Formed Steel Structural Members."

H. Electrical-Raceway Units: Provide UL-labeled cellular floor deck units complying with UL 209 and listed in UL's "Electrical Construction Equipment Directory" for use with standard header ducts and outlets for electrical distribution systems.

FM Listing: Provide steel roof deck evaluated by Factory Mutual and listed in FM's "Approval Guide, Building Materials" for Class 1 fire rating and Class 1-60 windstorm ratings.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Protect steel deck from corrosion, deformation, and other damage during delivery, storage, and handling.

B. Stack steel deck on platforms or pallets and slope to provide drainage. Protect with a waterproof covering and ventilate to avoid condensation.

1.6 COORDINATION

A. Coordinate installation of sound-absorbing insulation strips in topside ribs of acoustical deck with roofing installation specified as indicated on the drawings to ensure protection of insulation strips against damage from effects of weather and other causes.

B. Coordinate layout and installation of trench headers, preset inserts, duct fittings, and other components with installation of cellular metal floor deck.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Approved Manufacturers: Manufacturers offering products that may be incorporated into the Work should submit evidences of compliance with this Specification’s requirements.

2.2 ROOF DECK

A. Steel Roof Deck: Fabricate panels, without top-flange stiffening grooves, to comply with "SDI Specifications and Commentary for Steel Roof Deck," in SDI Publication No. 28, and the following:

1. Prime-Painted Steel Sheet: ASTM A 611, Grade C minimum, shop primed with gray or white baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
2. Galvanized Steel Sheet: ASTM A 653, Grade A, Z 275 zinc coated according to ASTM A 653M.
3. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
4. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, Structural Steel (SS), Grade 230 minimum, AZ150 aluminum-zinc alloy coating.
5. Deck Profile: Type NR, narrow rib.
6. Cellular Deck Profile: Type WR, wide rib or Type 3DR, deep rib, with bottom plate.
7. Profile Depth: 38 mm Unless indicated otherwise on drawings.
8. Design Uncoated-Steel Thickness: As indicated on drawings.
9. Span Condition: Triple span or more.
10. Side Joints: Overlapped or interlocking seam as indicated on Drawings. If not indicated on drawings, type of side joints shall be at the Contractor's option.

2.3 ACOUSTICAL ROOF DECK

A. Acoustical Steel Roof Deck: Fabricate panels, without top-flange stiffening grooves, to comply with "SDI Specifications and Commentary for Steel Roof Deck," in SDI Publication No. 29, and the following:

1. Prime-Painted Steel Sheet: ASTM A 611, Grade C minimum, shop primed with gray or white baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
2. Galvanized Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 or Z275 zinc coating.
3. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
4. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, Structural Steel (SS), Grade 230 minimum, AZ150 aluminum-zinc alloy coating.
5. Deck Profile: Type WR, wide rib or Type 3DR, deep rib.
6. Cellular Deck Profile: Type WR, wide rib or Type 3DR, deep rib, with bottom plate.
7. Profile Depth: As indicated on the drawings.
8. Design Uncoated-Steel Thickness: As indicated on the drawings.
9. Span Condition: As indicated on the drawings.

10. Side Laps: Overlapped or Interlocking seam as indicated. If not indicated, Overlapped or interlocking seam at the Contractor's option.
11. Acoustical Perforations: Cellular deck units with manufacturer's standard perforated flat-bottom plate welded to ribbed deck.
12. Sound-Absorbing Insulation: Manufacturer's standard premolded roll or strip glass fiber or mineral fiber.

    a. Factory install sound-absorbing insulation into cells of cellular deck according to manufacturer's written instructions.

13. Acoustical Performance: NRC 0.80, tested according to ASTM C 423.

2.4 COMPOSITE FLOOR DECK

A. Composite Steel Floor Deck: Fabricate panels, with integrally embossed or raised pattern ribs and interlocking side laps, to comply with "SDI Specifications and Commentary for Composite Steel Floor Deck," in SDI Publication No. 28, the minimum section properties indicated, and the following:

1. Prime-Painted Steel Sheet: ASTM A 611, Grade C minimum, with top surface phosphatized and unpainted and bottom surface shop primed with gray or white baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
2. Galvanized Steel Sheet: ASTM A 653, Grade A, Z 275 zinc coated according to ASTM A 653M.
3. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; with unpainted top and bottom surface cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
4. Profile Depth: 76 mm unless otherwise indicated on drawings.
5. Design Uncoated-Steel Thickness: As indicated on drawings.
6. Span Condition: As indicated on the drawings.

2.5 CELLULAR DECK FLOOR SYSTEMS WITH ELECTRICAL DISTRIBUTION

A. Cellular Deck Floor Systems with Electrical Distribution: Fabricate steel sheet cellular floor deck panels, consisting of a ribbed top section welded to a lower flat-bottom sheet with interlocking side laps, to comply with "SDI Specifications and Commentary for Cellular Deck Floor Systems with Electrical Distribution," in SDI Publication No. 29. Fabricate deck to the minimum section properties, width of panel, number and area of cells per panel indicated, and the following:

1. Cellular Deck Type: As indicated.
2. Galvanized Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 or Z275 zinc coating.
3. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; with bottom surface cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.

4. Profile Depth: As indicated.
5. Design Uncoated-Steel Thicknesses; Deck Unit/Bottom Plate: As indicated on the drawings.
6. Span Condition: As indicated.
7. Factory-punch holes, of size and arrangement indicated, into each deck cell at preset inserts and header duct locations.

2.6 NONCOMPOSITE FORM DECK

A. Noncomposite Steel Form Deck: Fabricate ribbed-steel sheet noncomposite form deck panels to comply with "SDI Specifications and Commentary for Noncomposite Steel Form Deck," in SDI Publication No. 29, the minimum section properties indicated, and the following:

1. Uncoated Steel Sheet: ASTM A 611, Grade C minimum.
2. Prime-Painted Steel Sheet: ASTM A 611, Grade C minimum, top and bottom surface shop primed with gray or white baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
3. Galvanized Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230 Z180 or Z275 zinc coating.
4. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
5. Profile Depth: As indicated.
6. Design Uncoated-Steel Thickness: As indicated on the drawings.
7. Span Condition: As indicated.
8. Side Laps: Overlapped or Interlocking seam as indicated. If not indicated, Overlapped or interlocking seam at the Contractor's option.

2.7 NONCOMPOSITE VENTED FORM DECK

A. Noncomposite Vented Steel Form Deck: Fabricate ribbed- and vented-steel sheet noncomposite form deck panels to comply with "SDI Specifications and Commentary for Noncomposite Steel Form Deck," in SDI Publication No. 29, and the following:

1. Galvanized Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230 zinc coating.
2. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 230, Z180 zinc coating; cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.
3. Profile Depth: As indicated.
4. Design Uncoated-Steel Thickness: As indicated on the darwings.
5. Span Condition: As indicated.
6. Side Laps: Overlapped or Interlocking seam as indicated. If not indicated, Overlapped or interlocking seam at the Contractor's option.
7. Vent Slot Area: Manufacturer's standard vent slots providing 1-1/2 percent open area.

2.8 ACCESSORIES

A. General: Provide manufacturer's standard accessory materials for steel deck that comply with requirements indicated, and recommendations of the steel deck manufacturer.

B. Mechanical Fasteners: Manufacturer Standard, Corrosion-resistant, low-velocity, power-actuated or pneumatically driven carbon-steel fasteners; or self-drilling, self-threading screws.

C. Side-Lap Fasteners: Manufacturer Standard, Corrosion-resistant, hexagonal washer head; self-drilling, carbon-steel screws, 4.8 mm minimum diameter.

D. Flexible Closure Strips: Vulcanized, closed-cell, synthetic rubber.

E. Miscellaneous Sheet Metal Deck Accessories: Steel sheet, minimum yield strength of 230 MPa, not less than 0.91 mm design uncoated thickness, of same material and finish as deck; of profile indicated or required for application.

F. Steel Sheet Accessories: ASTM A 653, 2180 coating class, galvanized according to ASTM A 653M.

G. Pour Stops and Girder Fillers: Steel sheet, minimum yield strength of 230 MPa, of same material and finish as deck, and of thickness and profile.

H. Column Closures, End Closures, Z-Closures, and Cover Plates: Steel sheet, of same material, finish, and thickness as deck, unless otherwise indicated.

I. Piercing Hanger Tabs: Piercing steel sheet hanger attachment devices for use with floor deck.

J. Weld Washers: Uncoated steel sheet, shaped to fit deck rib, 1.90 mm thick, with factory-punched hole of 9.5 mm minimum diameter.

K. Recessed Sump Pans: Single-piece steel sheet, 1.80 mm thick, of same material and finish as deck, with 76 mm wide flanges and level recessed pans of 38 mm minimum depth. For drains, cut holes in the field.

L. Flat Sump Plate: Single-piece steel sheet, 1.80 mm thick, of same material and finish as deck. For drains, cut holes in the field.

M. Shear Connectors: ASTM A 108, Grades 1010 through 1020 headed stud type, cold-finished carbon steel, AWS D1.1, Type B, with arc shields.

N. Galvanizing Repair Paint: SSPC-Paint 20 or DOD-P-21035, with dry film containing a minimum of 94 percent zinc dust by weight.

O. Repair Paint: Lead- and chromate-free rust-inhibitive primer complying with performance requirements of FS TT-P-664.

P. Preset Inserts: Manufacturer's standard, UL-labeled single-piece preset inserts, fabricated from either steel sheet galvanized according to ASTM A 653M, Z 180 coating class, or zinc sheet, with removable covers.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine supporting frame and field conditions for compliance with requirements for installation tolerances and other conditions affecting performance, of steel deck.

3.2 Preparation:

A. Do not place deck panels on concrete supporting structure until concrete has cured and is dry.

B. Locate decking bundles to prevent overloading of supporting members.

3.3 INSTALLATION, GENERAL

A. Install deck panels and accessories according to applicable specifications and commentary in SDI Publication No. 28, manufacturer's written instructions, and requirements in this Section.

B. Install temporary shoring before placing deck panels, if required to meet deflection limitations.

C. Locate decking bundles to prevent overloading of supporting members.

D. Place deck panels on supporting frame and adjust to final position with ends accurately aligned and bearing on supporting frame before being permanently fastened. Do not stretch or contract side-lap interlocks.

1. Align cellular deck panels for entire length of cell runs and align cells at ends of abutting panels.

E. Place deck panels flat and square and fasten to supporting frame without warp or deflection.

F. Cut and neatly fit deck panels and accessories around openings and other work projecting through or adjacent to decking.

G. Provide additional reinforcement and closure pieces at openings as required for strength, continuity of decking, and support of other work.

H. Comply with AWS requirements and procedures for manual shielded metal arc welding, appearance and quality of welds, and methods used for correcting welding work.

I. Mechanical fasteners may be used in lieu of welding to fasten deck. Locate mechanical fasteners and install according to deck manufacturer's written instructions.

3.4 ROOF DECK INSTALLATION

A. Fasten roof deck panels to steel supporting members by arc spot (puddle) welds of the surface diameter indicated or arc seam welds with an equal perimeter, but not less than 38 mm long, and as follows:

1. Weld Diameter: 19 mm, nominal.
2. Weld Spacing: Weld edge and interior ribs of deck units with a minimum of two welds per deck unit at each support. Space welds as indicated.
3. Weld Washers: Install weld washers at each weld location.

B. Side-Lap and Perimeter Edge Fastening: Fasten side laps and perimeter edges of panels between supports, at intervals not exceeding the lesser of 1/2 of the span or 450 mm, and as follows:

1. Mechanically fasten with self-drilling 4.8 mm diameter or larger carbon-steel screws.
2. Mechanically clinch or button punch.
3. Fasten with a minimum of 38 mm long welds.

C. End Bearing: Install deck ends over supporting frame with a minimum end bearing of 38 mm, with end joints as follows:

1. End Joints: Butted.

D. Roof Sump Pans and Sump Plates: Install over openings provided in roof decking and weld flanges to top of deck. Space welds not more than 305 mm apart with at least 1 weld at each corner.

E. Miscellaneous Roof Deck Accessories: Install ridge and valley plates, finish strips, cover plates, end closures, and reinforcing channels according to deck manufacturer's written instructions. Weld to substrate to provide a complete deck installation.

F. Flexible Closure Strips: Install flexible closure strips over partitions, walls, and where indicated. Install with adhesive according to manufacturer's written instructions to ensure complete closure.

G. Sound-Absorbing Insulation: Installation into topside ribs of deck as indicated on the drawings.

3.5 FLOOR DECK INSTALLATION

A. Fasten floor deck panels to steel supporting members by arc spot (puddle) welds of the surface diameter indicated and as follows:

1. Weld Diameter: 19 mm, nominal.
2. Weld Spacing: Space and locate welds as indicated.
3. Weld Washers: Install weld washers at each weld location.

B. Side-Lap and Perimeter Edge Fastening: Fasten side laps and perimeter edges of panels between supports, at intervals not exceeding the lesser of 1/2 of the span or 910 mm, and as follows:

1. Mechanically fasten with self-drilling 4.8 mm diameter or larger carbon-steel screws.
2. Mechanically clinch or button punch.
3. Fasten with a minimum of 38 mm long welds.

C. End Bearing: Install deck ends over supporting frame with a minimum end bearing of 38 mm, with end joints as follows:

1. End Joints: Butted.

D. Shear Connectors: Weld shear connectors through deck to supporting frame according to AWS D1.1 and manufacturer's written instructions. Butt end joints of deck panels; do not overlap. Remove and discard arc shields after welding shear connectors.

E. Pour Stops and Girder Fillers: Weld steel sheet pour stops and girder fillers to supporting structure according to SDI recommendations, unless otherwise indicated.

F. Floor Deck Closures: Weld steel sheet column closures, cell closures, and Z-closures to deck, according to SDI recommendations, to provide tight-fitting closures at open ends of ribs and sides of decking. Weld cover plates at changes in direction of floor deck panels, unless otherwise indicated.

G. Cellular-Raceway Deck Floor Systems with Electrical Distribution: Install cellular floor system with deck assembled from units indicated.

1. Maintain smooth cellular-raceway interiors free of welds or mechanical fasteners.

H. Install piercing hanger tabs not more than 355 mm apart in both directions, within 228 mm of walls at ends, and not more than 305 mm from walls at sides, unless otherwise indicated.

3.6 FIELD QUALITY CONTROL

A. Testing: The Employer will engage a qualified independent testing agency to perform field quality-control testing.

B. Field welds will be subject to inspection.

C. Shear connector stud welds will be inspected and tested according to AWS D1.1 for stud welding and as follows:

1. Shear connector stud welds will be visually inspected.
2. Bend tests will be performed if visual inspections reveal less than a full 360-degree flash or welding repairs to any shear connector stud.
3. Tests will be conducted on additional shear connector studs if weld fracture occurs on shear connector studs already tested according to AWS D1.1.

D. Testing agency will report test results promptly and in writing to the Contractor and Engineer.

E. Remove and replace work that does not comply with specified requirements.

F. Additional testing and inspecting, at the Contractor's expense, will be performed to determine compliance of corrected work with specified requirements.

3.7 REPAIRS AND PROTECTION

A. Galvanizing Repairs: Prepare and repair damaged galvanized coatings on both surfaces of deck with galvanized repair paint according to ASTM A 780 and manufacturer's written instructions.

B. Repair Painting: Wire brush and clean rust spots, welds, and abraded areas on both surfaces of prime-painted deck immediately after installation, and apply repair paint.

1. Apply repair paint, of same color as adjacent shop-primed deck, to bottom surfaces of deck exposed to view.

C. Provide final protection and maintain conditions to ensure that steel deck is without damage or deterioration at time of Substantial Completion.

END OF SECTION 05310

35.

SECTION 05400 - COLD-FORMED METAL FRAMING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Exterior load-bearing wall framing.
2. Interior load-bearing wall framing.
3. Exterior non-load-bearing curtain-wall framing.
4. Floor joist framing.
5. Roof trusses.
6. Roof rafter framing.
7. Ceiling joist framing.
8. Gypsum sheathing and air-infiltration barriers.

B. Related Sections include the following:

1. Steel Specification Section "Metal Fabrications" for masonry shelf angles and connections.
2. Woods & Plastics Specification Section "Rough Carpentry" for subflooring, wall sheathing, or roof sheathing using wood-based structural-use panels, particleboard, fibrous-felted board, and foam-plastic sheathing.

1.3 DEFINITIONS

A. Minimum Uncoated Steel Thickness: Minimum uncoated thickness of cold-formed framing delivered to the Project site shall be not less than 95 percent of the thickness used in the cold-formed framing design. Lesser thicknesses shall be permitted at bends due to cold forming.

B. Producer: Entity that produces steel sheet coil fabricated into cold-formed members.

1.4 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide cold-formed metal framing capable of withstanding design loads within limits and under conditions indicated.

1. Deflection Limits: Design framing systems to withstand design loads without deflections greater than the following:

35.

a. Exterior Load-Bearing Wall Framing: Horizontal deflection of 1/360 of the wall height.
b. Interior Load-Bearing Wall Framing: Horizontal deflection of 1/360 of the wall height.
c. Exterior Non-Load-Bearing Curtain-Wall Framing: Horizontal deflection of 1/360 of the wall height.
d. Floor Joist Framing: Vertical deflection of 1/360 of the span.
e. Roof Trusses: Vertical deflection of 1/360 of the span.
f. Scissor Roof Trusses: Horizontal deflection of 32 mm at reactions.
g. Roof Rafter Framing: Horizontal deflection of 1/360 of the horizontally projected span.
h. Ceiling Joist Framing: Vertical deflection of 1/360 of the span.

2. Design framing systems to provide for movement of framing members without damage or overstressing, sheathing failure, connection failure, undue strain on fasteners and anchors, or other detrimental effects when subject to a maximum ambient temperature change of 67 deg. C. (153 deg. F).
3. Design framing system to maintain clearances at openings, to allow for construction tolerances, and to accommodate live load deflection of primary building structure as follows:

a. Upward and downward movement of 25 mm.

B. Design exterior non-load-bearing curtain-wall framing to accommodate horizontal deflection without regard for contribution of sheathing materials.

C. Design roof trusses according to AISI's "Design Guide for Cold-Formed Steel Trusses."

1.5 SUBMITTALS

A. Product Data: For each type of cold-formed metal framing product and accessory indicated.

B. Shop Drawings: Show layout, spacings, sizes, thicknesses, and types of cold-formed metal framing; fabrication; and fastening and anchorage details, including mechanical fasteners. Show reinforcing channels, opening framing, supplemental framing, strapping, bracing, bridging, splices, accessories, connection details, and attachment to adjoining Work.

1. For cold-formed metal framing indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

C. Mill certificates signed by steel sheet producer or test reports from a qualified independent testing agency indicating steel sheet complies with requirements.

D. Welding Certificates: Copies of certificates for welding procedures and personnel.

E. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project

names and addresses, names and addresses of architects/engineers and owners, and other information specified.

F. Product Test Reports: From a qualified testing agency indicating that each of the following complies with requirements, based on comprehensive testing of current products:

1. Expansion anchors.
2. Mechanical fasteners.
3. Vertical deflection clips.
4. Miscellaneous structural clips and accessories.

G. Research/Evaluation Reports: Evidence of cold-formed metal framing's compliance with building code in effect for Project, from a model code organization acceptable to authorities having jurisdiction.

1.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Installer Qualifications: An experienced installer who has completed cold-formed metal framing similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

C. Engineering Responsibility: Engage a qualified professional engineer to prepare design calculations, Shop Drawings, and other structural data.

D. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of cold-formed metal framing that are similar to those indicated for this Project in material, design, and extent.

E. Mill certificates signed by steel sheet producer or test reports from a qualified independent testing agency indicating steel sheet complies with requirements, including uncoated steel thickness, yield strength, tensile strength, total elongation, chemical requirements, ductility, and galvanized-coating thickness.

F. Testing Agency Qualifications: An independent testing agency, acceptable to authorities having jurisdiction, qualified according to ASTM E 329 to conduct the testing indicated, as documented according to ASTM E 548.

G. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code--Steel," and AWS D1.3, "Structural Welding Code--Sheet Steel."

H. Fire-Test-Response Characteristics: Where metal framing is part of a fire-resistance-rated assembly, provide framing identical to that of assemblies tested for fire resistance per ASTM E 119 by a testing and inspecting agency acceptable to authorities having jurisdiction.

35.

1. Fire-Resistance Ratings: Indicated by GA File Numbers in GA-600, "Fire Resistance Design Manual," or by design designations from UL's "Fire Resistance Directory" or from the listings of another testing and inspecting agency.

I. AISI Specifications: Comply with AISI's "Specification for the Design of Cold-Formed Steel Structural Members" for calculating structural characteristics of cold-formed metal framing.

J. Comply with HUD's "Prescriptive Method for Residential Cold-Formed Steel Framing."

K. Preinstallation Conference: Conduct conference at Project site to comply with requirements in General Requirements Section "Project Management and Coordination."

1.7 DELIVERY, STORAGE, AND HANDLING

A. Protect cold-formed metal framing from corrosion, deformation, and other damage during delivery, storage, and handling.

B. Store cold-formed metal framing, protect with a waterproof covering, and ventilate to avoid condensation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Approved Manufacturers: Manufacturers offering products that may be incorporated into the Work should submit evidences of compliance with the Specification’s requirements.

2.2 MATERIALS

A. Steel Sheet: ASTM A 653/A 653M, structural steel, zinc coated, of grade and coating as follows:

1. Grade: As required by structural performance.
2. Coating: Z275.

B. Steel Sheet: ASTM A 1011/A 1011M, hot rolled or ASTM A 611, cold rolled; cleaned, pretreated, and primed with manufacturer's baked-on, lead- and chromate-free, rust-inhibitive primer complying with performance requirements in FS TT-P-664, of grade as follows:

1. Grade: As required by structural performance.

C. Steel Sheet: ASTM A 792/A 792M, structural steel, 55 percent aluminum-zinc-alloy coated, of grade and coating as follows:

1. Grade: As required by structural performance.
2. Coating: AZ180.

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2.3 LOAD-BEARING WALL FRAMING

A. Steel Studs: Manufacturer's standard C-shaped steel studs, of web depths indicated, punched, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by design.

B. Steel Track: Manufacturer's standard U-shaped steel track, of web depths indicated, unpunched, with straight flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: Matching steel studs.
2. Flange Width: As indicated.

2.4 NON-LOAD-BEARING CURTAIN-WALL FRAMING

A. Steel Studs: Manufacturer's standard C-shaped steel studs, of web depths indicated, punched, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by the design

B. Steel Track: Manufacturer's standard U-shaped steel track, of web depths indicated, unpunched, with unstiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: Matching steel studs.
2. Flange Width: As indicated.

C. Single Deflection Track: Manufacturer's single, deep-leg, U-shaped steel track; unpunched, with unstiffened flanges, of web depth to contain studs while allowing free vertical movement, with flanges designed to support horizontal and lateral loads, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.

D. Double Deflection Tracks: Manufacturer's double, deep-leg, U-shaped steel tracks, consisting of nested inner and outer tracks; unpunched, with unstiffened flanges.

1. Outer Track: Of web depth to allow free vertical movement of inner track, with flanges designed to support horizontal and lateral loads, and as follows:

   a. Minimum Uncoated-Steel Thickness: As indicated.
   b. Flange Width: As indicated

2. Inner Track: Of web depth indicated, and as follows:

   a. Minimum Uncoated-Steel Thickness: As indicated.

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b. Flange Width: As indicated

E. Vertical Deflection Clips: Manufacturer's standard bypass or head clips, capable of accommodating upward and downward vertical displacement of primary structure.

2.5 FLOOR JOIST FRAMING

A. Steel Joists: Manufacturer's standard C-shaped steel joists, of web depths indicated, **punched**, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by the design.

B. Steel Joist Track: Manufacturer's standard U-shaped steel joist track, of web depths indicated, unpunched, with unstiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: Matching steel joists.
2. Flange Width: As indicated.

2.6 ROOF TRUSSES

A. Roof Truss Members: Manufacturer's standard C-shaped steel sections, of web depths indicated, unpunched, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by the design.

2.7 ROOF-RAFTER FRAMING

A. Steel Rafters: Manufacturer's standard C-shaped steel sections, of web depths indicated, unpunched, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by the design.

B. Built-up Members: Built-up members of manufacturer's standard C-shaped steel section, with stiffened flanges, nested into a U-shaped steel section joist track, with unstiffened flanges; unpunched; of web depths indicated; complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: Matching steel rafters.
2. Flange Width: As indicated.

2.8 CEILING JOIST FRAMING

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A. Steel Ceiling Joists: Manufacturer's standard C-shaped steel sections, of web depths indicated, unpunched, with stiffened flanges, complying with ASTM C 955, and as follows:

1. Minimum Uncoated-Steel Thickness: As indicated.
2. Flange Width: As indicated.
3. Section Properties: As required by the design.

2.9 FRAMING ACCESSORIES

A. Fabricate steel-framing accessories of the same material and finish used for framing members, with a minimum yield strength of 230 MPa.

B. Provide accessories of manufacturer's standard thickness and configuration, unless otherwise indicated, as follows:

1. Supplementary framing.
2. Bracing, bridging, and solid blocking.
3. Web stiffeners.
4. End clips.
5. Foundation clips.
6. Gusset plates.
7. Stud kickers, knee braces, and girts.
8. Joist hangers and end closures.
9. Hole reinforcing plates.
10. Backer plates.

2.10 ANCHORS, CLIPS, AND FASTENERS

A. Steel Shapes and Clips: ASTM A 36/A 36M, zinc coated by hot-dip process according to ASTM A 123.

B. Anchor Bolts: ASTM F 1554, Grade 36, threaded carbon-steel hex-headed bolts and carbon-steel nuts; and flat, hardened-steel washers; zinc coated by hot-dip process according to ASTM A 153/A 153M, Class C or mechanically deposited according to ASTM B 695, Class 50.

C. Expansion Anchors: Fabricated from corrosion-resistant materials, with capability to sustain, without failure, a load equal to 5 times design load, as determined by testing per ASTM E 488 conducted by a qualified independent testing agency.

D. Mechanical Fasteners: Corrosion-resistant-coated, self-drilling, self-threading steel drill screws.

1. Head Type: Low-profile head beneath sheathing, manufacturer's standard elsewhere.

E. Welding Electrodes: Comply with AWS standards.

2.11 MISCELLANEOUS MATERIALS

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A. Galvanizing Repair Paint: SSPC-Paint 20, DOD-P-21035, or ASTM A 780.

B. Cement Grout: Portland cement, ASTM C 150, Type I; and clean, natural sand, ASTM C 404. Mix at ratio of 1 part cement to 2-1/2 parts sand, by volume, with minimum water required for placement and hydration.

C. Nonmetallic, Nonshrink Grout: Premixed, nonmetallic, noncorrosive, nonstaining grout containing selected silica sands, Portland cement, shrinkage-compensating agents, and plasticizing and water-reducing agents, complying with ASTM C 1107, with fluid consistency and 30-minute working time.

D. Thermal Insulation: ASTM C 665, Type I, unfaced mineral-fiber blankets produced by combining glass or slag fibers with thermosetting resins.

2.12 GYPSUM SHEATHING

A. Sheathing: Comply with requirements in Woods & Plastics Specification Section "Rough Carpentry.",

B. Paper-Surfaced Gypsum Sheathing Board with Water-Resistant Core: ASTM C 79, with water-resistant material incorporated into core and with water-repellent paper bonded to core's face, back, and long edges.

1. Type and Thickness: As indicated
2. Edge and End Configuration: Square.
3. Size: Manufacturer's standard.

C. Glass-Mat Gypsum Sheathing Board: ASTM C 1177/C 1177M.

1. Type and Thickness: As indicated.
2. Size: Manufacturer’s standard.

2.13 SHEATHING ACCESSORIES

A. Air-Infiltration Barrier: Proprietary building wrap with flame-spread and smoke-developed ratings of less than 25 and 450, respectively, when tested according to ASTM E 84. Provide one of the following products:

1. Polyethylene sheet; 0.097 to 0.163 mm thick; formed by spinning continuous strands of fine, high-density-polyethylene interconnected fibers and bonding them together by heat and pressure; incorporating an additive to provide ultraviolet light resistance for up to 120 days; and with a water-vapor transmission rate equaling 535 g through 1 sq. m of surface in 24 hours according to ASTM E 96, Desiccant Method (Method A).

2. Polypropylene sheet; 0.29 mm thick; consisting of spun-bonded polypropylene substrate with a polypropylene coating attached directly to one side; and with a water-vapor

transmission rate equaling 125 g through 1 sq. m of surface in 24 hours according to ASTM E 96, Desiccant Method (Method A).

3. Laminated polyethylene sheet; 0.08 mm thick; consisting of 2 plies of microperforated, cross-laminated, high-density-polyethylene sheet; and with a water-vapor transmission rate equaling 125 g through 1 sq. m of surface in 24 hours according to ASTM E 96, Desiccant Method (Method A).

4. Woven polyolefin sheet; 0.13 mm thick; with a water-vapor transmission rate equaling 63 g through 1 sq. m of surface in 24 hours according to ASTM E 96, Desiccant Method (Method A).

B. Fasteners: Steel drill screws, ASTM C 954, in length recommended by sheathing manufacturer for thickness of sheathing board to be attached, with organic-polymer or other corrosion-protective coating having a salt-spray resistance of more than 800 hours according to ASTM B 117.

C. Elastomeric Sealant: Medium-modulus, neutral-curing silicone sealant compatible with joint substrates formed by gypsum sheathing and other materials, recommended by sheathing manufacturers for application indicated,

D. Silicone Emulsion Sealant for Glass-Mat Gypsum Sheathing: Product complying with ASTM C 834, compatible with sheathing tape and gypsum sheathing, recommended by sheathing and tape manufacturers for use with glass-fiber sheathing tape and for covering exposed fasteners.

E. Glass-Fiber Sheathing Tape for Glass-Mat Gypsum Sheathing: Self-adhering glass-fiber tape, minimum 50 mm wide, 390 by 390 or 390 by 780 threads per meter, of type recommended by sheathing and tape manufacturers for use with silicone emulsion sealant in sealing joints in glass-mat gypsum sheathing board and with a history of successful in-service use.

2.14 FABRICATION

A. Fabricate cold-formed metal framing and accessories plumb, square, and true to line, and with connections securely fastened, according to manufacturer's written recommendations and requirements in this Section.

1. Fabricate framing assemblies using jigs or templates.
2. Cut framing members by sawing or shearing; do not torch cut.
3. Fasten cold-formed metal framing members by welding. Wire tying of framing members is not permitted. Comply with AWS D1.3 requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.
4. Fasten cold-formed metal framing members by welding or screw fastening, as standard with fabricator. Wire tying of framing members is not permitted.

    a. Comply with AWS D1.3 requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.
    b. Locate mechanical fasteners and install according to Shop Drawings, with screw penetrating joined members by not less than three exposed screw threads.

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5. Fasten other materials to cold-formed metal framing by welding, bolting, or screw fastening, according to Shop Drawings.

B. Reinforce, stiffen, and brace framing assemblies to withstand handling, delivery, and erection stresses. Lift fabricated assemblies to prevent damage or permanent distortion.

C. Fabrication Tolerances: Fabricate assemblies level, plumb, and true to line to a maximum allowable tolerance variation of 1:960 and as follows:

1. Spacing: Space individual framing members no more than plus or minus 3 mm from plan location. Cumulative error shall not exceed minimum fastening requirements of sheathing or other finishing materials.
2. Squareness: Fabricate each cold-formed metal framing assembly to a maximum out-of-square tolerance of 3 mm.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine supporting substrates and abutting structural framing for compliance with requirements for installation tolerances and other conditions affecting performance. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Before sprayed fire-resistive materials are applied, attach continuous angles, supplementary framing, or tracks to structural members indicated to receive sprayed fire-resistive materials.

B. After applying sprayed fire-resistive materials, remove only as much of these materials as needed to complete installation of cold-formed framing without reducing thickness of fire-resistive materials below that are required to obtain fire-resistance rating indicated. Protect remaining fire-resistive materials from damage.

C. Grout bearing surfaces uniform and level to ensure full contact of bearing flanges or track webs on supporting concrete or masonry construction.

3.3 INSTALLATION, GENERAL

A. Cold-formed metal framing may be shop or field fabricated for installation, or it may be field assembled.

B. Install cold-formed metal framing according to ASTM C 1007, unless more stringent requirements are indicated.

C. Install shop- or field-fabricated, cold-formed framing and securely anchor to supporting structure.

1. Bolt or weld wall panels at horizontal and vertical junctures to produce flush, even, true-to-line joints with maximum variation in plane and true position between fabricated panels not exceeding 1.6 mm.

D. Install cold-formed metal framing and accessories plumb, square, and true to line, and with connections securely fastened, according to manufacturer's written recommendations and requirements in this Section.

1. Cut framing members by sawing or shearing; do not torch cut.
2. Fasten cold-formed metal framing members by welding or screw fastening, as standard with fabricator. Wire tying of framing members is not permitted.

a. Comply with AWS D1.3 requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.
b. Locate mechanical fasteners and install according to Shop Drawings, with screw penetrating joined members by not less than three exposed screw threads.

E. Install framing members in one-piece lengths, unless splice connections are indicated for track or tension members.

F. Install temporary bracing and supports to secure framing and support loads comparable in intensity to those for which structure was designed. Maintain braces and supports in place, undisturbed, until entire integrated supporting structure has been completed and permanent connections to framing are secured.

G. Do not bridge building expansion and control joints with cold-formed metal framing. Independently frame both sides of joints.

H. Install insulation in built-up exterior framing members, such as headers, sills, boxed joists, and multiple studs at openings, which are inaccessible on completion of framing work.

I. Fasten hole reinforcing plate over web penetrations that exceed size of manufacturer's standard punched openings.

J. Erection Tolerances: Install cold-formed metal framing level, plumb, and true to line to a maximum allowable tolerance variation of 1:960 and as follows:

1. Space individual framing members no more than plus or minus 3 mm from plan location. Cumulative error shall not exceed minimum fastening requirements of sheathing or other finishing materials.

3.4 LOAD-BEARING WALL INSTALLATION

A. Install continuous top and bottom tracks sized to match studs. Align tracks accurately and securely anchor at corners and ends, and at spacings as follows:

1. Anchor Spacing: As shown on Shop Drawings.

B. Squarely seat studs against webs of top and bottom tracks. Fasten both flanges of studs to top and bottom tracks. Space studs as follows:

1. Stud Spacing: As indicated.

C. Set studs plumb, except as needed for diagonal bracing or required for nonplumb walls or warped surfaces and similar configurations.

D. Align studs vertically where wall-framing continuity is interrupted by floor framing. Where studs cannot be aligned, continuously reinforce track to transfer loads.

E. Align floor and roof framing over studs. Where framing cannot be aligned, continuously reinforce track to transfer loads.

F. Anchor studs abutting structural columns or walls, including masonry walls, to supporting structure as indicated.

G. Install headers over wall openings wider than stud spacing. Locate headers above openings as indicated. Fabricate headers of compound shapes indicated or required to transfer load to supporting studs, complete with clip-angle connectors, web stiffeners, or gusset plates.

1. Frame wall openings with not less than a double stud at each jamb of frame as indicated on Shop Drawings.
2. Install runner tracks and jack studs above and below wall openings. Anchor tracks to jamb studs with clip angles or by welding, and space jack studs same as full-height wall studs.

H. Install supplementary framing, blocking, and bracing in stud framing indicated to support fixtures, equipment, services, casework, heavy trim, furnishings, and similar work requiring attachment to framing.

1. If type of supplementary support is not indicated, comply with stud manufacturer's written recommendations and industry standards in each case, considering weight or load resulting from item supported.

I. Install horizontal bridging in stud system, spaced the dimension indicated on Shop Drawings apart. Fasten at each stud intersection.

1. Bridging: Cold-rolled steel channel, welded or mechanically fastened to webs of punched studs with a minimum of two screws into each flange of the clip angle.
2. Bridging: Combination of flat, taut, steel sheet straps of width and thickness indicated and stud-track solid blocking of width and thickness to match studs. Fasten flat straps to stud flanges and secure solid blocking to stud webs or flanges.

J. Install steel sheet diagonal bracing straps to both stud flanges, terminate at and fasten to reinforced top and bottom tracks. Fasten clip-angle connectors to multiple studs at ends of bracing and anchor to structure.

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K. Install miscellaneous framing and connections, including supplementary framing, web stiffeners, clip angles, continuous angles, anchors, and fasteners, to provide a complete and stable wall-framing system.

3.5 NON-LOAD-BEARING CURTAIN-WALL INSTALLATION

A. Install continuous tracks sized to match studs. Align tracks accurately and securely anchor to supporting structure as indicated.

B. Fasten both flanges of studs to top and bottom track, unless otherwise indicated. Space studs as follows:

1. Stud Spacing: As indicated.

C. Set studs plumb, except as needed for diagonal bracing or required for nonplumb walls or warped surfaces and similar requirements.

D. Isolate non-load-bearing steel framing from building structure to prevent transfer of vertical loads while providing lateral support.

1. Install single deep-leg deflection tracks and anchor to building structure.
2. Install double deep-leg deflection tracks and anchor outer track to building structure.
3. Connect vertical deflection clips to bypassing or infill studs and anchor to primary building structure.

E. Install horizontal bridging in curtain-wall studs, spaced in rows indicated on Shop Drawings but not more than 1370 mm apart. Fasten at each stud intersection.

1. Top Bridging for Single Deflection Track: Install row of horizontal bridging within 300 mm of single deflection track. Install a combination of flat, taut, steel sheet straps of width and thickness indicated and stud or stud-track solid blocking of width and thickness matching studs. Fasten flat straps to stud flanges and secure solid blocking to stud webs or flanges.

a. Install solid blocking at centers indicated.

2. Bridging: Cold-rolled steel channel, welded or mechanically fastened to webs of punched studs.
3. Bridging: Combination of flat, taut, steel sheet straps of width and thickness indicated and stud-track solid blocking of width and thickness to match studs. Fasten flat straps to stud flanges and secure solid blocking to stud webs or flanges.

F. Install miscellaneous framing and connections, including stud kickers, web stiffeners, clip angles, continuous angles, anchors, fasteners, and stud girts, to provide a complete and stable curtain-wall-framing system.

3.6 JOIST INSTALLATION

A. Install perimeter joist track sized to match joists. Align and securely anchor or fasten track to supporting structure at corners, ends, and spacings indicated on Shop Drawings.

B. Install joists bearing on supporting frame, level, straight, and plumb; adjust to final position, brace, and reinforce. Fasten joists to both flanges of joist track.

1. Install joists over supporting frame with a minimum end bearing of 38 mm.
2. Reinforce ends and bearing points of joists with web stiffeners, end clips, joist hangers, steel clip angles, or steel-stud sections as indicated on Shop Drawings.

C. Space joists not more than 51 mm from abutting walls, and as follows:

1. Joist Spacing: As indicated.

D. Frame openings with built-up joist headers consisting of joist and joist track, nesting joists, or another combination of connected joists if indicated.

E. Install joist reinforcement at interior supports with single, short length of joist section located directly over interior support, with lapped joists of equal length to joist reinforcement, or as indicated on Shop Drawings.

1. Install web stiffeners to transfer axial loads of walls above.

F. Install bridging at each end of joists and at intervals indicated on Shop Drawings. Fasten bridging at each joist intersection as follows:

1. Bridging: Cold-rolled steel channel, welded or mechanically fastened to bottom flange of joists.
2. Bridging: Joist-track solid blocking of width and thickness indicated, secured to joist webs.
3. Bridging: Combination of flat, taut, steel sheet straps of width and thickness indicated and joist-track solid blocking of width and thickness indicated. Fasten flat straps to bottom flange of joists and secure solid blocking to joist webs.

G. Secure joists to load-bearing interior walls to prevent lateral movement of bottom flange.

H. Install miscellaneous joist framing and connections, including web stiffeners, closure pieces, clip angles, continuous angles, hold-down angles, anchors, and fasteners, to provide a complete and stable joist-framing assembly.

3.7 TRUSS INSTALLATION

A. Install, bridge, and brace trusses according to Shop Drawings and requirements in this Section.

B. Truss Spacing: As indicated.

C. Do not alter, cut, or remove framing members or connections of trusses.

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D. Erect trusses with plane of truss webs plumb and parallel to each other, align, and accurately position at spacings indicated.

E. Erect trusses without damaging framing members or connections.

F. Align webs of bottom chords and load-bearing studs or continuously reinforce track to transfer loads to structure. Anchor trusses securely at all bearing points.

G. Install continuous bridging and permanently brace trusses as indicated on Shop Drawings.

3.8 GYPSUM SHEATHING INSTALLATION

A. General: Install gypsum sheathing to comply with GA-253 and manufacturer's written instructions.

B. Cut boards at penetrations, edges, and other obstructions of the work; fit tightly against abutting construction, except provide a 9 mm setback where non-load-bearing construction abuts structural elements.

C. Coordinate sheathing installation with flashing and joint sealant installation so that these materials are installed in the sequence and manner that prevent exterior moisture from passing through completed exterior wall assembly.

D. Apply fasteners so that screw heads bear tightly against face of sheathing boards but do not cut into facing.

E. Do not bridge building expansion joints with sheathing; cut and space edges to match spacing of structural support elements.

F. Horizontal Installation: Install 600 mm wide gypsum sheathing boards horizontally with V-grooved edge down and tongue edge up. Interlock tongue with groove to bring long edges in contact with edges of adjacent boards without forcing. Abut ends of boards over centers of stud flanges and stagger end joints of adjacent boards not less than one stud spacing. Screw-attach boards at perimeter and within field of board to each steel stud at approximately 200 mm o.c. and set back a minimum of 9 mm from edges and ends of boards.

G. Vertical Installation: Install 1200 mm wide gypsum sheathing boards vertically with vertical edges centered over flanges of steel studs. Abut ends and edges of each board with those of adjacent boards. Screw-attach boards at perimeter and within field of board to each steel stud at approximately 200 mm o.c. and set back a minimum of 9 mm from edges and ends of boards.

H. Air-Infiltration Barrier Application: Cover sheathing with air-infiltration barrier as follows:

1. Cut back air-infiltration barrier 13 mm on each side of break in supporting members at expansion- or control-joint locations.
2. Apply asphalt-saturated organic felt horizontally with 50 mm overlap and 150 mm end lap; fasten to sheathing with corrosion-resistant staples.
3. Apply proprietary building wrap to comply with manufacturer's written installation instructions.

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4. Apply air-infiltration barrier to cover vertical flashing with 100 mm overlap.

I. Sealing Sheathing Joints: Seal joints according to sheathing manufacturer's written recommendations and as follows:

1. Apply elastomeric sealant on joints and fasteners and trowel flat. Apply sufficient quantity of sealant to completely cover joints and fasteners after troweling. Seal other penetrations and openings.
2. Apply glass-fiber sheathing tape to glass-mat gypsum sheathing board joints, and apply and trowel silicone emulsion sealant to embed sealant in entire face of tape. Apply sealant to exposed fasteners with a trowel so fasteners are completely covered. Seal other penetrations and openings.

3.9 FIELD QUALITY CONTROL

A. Testing: The Employer will engage a qualified independent testing agency to perform field quality-control testing.

B. Field and shop welds will be subject to inspection and testing.

C. Testing agency will report test results promptly and in writing to the Contractor and the Engineer.

D. Remove and replace Work that does not comply with specified requirements.

E. Additional testing and inspecting, at the Contractor's expense, will be performed to determine compliance of corrected Work with specified requirements.

3.10 REPAIRS AND PROTECTION

A. Galvanizing Repairs: Prepare and repair damaged galvanized coatings on fabricated and installed cold-formed metal framing with galvanized repair paint according to ASTM A 780 and manufacturer's written instructions.

B. Touchup Painting: Wire brush, clean, and paint scarred areas, welds, and rust spots on fabricated and installed prime-painted, cold-formed metal framing. Paint framing surfaces with same type of shop paint used on adjacent surfaces.

C. Protect paper-surfaced gypsum sheathing that will be exposed to weather for more than 30 days by covering exposed exterior surface of sheathing with a securely fastened air-infiltration barrier. Apply covering immediately after sheathing is installed.

D. Protect cutouts, corners, and joints in sheathing by filling with a flexible sealant or by applying tape recommended by sheathing manufacturer at time sheathing is applied.

E. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and Installer, which ensure that cold-formed metal framing is without damage or deterioration at time of Substantial Completion.

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END OF SECTION 05400

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SECTION 05500 - METAL FABRICATIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following metal fabrications:

1. Ladders (including elevator pit ladders).
2. Ladder safety cages.
3. Ship's ladders.
4. Loose bearing and leveling plates.
5. Loose steel lintels.
6. Shelf and relieving angles.
7. Miscellaneous framing and supports for the following:

    a. Overhead doors.
    b. Suspended toilet partitions.
    c. Suspended operable partitions.
    d. Elevator hoisting machines and sheaves.
    e. Elevator door sills.
    f. Applications where framing and supports are not specified in other sections.

8. Miscellaneous steel trim, including the following:

    a. Steel angle corner guards.
    b. Edgings.
    c. Loading dock edge angles.

9. Structural steel door frames.
10. Floor plate and supports.
11. Tread plate and supports.
12. Cast nosings, treads, and thresholds.
13. Extruded nosings and treads.
14. Pipe guards.
15. Wheel guards.
16. Pipe bollards.

B. Related Sections include the following:

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1. Steel Specification Section "Structural Steel" for structural-steel framing system components.
2. Steel Specification Section "Metal Stairs" for metal-framed stairs with metal pan, metal plate, or grating treads.
3. Steel Specification Section "Gratings" for metal gratings.
4. Steel Specification Section "Pipe and Tube Railings" for metal pipe and tube handrails and railings.
5. Steel Specification Section "Ornamental Metal" for ornamental metal items fabricated from custom components.
6. Steel Specification Section "Ornamental Handrails and Railings" for ornamental metal handrails and railings fabricated from stock components.
7. Woods & Plastics Specification Section "Rough Carpentry" for metal framing anchors and other rough hardware.

1.3 SUBMITTALS

A. Product Data: For nonslip aggregates and nonslip aggregate surface finishes, cast nosings, treads and thresholds, steel floor plate, paint products, and grout.

B. Shop Drawings: Detail fabrication and erection of each metal fabrication indicated. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items.

1. Provide templates for anchors and bolts specified for installation under other Sections.

C. Samples representative of materials and finished products as may be requested by the Engineer.

D. Mill Certificates: Signed by manufacturers of stainless-steel sheet certifying that products furnished comply with requirements.

E. Welding Certificates signed by the Contractor certifying that welders comply with requirements specified under the "Quality Assurance" Article.

F. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

G. As-Built Drawings: At project close out, submit Record (As- Built) Drawings of completed work products, in accordance with requirements of the Specification as indicated in General Requirements.

1.4 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

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B. Fabricator Qualifications: A firm experienced in producing metal fabrications similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units, without delaying the work.

C. Welding: Qualify procedures and personnel according to the following:

1. AWS D1.1, "Structural Welding Code--Steel."
2. AWS D1.2, "Structural Welding Code--Aluminum."
3. AWS D1.3, "Structural Welding Code--Sheet Steel."
4. Certify that each welder has satisfactorily passed AWS qualification tests for welding processes involved and, if pertinent, has undergone recertification.

1.5 PROJECT CONDITIONS

A. Field Measurements: Where metal fabrications are indicated to fit walls and other construction, verify dimensions by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1. Where field measurements cannot be made without delaying the Work, guarantee dimensions and proceed with fabricating products without field measurements. Coordinate construction to ensure that actual dimensions correspond to guaranteed dimensions. Allow for trimming and fitting.

1.6 COORDINATION

A. Coordinate installation of anchorages for metal fabrications. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

PART 2 - PRODUCTS

2.1 METALS, GENERAL

A. Metal Surfaces, General: For metal fabrications exposed to view in the completed Work, provide materials with smooth, flat surfaces without blemishes. Do not use materials with exposed pitting, seam marks, roller marks, rolled trade names, or roughness.

2.2 FERROUS METALS

A. Steel Plates, Shapes, and Bars: ASTM A 36M.

B. Stainless-Steel Sheet, Strip, Plate, and Flat Bars: ASTM A 666, Type 304.

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C. Stainless Steel:

1. Grade and type designated below for each form required:

a. Tubing: ASTM A 554, Grade MT 316
b. Pipe: ASTM A 312M, Grade TP 316.
c. Castings: ASTM A 743M, Grade CF 8M.
d. Plate: ASTM A 167, Type 316.
e. Bar Stock: ASTM A 276.

2. Finish: Bright, directional polish; match AISI No. 4 finish.

D. Rolled-Steel Floor Plate: ASTM A 786M, rolled from plate complying with ASTM A 36/A 36M or ASTM A 283/A 283M, Grade C or D.

E. Rolled-Stainless-Steel Floor Plate: ASTM A 793.

F. Steel Tubing: Product type (manufacturing method) and as follows:

1. Cold-Formed Steel Tubing: ASTM A 500.

a. Provide tubing with hot-dip galvanized coating per ASTM A 53.

G. Steel Pipe: ASTM A 53, standard weight (Schedule 40), unless another weight is indicated or required by structural loads.

1. Provide galvanized finish for all steel pipe.

H. Slotted Channel Framing: Cold-formed metal channels with flange edges returned toward web and with 15 mm wide slotted holes in webs at 50 mm o.c.

1. Width of Channels: As indicated.
2. Depth of Channels: As indicated.
3. Metal and Thickness: Galvanized steel complying with ASTM A 653/A 653M, structural quality, Grade 230, with Z275 coating; as indicated nominal thickness.
4. Finish: As indicated.

I. Malleable-Iron Castings: ASTM A 47M, Grade 22010.

J. Gray-Iron Castings: ASTM A 48M, Class 200, unless another class is indicated or required by structural loads.

K. Cast-in-Place Anchors in Concrete: Anchors of type indicated below, fabricated from corrosion-resistant materials capable of sustaining, without failure, the load imposed within a safety factor of 4, as determined by testing per ASTM E 488, conducted by a qualified independent testing agency.

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1. Threaded or wedge type; galvanized ferrous castings, either ASTM A 47M malleable iron or ASTM A 27/A 27M cast steel. Provide bolts, washers, and shims as needed, hot-dip galvanized per ASTM A 153/A 153M.

L. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.

2.3 ALUMINUM

A. Aluminum Extrusions: ASTM B 221M, alloy 6063-T6.

B. Aluminum-Alloy Rolled Tread Plate: ASTM B 632M, alloy 6061-T6.

2.4 PAINT

A. Shop Primers: Provide primers that comply with requirements.

B. Shop Primer for Ferrous Metal: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with performance requirements in FS TT-P-664; selected for good resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated, and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure.

C. Shop Primer for Ferrous Metal: Organic zinc-rich primer, complying with SSPC-Paint 20 and compatible with topcoat.

D. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds in galvanized steel, with dry film containing not less than 94 percent zinc dust by weight, and complying with DOD-P-21035 or SSPC-Paint 20.

E. Bituminous Paint: Cold-applied asphalt mastic complying with SSPC-Paint 12, except containing no asbestos fibers, or cold-applied asphalt emulsion complying with ASTM D 1187.

2.5 FASTENERS

A. General: Provide Type 304 or 316 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633, Class Fe/Zn, where built into exterior walls. Select fasteners for type, grade, and class required.

B. Bolts and Nuts: Regular hexagon-head bolts, ASTM F 568M, Property Class 4.6; with hex nuts, ASTM A 563M; and, where indicated, flat washers.

C. Anchor Bolts: ASTM F 1554, Grade 36.

D. Machine Screws: ASME B18.6.7M.

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E. Lag Bolts: ASME B18.2.3.8M.

F. Wood Screws: Flat head, carbon steel, ASME B18.6.1.

G. Plain Washers: Round, carbon steel, ASME B18.22M.

H. Lock Washers: Helical, spring type, carbon steel, ASME B18.21.2M.

I. Expansion Anchors: Anchor bolt and sleeve assembly of material indicated below with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete, as determined by testing per ASTM E 488, conducted by a qualified independent testing agency.

1. Material: Carbon-steel components zinc-plated to comply with ASTM B 633, Class Fe/Zn 5.
2. Material: Alloy Group 1 or 2 stainless-steel bolts complying with ASTM F 738M and nuts complying with ASTM F 836M.

J. Toggle Bolts: FS FF-B-588, tumble-wing type, class and style as needed.

2.6 GROUT

A. General: Grout shall meet the requirements of Concrete Specification Section, “Cast-in-Place Concrete”

B. Nonshrink, Metallic Grout: Factory-packaged, ferrous-aggregate grout complying with ASTM C 1107, specifically recommended by the manufacturer for heavy-duty loading applications.

C. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by the manufacturer for interior and exterior applications.

D. Available Products: Subject to compliance with requirements, products that may be incorporated in the Work include, but are not limited to, the following:

1. Nonshrink, Nonmetallic Grouts:

a. B-6 Construction Grout; W. R. Bonsal Co.
b. Diamond-Crete Grout; Concrete Service Materials Co.
c. Supreme; Cormix Construction Chemicals.
d. Sure-grip High Performance Grout; Dayton Superior Corp.
e. Euco N-S Grout; Euclid Chemical Co.
f. Five Star Grout; Five Star Products.
g. Vibropruf #11; Lambert Corp.
h. Crystex; L & M Construction Chemicals, Inc.
i. Masterflow 928 and 713; Master Builders Technologies, Inc.
j. Sealtight 588 Grout; W. R. Meadows, Inc.

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k. Sonogrout 14; Sonneborn Building Products--ChemRex, Inc.
l. Kemset; The Spray-Cure Company.

2.7 CONCRETE FILL

Concrete Materials and Properties: Comply with requirements in Concrete Specification Section "Cast-in-Place Concrete" for normal-weight, air-entrained, ready-mix concrete with a minimum 28-day compressive strength of 20 MPa, unless otherwise indicated.

2.8 FABRICATION, GENERAL

A. Shop Assembly: Preassemble items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.

B. Shear and punch metals cleanly and accurately. Remove burrs.

C. Ease exposed edges to a radius of approximately 1 mm, unless otherwise indicated. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.

D. Weld corners and seams continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. At exposed connections, finish exposed welds and surfaces smooth and blended so that no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

E. Provide for anchorage of type indicated; coordinate with supporting structure. Fabricate and space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.

F. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.

G. Fabricate joints that will be exposed to weather in a manner so as to exclude water, or provide weep holes where water may accumulate.

H. Allow for thermal movement resulting from the following maximum change (range) in ambient and surface temperatures by preventing buckling, opening up of joints, overstressing of components, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

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1. Temperature Change (Range): 67 deg. C, ambient; 100 deg. C, material surfaces.

I. Form exposed work true to line and level with accurate angles and surfaces and straight sharp edges.

J. Remove sharp or rough areas on exposed traffic surfaces.

K. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners where possible. Use exposed fasteners of type indicated or, if not indicated, Phillips flat-head (countersunk) screws or bolts. Locate joints where least conspicuous.

L. Form metal fabrications from materials of size, thickness, and shapes indicated but not less than that needed to comply with performance requirements indicated. Work to dimensions indicated or accepted on shop drawings, using proven details of fabrication and support. Use type of materials indicated or specified for various components of each metal fabrication.

2.9 STEEL LADDERS

A. General: Fabricate ladders for locations shown, with dimensions, spacings, details, and anchorages as indicated.

1. Comply with ANSI A14.3, unless otherwise indicated.
2. For elevator pit ladders, comply with ASME A17.1.

B. Side-rails: As indicated on the drawings.

C. Bar Rungs: As indicated on the drawings.

D. Fit rungs in centerline of side rails; plug-weld and grind smooth on outer rail faces.

E. Support each ladder at top and bottom and not more than 1500 mm o.c. with welded or bolted steel brackets.
1. Size brackets to support design dead and live loads indicated and to hold centerline of ladder rungs clear of the wall surface by not less than 180 mm.
2. Extend side rails 1.1 m above top rung, and return rails to wall or structure unless other secure handholds are provided. If the adjacent structure does not extend above the top rung, goose-neck the extended rails back to the structure to provide secure ladder access.

F. Provide nonslip surfaces on top of each rung, either by coating rung with aluminum-oxide granules set in epoxy-resin adhesive or by using a type of manufactured rung filled with aluminum-oxide grout.

G. Provide nonslip surfaces on top of each rung by coating with abrasive material metallically bonded to rung by a proprietary process.

H. Galvanize ladders, including brackets and fasteners, in the following locations:

1. Exterior.

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2. Interior, where indicated.

2.10 LADDER SAFETY CAGES

A. General: Fabricate ladder safety cages to comply with ANSI A14.3. Assemble by welding or riveting.

B. Primary Hoops: 8-by-100 mm steel flat bar hoops; and for cages longer than 6 m, intermediate hoops spaced not more than 6 m o.c.

C. Provide at tops and bottoms of cages and spaced not more than 6 m o.c.

D. Secondary Intermediate Hoops: 8-by-50 mm steel flat bar hoops, spaced not more than 1200 mm o.c. between primary hoops.

E. Vertical Bars: 8-by-50 mm steel flat bars secured to each hoop, spaced approximately 230 mm o.c.

F. Fasten assembled safety cage to ladder rails and adjacent construction by welding or riveting, unless otherwise indicated.

G. Galvanize ladder safety cages, including fasteners, in the following locations:

1. Exterior.
2. Interior, where indicated.

2.11 LOOSE STEEL LINTELS

A. Unless otherwise indicated on Drawings, provide steel lintels as follows.

B. Fabricate loose structural-steel lintels from steel angles and shapes of size indicated for openings and recesses in masonry walls and partitions at locations indicated.

C. Weld adjoining members together to form a single unit where indicated.

D. Size loose lintels to provide bearing length at each side of openings equal to one-twelfth of clear span, but not less than 200 mm, unless otherwise indicated.

E. Galvanize loose steel lintels located in exterior walls.

2.12 MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Provide steel framing and supports that are not a part of structural-steel framework as necessary to complete the Work.

B. General: Provide steel framing and supports indicated and as necessary to complete the Work.

C. Fabricate units from structural-steel shapes, plates, and bars of welded construction, unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction retained by framing and supports. Cut, drill, and tap units to receive hardware, hangers, and similar items.

1. Fabricate units from slotted channel framing where indicated.
2. Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors 32 mm wide by 6 mm thick by 200 mm long at 600 mm o.c., unless otherwise indicated.
3. Furnish inserts if units must be installed after concrete is placed.

D. Fabricate supports for operable partitions as follows:

1. Beams: Continuous steel shapes of sizes indicated with attached bearing plates, anchors, and braces as indicated. Drill bottom flanges of beams to receive partition track hanger rods; locate holes where indicated on operable partition Shop Drawings.

E. Fabricate steel girders for wood frame construction from continuous steel shapes of sizes indicated.

1. Provide bearing plates welded to beams where indicated.
2. Drill girders and plates for field-bolted connections where indicated.
3. Where wood nailers are attached to girders with bolts or lag screws, drill holes at 600 mm o.c.

F. Galvanize miscellaneous framing and supports in the following locations:

1. Exterior locations.
2. Interior locations where indicated.

G. Fabricate support for suspended toilet partitions as follows:

1. Beams: Continuous steel shapes of size required to limit deflection to L/360 between hangers, but use not less than C8x11.5 channels or another shape with equivalent structural properties.
2. Hangers: Steel rods, 13 mm minimum diameter, spaced not more than 900 mm o.c. Thread rods to receive anchor and stop nuts. Fit hangers with wedge-shaped washers for full bearing on sloping flanges of support beam.
3. Braces and Angles: Steel angles of size required for rigid support of beam and for secure anchorage.

2.13 MISCELLANEOUS STEEL TRIM

A. Unless otherwise indicated, fabricate units from structural-steel shapes, plates, and bars of profiles shown with continuously welded joints, and smooth exposed edges. Miter corners and use concealed field splices where possible.

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B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work. Provide anchors, welded to trim, for embedding in concrete or masonry construction, spaced not more than 150 mm from each end, 150 mm from corners, and 600 mm o.c., unless otherwise indicated.

C. Galvanize miscellaneous steel trim in the following locations:

1. Exterior.
2. Interior, where indicated.

2.14 METAL FLOOR PLATE

A. Fabricate raised-pattern floor plates from rolled-steel floor plate of thickness indicated below:

1. Thickness: As indicated.
2. Pattern: As selected by the Engineer from manufacturer's standard patterns.

B. Fabricate raised-pattern floor plates from rolled-aluminum-alloy tread plate of thickness indicated below:

1. Thickness: As indicated.

C. Abrasive-Surface Floor Plate: Fabricate from steel plate, of thickness indicated below, with abrasive granules rolled into surface. Provide material with coefficient of friction of 0.6 or higher when tested according to ASTM C 1028.

D. Abrasive-Surface Floor Plate: Fabricate from steel plate, of thickness indicated below, with abrasive material metallically bonded to steel by a proprietary process. Provide material with coefficient of friction of 0.6 or higher when tested according to ASTM C 1028.

E. Include steel angle stiffeners, and fixed and removable sections as indicated.

1. Provide flush steel bar drop handles for lifting removable sections, one at each end of each section.

F. Include stainless-steel angle stiffeners, and fixed and removable sections as indicated.

1. Provide flush stainless-steel bar drop handles for lifting removable sections, one at each end of each section.

G. Include aluminum angle stiffeners, and fixed and removable sections as indicated.

1. Provide flush aluminum bar drop handles for lifting removable sections, one at each end of each section.

2.15 EXTRUDED NOSINGS AND TREADS

A. Fabricate units in sizes and configurations indicated and in lengths necessary to accurately fit openings or conditions. Provide extruded-aluminum units with abrasive filler consisting of aluminum oxide, silicon carbide, or a combination of both, in an epoxy-resin binder.

1. Provide ribbed units, with abrasive filler strips projecting 1.5 mm above aluminum extrusion.
2. Provide solid-abrasive-type units without ribs.

B. Configurations: Provide units in the following configurations, unless otherwise indicated:

1. Nosings: Units, 75 mm wide, for casting into concrete steps.
2. Treads: Square-back units, full depth of tread with 35 mm lip, for application over existing stairs.

C. Provide anchors for embedding units in concrete, either integral or applied to units, as standard with manufacturer.

D. Drill for mechanical anchors and countersink. Locate not more than 100 mm from ends and not more than 300 mm o.c., evenly spaced between ends, unless otherwise indicated. Provide closer spacing if recommended by the manufacturer.

1. Provide two rows of holes for units more than 125 mm wide, with two holes aligned at ends and intermediate holes staggered.

E. Apply clear lacquer to concealed bottoms, sides, and edges of units set into concrete.

2.16 PIPE GUARDS

A. Provide pipe guards of 76-by-76-by-8 mm steel angles, extending from floor to 1100 mm above floor, with 10 mm steel baseplates for bolting to floor. Provide at least two vertical angles at each location, except at internal corners where one may be used. Connect tops of angles and anchor to wall or column with 6-by-50 mm steel strap braces welded to angles and bolted to wall.

B. Galvanize pipe guards after fabrication.

2.17 WHEEL GUARDS

A. Provide wheel guards of 19 mm thick, hollow-core, gray-iron castings; of size and shape indicated. Provide holes for countersunk anchor bolts and grouting.

2.18 PIPE BOLLARDS

A. Fabricate pipe bollards from Schedule 40 steel pipe.

B. Fabricate pipe bollards from Schedule 80 steel pipe.

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1. Cap bollards with 6 mm minimum steel plate.

C. Fabricate bollards with 10 mm thick steel baseplates for bolting to concrete slab. Drill baseplates at all four corners for 19 mm anchor bolts.

1. Where bollards are to be anchored to sloping concrete slabs, angle baseplates for plumb alignment of bollards.

D. Fabricate sleeves for bollard anchorage from steel pipe with 6 mm thick steel plate welded to bottom of sleeve.

E. Fabricate internal sleeves for removable bollards from Schedule 40 steel pipe or 6 mm wall-thickness steel tubing with an OD 1.5 mm less than ID of bollards. Match drill sleeve and bollard for 12 mm steel machine bolt.

2.19 SHIP'S LADDERS

A. Provide ship's ladders where indicated. Fabricate of open type construction with structural steel channel or steel plate stringers, steel pipe handrails, and open steel grating treads, unless otherwise indicated. Provide brackets and fittings for installation.

1. Comply with applicable requirements of Section 05511, "Metal Stairs" for steel pipe railings.

B. Galvanize ladders, including brackets and fasteners, in the following locations:

1. Exterior locations.
2. Interior locations where indicated.

2.20 LOOSE BEARING AND LEVELING PLATES

A. Column base plates are not included in this category.

B. Provide loose bearing and leveling plates for steel items bearing on masonry or concrete construction, made flat, free from warps or twists, and of the required thickness and bearing area. Drill plates to receive anchor bolts and for grouting as required. Galvanize after fabrication.

2.21 LOOSE STEEL LINTELS

A. Unless otherwise indicated on Drawings, provide steel lintels as follows.

B. Fabricate loose structural steel lintels from steel angles and shapes of size indicated for openings and recesses in masonry walls and partitions at locations indicated.

C. Weld adjoining members together to form a single unit where indicated.

D. Size loose lintels for equal bearing of 85 mm per meter of clear span but not less than 200 mm bearing at each side of openings, unless otherwise indicated.

E. Galvanize loose steel lintels located in exterior walls.

2.22 SHELF AND RELIEVING ANGLES

A. Fabricate shelf and relieving angles from steel angles of sizes indicated and for attachment to concrete framing. Provide slotted holes to receive 19 mm bolts, spaced not more than 150 mm from ends and not more than 600 mm o.c., unless otherwise indicated.

B. For cavity walls, provide vertical channel brackets to support shelf/relieving angles from back-up masonry and concrete. Align expansion joints in angles with indicated control and expansion joints in cavity wall exterior wythe.

C. Galvanize shelf angles to be installed on exterior concrete framing.

D. Furnish wedge-type concrete inserts, complete with fasteners, to attach shelf angles to cast-in-place concrete.

2.23 STRUCTURAL STEEL DOOR FRAMES

A. Fabricate steel door frames from structural shapes and bars of size and to dimensions indicated, fully welded together, with 16-by-38 mm steel bar stops, unless otherwise indicated. Plug-weld built-up members and continuously weld exposed joints. Secure removable stops to frame with countersunk machine screws, uniformly spaced at not more than 250 mm o.c. Reinforce frames and drill and tap as required to accept finish hardware.

B. Provide steel strap anchors for securing door frames into adjoining concrete or masonry, using 3-by-50 mm straps of the length required for a minimum 200 mm embedment, unless otherwise indicated. Weld anchors to frame jambs no more than 300 mm from both bottom and head of frame and space anchors not more than 750 mm apart.

C. Extend bottom of frames to floor elevation indicated with steel angle clips welded to frames for anchoring frame to floor with expansion shields and bolts.

D. Galvanize frames and anchors in the following locations:

1. Exterior locations.
2. Interior locations where indicated.

2.24 CAST NOSINGS, TREADS, AND THRESHOLDS

A. Fabricate units of material, sizes, and configurations indicated. If not indicated, provide cast-iron units with an integral abrasive finish. Furnish in lengths as required to accurately fit each opening or conditions.

1. Cast units with an integral abrasive grit consisting of aluminum oxide, silicon carbide, or a combination of both.

B. Provide anchors for embedding units in concrete, either integral or applied to units, as standard with the manufacturer.

C. Drill for mechanical anchors with countersunk holes located not more than 100 mm from ends and not more than 300 mm o.c., evenly spaced between ends, unless otherwise indicated. Provide closer spacing if recommended by the manufacturer.

1. Provide 2 rows of holes for units over 125 mm wide, with 2 holes aligned at ends and intermediate holes staggered.

D. Apply black asphaltic coating to concealed bottoms, sides, and edges of cast-iron units set into concrete.

E. Provide a plain surface texture, except where fluted or cross-hatched surfaces are indicated.

2.25 FINISHES, GENERAL

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Finish metal fabrications after assembly.

2.26 STEEL AND IRON FINISHES

A. Galvanizing: For those items indicated for galvanizing, apply zinc coating by the hot-dip process complying with the following requirements:

1. ASTM A 153 for galvanizing iron and steel hardware.
2. ASTM A 123 for galvanizing both fabricated and unfabricated iron and steel products made of uncoated rolled, pressed, and forged shapes, plates, bars, and strip 0.76 mm thick or thicker.

B. Preparation for Shop Priming: Prepare uncoated ferrous-metal surfaces to comply with minimum requirements indicated below for SSPC surface-preparation specifications and environmental exposure conditions of installed metal fabrications:

1. Exteriors (SSPC Zone 1B): SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
2. Interiors (SSPC Zone 1A): SSPC-SP 3, "Power Tool Cleaning."

C. Apply shop primer to uncoated surfaces of metal fabrications, except those with galvanized finishes and those to be embedded in concrete, sprayed-on fireproofing, or masonry, unless

otherwise indicated. Comply with SSPC-PA 1, "Paint Application Specification No. 1," for shop painting.

1. Stripe paint corners, crevices, bolts, welds, and sharp edges.

2.27 STAINLESS-STEEL FINISHES

A. Remove tool and die marks and stretch lines or blend into finish.

B. Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

C. Bright, Directional Polish: No. 4 finish.

D. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

2.28 ALUMINUM FINISHES

A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.

B. As-Fabricated Finish: AA-M10 (Mechanical Finish: as-fabricated, unspecified).

C. Class I, Clear Anodic Finish: AA-M12C22A41 (Mechanical Finish: nonspecular as-fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, clear coating 0.018 mm or thicker) complying with AAMA 607.1.

PART 3 - EXECUTION

3.1 PREPARATION

A. Coordinate and furnish anchorages, setting drawings, diagrams, templates, instructions, and directions for installing anchorages, including concrete inserts, sleeves, anchor bolts, and miscellaneous items having integral anchors that are to be embedded in concrete or masonry construction. Coordinate delivery of such items to Project site.

B. Center nosings on tread widths with noses flush with riser faces and tread surfaces.

C. Set sleeves in concrete with tops flush with finish surface elevations. Protect sleeves from water and concrete entry.

3.2 INSTALLATION, GENERAL

A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing metal fabrications to in-place construction. Include threaded fasteners for concrete and masonry inserts, toggle bolts, through-bolts, lag bolts, wood screws, and other connectors.

B. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.

C. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.

D. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.

E. Field Welding: Comply with the following requirements:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. At exposed connections, finish exposed welds and surfaces smooth and blended so that no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

F. Corrosion Protection: Coat concealed surfaces of aluminum that will come into contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

3.3 SETTING BEARING AND LEVELING PLATES

A. Clean concrete and masonry bearing surfaces of bond-reducing materials, and roughen to improve bond to surfaces. Clean bottom surface of plates.

B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with grout.

1. Use nonshrink grout, either metallic or nonmetallic, in concealed locations where not exposed to moisture; use nonshrink, nonmetallic grout in exposed locations, unless otherwise indicated.
2. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.4 INSTALLING MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings, if any.

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B. Anchor supports for operable partitions securely to and rigidly brace from building structure.

C. Support steel girders on solid grouted masonry, concrete, or steel pipe columns. Secure girders with anchor bolts embedded in grouted masonry or concrete or with bolts through top plates of pipe columns.

1. Where grout space under bearing plates is indicated at girders supported on concrete or masonry, install as specified above for setting and grouting bearing and leveling plates.

3.5 INSTALLING NOSINGS, TREADS, AND THRESHOLDS

A. Install with anchorage system indicated to comply with manufacturer's written instructions.

B. Center nosings on tread widths.

C. For nosings embedded in concrete steps or curbs, align nosings flush with riser faces and level with tread surfaces.

D. Seal thresholds exposed to exterior with elastomeric sealant to provide a watertight installation.

3.6 INSTALLING PIPE GUARDS

A. Install pipe guards at exposed vertical pipes in parking garage where not protected by curbs or other barriers. Install by bolting to floor and wall or column with drilled-in expansion anchors.

3.7 INSTALLING WHEEL GUARDS

A. Anchor wheel guards to concrete or masonry construction to comply with manufacturer's written instructions. Fill cores solidly with concrete.

3.8 INSTALLING PIPE BOLLARDS

A. Anchor bollards in concrete with pipe sleeves preset and anchored into concrete. After bollards have been inserted into sleeves, fill annular space between bollard and sleeve solidly with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions. Slope grout up approximately 3 mm toward bollard.

B. Anchor bollards in concrete in formed or core-drilled holes not less than 200 mm deep and 19 mm greater than OD of bollard. After bollards have been inserted into holes, fill annular space surrounding bollard solidly with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions. Slope grout up approximately 3 mm toward bollard.

C. Anchor bollards in place with concrete footings. Support and brace bollards in position in footing excavations until concrete has been placed and cured.

D. Anchor bollards to existing construction with postinstalled anchors and bolts. Provide four 19 mm anchors at each bollard, unless otherwise indicated. Embed anchors at least 100 mm in existing concrete.

E. Anchor internal sleeves for removable bollards in concrete by inserting into pipe sleeves preset into concrete. After internal sleeves have been inserted, fill annular space between sleeves solidly with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions. Slope grout up approximately 3 mm toward internal sleeve.

F. Anchor internal sleeves for removable bollards in formed or core-drilled holes not less than 200 mm deep and 19 mm greater than OD of sleeve. After sleeves have been inserted into holes, fill annular space surrounding sleeves solidly with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions. Slope grout up approximately 3 mm toward sleeve.

G. Anchor internal sleeves for removable bollards in place with concrete footings. Support and brace sleeves in position in footing excavations until concrete has been placed and cured.

1. Place removable bollards over internal sleeves and secure with 12 mm machine bolts and nuts. After tightening nuts, drill holes in bolts for inserting padlocks. Furnish padlocks.

H. Fill bollards solidly with concrete, mounding top surface.

1. Do not fill removable bollards with concrete.

3.9 ADJUSTING AND CLEANING

A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

1. Apply by brush or spray to provide a minimum 0.05 mm dry film thickness.

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

END OF SECTION 05500

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SECTION 05521 - PIPE AND TUBE RAILINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Aluminum pipe and tube handrails and railings.
2. Stainless-steel pipe and tube handrails and railings.
3. Steel pipe and tube handrails and railings.

B. Related Sections include the following:

1. Steel Specification Section "Metal Stairs" for steel pipe handrails and railings included with metal stairs.
2. Steel Specification Section "Ornamental Metal" for ornamental metal handrails and railings fabricated from custom components.
3. Steel Specification Section "Ornamental Handrails and Railings" for ornamental metal handrails and railings fabricated from stock components.

1.3 PERFORMANCE REQUIREMENTS

A. General: In engineering handrails and railings to withstand structural loads indicated, determine allowable design working stresses of handrail and railing materials based on the following:

1. Aluminum: AA 30, "Specifications for Aluminum Structures."
2. Stainless Steel: ASCE 8, "Specification for the Design of Cold-Formed Stainless Steel Structural Members."
3. Structural Steel: AISC S335, "Specification for Structural Steel Buildings Allowable Stress Design and Plastic Design with Commentary."
4. Cold-Formed Structural Steel: AISI SG-673, Part I, "Specification for the Design of Cold-Formed Steel Structural Members."

B. Structural Performance of Handrails and Railings: Provide handrails and railings capable of withstanding structural loads required by ASCE 7 without exceeding allowable design working stresses of materials for handrails, railings, anchors, and connections.

C. Thermal Movements: Provide handrails and railings that allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures by preventing

buckling, opening of joints, overstressing of components, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change (Range): 67 deg. C, ambient; 100 deg. C, material surfaces.

D. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

1.4 SUBMITTALS

A. Product Data: For the following:

1. Manufacturer's product lines of mechanically connected handrails and railings.
2. Grout, anchoring cement, and paint products.

B. Shop Drawings: Show fabrication and installation of handrails and railings. Include plans, elevations, sections, component details, and attachments to other Work.

1. For installed handrails and railings indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

C. Samples for Initial Selection: Manufacturer's color charts showing the full range of colors available for products with factory-applied color finishes.

D. Samples for Initial Selection: Short sections of railing or flat, sheet metal samples showing available mechanical finishes.

E. Samples for Verification: For each type of exposed finish required, prepared on components indicated below and of same thickness and metal indicated for the Work. If finishes involve normal color and texture variations, include sample sets showing the full range of variations expected.

1. 150 mm long sections of each distinctly different linear railing member, including handrails, top rails, posts, and balusters.
2. Fittings and brackets.
3. Assembled sample of railing system, made from full-size components, including top rail, post, handrail, and infill. Show method of finishing members at intersections. Sample need not be full height.

F. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

G. Product Test Reports: From a qualified testing agency indicating products comply with requirements, based on comprehensive testing of current products.

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H. Product Test Reports: From a qualified testing agency indicating handrails and railings comply with ASTM E 985, based on comprehensive testing of current products.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Professional Engineer Qualifications: A professional engineer who is legally registered and qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of handrails and railings that are similar to those indicated for this Project in material, design, and extent.

C. Testing Agency Qualifications: An independent testing agency, acceptable to the Engineer, with the experience and capability to conduct the testing indicated, as documented according to ASTM E 548.

D. Source Limitations: Obtain each type of handrail and railing through one source from a single manufacturer.

1.6 STORAGE

A. Store handrails and railings in a dry, well-ventilated, weathertight place, and protect from damage.

1.7 PROJECT CONDITIONS

A. Field Measurements: Verify handrail and railing dimensions by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.8 COORDINATION

A. Coordinate installation of anchorages for handrails and railings. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.9 SCHEDULING

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A. Schedule installation so handrails and railings are mounted only on completed walls. Do not support temporarily by any means that does not satisfy structural performance requirements.

PART 2 - PRODUCTS

2.1 METALS

A. General: Provide metal free from pitting, seam marks, roller marks, stains, discolorations, and other imperfections where exposed to view on finished units.

B. Aluminum: Alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than the strength and durability properties of alloy and temper designated below for each aluminum form required.

1. Extruded Bar and Tube: ASTM B 221M, alloy 6063-T5/T52.
2. Extruded Structural Pipe and Tube: ASTM B 429, alloy 6063-T6.
3. Drawn Seamless Tube: ASTM B 210M, alloy 6063-T832.
4. Plate and Sheet: ASTM B 209M, alloy 6061-T6.
5. Die and Hand Forgings: ASTM B 247M, alloy 6061-T6.
6. Castings: ASTM B 26/B 26M, alloy A356-T6.

C. Stainless Steel: Grade or type designated below for each form required:

1. Tubing: ASTM A 554, Grade MT 304.
2. Pipe: ASTM A 312/A 312M, Grade TP 304.
3. Castings: ASTM A 743/A 743M, Grade CF 8 or CF 20.
4. CPlate: ASTM A 666, Type 304.
5. Plate: ASTM A 666, Type 316L.

D. Steel and Iron: Provide steel and iron in the form indicated, complying with the following requirements:

1. Steel Pipe: ASTM A 53; finish, type, and weight class as follows:

a. Black finish, unless otherwise indicated.
b. Galvanized finish for exterior installations and where indicated.
c. Type F, or Type S, Grade A, standard weight (Schedule 40), unless another grade and weight are required by structural loads.

2. Steel Tubing: Cold-formed steel tubing, ASTM A 500, Grade A, unless another grade is required by structural loads.
3. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
4. Iron Castings: Malleable iron complying with ASTM A 47M, Grade 22010.

E. Brackets, Flanges, and Anchors: Cast or formed metal of same type of material and finish as supported rails, unless otherwise indicated.

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2.2 WELDING MATERIALS, FASTENERS, AND ANCHORS

A. Welding Electrodes and Filler Metal: Provide type and alloy of filler metal and electrodes as recommended by producer of metal to be welded and as required for color match, strength, and compatibility in fabricated items.

B. Fasteners for Anchoring Handrails and Railings to Other Construction: Select fasteners of type, grade, and class required to produce connections suitable for anchoring handrails and railings to other types of construction indicated and capable of withstanding design loads.

1. For aluminum handrails and railings, use fasteners fabricated from Type 304 or Type 316 stainless steel.
2. For stainless-steel handrails and railings, use fasteners fabricated from Type 304 or Type 316 stainless steel.
3. For steel handrails, railings, and fittings, use plated fasteners complying with ASTM B 633, Class Fe/Zn 25 for electrodeposited zinc coating.

C. Fasteners for Interconnecting Handrail and Railing Components: Use fasteners fabricated from same basic metal as fastened metal, unless otherwise indicated. Do not use metals that are corrosive or incompatible with materials joined.

1. Provide concealed fasteners for interconnecting handrail and railing components and for attaching them to other work, unless otherwise indicated.
2. Provide concealed fasteners for interconnecting handrail and railing components and for attaching them to other work, unless exposed fasteners are unavoidable or are the standard fastening method for handrails and railings indicated.
3. Provide Phillips flat-head machine screws for exposed fasteners, unless otherwise indicated.

D. Cast-in-Place and Postinstalled Anchors: Anchors of type indicated below, fabricated from corrosion-resistant materials with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete, as determined by testing per ASTM E 488 conducted by a qualified independent testing agency acceptable to the Engineer.

1. Cast-in-place anchors.
2. Chemical anchors.
3. Expansion anchors.

2.3 PAINT

A. Shop Primer for Ferrous Metal: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with performance requirements in FS TT-P-664; selected for good resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated, and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure.

B. Shop Primer for Galvanized Steel: Zinc-dust, zinc-oxide primer formulated for priming zinc-coated steel and for compatibility with finish paint systems indicated, and complying with SSPC-Paint 5.

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C. Bituminous Paint: Cold-applied asphalt mastic complying with SSPC-Paint 12, except containing no asbestos fibers, or cold-applied asphalt emulsion complying with ASTM D 1187.

2.4 GROUT AND ANCHORING CEMENT

A. Nonshrink, Nonmetallic Grout: Premixed, factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by the manufacturer for interior and exterior applications.

B. Interior Anchoring Cement: Factory-packaged, nonshrink, nonstaining, hydraulic-controlled expansion cement formulation for mixing with water at Project site to create pourable anchoring, patching, and grouting compound. Use for interior applications only.

C. Erosion-Resistant Anchoring Cement: Factory-packaged, nonshrink, nonstaining, hydraulic-controlled expansion cement formulation for mixing with water at Project site to create pourable anchoring, patching, and grouting compound. Provide formulation that is resistant to erosion from water exposure without needing protection by a sealer or waterproof coating and that is recommended by the manufacturer for exterior use.

2.5 FABRICATION

A. General: Fabricate handrails and railings to comply with requirements indicated for design, dimensions, member sizes and spacing, details, finish, and anchorage, but not less than that required to support structural loads.

B. Assemble handrails and railings in the shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation. Use connections that maintain structural value of joined pieces.

C. Form changes in direction of railing members as follows:

1. As detailed.

D. Form simple and compound curves by bending members in jigs to produce uniform curvature for each repetitive configuration required; maintain cylindrical cross section of member throughout entire bend without buckling, twisting, cracking, or otherwise deforming exposed surfaces of handrail and railing components.

E. Welded Connections: Fabricate handrails and railings for connecting members by welding. Cope components at perpendicular and skew connections to provide close fit, or use fittings designed for this purpose. Weld connections continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove flux immediately.

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4. At exposed connections, finish exposed surfaces smooth and blended so that no roughness shows after finishing and welded surface matches contours of adjoining surfaces.

F. Nonwelded Connections: Fabricate handrails and railings by connecting members with concealed mechanical fasteners and fittings, unless otherwise indicated. Fabricate members and fittings to produce flush, smooth, rigid, hairline joints.

1. Fabricate splice joints for field connection using an epoxy structural adhesive where this is manufacturer's standard splicing method.

G. Welded Connections for Aluminum Pipe: Fabricate pipe handrails and railings to interconnect members with concealed internal welds that eliminate surface grinding, using manufacturer's standard system of sleeve and socket fittings.

H. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors to interconnect handrail and railing members to other work, unless otherwise indicated.

I. Provide inserts and other anchorage devices for connecting handrails and railings to concrete or masonry work. Fabricate anchorage devices capable of withstanding loads imposed by handrails and railings. Coordinate anchorage devices with supporting structure.

J. For railing posts set in concrete, provide preset sleeves of steel not less than 150 mm long with inside dimensions not less than 12 mm greater than outside dimensions of post, and steel plate forming bottom closure.

K. For removable railing posts, fabricate slip-fit sockets from steel tube whose ID is sized for a close fit with posts; limit movement of post without lateral load, measured at top, to not more than one-fortieth of post height. Provide socket covers designed and fabricated to resist being dislodged.

1. Provide chain with eye, snap hook, and staple across gaps formed by removable railing sections at locations indicated. Fabricate from same metal as railings.

L. Shear and punch metals cleanly and accurately. Remove burrs from exposed cut edges.

M. Ease exposed edges to a radius of approximately 1 mm, unless otherwise indicated. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing the Work.

N. Cut, reinforce, drill, and tap components, as indicated, to receive finish hardware, screws, and similar items.

O. Provide weep holes or other means to drain entrapped water in hollow sections of handrail and railing members that are exposed to exterior or to moisture from condensation or other sources.

P. Fabricate joints that will be exposed to weather in a watertight manner.

Q. Close exposed ends of handrail and railing members with prefabricated end fittings.

R. Provide wall returns at ends of wall-mounted handrails, unless otherwise indicated. Close ends of returns, unless clearance between end of railing and wall is 6 mm or less.

S. Toe Boards: Where indicated, provide toe boards at railings around openings and at edge of open-sided floors and platforms. Fabricate to dimensions and details indicated.

T. Fillers: Provide fillers made from steel plate, or other suitably crush-resistant material, where needed to transfer wall bracket loads through wall finishes to structural supports. Size fillers to suit wall finish thicknesses and to produce adequate bearing area to prevent bracket rotation and overstressing of substrate.

2.6 FINISHES, GENERAL

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in the same piece are not acceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

D. Provide exposed fasteners with finish matching appearance, including color and texture, of handrails and railings.

2.7 ALUMINUM FINISHES

A. Finish designations prefixed by AA comply with system established by the Aluminum Association for designating aluminum finishes.

B. Mechanical Finish: AA-M12 (Mechanical Finish: nonspecular as fabricated).

C. Class I, Color Anodic Finish: AA-M12C22A42/A44 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, integrally colored or electrolytically deposited color coating 0.018 mm or thicker) complying with AAMA 606.1 or AAMA 608.1.

1. Color: As selected by the Engineer from the full range of industry colors and color densities.

D. Polyester Powder Coating: approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce a hard, durable coating:

1. Surface Finish: As indicated.

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2. Minimum Dry Film Thickness: 60 microns.
3. Color and Surface Finish: As selected by the Engineer from manufacturer's full range.

2.8 STAINLESS-STEEL FINISHES

A. Remove or blend tool and die marks and stretch lines into finish.

B. Grind and polish surfaces to produce uniform, directionally textured polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

C. 180-Grit Polished Finish: Oil-ground, uniform, textured finish.

D. 320-Grit Polished Finish: Oil-ground, uniform, smooth finish.

E. Polished and Buffed Finish: Oil-ground, 180-grit finish followed by buffing.

F. Bright, Directional Polish: No. 4 finish.

G. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

2.9 STEEL FINISHES

A. Galvanized Handrails and Railings: Hot-dip galvanize exterior steel and iron handrails and railings to comply with ASTM A 123. Hot-dip galvanize hardware for exterior steel and iron handrails and railings to comply with ASTM A 153/A 153M.

B. Galvanizing: Hot-dip galvanize items as indicated to comply with applicable standard listed below:

1. ASTM A 123, for galvanizing steel and iron products.
2. ASTM A 153/A 153M, for galvanizing steel and iron hardware.

C. Fill vent and drain holes that will be exposed in finished Work, unless indicated to remain as weep holes, by plugging with zinc solder and filing off smooth.

D. For galvanized handrails and railings, provide galvanized fittings, brackets, fasteners, sleeves, and other ferrous components.

E. For nongalvanized steel handrails and railings, provide nongalvanized ferrous-metal fittings, brackets, fasteners, and sleeves, except galvanize anchors to be embedded in exterior concrete or masonry.

F. Preparation for Shop Priming: After galvanizing, thoroughly clean handrails and railings of grease, dirt, oil, flux, and other foreign matter, and treat with metallic-phosphate process.

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G. Preparation for Shop Priming: Prepare uncoated ferrous-metal surfaces to comply with minimum requirements indicated below for SSPC surface-preparation specifications and environmental exposure conditions of installed handrails and railings:

1. Exteriors (SSPC Zone 1B): SSPC-SP 6, "Commercial Blast Cleaning."
2. Interiors (SSPC Zone 1A): SSPC-SP 7, "Brush-off Blast Cleaning."

H. Apply shop primer to prepared surfaces of handrail and railing components, unless otherwise indicated. Comply with requirements in SSPC-PA 1, "Paint Application Specification No. 1," for shop painting. Primer need not be applied to surfaces to be embedded in concrete or masonry.

1. Do not apply primer to galvanized surfaces.
2. Stripe paint edges, corners, crevices, bolts, and welds.

I. Polyester Powder Coating: approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce produces a hard, durable coating:

1. Surface Finish: As indicated.
2. Minimum Dry Film Thickness: 60 microns.
3. Color and Surface Finish: As selected by the Engineer from manufacturer’s full range.

2.10 EXAMINATION

A. Examine plaster and gypsum board assemblies, where reinforced to receive anchors, to verify that locations of concealed reinforcements have been clearly marked for Installer. Locate reinforcements and mark locations if not already done.

2.11 INSTALLATION, GENERAL

A. Fit exposed connections together to form tight, hairline joints.

B. Perform cutting, drilling, and fitting required to install handrails and railings. Set handrails and railings accurately in location, alignment, and elevation; measured from established lines and levels and free from rack.

1. Do not weld, cut, or abrade surfaces of handrail and railing components that have been coated or finished after fabrication and that are intended for field connection by mechanical or other means without further cutting or fitting.
2. Set posts plumb within a tolerance of 2 mm in 1 m.
3. Align rails so that variations from level for horizontal members and from parallel with rake of steps and ramps for sloping members do not exceed 5 mm in 3 m.

C. Corrosion Protection: Coat concealed surfaces of aluminum that will be in contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

D. Adjust handrails and railings before anchoring to ensure matching alignment at abutting joints. Space posts at interval indicated, but not less than that required by structural loads.

E. Fastening to In-Place Construction: Use anchorage devices and fasteners where necessary for securing handrails and railings and for properly transferring loads to in-place construction.

2.12 RAILING CONNECTIONS

A. Nonwelded Connections: Use mechanical or adhesive joints for permanently connecting railing components. Use wood blocks and padding to prevent damage to railing members and fittings. Seal recessed holes of exposed locking screws using plastic cement filler colored to match finish of handrails and railings.

B. Welded Connections: Use fully welded joints for permanently connecting railing components. Comply with requirements for welded connections in "Fabrication" Article whether welding is performed in the shop or in the field.

C. Expansion Joints: Install expansion joints at locations indicated but not farther apart than required to accommodate thermal movement. Provide slip-joint internal sleeve extending 50 mm beyond joint on either side, fasten internal sleeve securely to one side, and locate joint within 150 mm of post.

2.13 ANCHORING POSTS

A. Use steel pipe sleeves preset and anchored into concrete for installing posts. After posts have been inserted into sleeves, fill annular space between post and sleeve with the following anchoring material, mixed and placed to comply with anchoring material manufacturer's written instructions:

B. Form or core-drill holes not less than 125 mm deep and 20 mm larger than OD of post for installing posts in concrete. Clean holes of loose material, insert posts, and fill annular space between post and concrete with the following anchoring material, mixed and placed to comply with anchoring material manufacturer's written instructions:

1. Nonshrink, nonmetallic grout or anchoring cement.

C. Cover anchorage joint with flange of same metal as post, attached to post as follows:

1. Welded to post after placing anchoring material.
2. By set screws.

D. Leave anchorage joint exposed; wipe off surplus anchoring material; and leave 3 mm build-up, sloped away from post.

E. Anchor posts to metal surfaces with oval flanges, angle type, or floor type as required by conditions, connected to posts and to metal supporting members as follows:

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1. For aluminum pipe railings, attach posts as indicated using fittings designed and engineered for this purpose.
2. For stainless-steel pipe railings, weld flanges to post and bolt to metal supporting surfaces.
3. For steel pipe railings, weld flanges to post and bolt to metal supporting surfaces.

F. Install removable railing sections, where indicated, in slip-fit metal sockets cast in concrete.

2.14 ANCHORING RAILING ENDS

A. Anchor railing ends into concrete and masonry with round flanges connected to railing ends and anchored into wall construction with postinstalled anchors and bolts.

B. Anchor railing ends to metal surfaces with flanges bolted to metal surfaces.

1. Weld flanges to railing ends.
2. Connect flanges to railing ends using nonwelded connections.

2.15 ATTACHING HANDRAILS TO WALLS

A. Attach handrails to wall with wall brackets. Provide bracket with 38 mm clearance from inside face of handrail and finished wall surface.

B. Locate brackets as indicated or, if not indicated, at spacing required to support structural loads.

C. Secure wall brackets to building construction as follows:

1. For concrete and solid masonry anchorage, use drilled-in expansion shields and hanger or lag bolts.
2. For hollow masonry anchorage, use toggle bolts.
3. For wood stud partitions, use hanger or lag bolts set into wood backing between studs. Coordinate with carpentry work to locate backing members.
4. For steel-framed gypsum board assemblies, fasten brackets directly to steel framing or concealed reinforcements using self-tapping screws of size and type required to support structural loads.

2.16 CLEANING

A. Clean aluminum and stainless steel by washing thoroughly with clean water and soap and rinsing with clean water.

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

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2.17 PROTECTION

A. Protect finishes of handrails and railings from damage during construction period with temporary protective coverings approved by railing manufacturer. Remove protective coverings at the time of Substantial Completion.

B. Restore finishes damaged during installation and construction period so that no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit, or provide new units.

END OF SECTION 05521

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SECTION 05530 – GRATINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

B. Related Sections include the following:

1. Steel Specification Section, "Structural Steel" for structural steel framing system components.
2. Woods & Plastics Specification Section "Cast Plastic Fabrications" for GRP gratings.

1.2 SUMMARY

A. This Section includes the following:

1. Metal bar gratings.
2. Heavy-duty metal bar gratings.
3. Expanded metal gratings.
4. Ductile cast iron grating.
5. Extruded-aluminum plank gratings.
6. Metal frames and supports for gratings.

1.3 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide gratings capable of withstanding the following structural loads without exceeding the allowable design working stress of the materials involved, including anchors and connections:
1. Floors: Capable of withstanding a uniform load of 11.97 kN/sq. m or a concentrated load of 13.40 kN, whichever produces the greater stress.
2. Walkways and Elevated Platforms Used as Exits: Capable of withstanding a uniform load of 4.79 kN/sq. m or a concentrated load of 1.33 kN on an area of 25.8 sq. cm, whichever produces the greater stress. Limit deflection to L/360 or 6.4 mm, whichever is less.
3. Sidewalks and Vehicular Driveways: Capable of withstanding a uniform load of 11.97 kN/sq. m or a concentrated load of 35.60 kN, whichever produces the greater stress.
4. Heavy Vehicular Driveways: Capable of withstanding a uniform load of 20 kN/sq. m or a truck wheel load of 130 kN, whichever produces the greater stress.

1.4 SUBMITTALS

A. Product Data: For the following:

1. Formed-metal plank gratings.
2. Extruded-aluminum plank gratings.
3. Clips and anchorage devices for gratings.
4. Paint products.

B. Shop Drawings: Show fabrication and installation details for gratings. Include plans, elevations, sections, and details of connections. Show anchorage and accessory items. Provide templates for anchors and bolts specified for installation under other Sections.

1. For installed products indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

C. Mill Certificates: Signed by manufacturers of stainless-steel sheet certifying that products furnished comply with requirements.

D. Welding Certificates: Copies of certificates for welding procedures and personnel, under the “Quality Assurance” article.

E. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

F. As-Built Drawings: At project close out, submit Record/As-Built Drawings of completed work products, in accordance with requirements of the Specification as indicated in General Requirements.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of gratings that are similar to those indicated for this Project in material, design, and extent.

C. Fabricator Qualifications: A firm experienced in producing gratings similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Metal Bar Grating Standards: Comply with applicable requirements of the following:

1. Non-Heavy-Duty Metal Bar Gratings: Comply with NAAMM MBG 531, "Metal Bar Grating Manual for Steel, Stainless Steel, and Aluminum Gratings and Stair Treads."
2. Heavy-Duty Metal Bar Gratings: Comply with NAAMM MBG 532, "Heavy-Duty Metal Bar Grating Manual."

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E. Welding: Qualify procedures and personnel according to relevant codes of Authorities having jurisdiction.

1. Certify that each welder has satisfactorily passed qualification tests by Authorities having jurisdiction for welding processes involved and, if pertinent, has undergone recertification.

1.6 PROJECT CONDITIONS

A. Field Measurements: Where gratings are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.7 COORDINATION

A. Coordinate installation of anchorages for gratings, grating frames, and supports. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

PART 2 - PRODUCTS

2.1 FERROUS METALS

A. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.

B. Wire Rod for Grating Crossbars: ASTM A 510M.

C. Uncoated Steel Sheet: ASTM A 1011/A 1011M, Grade 33.

D. Galvanized Steel Sheet: ASTM A 653/A 653M, structural quality, Grade 230, with Z275 coating.

E. Expanded Metal, Carbon Steel: ASTM F 1267, Class 1.

F. Expanded Metal, Galvanized: ASTM F 1267, Class 2, Grade A.

G. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy to be welded.

H. Steel Bars for Gratings: ASTM A 569M or ASTM A 36M.

2.2 STAINLESS STEEL

A. Sheet, Strip, Plate, and Flat Bars: ASTM A 666, Type 304.

B. Bars and Shapes: ASTM A 276, Type 304 or 316.

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C. Sheet, Strip, Plate, and Flat Bars: ASTM A 666, Type 316.

D. Expanded Metal: ASTM F 1267, Class 3.

2.3 ALUMINUM

A. Extruded Bars and Shapes: ASTM B 221M, alloys as follows:

1. 6061-T6 or 6063-T6, for bearing bars of gratings and shapes.
2. 6061-T1, for grating crossbars.

B. Aluminum Sheet: ASTM B 209M, alloy 5052-H32.

2.4 PAINT

A. Shop Primers: Provide epoxy zinc rich primers with minimum dry film coating of 40 microns.

B. Shop Primer for Ferrous Metal: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with performance requirements in FS TT-P-664; selected for good resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated, and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure.

C. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds in steel, complying with SSPC-Paint 20.

D. Bituminous Paint: Cold-applied asphalt mastic complying with SSPC-Paint 12, except containing no asbestos fibers, or cold-applied asphalt emulsion complying with ASTM D 1187.

E. Painting to Steel Surfaces:

1. Two coats of epoxy zinc rich primer shop applied.
2. One coat of high build epoxy enamel.
3. One finishing coat of high build polyurethane enamel.

2.5 FASTENERS

A. General: Provide Type 304 or 316 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633, Class Fe/Zn 5, where built into exterior walls. Select fasteners for type, grade, and class required.

B. Fasteners for Aluminum Gratings: Provide fasteners of aluminum, nonmagnetic stainless steel, zinc-plated steel, or other material warranted by the manufacturer to be compatible with aluminum gratings and other components.

C. Bolts and Nuts: Regular hexagon-head bolts, ASTM F 568M, Property Class 4.6; with hex nuts, ASTM A 563M; and, where indicated, flat washers.

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D. Plain Washers: Round, carbon steel, ASME B18.22M.

E. Lock Washers: Helical, spring type, carbon steel, ASME B18.21.2M.

F. Expansion Anchors: Anchor bolt and sleeve assembly of material indicated below with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete, as determined by testing per ASTM E 488, conducted by a qualified independent testing agency.

1. Material: Carbon-steel components zinc-plated to comply with ASTM B 633, Class Fe/Zn 5.
2. Material: Alloy Group 1 or 2 stainless-steel bolts complying with ASTM F 738M and nuts complying with ASTM F 836M.

2.6 FABRICATION

A. Shop Assembly: Fabricate grating sections in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.

B. Form from materials of size, thickness, and shapes indicated, but not less than that needed to support indicated loads.

C. Shear and punch metals cleanly and accurately. Remove burrs.

D. Ease exposed edges to a radius of approximately 1 mm, unless otherwise indicated.

E. Fit exposed connections accurately together to form hairline joints.

F. Welding: Comply with AWS recommendations and the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.

G. Provide for anchorage of type indicated; coordinate with supporting structure. Fabricate and space anchoring devices to secure gratings, frames, and supports rigidly in place and to support indicated loads.

1. Fabricate toeplates to fit grating units and weld to units in shop, unless otherwise indicated.
2. Fabricate toeplates for attaching in the field.
3. Toeplate Height: 100 mm, unless otherwise indicated.

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2.7 METAL BAR GRATINGS

A. Medium and heavy-duty gratings manufactured from Hot-dip galvanized steel, stainless steel grade 304, or stainless steel grade 316. Bearing bars and cross bars are to be of rectangular section with size and spacing as per manufacturer's written recommendation for the span and type of loading indicated on the Drawings. Cross bars are to be locked at right angles to and in the same plane as, the top surface of bearing bars. Properties shall be as follows:

1. Grating Mark: As indicated, but with bearing bar size not less than that required complying with structural performance requirements.
2. Type: Welded.
3. Traffic Surface for Aluminum Bar Gratings: Plain.
4. Traffic Surface for Steel Bar Gratings: Plain.
5. Aluminum Finish: As indicated.
6. Steel Finish: Hot-dip galvanized with a coating weight of not less than 550 g/sq. m of coated surface. Shop paint with epoxy primer and epoxy high build coating.

7. Cut outs: Fabricate cutouts in grating sections for penetrations indicated. Arrange cutouts to permit grating removal without disturbing items penetrating gratings.

    a. Edge band openings in grating that interrupt 4 or more bearing bars with bars of same size and material as bearing bars.
    b. Do not notch bearing bars at supports to maintain elevation.

8. Fabricate removable grating sections with banding bars attached by welding to entire perimeter of each section. Include anchors and fasteners of type indicated or, if not indicated, as recommended by manufacturer for attaching to supports.

    a. Provide not less than four weld lugs for each heavy-duty grating section, with each lug shop welded to two bearing bars.
    b. Furnish threaded bolts with nuts and washers for securing grating to supports.
    c. Provide not less than 4 saddle clips for each grating section composed of rectangular bearing bars 4.8 mm or less in thickness and spaced not less than 24 mm on centers, with each clip designed and fabricated to fit over 2 bearing bars.

B. Ductile Cast Iron Grating: Medium and heavy-duty gratings non-rock type, manufactured from ductile cast iron to ASTM A 536, grade 65-45-12 for medium duty and grade 80-55-06 for heavy duty. All castings shall be free from blow holes, shrinkage defects, swells, cracks and other defects. Heavy-duty gratings are to be capable of supporting 13 tons axle loads.

2.8 EXPANDED METAL GRATINGS

A. Provide expanded metal gratings in material, finish, style, size, thickness, weight, and type indicated or, if not indicated, as recommended by manufacturer for indicated applications and as needed to support indicated loads.

1. Material: Stainless Steel.
2. Stainless-Steel Finish: Mill, as fabricated.
3. Type: II, expanded and flattened.

B. Fabricate cutouts in grating sections for penetrations of sizes and at locations indicated. Cut openings neatly and accurately to size. Edge-band openings with bars having a thickness not less than overall grating thickness at contact points.

C. Where gratings are pierced by pipes, ducts, and structural members, cut openings neatly and accurately to size and weld a strap collar not less than 3 mm thick to the cut ends. Divide panels into sections only to the extent required for installation where grating platforms and runways are to be placed around previously installed pipe, ducts, and structural members.

2.9 EXTRUDED-ALUMINUM PLANK GRATINGS

A. Provide extruded-aluminum plank gratings in type, size, and finish indicated or, if not indicated, as recommended by manufacturer for indicated applications and as needed to support indicated loads.

1. Type: Extruded-aluminum planks approximately 152 mm wide with multiple flanges approximately 30 mm o.c., acting as bearing bars connected by a web that serves as a walking surface. Top surface has raised ribs to increase slip resistance.
2. Depth: As needed to support indicated loads.
3. Perforations: As fabricated.
4. Finish: Mill, as fabricated.

2.10 GRATING FRAMES AND SUPPORTS

A. General: Provide steel frames and supports, where indicated.

B. Steel Frames and Supports: Fabricate from structural-steel shapes, plates, and bars of welded construction to sizes, shapes, and profiles indicated and as necessary to receive gratings. Miter and weld connections for perimeter angle frames. Cut, drill, and tap units to receive hardware and similar items.

C. Aluminum Frames: Fabricate frames for aluminum gratings from extruded-aluminum shapes to sizes, shapes, and profiles indicated and as necessary to receive gratings. Miter and weld connections. Cut, drill, and tap units to receive hardware and similar items.

D. Equip units with integrally welded anchors for casting into concrete or building into masonry.

1. Unless otherwise indicated, space anchors 600 mm o.c. and provide minimum anchor units in the form of steel straps 32 mm wide by 6 mm thick by 200 mm long.

E. Galvanize frames and supports in the following locations:

1. Exterior.
2. Interior, where indicated.

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2.11 FINISHES

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Finish gratings, frames, and supports after assembly.

C. Galvanizing: For those items indicated for galvanizing, apply zinc coating by the hot-dip process complying with ASTM A 123.

D. Preparation for Shop Priming: Prepare uncoated ferrous-metal surfaces to comply with minimum requirements indicated below for SSPC surface-preparation specifications and environmental exposure conditions of installed items:

1. Exteriors (SSPC Zone 1B): SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
2. Interiors (SSPC Zone 1A): SSPC-SP 3, "Power Tool Cleaning."

E. Apply shop primer to uncoated surfaces of gratings, frames, and supports, except those with galvanized finishes and those to be embedded in concrete or masonry, unless otherwise indicated. Comply with SSPC-PA 1, "Paint Application Specification No. 1," for shop painting.

PART 3 - EXECUTION

3.1 PREPARATION

A. Coordinate and furnish anchorages, setting drawings, diagrams, templates, instructions, and directions for installation of anchorages, including sleeves, anchor bolts, and miscellaneous items having integral anchors that are to be embedded in concrete or masonry construction. Coordinate delivery of such items to Project site.

3.2 INSTALLATION, GENERAL

A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing gratings to in-place construction. Include threaded fasteners for concrete and masonry inserts, through-bolts, lag bolts, and other connectors.

B. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing gratings. Set units accurately in location, alignment, and elevation; measured from established lines and levels and free from rack.

C. Provide temporary bracing or anchors in formwork for items that are to be built into concrete or masonry.

D. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade the surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.

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E. Attach toeplates to gratings by welding at locations indicated.

F. Field Welding: Comply with the following requirements:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.

G. Corrosion Protection: Coat concealed surfaces of aluminum that will come into contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

3.3 INSTALLING METAL BAR GRATINGS

A. General: Install gratings to comply with recommendations of referenced metal bar grating standards that apply to grating types and bar sizes indicated, including installation clearances and standard anchoring details.

B. Attach removable units to supporting members with type and size of clips and fasteners indicated or, if not indicated, as recommended by grating manufacturer for type of installation conditions shown.

C. Attach non-removable units to supporting members by welding where both materials are the same; otherwise, fasten by bolting as indicated above.

D. Attach toe plates to gratings by welding at locations indicated.

3.4 INSTALLING METAL PLANK GRATINGS

A. General: Comply with manufacturer's written instructions for installing gratings. Use manufacturer's standard anchor clips and hold-down devices for bolted connections.

B. Attach removable units to supporting members by bolting at every point of contact.

C. Attach nonremovable units to supporting members by welding, unless otherwise indicated. Comply with manufacturer's written instructions for size and spacing of welds.

D. Attach aluminum units to steel supporting members by bolting at side channels at every point of contact and by bolting intermediate planks at each end on alternate sides. Bolt adjacent planks together at midspan.

3.5 ADJUSTING AND CLEANING

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A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with the same material as used for shop painting to comply with SSPC-PA 1 requirements for touching up shop-painted surfaces.

   1. Apply by brush or spray to provide a minimum 0.05 mm dry film thickness.

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

END OF SECTION 05530

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SECTION 05580 - FORMED-METAL FABRICATIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Closures and trim.
2. Filler panels.
3. Column covers.
4. Mezzanine floor deck fascia panels.
5. Lighting troughs.
6. Heating-cooling unit enclosures.

B. Related Sections include the following:

1. Steel Specification Section "Ornamental Metal" for items made primarily from plate, bars, extrusions, tubes, castings, and other forms of metal, but which may include sheet metal components.

1.3 PERFORMANCE REQUIREMENTS

A. Structural Performance: Design, engineer, and fabricate column covers and heating-cooling unit enclosures so that, when installed, they are capable of withstanding the following structural loads without exceeding the allowable design working stress of the materials, including anchors and connections, and without exhibiting permanent deformation including surface warping or oilcanning in any of the components making up the fabrications:

1. 9.6 kN/sq. m or 2.2 kN/m, whichever produces the greatest stress.

B. Corrosion Control: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

1.4 SUBMITTALS

A. Product Data: For the following:

1. Column covers.

2. Sound-deadening insulation.
3. Insulation adhesive.
4. Gaskets.
5. Structural anchors.
6. Nonstructural anchors.
7. Paint products.

B. Shop Drawings: Detail fabrication and installation of the following formed-metal fabrications. Include plans, elevations, sections, and details of components and their connections. Show anchorage and accessory items.

1. Closures and trim.
2. Filler panels.
3. Column covers.
4. For installed products indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

C. Coordination Drawings: For formed-metal fabrications housing items specified in other Sections. Show dimensions of housed items, including locations of housing penetrations and attachments, and necessary clearances.

D. Samples for Initial Selection: Manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available for each type of formed-metal fabrication indicated.

E. Samples for Verification: For each type of exposed finish required, prepared on 150 mm square samples of metal of same thickness and material indicated for the Work. If finishes involve normal color and texture variations, include sample sets showing the full range of variations expected.

F. Mill Certificates: Signed by manufacturers of stainless-steel sheet certifying that products furnished comply with requirements.

G. As-Built Drawings: At project close out, submit Record/As-Built Drawings of completed work products, in accordance with requirements of the Specification as indicated in General Requirements.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Professional Engineer Qualifications: A professional engineer who is legally registered and qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of heating-cooling unit enclosures that are similar to those indicated for this Project in material, design, and extent.

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C. Fabricator Qualifications: A firm experienced in producing formed-metal fabrications similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Source Limitations: Obtain formed-metal fabrications through one source from a single manufacturer.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver formed-metal fabrications wrapped in protective coverings and strapped together in suitable packs or in heavy-duty cartons. Remove protective coverings before they stain or bond to finished surfaces.

B. Store products on elevated platforms in a dry location.

1.7 PROJECT CONDITIONS

A. Field Measurements: Where formed-metal fabrications are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

2.1 SHEET METAL

A. General: Provide sheet metal without pitting, seam marks, roller marks, stains, discolorations, or other imperfections where exposed to view on finished units.

B. Aluminum Sheet: Flat sheet complying with ASTM B 209M, alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than strength and durability properties of alloy 5005-H15.

C. Galvanized Steel Sheet: ASTM A 653/A 653M, Z275 coating, either commercial quality or lock-forming quality, stretcher-leveled standard of flatness.

D. Electrolytic Zinc-Coated Steel Sheet: ASTM A 591/A 591M, Class C coating, commercial quality, stretcher-leveled standard of flatness.

E. Uncoated, Cold-Rolled Steel Sheet: ASTM A 366/A 366M, matte finish, stretcher-leveled standard of flatness. Electrolytic zinc-coated steel sheet complying with ASTM A 591/A 591M, Class C coating, commercial quality, stretcher-leveled standard of flatness, may be substituted at fabricator's option.

F. Stainless-Steel Sheet: ASTM A 167, Type 304, stretcher-leveled standard of flatness, with the following mill finish:

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1. Polished for surfaces requiring little or no polishing after fabrication to produce the uniform polished finish specified under the Article "Stainless-Steel Finishes".

G. Stainless-Steel Sheet: ASTM A 666, Type 316, stretcher-leveled standard of flatness.

H. Bronze Sheet: ASTM B 36/B 36M, alloy UNS No. C28000 (Muntz metal, 60 percent copper) or alloy UNS No. C23000 (red brass, 85 percent copper).

I. Brass Sheet: ASTM B 36/B 36M, alloy UNS No. C26000 (cartridge brass, 70 percent copper).

2.2 MISCELLANEOUS MATERIALS

A. Sound-Deadening Insulation: Unfaced, mineral-fiber blanket insulation complying with ASTM C 665, Type I, and passing ASTM E 136 test.

B. Insulation Adhesive: Duct thermal-insulation adhesive complying with ASTM C 916, Type I or Type III.

C. Flexible Cellular Neoprene Gaskets: ASTM D 1056, Type 1, Class A, grade as recommended by gasket manufacturer to obtain airtight seal for application indicated.

D. Joint Sealants for Concealed Joints: One-part, nonsag, solvent-release-curing, polymerized butyl sealant complying with ASTM C 1085 and formulated with minimum of 75 percent solids.

E. Filler Metal and Electrodes: Provide type and alloy of filler metal and electrodes as recommended by producer of metal to be welded or brazed and as necessary for strength, corrosion resistance, and compatibility in fabricated items.

1. Use filler metals that will match the color of metal being joined and will not cause discoloration.

F. Fasteners: Use fasteners fabricated from same basic metal and alloy as fastened metal, unless otherwise indicated. Do not use metals that are corrosive or incompatible with materials joined.

1. Provide concealed fasteners for interconnecting formed-metal fabrications and for attaching them to other work, unless otherwise indicated.
2. Provide concealed fasteners for interconnecting formed-metal fabrications and for attaching them to other work, unless exposed fasteners are unavoidable or are the standard fastening method.
3. Provide Phillips flat-head machine screws for exposed fasteners, unless otherwise indicated.

G. Structural Anchors: For applications indicated to comply with certain design loads, provide anchors of type indicated below, fabricated from corrosion-resistant materials with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete, as determined by testing per ASTM E 488 conducted by a qualified independent testing agency.

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1. Chemical anchors.
2. Torque-controlled expansion anchors.

H. Nonstructural Anchors: For applications not indicated to comply with design loads, provide anchors of type, size, and material necessary for type of load and installation indicated, as recommended by the manufacturer from options listed below, unless otherwise indicated. Use nonferrous-metal or hot-dip galvanized anchors for exterior installations and elsewhere as needed for corrosion resistance.

1. Metal expansion sleeve anchors.
2. Metal impact expansion anchors.

I. Fire-Resistive Mastic: Cold-applied synthetic elastomer adhesive complying with ASTM C 916.

2.3 PAINT

A. Shop Primers: Provide primers that comply with requirements.

B. Shop Primer for Ferrous Metal: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with performance requirements in FS TT-P-664; selected for good resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated, and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure.

C. Shop Primer for Galvanized Steel: Zinc-dust, zinc-oxide primer formulated for priming zinc-coated steel and for compatibility with finish paint systems indicated; complying with SSPC-Paint 5.

D. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds in steel, complying with SSPC-Paint 20.

E. Bituminous Paint: Cold-applied asphalt mastic complying with SSPC-Paint 12, except containing no asbestos fibers, or cold-applied asphalt emulsion complying with ASTM D 1187.

2.4 FABRICATION, GENERAL

A. General: Fabricate sheet metal to comply with requirements indicated for design, dimensions, materials, joinery, and performance.

B. Shop Assembly: Preassemble formed-metal fabrications in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.

C. Coordinate dimensions and attachment methods of formed-metal fabrications with those of adjoining construction to produce integrated assemblies with closely fitting joints and with edges and surfaces aligned, unless otherwise indicated.

D. Form metal to profiles indicated, in maximum lengths to minimize joints. Produce flat, flush surfaces without cracking or grain separation at bends. Fold back exposed edges of unsupported sheet metal to form a 12 mm wide hem on the concealed side, or ease edges to a radius of approximately 1 mm and support with concealed stiffeners.

E. Increase metal thickness or reinforce with concealed stiffeners, backing materials, or both, as needed to provide surface flatness specified for stretcher-leveled sheet metal and sufficient strength for indicated use.

1. Support joints with concealed stiffeners as needed to hold exposed faces of adjoining sheets in flush alignment.

F. Build in straps, plates, and brackets as needed to support and anchor fabricated items to adjoining construction. Reinforce formed-metal units as needed to attach and support other construction.

G. Provide support framing, mounting and attachment clips, splice sleeves, fasteners, and accessories needed to install formed-metal fabrications.

H. Where welding or brazing is indicated, weld or braze joints and seams continuously. Grind, fill, and dress to produce smooth, flush, exposed surfaces in which joints are not visible after finishing is completed.

I. Use welding and brazing procedures that will blend with and not cause discoloration of metal being joined.

2.5 CLOSURES AND TRIM

A. Form closures and trim from metal of type and thickness indicated below. Fabricate to fit tightly to adjoining construction.

1. Metal for Interior Installations: Steel sheet, 1.1 mm.
2. Metal for Exterior Installations: Galvanized steel sheet, 1.3 mm, with weathertight joints at exterior installations.
3. Closures and trim may be fabricated from prefinished metal sheet in lieu of finishing after fabrication provided unfinished edges are concealed from view and not exposed to weather.

B. Conceal fasteners where possible; otherwise, locate where they will be as inconspicuous as possible. Size fasteners to support closures and trim, with fasteners spaced to prevent buckling or waviness in finished surfaces.

C. Drill and tap holes needed for securing closures and trim to other surfaces.

D. Incorporate gaskets where indicated or needed for concealed, continuous seal at abutting surfaces.

E. Miter or cope trim members at corners and reinforce with bent metal splice plates to form tight joints.

2.6 FILLER PANELS

A. Form filler panels for closing ends of partition systems and for other applications indicated. Form from two sheets of metal of type and thickness indicated below, separated by channels formed from the same material, producing a panel of same thickness as partitions, unless otherwise indicated. Incorporate reveals, trim, and concealed anchorages for attaching to adjacent surfaces.

1. Galvanized Steel Sheet: 1.6 mm.
2. Steel Sheet: 1.3 mm.
3. Filler panels may be fabricated from prefinished metal sheet in lieu of finishing after fabrication provided unfinished edges are concealed from view.

B. Fill interior of panel with sound-deadening insulation permanently attached to inside panel faces.

C. Adhesively attach gaskets to filler panel edges that abut glass. Use 25 mm square material, unless otherwise indicated, set approximately 6 mm into channeled edge of filler panel.

D. Attach gaskets to all edges of panels that abut adjacent surfaces to form a continuous seal. Use compressible gaskets or mastic sealing tape, applied to center of panel edges to be concealed from view, unless otherwise indicated.

2.7 COLUMN COVERS

A. Spackled-Seam Type: Form column covers from 3.2 mm aluminum, rolled to radii indicated. Taper edges of adjoining pieces of column covers, for taping and spackling, to 2.4 mm thickness in approximately 25 mm of width. Punch tapered edges for gypsum board screws at 12 mm o.c., and mill grooves in tapered edge to improve bond with joint compound.

1. Support Framing: At vertical joints, provide 40-by-90 mm steel channel support posts formed from 1.0 mm galvanized steel.

B. Snap-Together Type: Form column covers to shapes indicated from metal of type and thickness indicated below. Return vertical edges and bend to form hook that will engage continuous mounting clips.

1. Aluminum Sheet: 3.2 mm thick.
2. Steel Sheet: 1.3 mm.
3. Stainless-Steel Sheet: 1.3 mm.
4. Bronze Sheet: 1.3 mm.
5. Brass Sheet: 1.3 mm.
6. Column covers may be fabricated from prefinished metal sheet in lieu of finishing after fabrication provided unfinished edges are concealed from view.
7. Form returns at vertical joints to provide 12 mm wide reveal at joints. Provide snap-in metal filler strips at reveals that leave reveals 12 mm deep.
8. Fabricate column covers without horizontal joints.
9. Apply a coating of heavy-build, sound-deadening mastic to backs of column covers.

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2.8 MEZZANINE FLOOR DECK FASCIA PANELS

A. Form fascia panels from sheet metal of type and minimum nominal thickness indicated below, with end closures.

1. Material: Galvanized steel sheet, minimum 1.3 mm thick.

B. Provide hat-shaped channel stiffeners on back surface of panels to prevent surface warping or oilcanning of panel face.

2.9 LIGHTING TROUGHS

A. Form lighting troughs from sheet metal of type and thickness indicated below. Coordinate size of troughs, location of cutouts for electrical wiring, light fixtures, mechanical ducts, registers, grills, structural duct supports, and method of attachment to adjoining construction.

1. Material: Galvanized steel sheet, 1.6 mm thick.

B. Provide hat-shaped channel stiffeners on back surface of panels as required to prevent surface warping or oilcanning of panel face.

2.10 HEATING-COOLING UNIT ENCLOSURES

A. Fabricate heating-cooling unit enclosures from sheet metal of type and thickness indicated below:

1. Material: Galvanized steel sheet, minimum thicknesses as follows:

a. Framing: 2.8 mm.
b. Sills and Stools: 2.0 mm.
c. Front Panels and Bases: 1.6 mm.
d. Concealed Panels and Trim: 1.0 mm.

B. Incorporate stiffeners or laminated backing using noncombustible materials as required for strength and rigidity. Fill space between stiffeners with sound-deadening insulation attached to face sheet with fire-resistive mastic, unless otherwise indicated.

C. Provide louvers and grilles of size, type, and materials indicated.

1. For removable grilles, use modular units, with recessed openings formed into surfaces of enclosures, and without blank filler panels between grilles, so that face panels and stools are continuous. Fabricate removable grilles and openings to precise tolerances to produce well-fitted assemblies free of warp or rattle, with grilles supported continuously along parallel edges, and with tops flush with top of enclosure.

D. Incorporate removable tops and fronts where indicated or required for access to heating-cooling units and to piping, ductwork, controls, and electrical service, with panels and the openings as follows:

1. Fabricate with a fitting tolerance of not less than 1 mm and not more than 2.5 mm at each edge, with face of panels flush with adjoining fixed surfaces of enclosure.
2. Form panels for easy removal without interfering with adjoining construction or fixed items. Hold panels in place with concealed clips and hardware that prevent warp and rattle.

E. Incorporate hinged access panels in enclosures for access to heating-cooling unit controls, either as separate elements or integrated with grille openings, as indicated or required. In areas accessible to the public, provide keyed locking device or special tool for securing hinged access panels.

F. Coordinate construction, configuration, and dimensions of enclosures with those of heating-cooling units. Provide support for heating-cooling units and controls where indicated. Provide blind knockouts and supports for piping, ductwork, control lines, electrical conduit, and wiring where indicated or required.

G. Locate fixed surfaces of enclosures to coordinate with window mullions and partition system terminations. Provide closures at ends of units, at recessed openings in base of units, and at other locations where required to conceal from view unfinished wall or floor surfaces, piping, conduit, ductwork, or heating-cooling units.

1. Provide built-in partitions (bulkheads) within enclosures between heating-cooling units, located to coincide with mullions and partition system terminations. Fabricate panels in a manner similar to exposed filler panels to prevent sound transmission.

H. Provide sound deadening for concealed faces of metal panels over 150 mm wide, consisting of a heavy coating of fire-resistive mastic applied at the minimum rate of 0.5 sq. m/L. Apply sound-deadening coating after completing shop finishing.

2.11 FINISHES, GENERAL

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Comply with SSPC-PA 1, "Paint Application Specification No. 1," for steel sheet finishes.

C. Complete mechanical finishes of flat sheet metal surfaces before fabrication where possible. After fabrication, finish all joints, bends, abrasions, and other surface blemishes to match sheet finish.

D. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

E. Apply organic and anodic finishes to formed metal after fabrication, unless otherwise indicated.

2.12 ALUMINUM FINISHES

A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.

B. Class I, Clear Anodic Finish: AA-M12C22A41 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, clear coating 0.018 mm or thicker) complying with AAMA 607.1.

C. Class I, Color Anodic Finish: AA-M12C22A42/A44 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, integrally colored or electrolytically deposited color coating 0.018 mm or thicker) complying with AAMA 606.1 or AAMA 608.1.

1. Color: As selected by the Engineer from the full range of industry colors and color densities.

D. Baked-Enamel Finish: AA-C12C42R1x (Chemical Finish: cleaned with inhibited chemicals; Chemical Finish: acid-chromate-fluoride-phosphate conversion coating; Organic Coating: as specified below). Apply baked enamel complying with paint manufacturer's written specifications for cleaning, conversion coating, and painting.

1. Organic Coating: Thermosetting, modified-acrylic enamel primer/topcoat system complying with AAMA 603.8 except with a minimum dry film thickness of 0.04 mm, medium gloss.
2. Color: As selected by the Engineer from manufacturer's full range.
3. Fluoropolymer 3-Coat Coating System: Manufacturer's standard 3-coat thermocured system composed of specially formulated inhibitive primer, fluoropolymer color coat, and clear fluoropolymer topcoat, with both color coat and clear topcoat containing not less than 70 percent polyvinylidene resin by weight; complying with AAMA 605.2.

    a. Color and Gloss: As selected by the Engineer from manufacturer's full range of choices for color and gloss.

E. Polyester Powder Coating: Approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce produces a hard, durable coating:

1. Surface Finish: As indicated.
2. Minimum Dry Film Thickness: 60 microns.
3. Color and Gloss: As selected by the Engineer from manufacturer's full range.

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2.13 GALVANIZED STEEL SHEET FINISHES

A. Surface Preparation: Clean surfaces with nonpetroleum solvent so surfaces are free of oil and other contaminants. After cleaning, apply a conversion coating suited to the organic coating to be applied over it. Clean welds, mechanical connections, and abraded areas, and apply galvanizing repair paint to comply with ASTM A 780.

1. Galvanizing Repair Paint: High-zinc-dust-content paint for regalvanizing welds in galvanized steel, with dry film containing not less than 94 percent zinc dust by weight, and complying with DOD-P-21035 or SSPC Paint 20.

B. Factory Priming for Field-Painted Finish: Where field painting after installation is indicated, apply air-dried primer immediately after cleaning and pretreating.

C. Baked-Enamel Finish: Immediately after cleaning and pretreating, apply manufacturer's standard two-coat, baked-enamel finish consisting of prime coat and thermosetting topcoat, with a minimum dry film thickness of 0.025 mm for topcoat. Comply with paint manufacturer's written instructions for applying and baking to achieve a minimum dry film thickness of 0.05 mm.

1. Color and Gloss: As selected by the Engineer from manufacturer's full range.

D. Polyester Powder Coating: Approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce a hard, durable coating:

1. Surface Finish: As indicated.
2. Minimum Dry Film Thickness: 60 microns.
3. Color and Gloss: As selected by the Engineer from manufacturer's full range.

E. High-Performance Organic Coating Finish: Apply the following system by coil-coating process on galvanized steel sheet as recommended by coating manufacturers and applicator.

1. Fluoropolymer 2-Coat Coating System: Manufacturer's standard 2-coat thermocured system composed of specially formulated inhibitive primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with AAMA 605.2.

a. Color and Gloss: As selected by the Engineer from manufacturer's full range of choices for color and gloss.

2.14 STEEL SHEET FINISHES

A. Surface Preparation: Clean surfaces to comply with SSPC-SP 1, "Solvent Cleaning," to remove dirt, oil, grease, or other contaminants that could impair paint bond. Remove mill scale and rust, if present, from uncoated steel, complying with SSPC-SP 5/NACE No. 1, "White Metal Blast Cleaning," or SSPC-SP 8, "Pickling."

B. Pretreatment: Immediately after cleaning, apply a conversion coating of type suited to organic coating applied over it.

C. Factory Priming for Field-Painted Finish: Immediately after cleaning and pretreating, apply shop primer.

D. Baked-Enamel Finish: Immediately after cleaning and pretreating, apply manufacturer's standard two-coat, baked-enamel finish consisting of prime coat and thermosetting topcoat. Comply with paint manufacturer's written instructions for applying and baking to achieve a minimum dry film thickness of 0.05 mm.

E. Polyester Powder Coating: approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce a hard, durable coating:

1. Surface Finish: As indicated.
2. Minimum Dry Film Thickness: 60 microns.
3. Color and Gloss: As selected by the Engineer from manufacturer's full range.

2.15 STAINLESS-STEEL FINISHES

A. Remove tool and die marks and stretch lines or blend into finish.

B. Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

C. Bright, Cold-Rolled, Unpolished Finish: No. 2B finish.

D. Bright, Directional Polish: No. 4 finish.

E. Satin, Directional Polish: No. 6 finish.

F. Satin, Reflective, Directional Polish: No. 7 finish.

G. Mirrorlike Reflective, Nondirectional Polish: No. 8 finish.

H. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

2.16 COPPER-ALLOY FINISHES

A. Finish designations prefixed by CDA comply with the system established by the Copper Development Association for designating copper-alloy finishes.

B. Buffed Finish, Lacquered: CDA M21-O6x (Mechanical Finish: buffed, smooth specular; Coating: clear organic, air drying, as specified below).

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1. Clear, Organic Coating: Clear, air-drying, acrylic lacquer specially developed for coating copper-alloy products, applied by air spray in two coats per manufacturer's written instructions, with interim drying, to a total thickness of 0.025 mm.

    a. Product: Subject to compliance with requirements, provide "Incralac" developed by the International Copper Research Corporation.

C. Hand-Rubbed Finish, Lacquered: CDA M31-M34-O6x (Mechanical Finish: directionally textured, fine satin; Mechanical Finish: directionally textured, hand rubbed; Coating: clear organic, air drying, as specified below).

1. Clear, Organic Coating: Clear, air-drying, acrylic lacquer specially developed for coating copper-alloy products, applied by air spray in two coats per manufacturer's written instructions, with interim drying, to a total thickness of 0.025 mm.

    a. Product: Subject to compliance with requirements, provide "Incralac" developed by the International Copper Research Corporation.

D. Statuary Conversion Coating over Satin Finish: CDA-M32-C55 (Mechanical Finish: directionally textured, fine satin; Chemical Finish: conversion coating, sulfide).

1. Color: Match the Engineer's sample.

PART 3 - EXECUTION

3.1 PREPARATION

A. Coordinate and furnish anchorages and setting drawings, diagrams, templates, instructions, and directions for installing items having integral anchors that are to be embedded in concrete or masonry construction. Coordinate delivery of such items to the Project site.

3.2 INSTALLATION

A. Locate and place formed-metal fabrications level, plumb, and in alignment with adjacent construction.

B. Use concealed anchorages where possible. Provide brass or lead washers fitted to screws where needed to protect metal surfaces and to make a weathertight connection.

C. Form tight joints with exposed connections accurately fitted together. Provide reveals and openings for sealants and joint fillers as indicated.

D. Install concealed gaskets, joint fillers, insulation, and flashings, as the Work progresses, to make exterior formed-metal fabrications weatherproof.

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E. Install concealed gaskets, joint fillers, and insulation, as the Work progresses, to make interior formed-metal fabrications soundproof or lightproof as applicable to the type of fabrication indicated.

F. Corrosion Protection: Coat concealed surfaces of aluminum, zinc-coated, and nonferrous metals that will come into contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

G. Drill and tap holes in substrates and other surfaces for securing closures and trim.

3.3 ADJUSTING

A. Restore finishes damaged during installation and construction period so that no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit or provide new units.

3.4 PROTECTION

A. Protect finishes of formed-metal fabrications from damage during construction period. Remove temporary protective coverings at time of Substantial Completion.

END OF SECTION 05580

SECTION 05700 - ORNAMENTAL METAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Ornamental railings fabricated from custom shapes.
2. Ornamental window grilles (security bars).
3. Ornamental mechanical grilles and frames.
4. Custom door pulls.
5. Metal reveals at wood paneling.
6. Cast-metal rosettes at marble joints.

B. Related Sections include the following:

1. Steel Specification Section "Metal Fabrications" for non-ornamental metal fabrications.
2. Steel Specification Section "Formed-Metal Fabrications" for metal fabrications made from sheet metal.
3. Steel Specification Section "Ornamental Handrails and Railings" for ornamental metal handrails and railings fabricated from stock components.

1.3 PERFORMANCE REQUIREMENTS FOR HANDRAILS AND RAILINGS

A. Structural Performance of Handrails and Railings: Provide handrails and railings complying with requirements of ASTM E 985 for structural performance, based on testing performed according to ASTM E 894 and ASTM E 935.

B. Thermal Movements: Provide handrails and railings that allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, overstressing of components, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change (Range): 67 deg. C, ambient; 100 deg. C, material surfaces.

C. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

1.4 SUBMITTALS

A. Product Data: For each product used in ornamental metal, including finishing materials and methods.

1. Include Product Data for grout and anchoring cement.

B. Shop Drawings: Show fabrication and installation of ornamental metal. Include plans, elevations, component details, and attachments to other Work. Indicate materials and profiles of each ornamental metal member, fittings, joinery, finishes, fasteners, anchorages, and accessory items.

1. Include setting drawings, templates, and directions for installing anchor bolts and other anchorages.

C. Patterns, models, or plaster castings made from proposed patterns for each design of custom casting required.

D. Samples for Initial Selection: Manufacturer's color charts showing the full range of colors and other finish characteristics available for each item indicated below:

1. Ornamental metal consisting of color-anodized aluminum components.
2. Ornamental metal with baked-enamel coating.
3. Ornamental metal with high-performance coating.
4. Ornamental metal with electrostatically applied polyester powder coating.

E. Samples for Verification: For each profile and pattern of fabricated metal and for each type of metal finish required, prepared on metal of same thickness and alloy indicated for the Work. If finishes involve normal color and texture variations, include sample sets, consisting of two or more units, showing the full range of variations expected.

1. Include 150 mm long samples of linear shapes.
2. Include 150 mm square samples of plates.
3. Include full-size samples of castings and forgings.

a. For custom castings, submit finished samples of similar previous work to show ability to reproduce detail, cast-metal color, and quality of finish.

F. Welding Certificates: Copies of certificates for welding procedures and personnel.

G. Qualification Data: For firms and persons specified in the "Quality Assurance" Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects/engineers and owners, and other information specified.

1.5 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Installer Qualifications: Arrange for installation of ornamental metal specified in this Section by the same firm that fabricated it.

C. Fabricator Qualifications: A firm experienced in producing ornamental metal similar to that indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Coating Applicator Qualifications: A firm experienced in successfully applying coatings of type indicated to aluminum extrusions and employing competent control personnel to conduct continuing, effective quality-control program to ensure compliance with requirements.

E. Anodic Finisher Qualifications: A firm experienced in successfully applying anodic finishes of type indicated and employing competent control personnel to conduct continuing, effective quality-control program to ensure compliance with requirements.

F. Testing Agency Qualifications: An independent testing agency acceptable to the Engineer with the experience and capability to conduct the testing indicated, as documented according to ASTM E 548.

G. Preconstruction Testing Service: Engage a qualified independent testing agency to test handrails and railings for compliance with specified requirements for performance and test methods. Conduct tests using specimens and assemblies representative of proposed materials and construction.

1. Fabricate and install test assemblies using personnel who will perform the same tasks for Project.
2. Select sizes and configurations of assemblies to adequately demonstrate capability of handrails and railings to comply with performance requirements.
3. Notify the Engineer seven days in advance of dates and times when assemblies will be constructed.
4. When testing is complete, remove assemblies; do not reuse materials on Project.

H. Welding Standards: Qualify procedures and personnel according to codes published by Authorities having jurisdiction..

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store ornamental metal inside a well-ventilated area, away from uncured concrete and masonry, and protected from weather, moisture, soiling, abrasion, extreme temperatures, and humidity.

B. Deliver and store cast-metal products in wooden crates surrounded by sufficient excelsior to ensure that products will not be cracked or otherwise damaged.

1.7 PROJECT CONDITIONS

A. Field Measurements: Where ornamental metal is indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate

measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1.8 COORDINATION

A. Coordinate installation of anchorages for ornamental metal items. Furnish Setting Drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Handrails and Railings:

2. Ornamental Mechanical Grilles:

3. Ornamental Castings and Forgings:

4. Stainless-Steel Wire Rope and Fittings:

2.2 METALS

A. General: Provide metals free from surface blemishes where exposed to view in finished unit. Exposed-to-view surfaces exhibiting pitting, seam marks, roller marks, stains, discolorations, or other imperfections on finished units are not acceptable.

B. Aluminum: Alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated, and with not less than the strength and durability properties of alloy and temper designated below for each aluminum form required:

1. Extruded Bars and Shapes: ASTM B 221M, alloy 6063-T6.
2. Extruded Pipe and Tubes: ASTM B 429, alloy 6063-T6.
3. Drawn Seamless Tubes: ASTM B 483M, alloy 6063-T832.
4. Plate and Sheet: ASTM B 209M, alloy 6061-T6.
5. Die and Hand Forgings: ASTM B 247M, alloy 6061-T6.
6. Castings: ASTM B 26/B 26M, alloy A356-T6.

C. Copper Alloys, Bronze: Provide copper alloy of type and form indicated to comply with the following requirements:

1. Extruded Shapes: ASTM B 455, alloy UNS No. C38500 (extruded architectural bronze).

2. Plate, Sheet, Strip, and Bars: ASTM B 36/B 36M, alloy UNS No. C28000 (Muntz metal, 60 percent copper).
3. Seamless Pipe: ASTM B 43, alloy UNS No. C23000 (red brass, 85 percent copper).
4. Seamless Tubes: ASTM B 135M, alloy UNS No. C23000 (red brass, 85 percent copper).
5. Composition Bronze Castings: ASTM B 62, alloy UNS No. C83600 (85-5-5-5 or composition bronze).
6. Sand Castings: ASTM B 584, alloy UNS No. C86500 (No. 1 manganese bronze).

D. Copper Alloys, Brass: Provide copper alloy of type and form indicated to comply with the following requirements:

1. Extruded Shapes: ASTM B 249M, alloy UNS C36000 (free-cutting brass).
2. Plate, Sheet, Strip, and Bars: ASTM B 36/B 36M, alloy UNS No. C26000 (cartridge brass, 70 percent copper).
3. Seamless Tubes: ASTM B 135M, alloy UNS No. C26000 (cartridge brass, 70 percent copper).
4. Sand Castings: ASTM B 584, alloy UNS No. C85200 (high-copper yellow brass).

E. Stainless Steel: Grade and type designated below for each form required:

1. Tubing: ASTM A 554, Grade MT 304 or 316.
2. Pipe: ASTM A 312/A 312M, Grade TP 304 or 316.
3. Castings: ASTM A 743/A 743M, Grade CF 8 or Grade CF 20.
4. Sheet, Strip, Plate, and Flat Bar: ASTM A 666, Type 304 or 316.
5. Bars and Shapes: ASTM A 276, Type 304 or 316.
6. Wire Rope: 1-by-19 wire rope made from wire complying with ASTM A 492, Type 316.

F. Steel and Iron: Provide steel and iron in form indicated to comply with the following requirements:

1. Tubing: Cold formed, ASTM A 500.
2. Steel Plate, Shapes, and Bars: ASTM A 36/A 36M.
3. Steel Sheet: Commercial-quality, cold-rolled, stretcher-leveled, carbon-steel sheet complying with ASTM A 366/A 366M, Class I, matte finish.
4. Gray-Iron Castings: ASTM A 48M, Class 200.
5. Malleable-Iron Castings: ASTM A 47M, grade as recommended by fabricator for type of use indicated.

2.3 MISCELLANEOUS MATERIALS

A. Welding Electrodes and Filler Metal: Type and alloy of filler metal and electrodes as recommended by producer of metal to be welded, complying with applicable AWS specifications, and as required for color match, strength, and compatibility in fabricated items.

B. Fasteners: Use fasteners of same basic metal as fastened metal, unless otherwise indicated. Do not use metals that are corrosive or incompatible with materials joined.

1. Provide concealed fasteners for interconnecting ornamental metal components and for attaching them to other work, unless otherwise indicated.

2. Provide Phillips flat-head machine screws for exposed fasteners, unless otherwise indicated.

C. Cast-in-Place and Postinstalled Anchors: Anchors of type indicated below, fabricated from corrosion-resistant materials with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and equal to four times the load imposed when installed in concrete as determined by testing per ASTM E 488 conducted by a qualified independent testing agency acceptable to the Engineer.

1. Chemical anchors.
2. Expansion anchors.

D. Wire-Rope Fittings: Connectors of types indicated, fabricated from stainless steel, and with capability to sustain, without failure, a load equal to minimum breaking strength of wire rope with which they are used.

E. Nonshrink, Nonmetallic Grout: Premixed, factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by the manufacturer for interior and exterior applications.

F. Shop Primer for Ferrous Metal: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with performance requirements of FS TT-P-664; selected for good resistance to normal atmospheric corrosion, compatibility with finish paint systems indicated, and capability to provide a sound foundation for field-applied topcoats despite prolonged exposure.

G. Shop Primer for Galvanized Steel: Zinc-dust, zinc-oxide primer formulated for priming zinc-coated steel and for compatibility with finish paint systems indicated, and complying with SSPC-Paint 5.

H. Bituminous Paint: Cold-applied asphalt mastic complying with SSPC-Paint 12, except containing no asbestos fibers.

2.4 FABRICATION, GENERAL

A. Form ornamental metal to required shapes and sizes, with true curves, lines, and angles. Provide components in sizes and profiles indicated, but not less than that needed to comply with requirements indicated for structural performance.

B. Provide necessary rebates, lugs, and brackets to assemble units and to attach to other work. Drill and tap for required fasteners, unless otherwise indicated. Use concealed fasteners where possible.

C. Comply with AWS for recommended practices in shop welding and brazing. Provide welds and brazes behind finished surfaces without distorting or discoloring exposed side. Clean exposed welded and brazed joints of all flux, and dress all exposed and contact surfaces.

D. Mill joints to a tight, hairline fit. Cope or miter corner joints. Form joints exposed to weather to exclude water penetration.

E. Provide castings that are sound and free of warp, cracks, blowholes, or other defects that impair strength or appearance. Grind, wire brush, sandblast, and buff castings to remove seams, gate marks, casting flash, and other casting marks.

F. Finish exposed surfaces to smooth, sharp, well-defined lines and arris.

G. Assemble items in the shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation. Use connections that maintain structural value of joined pieces.

H. Make up wire-rope assemblies in the shop to field-measured dimensions with fittings machine swaged. Minimize amount of turnbuckle take-up used for dimensional adjustment so that maximum amount is available for tensioning wire ropes. Tag wire-rope assemblies and fittings to identify installation locations and orientations for coordinated installation.

2.5 FABRICATING HANDRAILS AND RAILINGS

A. Nonwelded Connections: Fabricate handrails and railings to interconnect members with concealed mechanical fasteners and fittings, unless otherwise indicated. Fabricate members and fittings to produce flush, smooth, rigid, hairline joints.

1. Fabricate splice joints for field connection using an epoxy structural adhesive where this is fabricator's standard splicing method.

B. Welded Connections: Fabricate handrails and railings for connecting members by welding. Cope components at perpendicular and skew connections to provide close fit, or use fittings designed for this purpose. Weld connections continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove flux immediately.
4. At exposed connections, finish exposed surfaces smooth and blended so that no roughness shows after finishing and welded surface matches contours of adjoining surfaces.
5. Provide welded connections for stainless-steel handrails and railings.
6. Provide welded connections for ferrous handrails and railings.

C. Welded Connections: Fabricate aluminum-alloy handrails and railings to interconnect members with concealed internal welds that eliminate surface grinding, using fittings designed and fabricated for this purpose.

D. Brazed Connections: Fabricate copper-alloy handrails and railings for connecting members by brazing. For connections made during fabrication, braze corners and seams continuously to comply with the following:

1. Use materials and methods that match color of base metal, minimize distortion, and develop maximum strength and corrosion resistance.
2. Remove flux immediately.

3. At exposed connections, finish exposed surfaces smooth and blended so that no roughness shows after finishing and brazed surface matches contours of adjoining surfaces.

E. Form changes in direction of railing members as follows:

1. As detailed.

F. Form simple and compound curves by bending members in jigs to produce uniform curvature for each repetitive configuration required; maintain profile of member throughout entire bend without buckling, twisting, or otherwise deforming exposed surfaces of handrail and railing components.

G. For handrails and railings with nonwelded connections that are exposed to exterior or to moisture from condensation or other sources, provide weep holes or another means to drain water entrapped in hollow sections of railing members.

H. Provide wall returns at ends of wall-mounted handrails, unless otherwise indicated; close ends of returns.

I. Close exposed ends of handrail and railing members with manufacturer's standard prefabricated end fittings.

J. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors to interconnect handrail and railing members to other work, unless otherwise indicated.

1. Furnish inserts and other anchorage devices for connecting handrails and railings to concrete or masonry work. Fabricate anchorage devices capable of withstanding loads imposed by handrails and railings. Coordinate anchorage devices with supporting structure.
2. For railing posts set in concrete, provide preset sleeves of steel, not less than 150 mm long and inside dimensions not less than 13 mm greater than outside dimensions of post, with steel plate forming bottom closure.

2.6 FABRICATING ORNAMENTAL WINDOW GRILLES (SECURITY BARS)

A. General: Fabricate ornamental window grilles to designs indicated from steel bars and shapes of sizes and profiles indicated. Form steel bars by bending, forging, coping, mitering, and welding.

B. Form simple and compound curves by bending members in jigs to produce uniform curvature of radii indicated for each configuration required; maintain profile of member throughout entire bend without buckling, twisting, or otherwise deforming exposed surfaces of components.

C. Welding: Interconnect grille members with full-length, full-penetration welds, unless otherwise indicated. Use welding method that is appropriate for metal and finish indicated and that develops full strength of members joined. Finish exposed welds and surfaces smooth, flush, and blended to match adjoining surfaces.

D. Brackets, Fittings, and Anchors: Provide wall brackets, fittings, and anchors to connect ornamental window grilles to other work, unless otherwise indicated.

1. Furnish inserts and other anchorage devices to connect ornamental window grilles to concrete and masonry work. Coordinate anchorage devices with supporting structure.
2. Fabricate anchorage devices that are capable of withstanding loads indicated.

2.7 FABRICATING ORNAMENTAL MECHANICAL GRILLES AND FRAMES

A. Fabricate ornamental grilles from perforated brass sheet or plate of thickness, size, and pattern indicated. Form perforations by punching, cutting, or drilling to produce openings of sizes and shapes indicated. Roll, press, and grind perforated metal to flatten and to remove burrs and deformations.

1. Form perforations to match existing grilles.
2. Drawings indicate perforated metal patterns required and are based on products of one manufacturer. Perforated metal patterns produced by other manufacturers may be considered, provided deviations are minor and do not change design concept as judged solely by the Engineer. Burden of proof of equality is on proposer.

B. Drill and countersink grilles for oval-head mounting screws at 50 mm from corners and at 250 mm or less o.c. Provide units with brass oval-head wood screws.

C. Fabricate grille frames from extruded brass of the profiles, and to the sizes and shapes indicated. Miter frame members at corners and connect with concealed splice plates brazed to back of frames. Secure grilles in frames with 13 mm long brazing along perimeter of grilles at 100 mm o.c.

1. Provide frame profiles to match existing frames.
2. Drawings indicate frame profiles required and are based on products of one manufacturer. Similar frame profiles produced by other manufacturers may be considered, provided deviations are minor and do not change design concept as judged solely by the Engineer. Burden of proof of equality is on proposer.

D. Drill and countersink frames for oval-head mounting screws at 100 mm from corners and at 400 mm or less o.c. Provide units with brass oval-head wood screws.

2.8 FABRICATING CUSTOM DOOR PULLS

A. Fabricate custom door pulls from stainless-steel bar stock of profile indicated, fabricated to shapes indicated. Form curves by bending to produce uniform curvature of radii indicated; maintain profile of member throughout entire bend without buckling, twisting, or otherwise deforming exposed surfaces. Where radii of bends are too small to avoid buckling, grind bars after bending to restore original profile. Drill and tap door pulls to receive through bolts for attachment to doors.

B. Fabricate backing plates for custom door pulls from 3.5 mm stainless-steel sheet. Cut to shape indicated and bevel edges at a 45 degree angle for one-half thickness of metal. Drill and countersink holes where indicated for screws and bolts.

C. Provide units with oval-head through bolts for mounting pulls and with oval-head wood screws for mounting backing plates. Provide bolts and screws of metal to match that of door pulls.

2.9 FABRICATING METAL REVEALS

A. Fabricate metal reveals for wood paneling from 20-by-20-by-3 mm extruded-bronze channels. Drill for mounting screws 150 mm from ends of channels and not more than 600 mm o.c. Locate mounting screws at same heights for all channels. Provide black-finished, hex-socket, wafer-head wood screws for mounting reveals.

2.10 FABRICATING CAST-METAL ROSETTES

A. Fabricate cast-metal rosettes to design indicated from cast bronze. Drill and tap castings for threaded mounting studs.

1. Provide custom castings to match design indicated.
2. Manufacturer's stock castings may be considered, provided deviations are minor and do not change design concept as judged solely by the Engineer. Burden of proof of equality is on proposer.
3. Drawings indicate cast-metal rosette designs required and are based on products of one manufacturer. Castings produced by other manufacturers may be considered, provided deviations are minor and do not change design concept as judged solely by the Engineer. Burden of proof of equality is on proposer.

2.11 FINISHES, GENERAL

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

2.12 ALUMINUM FINISHES

A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.

B. Class I, Color Anodic Finish: AA-M12C22A42/A44 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, integrally colored or electrolytically deposited color coating 0.018 mm or thicker) complying with AAMA 606.1 or AAMA 608.1.

1. Color: As selected by the Engineer from the full range of industry colors and color densities.

C. Baked-Enamel Finish: AA-C12C42R1x (Chemical Finish: cleaned with inhibited chemicals; Chemical Finish: acid-chromate-fluoride-phosphate conversion coating; Organic Coating: as specified below). Apply baked enamel complying with paint manufacturer's specifications for cleaning, conversion coating, and painting.

1. Organic Coating: Thermosetting, modified-acrylic enamel primer/topcoat system complying with AAMA 603.8 except with a minimum dry film thickness of 0.04 mm, medium gloss.
2. Color: As selected by the Engineer from manufacturer's full range.

D. Polyester Powder Coating: Approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce a hard, durable coating:

1. Surface Finish: As indicated.
2. Minimum Dry Film Thickness: 60 microns.
3. Color and Gloss: As selected by the Engineer from manufacturer's full range.

2.13 COPPER-ALLOY FINISHES

A. Finish designations prefixed by CDA comply with the system established by the Copper Development Association for designating copper-alloy finish systems.

B. Buffed Finish: CDA M21 (Mechanical Finish: buffed, smooth specular).

C. Hand-Rubbed Finish, Lacquered: CDA M31-M34-O6x (Mechanical Finish: directionally textured, fine satin; Mechanical Finish: directionally textured, hand rubbed; Coating: clear organic, air drying, as specified below):

1. Clear, Organic Coating: Clear, air-drying, acrylic lacquer specially developed for coating copper-alloy products, applied by air spray in two coats per manufacturer's written instructions, with interim drying, to a total thickness of 0.025 mm.

a. Product: Subject to compliance with requirements, provide "Incralac" developed by International Copper Research Corporation.

D. Statuary Conversion Coating over Satin Finish: CDA-M32-C55 (Mechanical Finish: directionally textured, fine satin; Chemical Finish: conversion coating, sulfide).

1. Color: Match the Engineer's sample.

E. Patina Conversion Coating: CDA-M36-C12-C52 (Mechanical Finish: directionally textured, uniform; Chemical Finish: nonetched cleaned, degreased; Chemical Finish: conversion coating, ammonium sulfate).

1. Color: Match the Engineer's sample.

F. Statuary Conversion Coating, Bright Relieved and Lacquered: CDA-M12-C55-M2x-O6x (Mechanical Finish: matte finish, as cast; Chemical Finish: conversion coating, sulfide; Mechanical Finish: buffed, as specified; Coating: clear, organic, air drying, as specified below):

1. Color and Buffing: Match the Engineer's sample.
2. Clear, Organic Coating: Clear, air-drying, acrylic lacquer specially developed for coating copper-alloy products, applied by air spray in two coats per manufacturer's written instructions, with interim drying, to a total thickness of 0.025 mm.

    a. Product: Subject to compliance with requirements, provide "Incralac" developed by International Copper Research Corporation.

G. Blackened, Bright Relieved, and Lacquered: CDA-M33-O60-M2x-O6x (Mechanical Finish: directionally textured, coarse satin; Coating: black, air drying; Mechanical Finish: buffed, as specified; Coating: clear, organic, air drying, as specified below):

1. Blackening and Buffing: Match the Engineer's sample.
2. Clear, Organic Coating: Clear, air-drying, acrylic lacquer specially developed for coating copper-alloy products, applied by air spray in two coats per manufacturer's written instructions, with interim drying, to a total thickness of 0.025 mm.

    a. Product: Subject to compliance with requirements, provide "Incralac" developed by International Copper Research Corporation.

2.14 STAINLESS-STEEL FINISHES

A. Remove or blend tool and die marks and stretch lines into finish.

B. Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

C. Bright, Cold-Rolled, Unpolished Finish: No. 2B finish.

D. Bright, Directional Polish: No. 4 finish.

E. Satin, Directional Polish: No. 6 finish.

F. Satin, Reflective, Directional Polish: No. 7 finish.

G. Mirrorlike Reflective, Nondirectional Polish: No. 8 finish.

H. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

2.15 IRON AND STEEL FINISHES

A. Galvanizing: Hot-dip galvanize products made from rolled, pressed, and forged steel shapes, castings, plates, bars, and strips indicated to be galvanized to comply with ASTM A 123.

1. Hot-dip galvanize iron and steel hardware indicated to be galvanized to comply with ASTM A 153/A 153M.

B. Fill vent and drain holes that will be exposed in finished Work, unless indicated to remain as weep holes, by plugging with zinc solder and filing off smooth.

C. Preparation for Shop Priming: After galvanizing, thoroughly clean ornamental metal of grease, dirt, oil, flux, and other foreign matter, and treat with metallic-phosphate process.

D. Preparation for Shop Priming: Prepare uncoated ferrous-metal surfaces to comply with minimum requirements indicated below for SSPC surface-preparation specifications and environmental exposure conditions of installed ornamental metal:

1. Exteriors (SSPC Zone 1B): SSPC-SP 6, "Commercial Blast Cleaning."
2. Interiors (SSPC Zone 1A): SSPC-SP 7, "Brush-off Blast Cleaning."

E. Factory-Primed Finish: Apply air-dried primer immediately after cleaning and pretreatment, to provide a minimum dry film thickness of 0.05 mm per applied coat, to surfaces that will be exposed after assembly and installation, and to concealed, nongalvanized surfaces.

F. Polyester Powder Coating: Approved type to meet the requirements of AAMA 603 consisting of powder particles of resinous material and additives to improve performance. The coating is to be electrostatically sprayed on the object to produce a hard, durable coating:

1. Prepare galvanized metal by thoroughly removing grease, dirt, oil, flux, and other foreign matter.
2. Surface Finish: As indicated.
3. Minimum Dry Film Thickness: 60 microns.
4. Color and Gloss: As selected by the Engineer from manufacturer's full range.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Provide anchorage devices and fasteners where necessary for securing ornamental metal to in-place construction.

B. Perform cutting, drilling, and fitting required to install ornamental metal. Set products accurately in location, alignment, and elevation; measured from established lines and levels. Provide temporary bracing or anchors in formwork for items to be built into concrete, masonry, or similar construction.

C. Fit exposed connections accurately together to form tight, hairline joints or, where indicated, with uniform reveals and spaces for sealants and joint fillers. Where cutting, welding, and grinding

are required for proper shop fitting and jointing of ornamental metal, restore finishes to eliminate any evidence of such corrective work.

D. Do not cut or abrade finishes that cannot be completely restored in the field. Return items with such finishes to the shop for required alterations, followed by complete refinishing, or provide new units as required.

E. Install concealed gaskets, joint fillers, insulation, and flashings as work progresses.

F. Restore protective coverings that have been damaged during shipment or installation. Remove protective coverings only when there is no possibility of damage from other work yet to be performed at same location.

1. Retain protective coverings intact; remove coverings simultaneously from similarly finished items to preclude nonuniform oxidation and discoloration.

G. Field Welding: Comply with applicable AWS specification for procedures of manual shielded metal arc welding, for appearance and quality of welds, and for methods used in correcting welding work. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Grind exposed welded joints smooth and restore finish to match finish of adjacent surfaces.

H. Corrosion Protection: Coat concealed surfaces of aluminum that will be in contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

3.2 INSTALLING HANDRAILS AND RAILINGS

A. Adjust handrails and railings before anchoring to ensure alignment at abutting joints.

B. Concrete-Anchored Posts in Sleeves: Insert posts in preset sleeves, cast into concrete, and fill annular space between posts and sleeve with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions.

C. Concrete-Anchored Posts in Core-Drilled Holes: Core-drill concrete to produce holes with a diameter at least 20 mm larger than OD of post and not less than 125 mm deep. Clean holes of loose material, insert posts, and fill annular space between post and concrete with nonshrink, nonmetallic grout, mixed and placed to comply with grout manufacturer's written instructions.

1. Cover anchorage joint with flange or escutcheon plate attached to post after filling annular space.
2. Leave anchorage joint exposed; wipe off excess grout; and leave a 3.5 mm build-up, sloped away from post.

D. Anchor posts to metal surfaces with fittings designed for this purpose.

E. Nonwelded Connections: Use mechanical or adhesive joints for permanently connecting railing components. Use wood blocks and padding to prevent damage to railing members and fittings. Seal recessed holes of exposed locking screws using plastic cement filler colored to match finish of handrails and railings.

F. Welded Connections: Use fully welded joints for permanently connecting railing components by welding. Cope or butt components to provide 100 percent contact or use fittings designed for this purpose.

G. Anchor railing ends into concrete or masonry with fittings designed for this purpose.

H. Anchor railing ends to metal surfaces with fittings using concealed fasteners.

I. Anchor railing ends to metal surfaces by welding.

J. Expansion Joints: Provide expansion joints at locations indicated or, if not indicated, at intervals not to exceed 12 m. Provide slip-joint internal sleeve extending 50 mm beyond joint on either side, fasten internal sleeve securely to one side, and locate joint within 150 mm of post.

3.3 INSTALLING ORNAMENTAL WINDOW GRILLES (SECURITY BARS)

A. Secure window grilles to concrete and masonry walls with cast-in-place or postinstalled anchors. Peen exposed threads of anchors to prevent removal of grilles.

3.4 INSTALLING ORNAMENTAL MECHANICAL GRILLES

A. Mount ornamental mechanical grilles at heights and in positions indicated, adjusting ductwork to be centered on grilles. Secure to wall framing with wood screws. On marble, brick, and other solid surfaces, secure with wood screws in lead plugs.

3.5 INSTALLING CUSTOM DOOR PULLS

A. Install at heights and locations shown. Install with backing plates behind pulls on outside of doors and with additional backing plates on inside of doors. Fasten backing plates to doors with oval-head wood screws and secure pulls through doors and backing plates with oval-head bolts.

3.6 INSTALLING METAL REVEALS AT WOOD PANELING

A. Install metal reveals between wood panels as paneling is installed. Secure to wood grounds with specified screws.

3.7 INSTALLING CAST-METAL ROSETTES AT MARBLE JOINTS

A. Install cast-metal rosettes at intersections of marble joints where indicated. Install only after marble work is complete and joints are grouted. Secure to wall by drilling a 20 mm round hole at intersection of marble joints and by filling hole with molding plaster into which threaded stud is embedded. Angle drill and rotate so that bottom of hole is larger than at surface.

1. Secure rosettes in place with masking tape until plaster sets. After plaster has set, remove masking tape and adhesive residue.

3.8 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified independent testing and inspecting agency acceptable to the Engineer to perform field tests and inspections and to prepare test reports.

B. Extent and Testing Methodology: Testing agency will randomly select completed handrail and railing assemblies for testing that are representative of different railing designs and conditions in the completed Work. Handrails and railings will be tested according to ASTM E 894 and ASTM E 935 for compliance with ASTM E 985.

C. Testing agency will report test results promptly and in writing to the Contractor and Engineer.

D. Remove and replace handrails and railings where test results indicate that they do not comply with specified requirements, unless they can be repaired in a manner satisfactory to the Engineer and will comply with specified requirements.

E. Additional Testing: Where handrails and railings are removed and replaced or are repaired, additional testing will be performed to determine compliance of replaced or additional work with specified requirements.

3.9 CLEANING

A. Clean aluminum and stainless steel by washing thoroughly with clean water and soap and rinsing with clean water.

B. Clean copper alloys according to metal finisher's written instructions in a manner that leaves an undamaged and uniform finish matching approved Sample.

C. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with same material.

D. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

3.10 PROTECTION

A. Protect finishes of ornamental metal from damage during construction period with temporary protective coverings approved by ornamental metal fabricator. Remove protective covering at the time of Substantial Completion.

B. Restore finishes damaged during installation and construction period so that no evidence remains of correction work. Return items that cannot be refinished in the field to the shop; make required alterations and refinish entire unit, or provide new units.

END OF SECTION 05700

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SECTION 05811 - ARCHITECTURAL JOINT SYSTEMS

PART 4 - GENERAL

4.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including Conditions of Contract and General Requirements Specification Sections, apply to this Section.

4.2 SUMMARY

A. Types of joints for which architectural joint systems are specified include the following:

1. Exterior pedestrian and vehicular traffic joints.
2. Exterior wall and ceiling joints.
3. Exterior soffit joints.
4. Interior pedestrian and vehicular traffic joints.
5. Interior wall and ceiling joints.
6. Interior soffit joints.

B. Related Sections include the following:

1. Concrete Specification Section "Cast-in-Place Concrete" for block-outs and cast-in anchorage and frames for architectural joint systems in concrete floors, parking decks, and walls.

4.3 DEFINITIONS

A. Architectural Joint System: Any filler or cover used to span, fill, cover, or seal a joint, except expanding foam seals and poured or foamed-in-place sealants.

B. Construction joint: The surface where two successive placements of concrete meet, across which it may be desirable to achieve bond and through which reinforcement may be continuous.

C. Contraction Joint or Control joint: Formed, sawed, or tooled groove in a concrete structure to create a weakened plane and regulate the location of cracking resulting from the dimensional change of different parts of the structure.

D. Expansion Joint: (1) A separation provided between adjoining parts of a structure to allow movement where expansion is likely to exceed contraction; (2) a separation between pavement slabs on grade, filled with a compressible filler material; (3) an isolation joint intended to allow independent movement between adjoining parts.

E. Isolation Joint: A separation between adjoining parts of a concrete structure, usually a vertical plane, at a designated location such as to interfere least with performance of the structure, yet

such as to allow relative movement in three directions and avoid formation of cracks elsewhere in the concrete and through which all or part of the bonded reinforcement is interrupted.

F. Cyclic Movement: Periodic change between widest and narrowest joint widths in an automatically mechanically controlled system.

G. Fire Barriers: Any material or material combination, when fire tested after cycling, designated to resist passage of flame and hot gases through a movement joint.

H. Maximum Joint Width: Widest linear gap a joint system tolerates and performs its designed function without damaging its functional capabilities.

I. Minimum Joint Width: Narrowest linear gap a joint system tolerates and performs its designed function without damaging its functional capabilities.

J. Movement Capability: Value obtained from the difference between widest and narrowest widths of a joint opening typically expressed in numerical values (mm or inches) or a percentage of nominal value of joint width.

K. Nominal Joint Width: Width of linear gap indicated as representing the conditions existing when architectural joint systems will be installed or, if no nominal joint width is indicated, a width equal to the sum of maximum and minimum joint widths divided by two. For joint widths refer to TCA Handbook for Ceramic Tile Installation.

4.4 PERFORMANCE REQUIREMENTS

A. General: Provide factory-fabricated architectural joint systems capable of withstanding the types of loads and of accommodating the kinds of movement, and the other functions for which they are designed including those specified below, without failure. Types of failure include those listed in Appendix X3 of ASTM E 1399.

1. Vehicular Traffic Joints: Support vehicular traffic across joint.
2. Pedestrian Traffic Joints: Support pedestrian traffic across joint.
3. Exterior Joints: Maintain continuity of weather enclosure.
4. Joints in Fire-Resistance-Rated Assemblies: Maintain fire-resistance ratings of assemblies.
5. Joints in Smoke Barriers: Maintain integrity of smoke barrier.
6. Joints in Acoustically Rated Assemblies: Inhibit passage of airborne noise.
7. Other Joints: Where indicated, provide joint systems that prevent penetration of water, moisture, and other substances deleterious to building components or content.
8. Seismic Joints: Remain in place on exposure to seismic activity (movement).
9. Joints in Surfaces with Architectural Finishes: Serve as finished architectural joint closures.

B. Movement Joints: Provide 10 mm wide, non-structural, peripheral movement joints in floor tiling, based on TCA handbook, extending to the full depth of the finishing layer, around the perimeter of all tiled areas:

1. exceeding 6 m in any direction for ceramic tiles bedded in semi dry mix
2. exceeding 15 m in any direction for other types of rigid floor tiles or slabs.

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C. Movement Joints: Provide 10 mm wide, non-structural, intermediate movement joint in floor tiling of all tiled areas, based on TCA handbook, extending to the full depth of the finishing layer:

1. at 9 m in any direction for ceramic tiles bedded in semi dry mix
2. at 15 m in any direction for other types of rigid floor tiles or slabs.

4.5 SUBMITTALS

A. Product Data: Include manufacturer's product specifications, construction details, material and finish descriptions, and dimensions of individual components and seals.

B. Shop Drawings: For each joint system specified, provide the following:

1. Placement Drawings: Include line diagrams showing entire route of each joint system, plans, elevations, sections, details, joints, splices, locations of joints and splices, and attachments to other Work. Where joint systems change planes, provide Isometric Drawings depicting how components interconnect to achieve continuity of joint covers and fillers.

C. Samples for Initial Selection: Manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available for each exposed metal and elastomeric material of joint system indicated.

1. Include similar Samples of material for joints and accessories involving color selection.

D. Samples for Verification: Full-size units 150 mm long of each type of joint system indicated; in sets for each finish, color, texture, and pattern specified, showing the full range of variations expected in these characteristics.

E. Product Test Reports: From a qualified testing agency indicating architectural joint systems comply with requirements, based on comprehensive testing of current products.

F. Research/Evaluation Reports: Evidence of architectural joint system's compliance with building code in effect for Project, from a model code organization acceptable to authorities having jurisdiction.

4.6 QUALITY ASSURANCE

A. Quality System: Comply with ISO 9001/9002 Quality System as a minimum. Incorporate all the standard procedures supplied by the Engineer and the Employer.

B. Source Limitations: Obtain architectural joint systems through one source from a single manufacturer. Coordinate compatibility with adjoining joint systems specified in other Sections.

C. Fire-Test-Response Characteristics: Where indicated, provide joint systems incorporating fire barriers that are identical to those of assemblies tested for fire resistance per ASTM E 119, ASTM E 814, and UL 2079, including hose-stream test of vertical wall assemblies, by a testing and inspecting agency acceptable to authorities having jurisdiction.

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D. Product Options: Drawings indicate size, profiles, and dimensional requirements of architectural joint systems and are based on the specific systems indicated. Other manufacturers' systems complying with requirements may be considered. Refer to General Requirements Section "Product Requirements."

1. Do not modify intended aesthetic effects, as judged solely by the Engineer, except with the Engineer's approval. If modifications are proposed, submit comprehensive explanatory data to the Engineer for review.

E. Water-Tight Joints: for joints in wet areas or areas that may become exposed to water, do not rely on sealant only, install water-stops, or extend waterproofing membrane across the joint with an expansion loop.

4.7 Delivery, Storage and Handling

A. Exercise proper care in the handling of all work so as not to injure the finished surface, and take proper precautions to protect the work from damage after it is in place.

B. Deliver materials to the job site ready for use, and fabricated in as large sections and assemblies as practical. Assemblies shall be identical to submitted and reviewed shop drawings, samples and certificates.

C. Store materials under cover in a dry and clean location off the ground. Remove materials that are damaged or otherwise not suitable for installation from the job site and replace with acceptable materials at no additional cost.

PART 5 - PRODUCTS

5.1 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, those indicated in Part 2 "Architectural Joint Systems" Article.

5.2 MATERIALS

A. Aluminum: ASTM B 221M, alloy 6063-T5 for extrusions; ASTM B 209M, alloy 6061-T6 for sheet and plate.

1. Apply manufacturer's standard protective coating on aluminum surfaces to be placed in contact with cementitious materials.

B. Bronze: ASTM B 455, alloy C38500 for extrusions; alloy C23000 Red Brass for plates.

C. Brass: ASTM B 36/B 36M, UNS alloy C26000 for half hard sheet and coil.

D. Stainless Steel: ASTM A 666, Type 316L, with No. 2B finish, unless otherwise indicated, for plates, sheet, and strips.

E. Preformed Seals: Single or multicellular extruded elastomeric seals designed with or without continuous, longitudinal, internal baffles. Formed to be installed in frames or with anchored flanges, in color indicated or, if not indicated, as selected by the Engineer from manufacturer's standard colors.

F. Strip Seals: Elastomeric membrane or tubular extrusions with a continuous longitudinal internal baffle system throughout complying with ASTM E 1783; used with compatible frames, flanges, and molded-rubber anchor blocks.

G. Compression Seals: Preformed, elastomeric extrusions having internal baffle system complying with ASTM E 1612 in sizes and profiles indicated or as recommended by manufacturer.

H. Preformed Cellular Foams: Nonextruded, low-density, crosslinked, nitrogen-blown ethylene-vinyl-acetate copolymer, Neoprene, or polyurethane extruded, compressible foam.

I. Covers: stainless steel capping grade 316L to protect the rubber seal.

J. Support plate: provides additional support to the seal for use under heavy traffic.

K. Fire Barriers: Any material or material combination, when fire tested after cycling, designated to resist the passage of flame and hot gases through a movement joint.

L. Accessories: Manufacturer's standard anchors, clips, fasteners, set screws, spacers, flexible moisture barrier and filler materials, drain tubes, lubricants, adhesives, and other accessories compatible with material in contact, as indicated or required for complete installations.

5.3 ARCHITECTURAL JOINT SYSTEMS

A. General: Provide joint systems of design, basic profile, materials, and operation indicated. Provide units with the capability to accommodate joint widths indicated and variations in adjacent surfaces.

1. Furnish units in longest practicable lengths to minimize number of end joints. Provide hairline mitered corners where joint changes directions or abuts other materials.
2. Include closure materials and transition pieces, tee-joints, corners, curbs, cross-connections, and other accessories as required to provide continuous joint systems.
3. Frames for Strip Seals: Designed with semiclosed cavity that provides a mechanical lock for seals of type indicated.
4. Public Arena Seals: Non-slip seals designed for installation on treads and risers and to lie flat with adjacent surfaces, and complying with ADA guidelines for public areas.

B. Architectural Joint System: Metal frames and covers for exterior, interior, vehicular, and pedestrian traffic joints.

1. Available Products: Top quality.
2. Nominal Joint Width: 10 mm.

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3. Type of Movement Capability: Architectural Expansion and contraction.
4. Cyclic-Movement-Test-Response Characteristics: No evidence of visual fatigue, inability to cycle between designated joint widths, or other types of failure as determined by testing products identical to those indicated per ASTM E 1399 including Appendix X3.
5. Exposed Frame Material: As indicated.

5.4 FINISHES, GENERAL

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

5.5 ALUMINUM FINISHES

A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.

B. Class I, Color Anodic Finish: AA-M12C22A42/A44 (Mechanical Finish: nonspecular as fabricated; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, integrally colored or electrolytically deposited color coating 0.018 mm or thicker) complying with AAMA 606.1 or AAMA 608.1.

1. Color: As selected by the Engineer from the full range of industry colors and color densities.

C. High-Performance Organic Finish: AA-C12C42R1x (Chemical Finish: cleaned with inhibited chemicals; Chemical Finish: acid-chromate-fluoride-phosphate conversion coating; Organic Coating: as specified below). Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

1. Fluoropolymer Three-Coat System: Manufacturer's standard three-coat, thermocured system consisting of specially formulated inhibitive primer, fluoropolymer color coat, and clear fluoropolymer topcoat, with both color coat and clear topcoat containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with AAMA 2605.

a. Color and Gloss: As selected by the Engineer from manufacturer's full range. To obtain exact finish, insert names of coating products and manufacturers.

5.6 STAINLESS-STEEL FINISHES

A. Remove tool and die marks and stretch lines or blend into finish.

B. Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

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C. Bright, Cold-Rolled, Unpolished Finish: No. 2B finish.

D. Bright, Directional Polish: No. 4 finish.

E. Satin, Directional Polish: No. 6 finish.

F. Mirrorlike Reflective, Nondirectional Polish: No. 8 finish.

G. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

H. Finish designations prefixed by CDA comply with the system established by the Copper Development Association for designating copper-alloy finish systems, as defined in NAAMM's "Metal Finishes Manual for Architectural and Metal Products."

1. Remove tool and die marks and stretch lines or blend into finish.
2. Grind and polish surfaces to produce uniform, directionally textured polished finish indicated, free of cross scratches. Run grain with long dimension of each piece.

I. Standard Finish Designation: CDA M32 (Mechanical Finish: directionally textured, medium satin).

PART 6 - EXECUTION

6.1 PREPARATION

A. Prepare substrates according to architectural joint system manufacturer's written instructions.

B. Coordinate and furnish anchorages, Placement Drawings, and instructions for installing joint systems to be embedded in or anchored to concrete or to have recesses formed into edges of concrete slab for later placement and grouting-in of frames.

C. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary to secure joint systems to in-place construction, including threaded fasteners with drilled-in expansion shields for masonry and concrete where anchoring members are not embedded in concrete. Provide fasteners of metal, type, and size to suit type of construction indicated and to provide for secure attachment of joint systems.

6.2 INSTALLATION

A. Comply with manufacturer's written instructions for handling and installing architectural joint assemblies and materials, unless more stringent requirements are indicated.

B. Coordinate installation of architectural joint assembly materials and associated work so complete assemblies comply with assembly performance requirements.

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C. Terminate exposed ends of exterior architectural joint assemblies with factory-fabricated termination devices to maintain waterproof system.

D. Install factory-fabricated transitions between building expansion-joint cover assemblies and roof expansion-joint assemblies to provide continuous, uninterrupted, watertight construction.

E. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required to install joint systems.

1. Install joint cover assemblies in true alignment and proper relationship to joints and adjoining finished surfaces measured from established lines and levels.
2. Allow adequate free movement for thermal expansion and contraction of metal to avoid buckling.
3. Set covers in horizontal surfaces at elevations that place exposed surfaces flush with adjoining finishes.
4. Locate wall, ceiling, and soffit covers in continuous contact with adjacent surfaces.
5. Securely attach in place with required accessories.
6. Locate anchors at interval recommended by manufacturer, but not less than 75 mm from each end and not more than 600 mm o.c.

F. Continuity: Maintain continuity of joint systems with a minimum number of end joints and align metal members. Cut and fit ends to produce joints that will accommodate thermal expansion and contraction of metal to avoid buckling of frames. Adhere flexible filler materials, if any, to frames with adhesive or pressure-sensitive tape as recommended by manufacturer.

G. Extruded Preformed Seals: Install seals to comply with manufacturer's written instructions and with minimum number of end joints.

1. For straight sections, provide preformed seals in continuous lengths.
2. Vulcanize or heat-weld field splice joints in preformed seal material to provide watertight joints using procedures recommended by the manufacturer.
3. Apply adhesive, epoxy, or lubricant adhesive approved by the manufacturer to both frame interfaces before installing preformed seals.
4. Seal transitions according to the manufacturer's written instructions.
5. Install foam seals with adhesive recommended by the manufacturer and heat seal all splices.

H. Joint Systems with Seals: Seal end joints within continuous runs and joints at transitions according to manufacturer's written instructions to provide a watertight installation.

I. Seismic Seals: Install interior seals in continuous lengths. Install exterior seal in standard lengths and vulcanize or heat-weld field splice joints to provide watertight joints using manufacturer's recommended procedures. Seal transitions and end joints according to manufacturer's written instructions.

J. Fire Barriers: Install fire barriers to provide continuous, uninterrupted fire resistance throughout length of joint, including transitions and end joints.

6.3 CLEANING AND PROTECTION

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A. Do not remove protective covering until finish work in adjacent areas is complete. When protective covering is removed, clean exposed metal surfaces to comply with manufacturer's written instructions.

END OF SECTION 05811

TABLE — ARCHITECTURAL JOINT SYSTEMS

EXTERIOR JOINTS IN PARKING AND PLAZA DECKS							
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**SEAL OR GLAND**	**EXPOSED COVER**	**JOINT WIDTH (inches)**	**MOVEMENT**
Compression seal	HD traffic Pedestrian & vehicular	Watson Bowman Acme	Wabo WA Wabo WE	Neoprene Santoprene		1-1/8 to 3-1/2 1-3/8 to 2-7/8	1-3/16 to 3-1/8" 1-1/4 to 3-13/16"
Compression seal	Pavement seal Bridges & structures Pedestrian & low-stress traffic	D. S. Brown Company	Series E & V Series CV, CA & H Series CS, AF, AW & K	Neoprene Neoprene Neoprene		0.187 to 1.26 min. 0.18 to 3.6 1.35 to 4.2	4" max 0.50 to 3" 0.7 to 2.63"
Elastomeric membrane system	Parking decks Parking decks	Balco Metalines	Model EM Model ECR	Elastomeric Elastomeric		1-1/2 to 3 1-1/8 to 1-3/8	1-1/2 to 3" 1 to 2"
Elastomeric membrane system	Parking & plaza decks Parking & plaza decks	Watson Bowman Acme	Wabocrete 101 Wabocrete 201	Thermoplastic Thermoplastic		1-1/2 to 2 1-1/2 to 3-1/2	2" & 3-3/16" 1-1/4 to 4"
Factory-molded system	Wheel & foot traffic	Watson Bowman Acme	Wabo "T" Joint System	Polyurethane		1/2 to 3-3/4	1/2 to 3-3/4"
Inflated tubular seal	Parking & plaza decks Parking & plaza decks	Watson Bowman Acme	Jeene 25W to 100W Jeene 25FW to 125FW	Neoprene Neoprene		1 to 4 1 to 5	3/4 to 4" 5/8 to 3-1/4"
Preformed cellular foam	Parking & plaza decks	E-Poxy Industries	E 1.00 to E 5.00	Evazote 380		3/4 to 4	+30 to -60%
Preformed cellular foam	Parking & plaza decks	Watson Bowman Acme	Wabo Inverseal Mod IV	Neoprene		1 to 4	+25 to -55%
Public arena system	Stair treads & risers	Watson Bowman Acme	StadiaFlex SDF	Santoprene		1 to 4	1 to 4"
Self-centering metal cover	Seismic parking system	Watson Bowman Acme	Seismic pkg. APS & APX		Aluminum & SS	2 to 18	1-5/8 to 25-5/8"

Strip seal	Parking decks & HD traffic Parking decks	Watson Bowman Acme	Steel strip seal sys. SEC Steel strip seal sys. SE	Elastomeric Elastomeric		2-3/4 to 4 1-1/2 to 4-1/2	2-1/4 to 4" 3 to 8"
Strip seal	Parking decks	Balco Metalines	Model ETM	Elastomeric		1 to 3/4	2-1/4 to 3-1/4"
Strip seal	Parking decks	D. S. Brown Company	A2R, L2 & O	Neoprene		2	4"
Strip seal	Floors & plaza decks	Watson Bowman Acme	Wabo Series AS	Santoprene		1-1/2 to 4-1/2	1-1/4 to 8"
Strip seal	Bridges & structures Bridges & structures	E-Poxy Industries	S 300, LS 300 S 400, SC 400	Neoprene		1-5/8 to 2-1/4 2-1/8	3" 4"
Strip seal with anchor block	HD traffic	Watson Bowman Acme	Elastoflex EFJ	Santoprene		1-1/2 to 3-3/4	1-5/16 to 4"

Legend:
EVAEthylene vinyl acetate
HDHeavy duty
SSStainless steel

EXTERIOR WALL JOINTS

TYPE OF PRODUCT	RECOMMENDED USE OR LOCATION	MANUFACTURER	MODEL, PRODUCT, OR CATALOG NO.	SEAL OR GLAND	EXPOSED COVER	JOINT WIDTH (inches)	MOVEMENT
Preformed seal	Seismic exterior walls	Architectural Art Mfg.	R Series	Silicone		2 to 6	100% of joint width
Preformed seal	Walls & ceilings Walls & ceilings	Balco Metalines	Type FCVS Type FCWW	Silicone or Santoprene		2 to 18 3 to 18	+3,-2" to +18,-9" 1 to 10"
Preformed seal	Seismic walls & soffits	Watson Bowman Acme	WeatherSeam WSW	Santoprene		2 to 24	2-5/8 to 38"
Self-centering metal cover	Seismic wall systems	Watson Bowman Acme	Seismic Wall WFX		Aluminum	6 to 24	9-3/4 to 45-3/4"

Legend:
EVA: Ethylene vinyl acetate

EXTERIOR SOFFIT JOINTS							
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**SEAL OR GLAND**	**EXPOSED COVER**	**JOINT WIDTH (inches)**	**MOVEMENT**
Preformed seal Walls & ceilings	Walls & ceilings Walls & ceilings	Balco Metalines	Type FCVS Type FCWW	Silicone or Santoprene		2 to 18 3 to 18	+3,-2" to +18,-9" 1 to 10"
Preformed seal	Seismic walls & soffits	Watson Bowman Acme	WeatherSeam WSW	Santoprene		2 to 24	2-5/8 to 38"

INTERIOR FLOOR JOINTS							
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**SEAL OR GLAND**	**EXPOSED COVER**	**JOINT WIDTH (inches)**	**MOVEMENT**
High load system	Floors	Watson Bowman Acme	Wabo VIP	Santoprene		3/4 to 3	3/4 to 3"
Metal cover	Floors	Watson Bowman Acme	Economy Floor FJF		Aluminum or SS	1 to 6	1 to 6"
Metal cover	Floors	Architectural Art Mfg.	K Series		Aluminum	1 to 2	+75; -100% joint width
Metal cover w/ insert	Floors, seismic	Watson Bowman Acme	SeisMaxPlus NBS	Santoprene		1 to 24	3 to 35"
Metal cover w/ insert	Floors	Architectural Art Mfg.	K Series	Neoprene	Aluminum	1 to 2	+/-25% joint width
Metal hidden sightline	Floors & walls	Balco Metalines	Type CR			2 to 3	1 to 2"
Metal hidden sightline	Floors & walls	Watson Bowman Acme	Allure AFJ & AWJ			1 to 6	1 to 6"
Preformed seal	Floors, walls, & ceilings	Architectural Art Mfg.	C Series	Vinyl or polyurethane		1 to 2	+/-25% joint width
Preformed seal	Floors, walls, & ceilings	Architectural Art Mfg.	B Series	Dual vinyl	Aluminum	1 to 2	+/-50%
Preformed seal	Walls, soffits, & ceilings	Watson Bowman Acme	Accent ADA Model ACF Econowrap LPF, LPP, LPC	Santoprene Vinyl		1 to 4 1 to 3	2 to 4" 3/4 to 1-1/2"
Preformed seal	Floors, walls, & ceilings Floors, walls, & ceilings	Watson Bowman Acme	Model LP-, CW-, CE- Model SEC, LE, EFE & SE	Vinyl Santoprene		1 to 4 1-1/2 to 4-1/2	3/4 to 4" 1-1/4 to 8"

Preformed seal	Floors, walls, & ceilings Floors, walls, & ceilings	Balco Metalines	7500 Series Type A & G (Vinylines)	Santoprene Vinyl		1 & 2 1, 2, & 3	+/-30% nominal width 100% of nominal width
Preformed seal	Floors, walls, & ceilings	MM Systems Corp.	Series FS & VF	Elastomeric		1 to 3	20 to 50%
Preformed seal	Flush & surfaces Flush & recessed	Conspec Systems	Thinline GFT, GFPS & GFS Twinline GFR & GFP	TPE 2-gaskets		1 to 2 2 & 4	+/-25% +/-50%
Self-centering metal	Interior floors Interior floors	Watson Bowman Acme	Wabo Floor Cover FJG, FJC Wabo Floor Cover FJX		Aluminum SS & brass	2 to 24 2 to 18	2 to 35" 4 to 25"
Self-centering metal	Floors Floors Dual elastomeric floors	Balco Metalines	Type DS &DT Type RD & NB 9500 Series	 Santoprene	Aluminum Aluminum Aluminum	1 to 4 1 to 6 2, 3, & 4	100% joint width 100% joint width 2"
Solid urethane seal	Floors Floors, walls, & ceilings	Watson Bowman Acme	Wabo Flatline URF Wabo Flatline URE & URG	Polyurethane Polyurethane		2 to 4 1 to 2	3/4 to 2-3/4" 3/4 to 1-1/2"
Legend: HDHeavy duty Stainless steel							

INTERIOR WALL JOINTS							
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**SEAL OR GLAND**	**EXPOSED COVER**	**JOINT WIDTH (inches)**	**JOINT MOVEMENT**
Floating metal center	Walls & ceilings	Watson Bowman Acme	Contour CTR & CTC		Aluminum	2 to 24	2 to 24"
Floating metal center	Walls & ceilings	Architectural Art Mfg.	H Series		Aluminum	1 to 12	+/-50% joint width
Hidden sightline	Floors & walls	Watson Bowman Acme	Allure AFJ & AWJ			1 to 6	1 to 6"
Hidden sightline	Floors & walls	Balco Metalines	Type CR			2 to 3	1 to 2"
Metal cover w/ insert	Seismic walls & ceilings	Watson Bowman Acme	SeisMaxPlus EWH	Santoprene	Aluminum	1 to 24	3 to 35"
Preformed seal	Walls & ceilings	Architectural Art Mfg.	J Series	Vinyl		1 to 2	+100; -25% joint width
Preformed seal	Floors, walls, & ceilings	Architectural Art Mfg.	C Series	Vinyl or polyurethane		1 to 2	+/-25%
Preformed seal	Walls, soffits, & ceilings	Watson Bowman Acme	Accent ADA - ACW & ACC Econowrap CWC, CWS & CWF	Santoprene Vinyl		2 to 4 1 to 4	2 to 4" 3/4 to 4"
Preformed seal	Walls & ceilings Walls & ceilings	Balco Metalines	Type FCVS Type FCWW	Silicone or Santoprene		2 to 18 3 to 18	+3,-2" to +18,-9" 1 to 10"
Preformed seal	Seismic walls & soffits	Watson Bowman Acme	WeatherSeam WSW	Santoprene		2 to 24	2-5/8 to 38"

Self-centering metal	Seismic wall systems Exterior walls	Balco Metalines	Type CM Type 9W		Aluminum Aluminum	4 to 12 2 to 6	100% joint width 100% joint width
Self-centering metal	Seismic walls & ceilings	Watson Bowman Acme	Seismic Model WFX		Aluminum	6 to 24	9-3/4 to 45-3/4"
Snap-on metal cover	Wall systems	Nystrom, Inc.	Wall expansion joints		Aluminum	3 to 5-3/4	1-1/2 to 3"
Snap-on metal cover	Walls & ceilings	Watson Bowman Acme	Economy Wall EWH & EWF		Aluminum	1 to 3	up to 3"
Snap-on metal cover	Walls & ceilings Walls & ceilings	Balco Metalines	Type 1W1 & 1C1 Type C1W1 & C1C1		Aluminum Aluminum	1/2 to 2 1/2 to 2	100% of nominal width 50% of nominal width
Snap-on metal cover	Walls & ceilings	Balco Metalines	Type WD		Aluminum	1 to 4	+/-50%
Snap-on metal cover	Walls & ceilings	Architectural Art Mfg.	G Series		Aluminum	1 to 2	+/-50%
Solid urethane seal	Floors Floors, walls, & ceilings	Watson Bowman Acme	Wabo Flatline URF Wabo Flatline URE & URG	Polyurethane Polyurethane		2 to 4 1 to 2	3/4 to 2-3/4" 3/4 to 1-1/2"

INTERIOR CEILING AND SOFFIT JOINTS							
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**SEAL OR GLAND**	**EXPOSED COVER**	**JOINT WIDTH (inches)**	**MOVEMENT**
Floating metal center	Walls & ceilings	Architectural Art Mfg.	H Series		Aluminum	1 to 12	+/-50% joint width
Preformed seal	Floors, walls, & ceilings	Architectural Art Mfg.	B Series	Dual vinyl	Aluminum	1 to 2	+/-50%
Preformed seal	Floors, walls, & ceilings	Architectural Art Mfg.	C Series	Vinyl or polyurethane		1 to 2	+/-25%
Preformed seal	Soffits & ceilings Soffits & ceilings	Watson Bowman Acme	Accent ADA - ACW & ACC Econowrap CWC, CWS, CWF, CEB & CES	Santoprene Vinyl		2 to 4 1 to 4	2 to 4" 3/4 to 4"
Preformed seal	Walls & ceilings	Architectural Art Mfg.	J Series	Vinyl		1 to 2	+100; -25% joint width
Preformed seal	Floors, walls, & ceilings	Balco Metalines	7500 Series Type A & G (Vinylines)	Santoprene Vinyl		1 & 2 1, 2, & 3	+/-30% nominal width 100% of nominal width
Preformed seal	Walls & ceilings Walls & ceilings	Balco Metalines	Type FCVS Type FCWW	Silicone or Santoprene		2 to 18 3 to 18	+3,-2" to +18,-9" 1 to 10"
Solid urethane seal	Floors Floors, walls, & ceilings	Watson Bowman Acme	Wabo Flatline URF Wabo Flatline URE & URG	Polyurethane Polyurethane		2 to 4 1 to 2	3/4 to 2-3/4" 3/4 to 1-1/2"
Snap-on metal cover	Ceilings & soffits	Watson Bowman Acme	Economy Wall Model EWH		Aluminum	1 to 3	up to 3"
Snap-on metal cover	Walls & ceilings	Architectural Art Mfg.	G Series		Aluminum	1 to 2	+/-50%

Snap-on metal cover Walls & ceilings	Walls & ceilings Walls & ceilings	Balco Metalines	Type 1W1 & 1C1 Type C1W1 & C1C1		Aluminum Aluminum	1/2 to 2 1/2 to 2	100% of nominal width 50% of nominal width
Snap-on metal cover	Walls & ceilings	Balco Metalines	Type WD		Aluminum	1 to 4	+/-50%

FIRE-BARRIER INSERTS						
TYPE OF PRODUCT	**RECOMMENDED USE OR LOCATION**	**TESTED ACCORDING TO THE FOLLOWING TEST METHODS**	**MANUFACTURER**	**MODEL, PRODUCT, OR CATALOG NO.**	**JOINT WIDTH (inches)**	**MOVEMENT**
Fire-barrier insert	2-hour rating	UL 2079	Architectural Art Mfg.	P Series Fire Insert	1 to 18	+/-50%
Fire-barrier insert	2-hour walls & floors 2-hour walls & floors	UL 2079 and ASTM E 119	Balco Metalines	Metaflex FBR Systems Pan Systems	Up to 18 for wall and floor Up to 24 for wall & 36 for floor Up to 48 for wall	
Fire-barrier insert	2-, 3-, & 4-hour fire resistive	ASTM E 119	MM Systems Corp.	Pyro-Flex Shield	Compatible with cover	
Fire-barrier insert	Fire rated up to 4 hours Tested for seismic joints	UL 263 (ASTM E 119 & UBC Standard 7-1) tested for 500 cycles at 30 rpm.	Watson Bowman Acme	Thermoshield Flameguard	1 to 3 4 to 24	+/-50%